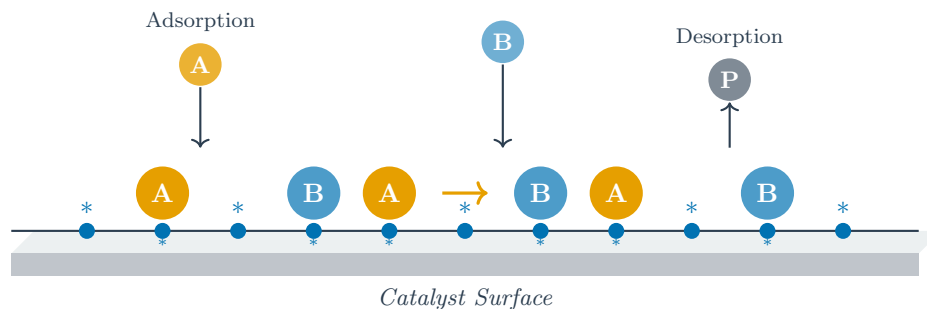


Surface Phenomena & Catalysis

Lecture Notes

by Viwat Vchirawongkwin



$$r = \frac{kK_A K_B P_A P_B}{(1 + K_A P_A + K_B P_B)^2}$$

Langmuir-Hinshelwood Rate

$$\theta_A = \frac{K_A P_A}{1 + K_A P_A}$$

Langmuir Isotherm

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Chapter 1

Bridge Module: Essential Notation and Tools

1.1 Introduction

This module establishes the foundational notation, units, and computational tools you will use throughout Surface Phenomena & Catalysis. Before we explore adsorption isotherms, surface reaction mechanisms, and reactor design, we must speak the same language—both mathematically and computationally.

Catalysis is not a niche specialism—it is the backbone of modern chemistry. More than 80 % of industrial chemical processes rely on at least one catalytic step, from petroleum refining and polymer production to pharmaceutical synthesis and environmental remediation. Catalytic routes replace wasteful stoichiometric reagents with recyclable catalysts, dramatically improving *atom economy*—a core principle of green chemistry that asks: how much of what you put in ends up in the product?

The importance of catalysis is reflected in over a century of Nobel Prizes: Fritz Haber (1918) for ammonia synthesis that feeds billions; Ziegler and Natta (1963) for polymerisation catalysis; Gerhard Ertl (2007) for elucidating surface chemistry at the molecular level; Heck, Negishi, and Suzuki (2010) for palladium-catalysed cross-coupling; and List and MacMillan (2021) for asymmetric organocatalysis. Each breakthrough rested on understanding how strongly molecules bind, how fast they react, and how reactor design translates molecular events into industrial output—the very quantities you will learn in this chapter.

Remark 1.1 (Five Steps of Heterogeneous Catalysis). Every heterogeneous catalytic reaction follows a five-step cycle that provides a roadmap for this entire course:

1. **Diffusion to the surface** — Reactants travel from the bulk fluid to the external catalyst surface (Chapter 4).
2. **Adsorption** — Reactant molecules bind to active surface sites (Chapter 2).
3. **Surface reaction** — Adsorbed species undergo chemical transformation on the surface (Chapter 2).
4. **Desorption** — Products detach from the surface (Chapter 2).
5. **Diffusion away** — Product molecules travel from the surface back to the bulk fluid (Chapter 4).

Any one of these steps can be rate-limiting. Identifying which step controls the overall rate is a central challenge in catalysis research. Steps 2–4 are *intrinsic* kinetic steps, while steps 1 and 5 involve *transport* phenomena.

Learning Objectives

After completing this module, you will be able to:

Note: The notation in Objective 1 may look overwhelming at first—you do not need to memorise all symbols now. This chapter introduces them as a reference; you will internalise each symbol naturally as it appears in context in later chapters.

- LO1. Interpret** the standard notation used throughout this course, including X (conversion), τ (space time), r (rate), θ (coverage), K (equilibrium constant), $k(T)$ (rate constant), ϕ (Thiele modulus), and η (effectiveness factor).
- LO2. Verify** dimensional consistency in kinetic equations using SI units.
- LO3. Use** Python (NumPy, pandas, matplotlib, SciPy) to load CSV data, create properly formatted plots, and perform linear regression.
- LO4. Evaluate** model quality using residual analysis to identify systematic deviations.

1.2 Essential Notation and Equations

Consistent notation is your first defence against errors. This section defines every symbol used in the course and collects the key equations you will encounter. Treat it as a reference card—skim now, return often.

1.2.1 Notation Reference

Table 1.1 lists the primary variables. All symbols correspond to macros in the document preamble, so their typographical style is always consistent.

Table 1.1: Primary notation for Surface Phenomena & Catalysis

Symbol	Name	Units	Description
X	Conversion	—	Fraction of limiting reactant consumed
τ	Space time	s	Average time fluid spends in the reactor
r	Reaction rate	mol/(kg _{cat} s)	Moles reacted per catalyst mass per time
θ	Surface coverage	—	Fraction of active sites occupied by adsorbate
K	Equilibrium const.	varies	Adsorption or reaction equilibrium const.
k	Rate constant	varies	Kinetic rate constant (temp. dependent)
ϕ	Thiele modulus	—	Reaction rate vs. diffusion rate (Ch. 4)
η	Effectiveness factor	—	Fraction of catalyst interior utilised (Ch. 4)
TOF	Turnover frequency	s ⁻¹	Moles converted per active site per second
TON	Turnover number	—	Total moles converted per active site
$P_{1/2}$	Half-coverage pressure	Pa or bar	Pressure at which $\theta = 0.5$; equals $1/K$

TOF and TON deserve a brief explanation. Turnover frequency measures catalyst *activity*: how many molecules one active site converts per second. Turnover number measures catalyst *productivity*: the total number of catalytic cycles one site completes before the catalyst is deactivated or the reaction is stopped. Both are essential for comparing different catalysts on a fair, per-site basis; we develop them rigorously in Chapter 3.

Remark 1.2 (k (lowercase) vs K (uppercase): A crucial distinction). The **rate constant** k (lowercase) is a *kinetic* quantity—it tells you *how fast* a reaction proceeds and always increases with temperature via the Arrhenius equation. The **equilibrium constant** K (uppercase) is a *thermodynamic* quantity—it tells you *how far* a reaction proceeds and may increase or decrease with temperature depending on the sign of ΔH . For an elementary step the two are

linked:

$$K = \frac{k_{\text{fwd}}}{k_{\text{rev}}} \quad (1.1)$$

Confusing k and K is among the most common—and consequential—errors in kinetic analysis.

1.2.2 Key Equations Reference

The equations below are presented without derivation. Each will be derived rigorously in its respective chapter; for now, focus on the mathematical structure and physical meaning.

Conversion. For species A:

$$X = \frac{n_{A,0} - n_A}{n_{A,0}} \xrightarrow{\text{const. density}} C_A = C_{A,0}(1 - X) \quad (1.2)$$

The mole-based definition (left) is general; the concentration form (right) requires constant fluid density.

Reaction rate. In heterogeneous catalysis the reaction occurs *on the catalyst surface*, not throughout the bulk fluid. Normalising by catalyst mass W (rather than reactor volume V) captures the amount of surface available for reaction:

$$r_{\text{homogeneous}} = -\frac{1}{V} \frac{dn_A}{dt} \quad [\text{mol}/(\text{m}^3 \text{ s})] \quad (1.3)$$

$$r_{\text{heterogeneous}} = -\frac{1}{W} \frac{dn_A}{dt} \quad [\text{mol}/(\text{kg}_{\text{cat}} \text{ s})] \quad (1.4)$$

The negative sign ensures $r > 0$ when reactant A is consumed ($dn_A/dt < 0$).

Site balance (foundation of all Langmuir-based models):

$$\theta_A + \theta_B + \cdots + \theta_v = 1 \quad (1.5)$$

Think of the surface as a fixed grid of “parking spaces.” Each space holds exactly one molecule. If 60 % are occupied by A and 10 % by B, then 30 % remain vacant ($\theta_v = 0.30$). This conservation constraint closes every Langmuir-based derivation.

Arrhenius equation.

$$k = A \exp\left(-\frac{E_a}{RT}\right) \xrightarrow{\ln} \ln k = \ln A - \frac{E_a}{R} \cdot \frac{1}{T} \quad (1.6)$$

A plot of $\ln k$ vs. $1/T$ gives slope = $-E_a/R$ and intercept = $\ln A$.

Eyring equation (transition-state theory):

$$k = \frac{k_B T}{h} \exp\left(\frac{\Delta S^\ddagger}{R}\right) \exp\left(-\frac{\Delta H^\ddagger}{RT}\right) \quad (1.7)$$

A plot of $\ln(k/T)$ vs. $1/T$ gives slope $= -\Delta H^\ddagger/R$ and intercept $= \ln(k_B/h) + \Delta S^\ddagger/R$. You may recognise this from your physical chemistry course on transition-state theory: the prefactor $k_B T/h$ sets a universal attempt frequency ($\approx 6 \times 10^{12} \text{ s}^{-1}$ at 300 K), and the exponential terms determine what fraction of those attempts succeed in crossing the activation barrier.

Remark 1.3 (Quick Check: Linking Symbols to Physical Meaning). Test your understanding of the notation introduced above:

1. If K increases at fixed P , does θ increase or decrease?
2. If E_a doubles, does the rate constant k increase or decrease at fixed T ?
3. What are the units of the exponent E_a/RT —and why must it be dimensionless?
4. In the site balance, if $\theta_A = 0.7$ and $\theta_B = 0.1$, what is θ_v ?

Answers: (1) θ increases—a larger equilibrium constant means stronger adsorption, so more sites are occupied. (2) k decreases—the exponential $\exp(-E_a/RT)$ becomes smaller when E_a increases. (3) E_a/RT is dimensionless because E_a has units of J mol^{-1} and R has units of $\text{J mol}^{-1} \text{ K}^{-1}$, so $E_a/(RT)$ has units of $\text{K}/\text{K} = 1$. Exponents must always be dimensionless. (4) $\theta_v = 1 - 0.7 - 0.1 = 0.2$ —20 % of sites are vacant.

1.2.3 Unit Conversion Quick Reference

Table 1.2: Essential unit conversions for catalysis calculations

Conversion	Value
1 mol/L	$= 1000 \text{ mol/m}^3$
1 bar	$= 10^5 \text{ Pa} \approx 1 \text{ atm}$
R (gas constant)	$= 8.314 \text{ J/(mol K)}$
$T(\text{K})$	$= T(^{\circ}\text{C}) + 273.15$
1 kJ	$= 1000 \text{ J}$

Example 1.1: Unit Verification: Arrhenius Equation

Problem: A second-order reaction has $A = 2.5 \times 10^8 \text{ m}^3/(\text{mol s})$ and $E_a = 65 \text{ kJ/mol}$. Verify that $k(T)$ at $T = 500 \text{ K}$ has the correct units for a second-order rate constant. Then convert A to $\text{L}/(\text{mol s})$.

Solution (Dimensional Analysis):

Step 1: Check the exponential argument. The argument of the exponential must be dimensionless:

$$\frac{E_a}{RT} = \frac{65 \times 10^3 \text{ J/mol}}{8.314 \text{ J}/(\text{mol K}) \times 500 \text{ K}} = 15.64 \quad \checkmark \text{ (dimensionless)}$$

Step 2: Evaluate k .

$$k = 2.5 \times 10^8 \frac{\text{m}^3}{\text{mol s}} \times \exp(-15.64) = 2.5 \times 10^8 \times 1.62 \times 10^{-7} = 40.5 \frac{\text{m}^3}{\text{mol s}}$$

Step 3: Convert to $\text{L}/(\text{mol s})$. Since $1 \text{ m}^3 = 10^3 \text{ L}$:

$$A = 2.5 \times 10^8 \frac{\text{m}^3}{\text{mol s}} \times \frac{10^3 \text{ L}}{1 \text{ m}^3} = 2.5 \times 10^{11} \frac{\text{L}}{\text{mol s}}$$

Answer:

$$k(500 \text{ K}) = 40.5 \text{ m}^3/(\text{mol s}); \quad A = 2.5 \times 10^{11} \text{ L}/(\text{mol s})$$

Physical interpretation: The exponential argument is dimensionless, confirming correct use of SI units throughout. The factor-of- 10^3 conversion between m^3 and L is the most common source of error—always track volume units explicitly.

Remark 1.4 (The factor-of-1000 trap). The conversion $1 \text{ mol/L} = 1000 \text{ mol/m}^3$ is a frequent source of error. For a second-order rate law $r = k C_A^2$, converting C_A to SI without converting k accordingly produces a rate that is wrong by a factor of 10^3 . **Rule:** pick one unit system and use it throughout the entire calculation. Similarly, **always use Kelvin** (not Celsius) in the Arrhenius, Eyring, and van't Hoff

equations—using Celsius at 100 °C with a typical E_a of 80 kJ/mol introduces an error of roughly **30 orders of magnitude**.

Remark 1.5 (Concept Check: Can you spot the error?). A student calculates the rate constant at 200 °C using $k = 10^8 \exp\left[-\frac{80\,000}{8.314 \times 200}\right]$. What went wrong?

Answer: The student used $T = 200$ (Celsius) instead of $T = 473.15$ K. With Celsius, $E_a/RT = 48.1$, but the correct value is $E_a/RT = 20.3$. The rate constant is wrong by a factor of $\exp(48.1 - 20.3) \approx 10^{12}$.

1.3 Python Quick Reference

The notation and equations above describe catalytic systems mathematically; now we introduce the computational tools for applying them to real data.

Every chapter is accompanied by a Jupyter notebook (NB1–NB6, plus NB-S). You will use four libraries throughout the course:

- **NumPy** – numerical arrays and mathematical functions
- **pandas** – reading CSV data into DataFrames
- **Matplotlib** – publication-quality plotting
- **SciPy** – linear regression, optimisation, ODE integration

The listing below demonstrates the complete workflow you will repeat in every notebook: load data, plot, fit a linear model, and check residuals. Study the pattern; the only things that change from chapter to chapter are the x and y variables.

```
1 df = pd.read_csv('kinetic_data.csv')
2 x, y = 1.0 / df['Temperature_K'].values,
   np.log(df['k_per_kg_cat'].values)
3 result = stats.linregress(x, y)
4 Ea = -result.slope * 8.314 # J/mol
5 residuals = y -(result.intercept + result.slope * x)
```

Listing 1.1: Core workflow: CSV → Linear fit → Residuals (snippet)

Full implementation: see NB1, Section 1.

Remark 1.6 (Expected terminal output). Running the code above with the sample dataset produces output similar to:

```
Ea = 75.3 +/- 2.1 kJ/mol
R^2 = 0.99847
```

The exact values depend on the dataset. Always check that $R^2 > 0.99$ for a good linear fit; lower values suggest non-Arrhenius behaviour or systematic error in the data.

1.3.1 Residual Analysis: Why R^2 Is Not Enough

A high R^2 does not guarantee that your model is appropriate. The *residual plot* is the definitive diagnostic. A residual is the difference between an observed value and its prediction:

$$\text{residual}_i = y_{\text{obs},i} - y_{\text{pred},i}$$

For a valid model, residuals should show **random scatter** around zero with no systematic pattern. Common warning signs include:

- **Systematic curvature** (U-shape) — the model form is wrong; e.g., a linear fit applied to Langmuir–Hinshelwood data.
- **Increasing spread** (fan shape) — variance depends on x ; consider weighted regression or a different transformation.
- **Trend with x** — a missing variable or interaction effect.

In Chapter 2, you will use residual analysis to distinguish between competing kinetic models. The model that produces random residuals is the appropriate mechanistic description.

Remark 1.7 (Concept Check: Residual Interpretation). You fit a Langmuir isotherm (θ vs. P) using a linearised form and obtain $R^2 = 0.997$. The residual plot shows a clear U-shape. Is the model adequate?

Answer: No. Despite the high R^2 , the U-shaped residuals indicate systematic deviation—the linear model does not capture the true functional form. You should try a different model (e.g., competitive Langmuir or BET).

Remark 1.8 (Colorblind-safe palette (Wong, 2011)). All plots in this course use colours that are distinguishable by readers with the most common forms of colour-vision deficiency. Always combine colour with distinct **markers** (o, s, ^) and **line styles** (--, :, -.).

Name	Hex	RGB	Typical use
Blue	#0072B2	(0, 114, 178)	Primary data
Orange	#E69F00	(230, 159, 0)	Fitted curves
Green	#009E73	(0, 158, 115)	Tertiary / remarks
Vermillion	#D55E00	(213, 94, 0)	Warnings / residuals
Sky blue	#56B4E9	(86, 180, 233)	Backgrounds
Purple	#CC79A7	(204, 121, 167)	Additional series

See Notebook `NB1.ipynb` (20–30 min) for hands-on practice with this workflow. The same pattern—*data* → *visualisation* → *linearisation* → *fitting* → *residual check*—recurs in every subsequent notebook.

1.4 Summary

- **Notation consistency** is essential—this course uses standardised symbols (X , τ , r , θ , K , k) defined in Table 1.1.
- **Distinguish k from K** : lowercase k is a rate constant (kinetic, *how fast*), uppercase K is an equilibrium constant (thermodynamic, *how far*).
- **Dimensional analysis** verifies equation correctness; rate constant units depend on reaction order. **Always use Kelvin** for Arrhenius, Eyring, and van’t Hoff equations.
- **Python workflow**: *data* → *visualisation* → *linearisation* → *regression* → *residual analysis*.
- **Residual plots** reveal whether your model form is appropriate—random scatter is good; systematic patterns indicate model inadequacy.

Table 1.3 provides a comprehensive reference for all symbols used throughout the course.

Table 1.3: Complete notation reference for Surface Phenomena & Catalysis

Symbol	Name	Units	First Major Use
<i>Primary Variables</i>			
X	Conversion	—	Ch. 3
τ	Space time	s	Ch. 3
r	Reaction rate	mol/(kg _{cat} s)	Ch. 2

continued on next page

Table 1.3 – *continued from previous page*

Symbol	Name	Units	First Major Use
θ	Surface coverage	—	Ch. 2
K	Equilibrium constant	varies	Ch. 2
k	Rate constant	varies	Ch. 2
ϕ	Thiele modulus	—	Ch. 4
η	Effectiveness factor	—	Ch. 4
TOF	Turnover frequency	s^{-1} or h^{-1}	Ch. 3
TON	Turnover number	— (mol/mol)	Ch. 3
<i>Thermodynamic & Kinetic Parameters</i>			
E_a	Activation energy	J/mol	Ch. 2
A	Pre-exponential factor	same as k	Ch. 2
$\Delta H^\ddagger$	Activation enthalpy	J/mol	Ch. 2
$\Delta S^\ddagger$	Activation entropy	J/(mol K)	Ch. 2
<i>Concentrations & Pressures</i>			
C_A	Concentration of A	mol/m ³	Throughout
$C_{A,0}$	Inlet concentration	mol/m ³	Ch. 3
P_A	Partial pressure of A	Pa or bar	Ch. 2
<i>Reactor Parameters</i>			
V	Reactor volume	m ³	Ch. 3
$\dot{V}_0$	Volumetric flow rate	m ³ /s	Ch. 3
W	Catalyst mass	kg	Ch. 4
<i>Transport Parameters</i>			
D_{eff}	Effective diffusivity	m ² /s	Ch. 4
k_c	Mass transfer coeff.	m/s	Ch. 4
<i>Selectivity Parameters</i>			
S	Selectivity	—	Ch. 5
Y	Yield	—	Ch. 5

Looking Ahead

With notation and computational tools in hand, you are ready to begin exploring catalytic surfaces. Chapter 2 introduces the Langmuir isotherm—the foundational model connecting gas-phase pressure to surface coverage θ —and establishes the site balance that underlies every kinetic derivation in this course. From there, the Python workflow you practised in NB1 (CSV $\rightarrow$ plot $\rightarrow$ fit $\rightarrow$ residuals) will recur in every chapter, applied to progressively richer kinetic models: Langmuir–Hinshelwood mechanisms, Arrhenius parameter extraction, reactor design equations, transport limitations, and selectivity optimisation.

As you move through the course, you will discover that the mathematical tools from this bridge module—linear fitting, exponential functions, and logarithmic transformations—appear in almost every analysis. In Chapter 2 you will use Python to visualise how molecules compete for surface sites and learn why one species can poison a catalyst by blocking active sites. Chapter 2 reveals how temperature dependence encodes activation barriers, linking the Arrhenius and Eyring equations you previewed here to transition state theory and the molecular-level energy landscape. The interplay of kinetics and thermodynamics—how fast *versus* how far a reaction proceeds—determines catalyst performance in every real system.

Think of this course as your physical chemistry knowledge—Gibbs energy, the Boltzmann distribution, reaction kinetics—applied to real catalytic systems: industrial reactors, porous pellets, zeolite cages, and nanostructured supports. The notation and Python skills you have built here are not prerequisites to be forgotten; they are the working tools you will reach for in every chapter ahead.

Chapter 2

Computational Perspectives on Surface Catalysis

Learning Objectives

By the end of this chapter, you will be able to:

1. Use Python to visualize competitive Langmuir adsorption and quantify catalyst poisoning sensitivity
2. Implement the dissociative Langmuir isotherm and compare it with the molecular form computationally
3. Construct Python rate maps for the Mars–van Krevelen mechanism and identify reduction- vs. reoxidation-controlled regimes
4. Apply kinetic isotope effect (KIE) analysis and interpret operando spectroscopy results for mechanism elucidation
5. Perform side-by-side Arrhenius and Eyring regression to extract activation parameters (E_a , $\Delta H^\ddagger$, $\Delta S^\ddagger$) and interpret their physical meaning
6. Generate curved Arrhenius plots to distinguish apparent from intrinsic activation energies across coverage regimes

2.1 Introduction

In Chapter 1 you established the notation, units, and Python tools used throughout this course. This chapter puts those tools to work on three interconnected problems in surface catalysis:

1. **Adsorption models** (Sections 2.3 to 2.5): How do reactants, inhibitors, and lattice oxygen compete for surface sites? We extend the

single-component Langmuir isotherm to competitive adsorption, dissociative adsorption, and the Mars–van Krevelen redox mechanism.

2. **Experimental diagnostics** (Section 2.6): How do we determine which mechanism operates? Kinetic isotope effects, rate-vs-pressure fingerprints, and operando spectroscopy provide complementary evidence.
3. **Temperature analysis** (Section 2.7, Section 2.8): How do we extract activation parameters—and why does the “observed” activation energy depend on coverage? The Eyring equation separates enthalpic and entropic barriers, and the apparent-vs-intrinsic E_a framework reveals how adsorption thermodynamics couple to kinetics.

The common thread is **computational exploration**: every topic is accompanied by Python code that generates the diagnostic plots, parameter sweeps, and regression analyses you will use in your own kinetic studies. The foundational equations—Langmuir isotherm, LH and ER rate laws, Arrhenius equation—were covered in the co-instructor’s lectures and are summarised in the review section that follows. This chapter extends them to situations where computational tools are essential for extracting physical insight from experimental data.

2.2 Review of Key Equations

This section provides a concise reference for the core equations covered in the co-instructor’s lectures on adsorption, surface kinetics, and temperature dependence. Derivations and detailed discussions are given in those lectures; here we collect the final results needed for the computational explorations in this chapter.

Remark 2.1 (Adsorption Fundamentals).

For single-component molecular adsorption on a surface with identical, non-interacting sites, the site balance requires

$$\theta_A + \theta_* = 1 \quad (2.1)$$

where θ_A is the fractional coverage of species A and θ_* is the fraction of empty sites. Equating adsorption and desorption rates at equilibrium yields the **Langmuir isotherm**:

$$\theta_A = \frac{K_A P_A}{1 + K_A P_A} \quad (2.2)$$

Limiting cases. At low pressure ($K_A P_A \ll 1$), coverage grows linearly: $\theta_A \approx K_A P_A$ (Henry’s law regime). At high pressure ($K_A P_A \gg 1$), coverage approaches unity: $\theta_A \approx 1$ (Langmuir limit).

1), the surface saturates: $\theta_A \rightarrow 1$. Half-coverage occurs at $P_{1/2} = 1/K_A$.

The adsorption equilibrium constant varies with temperature via the **van't Hoff relation**:

$$K(T) = K_0 \exp\left(-\frac{\Delta H_{\text{ads}}}{RT}\right) \quad (2.3)$$

Since chemisorption is exothermic ($\Delta H_{\text{ads}} < 0$), raising the temperature reduces K and lowers coverage at fixed pressure.

For multilayer physisorption, the **BET isotherm** extends the Langmuir model by allowing adsorbate layers to stack; it is the standard method for measuring total surface area (see Chapter 3).

Derivation and detailed discussion: see co-instructor's lectures.

Remark 2.2 (Sticking Coefficient: The Kinetic View of Adsorption).

The **sticking coefficient** S is the probability that a gas-phase molecule striking the surface actually adsorbs. In the Langmuir model, $S = 1$ on empty sites and $S = 0$ on occupied sites, giving

$$S = S_0 (1 - \theta) \quad (2.4)$$

where S_0 is the **initial (bare-surface) sticking coefficient**. For most chemisorption systems, S_0 ranges from 10^{-2} to 1 (e.g., $S_0 \approx 0.5$ for CO on Pt).

The flux of molecules hitting the surface per unit area per unit time is given by the **Hertz–Knudsen impingement rate**:

$$\Phi = \frac{P}{\sqrt{2\pi m k_B T}} \quad (2.5)$$

where m is the molecular mass. The adsorption rate is then $r_{\text{ads}} = S_0 (1 - \theta) \Phi$, which connects the macroscopic adsorption rate constant k_{ads} to the microscopic quantities S_0 and Φ . Setting $r_{\text{ads}} = r_{\text{des}}$ at equilibrium recovers the Langmuir isotherm—the sticking coefficient viewpoint is simply the kinetic derivation recast in molecular-level language.

Remark 2.3 (Surface Kinetics: LH and ER Mechanisms).

When the surface reaction between co-adsorbed A^* and B^* is the rate-determining step (Langmuir–Hinshelwood mechanism), the overall rate is

$$r = \frac{kK_A P_A K_B P_B}{(1 + K_A P_A + K_B P_B)^2} \quad (2.6)$$

The squared denominator arises because the rate is proportional to $\theta_A \cdot \theta_B$ (the probability that an A-occupied site neighbors a B-occupied site). Since both coverages share the same denominator from the site balance, their product produces the squared denominator.

When only one reactant adsorbs and the other reacts directly from the gas phase (Eley–Rideal mechanism), the rate becomes

$$r = \frac{kK_A P_A P_B}{1 + K_A P_A} \quad (2.7)$$

Key fingerprints for model discrimination:

Feature	LH	ER
Rate at high P_A	Decreases ($\sim 1/P_A$)	Saturates to kP_B
Maximum in r vs P_A	Yes	No
Denominator exponent	2	1

Derivation and detailed discussion: see co-instructor's lectures.

Remark 2.4 (Temperature Dependence: Arrhenius Equation).

The rate constant for an elementary surface reaction follows the **Arrhenius equation**:

$$k(T) = A \exp\left(-\frac{E_a}{RT}\right) \quad (2.8)$$

where A is the pre-exponential factor (frequency factor) and E_a is the activation energy.

Taking the natural logarithm gives the linearised form used for parameter extraction:

$$\ln k = \ln A - \frac{E_a}{R} \cdot \frac{1}{T} \quad (2.9)$$

A plot of $\ln k$ versus $1/T$ (the **Arrhenius plot**) yields a straight line with slope $= -E_a/R$ and intercept $= \ln A$.

Derivation and detailed discussion: see co-instructor's lectures.

2.3 Competitive Adsorption and Poisoning

Real catalytic systems rarely involve a single adsorbate. Reactants compete with each other—and with impurities—for the same surface sites. This competition has profound consequences for catalytic activity.

2.3.1 Two-Component Langmuir Model

Consider two species, A (the desired reactant) and I (an inhibitor or poison), both adsorbing on the same sites:



The site balance now includes both species:

$$\theta_* + \theta_A + \theta_I = 1 \quad (2.12)$$

Following the same kinetic equilibrium approach as before (setting adsorption rate equal to desorption rate for each species), we derive the competitive Langmuir expressions.

Definition 2.1 (Derivation: Competitive Langmuir Isotherm). **Physical Assumptions:**

- (i) Both species A and I adsorb on the same type of sites
- (ii) Each site accommodates at most one molecule (either A or I , not both)
- (iii) No lateral interactions between adsorbed molecules
- (iv) Both adsorption–desorption processes are in dynamic equilibrium

Step 1: Write Equilibrium Conditions for Each Species

For species A :

$$\begin{aligned} r_{\text{ads},A} &= r_{\text{des},A} \\ k_{\text{ads},A} P_A \theta_* &= k_{\text{des},A} \theta_A \end{aligned} \quad (2.13)$$

For species I :

$$\begin{aligned} r_{\text{ads},I} &= r_{\text{des},I} \\ k_{\text{ads},I} P_I \theta_* &= k_{\text{des},I} \theta_I \end{aligned} \quad (2.14)$$

Step 2: Express Coverages in Terms of Empty Sites

Define equilibrium constants as before:

$$K_A = \frac{k_{\text{ads},A}}{k_{\text{des},A}}, \quad K_I = \frac{k_{\text{ads},I}}{k_{\text{des},I}} \quad (2.15)$$

Rearranging Eqs. (2.13) and (2.14):

$$\theta_A = K_A P_A \theta_* \quad (2.16)$$

$$\theta_I = K_I P_I \theta_* \quad (2.17)$$

These express each species' coverage as proportional to the empty-site fraction θ_* .

Step 3: Apply the Site Balance

From Eq. (2.12), all coverages must sum to unity:

$$\theta_* + \theta_A + \theta_I = 1 \quad (2.18)$$

Substitute the expressions from Step 2:

$$\begin{aligned} \theta_* + K_A P_A \theta_* + K_I P_I \theta_* &= 1 \\ \theta_* (1 + K_A P_A + K_I P_I) &= 1 \end{aligned} \quad (2.19)$$

Step 4: Solve for Empty-Site Fraction

Isolating θ_* :

$$\theta_* = \frac{1}{1 + K_A P_A + K_I P_I} \quad (2.20)$$

Step 5: Obtain Individual Coverages

Substituting θ_* back into Eqs. (2.16) and (2.17):

$$\theta_A = \frac{K_A P_A}{1 + K_A P_A + K_I P_I} \quad (2.21)$$

$$\theta_I = \frac{K_I P_I}{1 + K_A P_A + K_I P_I} \quad (2.22)$$

Limiting Case Verification:

- When $P_I = 0$: Reduces to single-component Langmuir for A ✓
- When $P_A = 0$: Reduces to single-component Langmuir for I ✓
- As $K_I P_I \rightarrow \infty$: $\theta_A \rightarrow 0$ (complete poisoning) ✓
- Site balance: $\theta_* + \theta_A + \theta_I = 1$ is satisfied identically ✓

Key Structural Insight:

All three coverage expressions share the *same denominator*:

$$\text{Denominator} = 1 + \sum_i K_i P_i \quad (2.23)$$

This structure generalizes to n competing species. For the numerator: each coverage θ_j has $K_j P_j$; the empty-site fraction has 1. This pattern reappears in the LH and ER rate expressions summarised in Section 2.2.

The derivation yields the **competitive Langmuir isotherms**:

$$\theta_A = \frac{K_A P_A}{1 + K_A P_A + K_I P_I} \quad (2.24)$$

$$\theta_I = \frac{K_I P_I}{1 + K_A P_A + K_I P_I} \quad (2.25)$$

$$\theta_* = \frac{1}{1 + K_A P_A + K_I P_I} \quad (2.26)$$

Notice that each coverage expression has the *same denominator*—this is a general feature of competitive Langmuir adsorption and reappears in the rate expressions reviewed in Section 2.2.

2.3.2 The Physics of Poisoning

The competitive adsorption equations reveal why trace impurities can devastate catalytic activity. Consider what happens when $K_I \gg K_A$ —that is, the poison binds much more strongly than the reactant.

Even if the poison pressure P_I is tiny (say, parts per million), the product $K_I P_I$ can be comparable to or larger than $K_A P_A$. This makes the denominator large, suppressing θ_A and θ_* .

Remark 2.5 (Practical implications for the laboratory). This competitive adsorption phenomenon explains several observations familiar to synthetic chemists:

- **Catalyst poisoning by sulfur:** Thiols and sulfides bind extremely strongly to Pd and Pt surfaces, deactivating hydrogenation catalysts even at trace levels. This is why sulfur-containing substrates often require special catalyst formulations.
- **CO inhibition:** CO binds very strongly to precious metal surfaces (K_{CO} is large) via the **Blyholder model**: σ -donation from the carbon lone pair (5σ orbital) into empty metal d-orbitals, combined with π -backdonation from filled metal d-orbitals into the CO $2\pi^*$ antibonding orbital. This dual bonding mode creates an exceptionally strong M–CO bond. Despite being an inhibitor, this strong binding is exploited in **CO chemisorption** to count active metal sites—a standard characterization technique covered in Chapter 3.
- **Solvent effects:** Strongly adsorbing solvents can compete with substrates for active sites, reducing reaction rates. This is why solvent choice matters in heterogeneous catalysis.
- **Industrial mitigation—guard beds:** In industrial practice,

guard beds of ZnO are placed upstream of the main reactor to chemisorb H_2S before the feed contacts the precious metal catalyst. The reaction $\text{ZnO} + \text{H}_2\text{S} \rightarrow \text{ZnS} + \text{H}_2\text{O}$ reduces sulfur to sub-ppm levels, protecting downstream Pt or Pd catalysts from poisoning. The guard bed is periodically replaced or regenerated, a far cheaper solution than replacing the main catalyst.

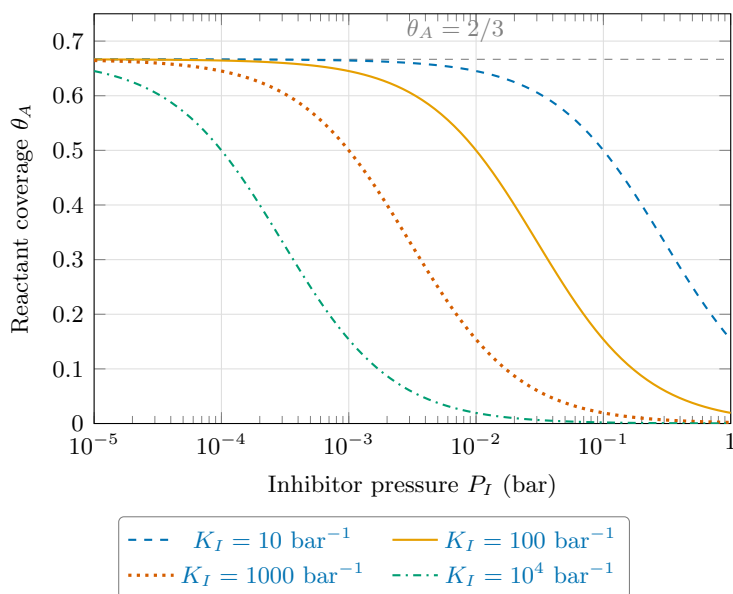


Figure 2.1: Effect of competitive adsorption (poisoning) on reactant coverage. At fixed reactant conditions ($K_A = 2 \text{ bar}^{-1}$, $P_A = 1 \text{ bar}$), increasing inhibitor pressure P_I reduces θ_A . Stronger-binding poisons (larger K_I) cause coverage to drop at lower inhibitor concentrations. Even trace amounts (ppm levels) of a strong poison can significantly suppress reactant coverage.

Interpreting the Poisoning Curves

Fig. 2.1 provides a quantitative picture of how competitive adsorption leads to poisoning. Let us analyze this figure systematically.

The Unpoisoned Baseline. The horizontal dashed line in Fig. 2.1 represents the reactant coverage in the *absence* of any inhibitor ($P_I = 0$). With the given parameters ($K_A = 2 \text{ bar}^{-1}$, $P_A = 1 \text{ bar}$), the competitive Langmuir equation (Eq. (2.24)) reduces to the single-component form:

$$\theta_A^\circ = \frac{K_A P_A}{1 + K_A P_A} = \frac{(2)(1)}{1 + (2)(1)} = \frac{2}{3} \approx 0.667 \quad (2.27)$$

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This value $\theta_A^\circ = 2/3$ serves as our reference point—any reduction below this baseline is caused by the competing inhibitor.

Effect of Poison Binding Strength. The four curves in Fig. 2.1 correspond to poisons with increasing binding strength: $K_I = 10, 10^2, 10^3$, and 10^4 bar^{-1} . Three key observations emerge:

1. **Steeper drop at lower P_I :** As K_I increases, the coverage curve shifts to the *left* on the logarithmic P_I axis. This means stronger poisons cause significant coverage loss at much lower concentrations.
2. **Self-similarity:** On a log scale, the curves have the same sigmoidal shape—they are simply translated horizontally. Each ten-fold increase in K_I shifts the curve one decade to the left.
3. **Ultimate suppression:** All curves approach $\theta_A \rightarrow 0$ as P_I becomes large, regardless of K_I . Given enough poison, coverage is always completely suppressed.

The Critical Poison Pressure. A useful concept is the **critical poison pressure** P_I^{crit} —the inhibitor pressure at which the reactant coverage drops to half its unpoisoned value. Setting $\theta_A = \theta_A^\circ/2 = 1/3$ and solving:

$$\frac{1}{3} = \frac{K_A P_A}{1 + K_A P_A + K_I P_I^{\text{crit}}} = \frac{2}{1 + 2 + K_I P_I^{\text{crit}}} \quad (2.28)$$

Solving for the denominator: $1 + 2 + K_I P_I^{\text{crit}} = 6$, hence:

$$\boxed{P_I^{\text{crit}} = \frac{3}{K_I}} \quad (2.29)$$

This result has a clear physical interpretation: *the critical poison pressure is inversely proportional to the poison's binding constant*. The stronger the poison binds, the less of it is needed to halve the coverage.

Remark 2.6 (General form of critical poison pressure). For arbitrary reactant conditions, the critical poison pressure that halves the coverage is:

$$P_I^{\text{crit}} = \frac{1 + K_A P_A}{K_I}$$

With the figure parameters ($K_A P_A = 2$), this simplifies to $P_I^{\text{crit}} = 3/K_I$.

Remark 2.7 (Reading the critical pressure from the figure). You can verify Eq. (2.29) directly from Fig. 2.1 by finding where each curve

crosses $\theta_A = 1/3$:

K_I (bar ⁻¹)	$P_I^{\text{crit}} = 3/K_I$ (bar)	Equivalent concentration
10^1	3×10^{-1}	30% of 1 bar
10^2	3×10^{-2}	3% (30,000 ppm)
10^3	3×10^{-3}	0.3% (3,000 ppm)
10^4	3×10^{-4}	300 ppm

For the strongest poison ($K_I = 10^4$ bar⁻¹), only 300 ppm of inhibitor reduces reactant coverage by half!

Quantitative Examples. To build physical intuition, let us calculate θ_A for specific cases:

Case 1: Moderate poison at moderate concentration. With $K_I = 100$ bar⁻¹ and $P_I = 0.01$ bar (1% of total pressure):

$$\theta_A = \frac{2}{1 + 2 + (100)(0.01)} = \frac{2}{1 + 2 + 1} = \frac{2}{4} = 0.50$$

The reactant coverage drops from 0.667 to 0.50—a 25% reduction caused by just 1% poison in the feed.

Case 2: Strong poison at trace concentration. With $K_I = 10^4$ bar⁻¹ and $P_I = 10^{-4}$ bar (100 ppm):

$$\theta_A = \frac{2}{1 + 2 + (10^4)(10^{-4})} = \frac{2}{1 + 2 + 1} = \frac{2}{4} = 0.50$$

The same 25% coverage reduction occurs, but now caused by only 100 ppm of poison! This dramatic effect arises because the product $K_I P_I = 1$ is identical in both cases.

Case 3: Devastating poisoning by ppm-level impurities. For sulfur poisoning of precious metals, K_S can exceed 10^5 bar⁻¹. At just $P_S = 10^{-5}$ bar (10 ppm):

$$K_S P_S = (10^5)(10^{-5}) = 1$$

Even this trace amount contributes as much to the denominator as the reactant term ($K_A P_A = 2$), reducing coverage substantially. This explains why industrial feed streams must be rigorously desulfurized before contacting precious metal catalysts.

Example 2.1: CO Poisoning in Propane Dehydrogenation

Problem: A Pt catalyst is used for propane dehydrogenation. The reactant (propane) has $K_{C_3} = 3 \text{ bar}^{-1}$ and is fed at $P_{C_3} = 0.5 \text{ bar}$. The feed contains trace CO with $K_{CO} = 1000 \text{ bar}^{-1}$.

- What is the propane coverage in the absence of CO?
- If the CO content is 500 ppm ($P_{CO} = 5 \times 10^{-4} \text{ bar}$), what is the new propane coverage?
- What fractional loss in coverage (and hence approximate fractional loss in reaction rate) results from this CO impurity?

Solution (Competitive Langmuir):

(a) Without CO ($P_{CO} = 0$):

$$\theta_{C_3}^{\circ} = \frac{K_{C_3} P_{C_3}}{1 + K_{C_3} P_{C_3}} = \frac{(3)(0.5)}{1 + (3)(0.5)} = \frac{1.5}{2.5} = 0.60$$

(b) With 500 ppm CO:

$$\theta_{C_3} = \frac{1.5}{1 + 1.5 + (1000)(5 \times 10^{-4})} = \frac{1.5}{1 + 1.5 + 0.5} = \frac{1.5}{3.0} = 0.50$$

(c) Fractional coverage loss:

$$\text{Fractional loss} = \frac{\theta_{C_3}^{\circ} - \theta_{C_3}}{\theta_{C_3}^{\circ}} = \frac{0.60 - 0.50}{0.60} = 0.167 \approx 17\%$$

Answer:

$$\theta_{C_3}^{\circ} = 0.60; \quad \theta_{C_3} = 0.50; \quad \text{Fractional loss} = 17\%$$

Physical interpretation: A mere 500 ppm of CO causes a 17% reduction in propane coverage. For a reaction that is first-order in surface coverage, this translates directly to a 17% reduction in reaction rate. This example illustrates why CO removal is essential in many catalytic processes, even when CO is present at only trace levels.

General Design Principle. The analysis above reveals a powerful design rule: *to assess poison sensitivity, compare the products $K_I P_I$ and $K_A P_A$.* When these are comparable, significant poisoning occurs. The dimensionless ratio

$$\alpha = \frac{K_I P_I}{K_A P_A} \quad (2.30)$$

quantifies the “poisoning severity.” When $\alpha \ll 1$, poisoning is negligible; when $\alpha \sim 1$, moderate poisoning occurs; when $\alpha \gg 1$, the poison dominates the surface and the catalyst is severely deactivated.

This framework connects directly to the Langmuir–Hinshelwood kinetics reviewed in Section 2.2.

Remark 2.8 (Concept Check: Poisoning Severity). Without calculating, predict: if you halve the poison’s partial pressure P_I but double its binding constant K_I , does the poisoning get better, worse, or stay the same?

Hint: Use the dimensionless ratio $\alpha = K_I P_I / (K_A P_A)$ to reason through this.

Answer: Doubling K_I while halving P_I leaves $\alpha = K_I P_I / (K_A P_A)$ unchanged, so the poisoning severity is identical. The product $K_I P_I$ is what matters, not the individual values—a tightly binding poison at low concentration has the same effect as a weakly binding one at high concentration.

Remark 2.9 (Beyond Inhibition: Promotional Adsorption).

The competitive Langmuir model *always* predicts that a co-adsorbate reduces θ_A —every additional species in the denominator of Eq. (2.24) makes it larger. Yet in industrial practice, some additives *increase* the rate. The classic example is ammonia synthesis on Fe: adding small amounts of alkali metals (K, Cs) increases the rate of N_2 dissociation by a factor of 2–3, even though the promoter occupies surface sites. The mechanism is **electronic promotion**: the electropositive alkali donates electron density to neighboring Fe atoms, lowering the activation barrier for $N \equiv N$ bond breaking. This effect cannot be captured by Langmuir’s assumption of non-interacting, identical sites. The distinction between “poison” and “promoter” depends on whether the co-adsorbate modifies the *electronic structure* of neighboring active sites—a phenomenon beyond simple site blocking.

Python Implementation: Exploring Poisoning Effects

The following Python code reproduces Fig. 2.1 and allows you to explore how different parameters affect catalyst poisoning. All parameters are adjustable for experimentation.

```
1 def competitive_langmuir(P_A, K_A, P_I, K_I):
2     return K_A * P_A / (1 + K_A*P_A + K_I*P_I)
3
4 P_I = np.logspace(-5, 0, 200)
5 for K_I in [10, 100, 1000, 10000]:
6     theta = competitive_langmuir(P_A, K_A, P_I, K_I)
```

Listing 2.1: Competitive Langmuir adsorption (snippet)

Full implementation: see NB2.

2.4 Dissociative Adsorption

Not all molecules adsorb intact. Many diatomic and polyatomic molecules *dissociate* upon adsorption, forming two or more adsorbed fragments that occupy separate surface sites. This **dissociative adsorption** is fundamental to many catalytic processes and requires a modified treatment of the Langmuir isotherm.

2.4.1 The Dissociative Adsorption Process

Consider the adsorption of a diatomic molecule A_2 (such as H_2 , O_2 , or N_2) that dissociates into two adsorbed atoms:



This process requires *two adjacent empty sites*—the molecule must find a pair of neighboring sites to land on and split apart. This is fundamentally different from molecular (non-dissociative) adsorption, which requires only a single site.

Remark 2.10 (Chemical examples of dissociative adsorption).

- **Hydrogen on metals:** H_2 dissociates on Pd, Pt, Ni, and most transition metals. The resulting adsorbed H atoms are the active species in catalytic hydrogenation.
- **Oxygen on metals:** O_2 dissociates on Ag (ethylene epoxidation), Pt (combustion), and many oxide surfaces.
- **Nitrogen on iron:** N_2 dissociation on Fe is the rate-determining step in ammonia synthesis (Haber–Bosch). The strong $N \equiv N$

triple bond makes this step particularly challenging.

- **Alkane C–H activation:** On certain metal surfaces, alkanes undergo dissociative chemisorption of a C–H bond (sometimes described by analogy with oxidative addition in homogeneous catalysis), with both fragments remaining adsorbed.

Remark 2.11 (Why Only Certain Metals Dissociate H₂: The d-Band Model).

The ability of a transition metal to dissociate H₂ correlates with the position of its **d-band center** relative to the Fermi level—a concept developed by Hammer and Nørskov. When H₂ approaches a metal surface, its occupied σ bonding orbital interacts with the metal d-states, while the empty σ^* antibonding orbital can accept electron density from the metal. If the d-band center lies close to the Fermi level (as in Pd, Pt, Ni), the metal d-electrons are at sufficiently high energy to populate the σ^* orbital of H₂, weakening the H–H bond and facilitating dissociation.

In contrast, metals such as Au and Cu have nearly filled d-bands pushed well below the Fermi level by strong d–d interactions. Their d-electrons lack the energy to donate effectively into $\sigma^*(\text{H}_2)$, so dissociation is either activated or does not occur at all. This is why gold is catalytically inert toward H₂ under most conditions, despite being a fellow Group 10/11 transition metal.

The general trend among the most commonly used catalytic metals (Au → Pt → Pd → Ni → Fe—spanning periods 4–6) is that the d-band center shifts upward toward the Fermi level, and dissociative adsorption becomes progressively more favorable. Within a single period (e.g., Cu → Ni → Co → Fe in the 4th period), this trend is particularly clean. Students who have studied crystal field theory or ligand field theory in inorganic chemistry will recognise the underlying principle: the filling and energy of d-orbitals govern how strongly a metal interacts with incoming ligands—or, in this case, with adsorbate molecules.

Remark 2.12 (Activated vs. Non-Activated Dissociation).

Dissociative adsorption falls into two distinct regimes depending on whether the molecule encounters an energy barrier as it approaches the surface:

- **Non-activated:** No barrier beyond the thermodynamic requirement. The molecule dissociates spontaneously upon con-

tact with the surface. Examples include H_2 on Pd, Pt, and Ni, with initial sticking coefficients $S_0 \approx 0.1\text{--}1.0$ that are nearly temperature-independent.

- **Activated:** An additional energy barrier must be overcome before dissociation can occur. Examples include N_2 on Fe ($E_a \approx 40 \text{ kJ mol}^{-1}$), CH_4 on Ni, and O_2 on certain metal surfaces. The sticking coefficient $S_0 \ll 1$ at low temperature but increases exponentially with T following an Arrhenius-like dependence.

This distinction explains why ammonia synthesis requires temperatures above 400°C despite being thermodynamically favorable at much lower temperatures—the activation barrier for N_2 dissociation on Fe is the kinetic bottleneck. Molecular beam experiments confirm that the sticking probability for N_2 on Fe increases sharply with both gas temperature and kinetic energy, a hallmark of activated dissociation. In contrast, H_2 dissociation on the same metals proceeds readily even at cryogenic temperatures.

2.4.2 Derivation of the Dissociative Langmuir Isotherm

For dissociative adsorption of A_2 , we write the adsorption and desorption rates:

$$r_{\text{ads}} = k_{\text{ads}} P_{\text{A}_2} \theta_*^2 \quad (2.32)$$

$$r_{\text{des}} = k_{\text{des}} \theta_A^2 \quad (2.33)$$

The adsorption rate is proportional to θ_*^2 because two empty sites are required. The desorption rate is proportional to θ_A^2 because two adsorbed atoms must find each other and recombine.

At equilibrium, $r_{\text{ads}} = r_{\text{des}}$:

$$k_{\text{ads}} P_{\text{A}_2} \theta_*^2 = k_{\text{des}} \theta_A^2 \quad (2.34)$$

Defining the equilibrium constant $K = k_{\text{ads}}/k_{\text{des}}$ and applying the site balance $\theta_A + \theta_* = 1$:

$$K P_{\text{A}_2} = \frac{\theta_A^2}{(1 - \theta_A)^2} \quad (2.35)$$

Taking the square root of both sides:

$$\sqrt{K P_{\text{A}_2}} = \frac{\theta_A}{1 - \theta_A} \quad (2.36)$$

Solving for θ_A :

$$\theta_A = \frac{\sqrt{K P_{\text{A}_2}}}{1 + \sqrt{K P_{\text{A}_2}}} \quad (2.37)$$

Remark 2.13 (Comparison with molecular adsorption). Compare the dissociative isotherm (Eq. (2.37)) with the molecular Langmuir isotherm (Eq. (2.2)):

Type	Coverage expression	Half-coverage pressure
Molecular	$\frac{KP}{1 + KP}$	$P_{1/2} = 1/K$
Dissociative	$\frac{\sqrt{KP}}{1 + \sqrt{KP}}$	$P_{1/2} = 1/K$

The key difference is that KP appears under a *square root* for dissociative adsorption. This means coverage rises more slowly with pressure at low pressures and saturates more quickly at high pressures.

Remark 2.14 (Concept Check: Molecular or Dissociative?). An experimentalist measures coverage vs. pressure for CO adsorption on a metal surface. At $P = 0.01$ bar, $\theta = 0.30$.

1. If this follows *molecular* Langmuir, what value of K does this imply?
2. If instead it follows *dissociative* Langmuir, what K ?
3. Suggest a simple graphical test to distinguish the two cases from experimental data.

Answer: (1) Molecular: $0.30 = K(0.01)/(1 + K(0.01))$, so $K = 0.30/(0.01 \times 0.70) = 42.9 \text{ bar}^{-1}$. (2) Dissociative: $0.30 = \sqrt{K(0.01)}/(1 + \sqrt{K(0.01)})$, so $\sqrt{0.01 K} = 0.30/0.70 = 0.429$, giving $K = 18.4 \text{ bar}^{-1}$. (3) Plot θ vs P (linear at low P for molecular) or θ vs $\sqrt{P}$ (linear for dissociative). A straight line through the origin identifies the mode.

2.4.3 Limiting Behavior

Low-pressure limit ($KP_{A_2} \ll 1$):

$$\theta_A \approx \sqrt{K P_{A_2}} \quad (2.38)$$

Coverage depends on $\sqrt{P}$, not P . This “half-order” dependence is a diagnostic fingerprint of dissociative adsorption.

High-pressure limit ($KP_{A_2} \gg 1$):

$$\theta_A \rightarrow 1 \quad (2.39)$$

The surface saturates, just as in molecular adsorption.

Fig. 2.2 provides a visual summary of the key differences between molecular and dissociative adsorption. The top panels illustrate the fundamental site requirement for each process. In molecular adsorption (left panel), an intact molecule A (orange sphere) lands on a single active site (blue dot) on the gray surface slab. The process requires only one available site, and the molecule remains whole after adsorption. In contrast, dissociative adsorption (right panel) shows a diatomic molecule A_2 splitting into two fragments upon contact with the surface. This process *requires two adjacent empty sites*—if the molecule cannot find a neighboring pair, dissociation cannot occur. The orange spheres in the schematic emphasize how the two resulting adsorbed atoms occupy separate sites while maintaining proximity on the surface.

The bottom panel of Fig. 2.2 compares the corresponding isotherms and reveals the mathematical consequences of the two-site requirement. The blue solid curve represents the molecular Langmuir isotherm from Eq. (2.2), while the orange dashed curve shows the dissociative isotherm from Eq. (2.37). Notice how both isotherms reach half-coverage ($\theta = 0.5$) at the same pressure $P = 1/K$, as indicated by the annotation. However, the *approach* to this point differs dramatically. At low pressures (see the annotations near the origin), molecular adsorption shows linear behavior ($\theta \approx KP$), while dissociative adsorption follows square-root dependence ($\theta \approx \sqrt{KP}$). This means dissociative coverage rises more steeply at very low pressures—the curve has infinite slope at the origin (though this is visible only at extremely small pressures)—but then “bends over” more gradually as pressure increases.

This difference in low-pressure curvature serves as a powerful diagnostic tool. If you measure coverage versus pressure experimentally and plot the data, you can distinguish molecular from dissociative adsorption by examining the functional form at low coverage. A linear plot of θ vs P extrapolating through the origin indicates molecular adsorption; if instead θ vs $\sqrt{P}$ yields a straight line, dissociative adsorption is occurring. At high pressures, both isotherms saturate toward $\theta \rightarrow 1$ as all sites become occupied—the approach to saturation is steeper for the dissociative case.

2.4.4 Python Implementation

The following code implements the dissociative Langmuir isotherm and compares it with the molecular isotherm:

```
1 def dissociative_langmuir(P, K):
2     sqrt_KP = np.sqrt(K * P)
3     return sqrt_KP / (1 + sqrt_KP)
4
5 theta_mol = K*P / (1 + K*P)
6 theta_dis = dissociative_langmuir(P, K)
```

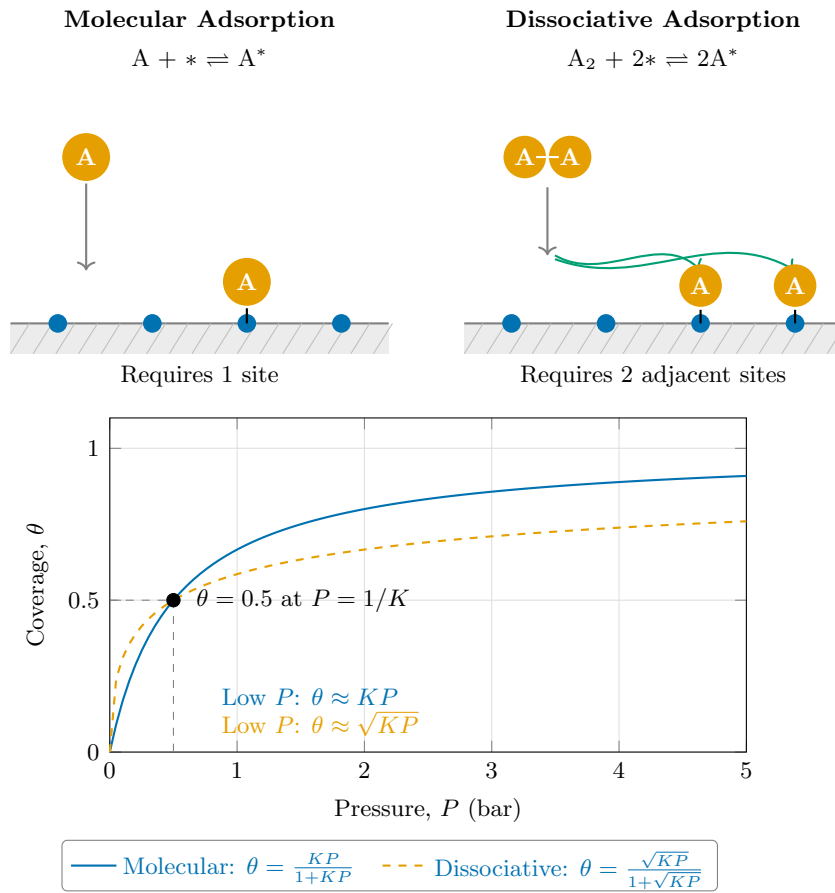



Figure 2.2: Comparison of molecular and dissociative adsorption. (Top) Molecular adsorption: a single molecule A adsorbs intact on one site. Dissociative adsorption: a diatomic molecule A_2 dissociates upon adsorption, with each atom occupying an adjacent site. (Bottom) Isotherm comparison: dissociative adsorption follows $\theta = \sqrt{KP}/(1 + \sqrt{KP})$, showing a square-root pressure dependence at low coverage rather than the linear behavior of molecular Langmuir adsorption.

Listing 2.2: Dissociative vs. molecular Langmuir isotherms (snippet)

Full implementation: see NB2.

2.4.5 Significance for Catalytic Kinetics

Understanding dissociative adsorption is essential for the rate expressions reviewed in Section 2.2. Consider catalytic hydrogenation:

1. H_2 adsorbs dissociatively: $H_{2(g)} + 2* \rightleftharpoons 2H^*$

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2. The organic substrate adsorbs molecularly: $S_{(g)} + * \rightleftharpoons S^*$
3. Adsorbed H atoms react with adsorbed substrate in a surface reaction

The rate depends on the coverages of both H^* and S^* , and the dissociative nature of H_2 adsorption affects how θ_H depends on P_{H_2} . In many hydrogenation rate expressions, you will see $\sqrt{K_{H_2}P_{H_2}}$ terms rather than $K_{H_2}P_{H_2}$ —a direct consequence of dissociative adsorption.

Remark 2.15 (Connection to characterization). The dissociative adsorption of H_2 and CO is exploited in catalyst characterization (Chapter 3). **H_2 chemisorption** titrates the number of surface metal atoms: since H_2 dissociates, each H_2 molecule consumed corresponds to two surface metal sites. Knowing the total H_2 uptake allows calculation of metal dispersion and active site counts for TOF determination.

Example 2.2: Pd-catalyzed hydrogenation: Dissociative H_2 adsorption

Problem: In Pd-catalyzed hydrogenation of alkenes and nitro compounds, hydrogen must first adsorb on the palladium surface. At 350 K, H_2 adsorbs *dissociatively* with $K_{H_2} = 50 \text{ bar}^{-1}$.

- (a) Calculate the coverage of adsorbed hydrogen atoms at $P_{H_2} = 0.1, 1.0$, and 3.0 bar .
- (b) At what H_2 pressure is half-coverage achieved?

Solution (Dissociative Langmuir Isotherm):

Since H_2 adsorbs dissociatively, we use the dissociative Langmuir isotherm (Eq. (2.37)):

$$\theta_H = \frac{\sqrt{K P_{H_2}}}{1 + \sqrt{K P_{H_2}}}$$

(a) At $P_{H_2} = 0.1 \text{ bar}$:

$$\theta_H = \frac{\sqrt{(50)(0.1)}}{1 + \sqrt{(50)(0.1)}} = \frac{\sqrt{5}}{1 + \sqrt{5}} = \frac{2.24}{3.24} = 0.69$$

At $P_{H_2} = 1.0$ bar:

$$\theta_H = \frac{\sqrt{(50)(1.0)}}{1 + \sqrt{(50)(1.0)}} = \frac{7.07}{8.07} = 0.88$$

At $P_{H_2} = 3.0$ bar:

$$\theta_H = \frac{\sqrt{(50)(3.0)}}{1 + \sqrt{(50)(3.0)}} = \frac{12.2}{13.2} = 0.92$$

(b) Setting $\theta_H = 0.5$:

$$0.5 = \frac{\sqrt{KP_{1/2}}}{1 + \sqrt{KP_{1/2}}} \Rightarrow \sqrt{KP_{1/2}} = 1$$

$$\Rightarrow P_{1/2} = \frac{1}{K} = \frac{1}{50} = 0.02 \text{ bar}$$

Python verification:

```
import numpy as np
K = 50.0
for P in [0.1, 1.0, 3.0]:
    theta = np.sqrt(K*P) / (1 + np.sqrt(K*P))
    print(f"P = {P:.1f} bar: theta = {theta:.4f}")
P_half = 1 / K
print(f"Half-coverage pressure: {P_half:.4f} bar")
```

Answer:

$\theta_H = 0.69, 0.88, 0.92$ at $P = 0.1, 1.0, 3.0$ bar; $P_{1/2} = 0.02$ bar

Physical interpretation: The high coverage even at moderate H_2 pressures explains why Pd-catalyzed hydrogenations work well at atmospheric pressure—there is plenty of adsorbed hydrogen available. In competitive adsorption with an organic substrate, hydrogen's strong dissociative adsorption ensures adequate H coverage for reaction.

Safety note: Dry Pd/C saturated with H_2 is pyrophoric. Always handle Pd/C catalysts under inert atmosphere or as a wet paste.

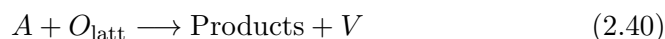
2.5 Mars–van Krevelen (MVK) Redox Mechanism

The Mars–van Krevelen mechanism applies to oxidation reactions where the catalyst participates through a **redox cycle**. Rather than adsorbing molecular oxygen, the catalyst donates *lattice oxygen* to the reactant, creating oxygen vacancies that are subsequently refilled.

2.5.1 The Redox Cycle

Consider oxidation of a substrate A over a metal oxide catalyst. The mechanism involves two half-cycles:

Reduction step (lattice oxygen consumed):



where O_{latt} is a lattice oxygen site and V is a vacancy.

Reoxidation step (vacancy refilled):

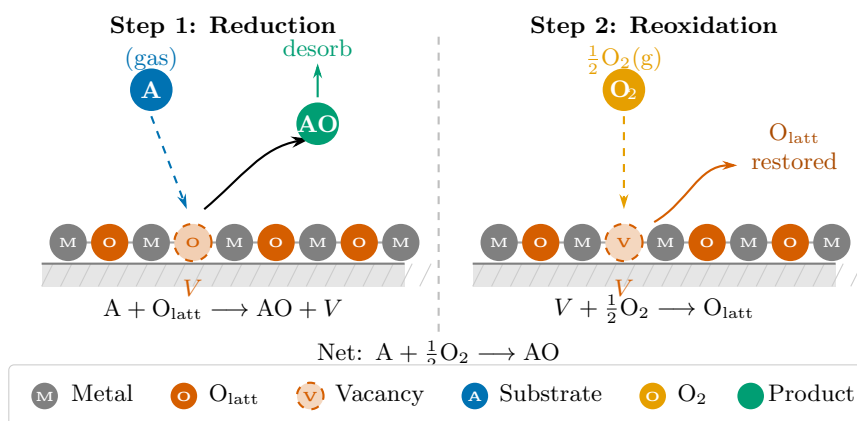


Figure 2.3: Schematic of the Mars–van Krevelen redox cycle. The catalyst alternates between oxidized and reduced states: (1) the substrate A extracts lattice oxygen O_{latt} to form oxidized product, leaving a vacancy V ; (2) gas-phase O_2 refills the vacancy, regenerating the active oxide surface. The cycle continues as long as both A and O_2 are supplied.

Fig. 2.3 illustrates the defining feature of Mars–van Krevelen kinetics: the catalyst itself participates chemically in the reaction through a redox cycle. Unlike LH and ER mechanisms where the surface merely provides a meeting place for reactants, in MVK the catalyst *donates* lattice oxygen to the substrate. This creates an oxygen vacancy that must be refilled by gas-phase O_2 before the cycle can repeat. The catalyst thus oscillates between

an oxidized state (rich in lattice oxygen, θ_O high) and a reduced state (rich in vacancies, θ_V high). At steady state, the rates of reduction and reoxidation must be equal, which leads to the characteristic MVK rate expression.

2.5.2 Deriving the MVK Rate Expression

Define θ_O as the fraction of lattice sites occupied by oxygen and $\theta_V = 1 - \theta_O$ as the vacancy fraction. The rates of the two steps are:

$$r_{\text{red}} = k_{\text{red}} P_A \theta_O \quad (2.42)$$

$$r_{\text{ox}} = k_{\text{ox}} P_{O_2}^n \theta_V \quad (2.43)$$

where n is typically 1/2 or 1 depending on the mechanism of oxygen incorporation. When O_2 dissociates on the surface before filling vacancies (as in many metal oxide catalysts), $n = 1/2$ by analogy with the dissociative Langmuir isotherm (Section 2.4). When molecular O_2 fills a single vacancy without prior dissociation, $n = 1$.

At steady state, the rates of reduction and reoxidation must be equal:

$$r_{\text{red}} = r_{\text{ox}} = r \quad (2.44)$$

Solving for θ_O :

$$k_{\text{red}} P_A \theta_O = k_{\text{ox}} P_{O_2}^n (1 - \theta_O) \quad (2.45)$$

$$\theta_O = \frac{k_{\text{ox}} P_{O_2}^n}{k_{\text{red}} P_A + k_{\text{ox}} P_{O_2}^n} \quad (2.46)$$

Substituting into either rate expression:

$$r = \frac{k_{\text{red}} k_{\text{ox}} P_A P_{O_2}^n}{k_{\text{red}} P_A + k_{\text{ox}} P_{O_2}^n} \quad (2.47)$$

Remark 2.16 (Two resistances in series: Physical intuition). This expression has the form of “two resistances in series.” The numerator is the product of the two rates; the denominator is their sum. This structure appears frequently in transport-limited kinetics (Chapter 4) and provides useful physical intuition. Just as electrical current through two series resistors is limited by the larger resistance, the overall catalytic rate is limited by whichever step—reduction or reoxidation—is slower under the given conditions.

2.5.3 Fingerprints of MVK Kinetics

Oxygen-rich limit ($k_{\text{ox}} P_{\text{O}_2}^n \gg k_{\text{red}} P_A$):

Reoxidation is fast, $\theta_{\text{O}} \rightarrow 1$, and:

$$r \approx k_{\text{red}} P_A \quad (2.48)$$

The reaction is **first order in A, zero order in O₂** (reduction-controlled).

Oxygen-lean limit ($k_{\text{ox}} P_{\text{O}_2}^n \ll k_{\text{red}} P_A$):

Reoxidation is slow, $\theta_{\text{O}} \rightarrow 0$, and:

$$r \approx k_{\text{ox}} P_{\text{O}_2}^n \quad (2.49)$$

The reaction is **zero order in A, order n in O₂** (reoxidation-controlled).

Remark 2.17 (Concept Check: MVK Regime Identification). A researcher doubles the oxygen partial pressure while holding P_A constant, and observes that the rate barely changes.

1. Is the catalyst in the reduction-controlled or reoxidation-controlled regime?
2. What would happen to the rate if P_A were doubled instead?

Answer: (1) Reduction-controlled. In this regime, the reoxidation is already fast ($k_{\text{ox}} P_{\text{O}_2}^n \gg k_{\text{red}} P_A$), so adding more O₂ has little effect; the rate is controlled by $k_{\text{red}} P_A$. (2) Doubling P_A would approximately double the rate, since $r \approx k_{\text{red}} P_A$ in this regime.

Remark 2.18 (Industrial MVK Reactions). The Mars–van Krevelen mechanism governs many of the largest-scale selective oxidation processes in the chemical industry. The common thread is a **reducible metal oxide** whose cation cycles between oxidation states:

1. **SO₂ oxidation on V₂O₅** (sulfuric acid production): Lattice oxygen from V₂O₅ oxidises SO₂ to SO₃, reducing V⁵⁺ to V⁴⁺. Gas-phase O₂ then reoxidises the V₂O₄ sites back to V₂O₅. This process produces over 200 Mt/yr of H₂SO₄ worldwide.
2. **Propane ODH on V₂O₅/TiO₂**: Lattice oxygen abstracts hydrogen from propane to produce propene and water, with propene yields of 15–20% reported for vanadium-containing microporous materials.
3. ***n*-Butane → maleic anhydride on (VO)₂P₂O₇ (VPO)**: A remarkable 14-electron selective oxidation using lattice oxygen—one of the most complex single-step industrial oxidation pro-

cesses.

4. **CO oxidation on CeO₂-based catalysts:** The Ce⁴⁺/Ce³⁺ redox couple provides lattice oxygen for CO oxidation at moderate temperatures, exploiting the exceptional oxygen storage capacity of ceria.

In each case, the MVK rate expression (Eq. (2.47)) applies, with the balance between k_{red} and k_{ox} determining whether the reduction or reoxidation half-cycle limits the overall rate.

Safety note: V₂O₅ dust is toxic by inhalation (TLV-TWA = 0.05 mg/m³ for respirable fraction) and is classified as a possible carcinogen (IARC Group 2B). Handle with appropriate respiratory protection in a fume hood.

Remark 2.19 (Tracking Oxidation States: Spectroscopic Evidence for MVK). The defining feature of the MVK mechanism—the catalyst cycling between oxidation states—can be observed directly with spectroscopic techniques that Chemistry students encounter in analytical and inorganic courses:

- **XPS (X-ray Photoelectron Spectroscopy):** The V 2p_{3/2} binding energy shifts from 517.2 eV (V⁵⁺) to 515.8 eV (V⁴⁺) during the reduction half-cycle, and back upon reoxidation. This 1.4 eV shift is easily resolved with modern instruments, providing a direct “snapshot” of the catalyst oxidation state.
- **UV-Vis Diffuse Reflectance:** The Ce⁴⁺/Ce³⁺ redox couple produces a visible colour change—pale yellow (Ce⁴⁺, 4f⁰) to blue-grey (Ce³⁺, 4f¹). In operando UV-Vis cells, this colour oscillation can be tracked in real time during catalytic CO oxidation on CeO₂.
- **Isotope labelling with ¹⁸O₂:** The most definitive MVK evidence comes from feeding ¹⁸O₂ while monitoring the product oxygen isotope composition by mass spectrometry. In a true MVK mechanism, the *first* product molecules contain lattice ¹⁶O, and ¹⁸O incorporation appears only after lattice vacancies are refilled. This sequential isotope pattern distinguishes MVK from LH/ER mechanisms, where isotope scrambling occurs immediately.

These techniques connect directly to what you learn in analytical chemistry (XPS, UV-Vis) and illustrate how spectroscopy provides molecular-level evidence for the mechanistic models derived from kinetic data alone.

2.5.4 Python Visualization: MVK Rate Surfaces

The following code explores the MVK rate expression (Eq. (2.47)) by plotting the rate as a function of one partial pressure while holding the other fixed at several values. This reveals how the controlling step shifts from reduction-limited to reoxidation-limited as the feed composition changes.

```

1 def mvk_rate(P_A, P_O2, k_red, k_ox, n):
2     red_term = k_red * P_A
3     ox_term = k_ox * P_O2**n
4     return red_term * ox_term / (red_term + ox_term)
5
6 r = mvk_rate(P_A, P_O2=0.5, k_red=2.0, k_ox=0.5, n=0.5)

```

Listing 2.3: Mars–van Krevelen rate expression (snippet)

Full implementation: see NB2.

2.6 Experimental Tools for Mechanism Elucidation

While kinetic fingerprints provide valuable clues about reaction mechanisms, conclusive identification often requires complementary experimental techniques. This section introduces three powerful approaches used by catalysis researchers to probe reaction mechanisms.

2.6.1 Kinetic Isotope Effects (KIE)

The **kinetic isotope effect** exploits the mass difference between isotopes (typically hydrogen and deuterium) to reveal information about the rate-determining step.

Definition 2.2 (Kinetic Isotope Effect). The kinetic isotope effect is defined as the ratio of rate constants for reactions involving different isotopes:

$$\text{KIE} = \frac{k_H}{k_D} \quad (2.50)$$

where k_H is the rate constant with the light isotope (e.g., ^1H) and k_D is the rate constant with the heavy isotope (e.g., ^2H or D).

Physical origin: Heavier isotopes have lower zero-point vibrational energies. When a bond involving the isotopic atom is broken or formed in the RDS, this energy difference manifests as different activation barriers, leading to measurably different rate constants.

Primary KIE: Occurs when the isotopically labeled bond is *broken or formed* in the rate-determining step. For C–H versus C–D bond cleavage:

$$\text{Primary KIE} \approx 2\text{--}7 \quad (\text{typically } 3\text{--}5 \text{ at room temperature}) \quad (2.51)$$

Secondary KIE: Occurs when the isotopic substitution is *adjacent to* (but not directly involved in) the bond-breaking step. Secondary KIEs are much smaller:

$$\text{Secondary KIE} \approx 0.8\text{--}1.2 \quad (2.52)$$

Remark 2.20 (Interpreting KIE Magnitude). A KIE value of 1.0 indicates that the isotopically labeled bond is not involved in the RDS. Values significantly greater than 2 strongly suggest that bond cleavage involving the labeled atom occurs in the rate-determining step.

Application to mechanism elucidation: Consider a surface reaction where you suspect C–H activation is rate-limiting. By comparing the rates using a substrate with C–H bonds versus C–D bonds:

- If $\text{KIE} \approx 4\text{--}6$: C–H bond breaking is likely the RDS
- If $\text{KIE} \approx 1$: C–H cleavage occurs in a fast step, not the RDS

2.6.2 Python Visualization: KIE Analysis

The following code creates a bar chart comparing rate constants k_H and k_D for three hypothetical reactions, illustrating how the KIE magnitude distinguishes primary from secondary isotope effects.

```
1 k_H = np.array([5.2, 2.8, 8.1])
2 k_D = np.array([1.3, 2.5, 1.2])
3 KIE = k_H / k_D # Primary if KIE > 2; secondary if ~1
```

Listing 2.4: Kinetic isotope effect (KIE) analysis (snippet)

Full implementation: see NB2.

2.6.3 Mechanism Fingerprint Summary

Table 2.1 consolidates the kinetic fingerprints for the three major surface reaction mechanisms discussed in this chapter. Use this table as a diagnostic checklist when analyzing experimental rate data. Note that the low- P_A behavior is identical for all three mechanisms—it is the high- P_A regime and the presence or absence of a rate maximum that most effectively discriminate between them.

The last row deserves special comment: in MVK reactions, $^{18}\text{O}_2$ labelling experiments show that lattice oxygen appears in the product *before* gas-phase oxygen is incorporated into the lattice, producing a sequential isotope pattern. In LH and ER mechanisms, where both reactants co-adsorb on the same surface, isotope scrambling is typically observed. This provides a

2. Computational Perspectives on Surface Catalysis

Table 2.1: Kinetic fingerprints for mechanism discrimination. LH = Langmuir–Hinshelwood (bimolecular, competitive adsorption); ER = Eley–Rideal; MVK = Mars–van Krevelen.

Diagnostic	LH	ER	MVK
r vs P_A at low P_A	$\propto P_A$	$\propto P_A$	$\propto P_A$
r vs P_A at high P_A	$\propto 1/P_A$	Saturates	Saturates
Maximum in $r(P_A)$?	Yes	No	No
Denominator exponent	2	1	1
E_{app} (low θ)	$E_a + \sum \Delta H_{\text{ads}}$	$E_a + \Delta H_{\text{ads,A}}$	E_a
E_{app} (high θ)	E_a	E_a	Complex
Product inhibition	Strong (in denom.)	Moderate	Weak
Isotope labelling of O_2	Scrambles	Scrambles	Sequential

definitive experimental test that complements the purely kinetic diagnostics above.

2.6.4 In-situ Spectroscopy for Surface Studies

Kinetic measurements tell us *how fast* reactions proceed, but spectroscopic techniques reveal *what species are present* on the catalyst surface during reaction. This molecular-level information is invaluable for mechanism elucidation.

FTIR of Adsorbed Species: Fourier-transform infrared spectroscopy can identify surface intermediates through their characteristic vibrational frequencies:

- **CO adsorption:** Linear CO on metal surfaces shows stretching frequencies around $2000\text{--}2100\text{ cm}^{-1}$; bridge-bonded CO appears at $1800\text{--}1900\text{ cm}^{-1}$
- **Adsorbed hydrocarbons:** C–H stretching modes ($2800\text{--}3100\text{ cm}^{-1}$) and deformation modes reveal molecular orientation
- **Surface hydroxyl groups:** O–H stretches ($3200\text{--}3700\text{ cm}^{-1}$) indicate the presence and environment of surface OH species

Operando techniques: Modern “operando” (Latin for “working”) spectroscopy performs measurements *under actual reaction conditions*—at elevated temperatures and pressures, with flowing reactants, while simultaneously measuring catalytic activity. This approach reveals which surface species are present during steady-state catalysis, not just during adsorption experiments.

Remark 2.21 (Surface Species–Rate Correlation). The correlation between spectroscopically observed surface species and measured reaction rates provides powerful evidence for reaction mechanisms. If a proposed intermediate does not appear in operando spectra, or if its coverage does not correlate with the reaction rate, the proposed mechanism must be revised.

Complementary techniques: Other surface-sensitive methods include the following:

- Temperature-programmed desorption (TPD): Reveals binding energies and site distributions
- X-ray photoelectron spectroscopy (XPS): Identifies oxidation states and surface composition
- Raman spectroscopy: Particularly useful for metal oxide catalysts

Remark 2.22 (Operando Spectroscopy in Practice). Modern operando setups combine FTIR or Raman spectroscopy with online product analysis (GC or MS), providing direct evidence for proposed mechanisms by simultaneously measuring surface species and reaction rates. For example, tracking the intensity of a surface intermediate band versus the rate reveals whether that species is kinetically relevant. If an intermediate accumulates while the rate stays constant, it is likely a spectator rather than a true intermediate.

Remark 2.23 (Nobel Connections and Laboratory Safety). **Nobel connections.** The experimental techniques described in this section—TPD, LEED, XPS, and operando spectroscopy—are precisely the tools that earned **Gerhard Ertl** the 2007 Nobel Prize in Chemistry “for his studies of chemical processes on solid surfaces.” Ertl combined TPD, LEED, and XPS to build a molecular-level picture of CO oxidation on Pt(111) and N₂ dissociation on Fe(111). His work on iron surfaces provided the first atomistic understanding of the rate-limiting step in the **Haber–Bosch process**—for which Fritz Haber received the 1918 Nobel Prize. The activated dissociative adsorption of N₂ on Fe, with $E_a \approx 40 \text{ kJ mol}^{-1}$, was directly measured using the surface science techniques presented here.

Safety notes. Many probe molecules and reactants in catalysis research pose serious hazards:

- **Carbon monoxide (CO):** Odourless and highly toxic. The

occupational exposure limit is $\text{TLV-TWA} = 25 \text{ ppm}$; the IDLH concentration is 1200 ppm. Always use CO in well-ventilated fume hoods with continuous CO monitors.

- **Hydrogen (H_2):** Forms explosive mixtures with air over an exceptionally wide range ($\text{LEL} = 4\%$, $\text{UEL} = 75\%$ by volume). Use leak detectors, eliminate ignition sources, and ensure adequate ventilation when working with H_2 flow systems.
- **General practice:** Catalysis laboratories handling toxic or flammable gases must have fixed gas detectors, emergency shut-off valves, and clearly posted safety data sheets (SDS) for all gases in use.

Remark 2.24 (Connections to Homogeneous Catalysis). Students pursuing organometallic chemistry will find that LH and ER concepts have direct analogs in homogeneous catalysis:

- **Oxidative addition** $\leftrightarrow$ dissociative adsorption on a surface
- **Reductive elimination** $\leftrightarrow$ associative desorption
- **Migratory insertion** $\leftrightarrow$ surface reaction between co-adsorbed species
- The metal center $\leftrightarrow$ the active site; coordination $\leftrightarrow$ adsorption; ligand dissociation $\leftrightarrow$ vacant site availability (θ_*)

These parallels allow you to transfer mechanistic intuition between subfields.

Having established kinetic fingerprints and experimental tools to identify surface reaction mechanisms, we now turn to extracting quantitative activation parameters from temperature-dependent rate data. The Arrhenius equation reviewed in Section 2.2 provides E_a , but the Eyring equation offers deeper mechanistic insight by separating enthalpic and entropic contributions to the barrier.

2.7 Eyring Equation and Transition State Theory

While the Arrhenius equation provides a practical framework for analyzing temperature-dependent kinetics, it is fundamentally empirical—it tells us *what* the temperature dependence is but not *why*. The Eyring equation, derived from transition state theory (TST), provides a more fundamental description that separates the temperature dependence into enthalpic and entropic contributions, offering deeper mechanistic insight.

2.7.1 The Eyring Equation

Transition state theory assumes that reactants and the transition state (or activated complex) are in quasi-equilibrium. The rate constant is then:

$$k(T) = \frac{k_B T}{h} \exp\left(\frac{\Delta S^\ddagger}{R}\right) \exp\left(-\frac{\Delta H^\ddagger}{RT}\right) \quad (2.53)$$

where:

- $k_B = 1.381 \times 10^{-23}$ J/K is the Boltzmann constant
- $h = 6.626 \times 10^{-34}$ J·s is Planck's constant
- $\Delta H^\ddagger$ is the **enthalpy of activation** (J/mol or kJ/mol)
- $\Delta S^\ddagger$ is the **entropy of activation** (J/(mol·K))

The prefactor $k_B T/h$ represents the universal frequency at which activated complexes cross the barrier, approximately $6 \times 10^{12} \text{ s}^{-1}$ at 300 K. The full Eyring equation includes a transmission coefficient $\kappa \leq 1$ accounting for trajectories that recross the barrier; in most textbook treatments (and throughout this course), $\kappa = 1$ is assumed.

2.7.2 Relationship to Gibbs Free Energy of Activation

The Eyring equation can be written more compactly using the Gibbs free energy of activation:

$$\Delta G^\ddagger = \Delta H^\ddagger - T\Delta S^\ddagger \quad (2.54)$$

Substituting into Eq. (2.53):

$$k(T) = \frac{k_B T}{h} \exp\left(-\frac{\Delta G^\ddagger}{RT}\right) \quad (2.55)$$

This form emphasizes that the rate constant depends on the *free energy* barrier to the transition state, not just the energy barrier. A reaction can be fast despite a high $\Delta H^\ddagger$ if it has a sufficiently positive $\Delta S^\ddagger$ (large entropic compensation).

2.7.3 Linearized Eyring Form

To extract $\Delta H^\ddagger$ and $\Delta S^\ddagger$ from experimental data, we divide by T and take logarithms:

$$\ln\left(\frac{k}{T}\right) = \ln\left(\frac{k_B}{h}\right) + \frac{\Delta S^\ddagger}{R} - \frac{\Delta H^\ddagger}{R} \cdot \frac{1}{T} \quad (2.56)$$

This has the form $y = b + mx$, where:

- $y = \ln(k/T)$
- $x = 1/T$
- slope $m = -\Delta H^\ddagger/R$
- y -intercept $b = \ln(k_B/h) + \Delta S^\ddagger/R = 23.76 + \Delta S^\ddagger/R$

Note that $\ln(k_B/h) = \ln(1.381 \times 10^{-23}/6.626 \times 10^{-34}) = 23.76$ is a universal constant.

2.7.4 Extracting Eyring Parameters

The procedure parallels Arrhenius analysis:

1. **Collect data:** Determine k at multiple temperatures
2. **Prepare linearization data:** Calculate $1/T$ and $\ln(k/T)$ for each measurement
3. **Linear regression:** Plot $\ln(k/T)$ versus $1/T$ and fit a straight line
4. **Extract parameters:**

$$\Delta H^\ddagger = -(\text{slope}) \times R \quad (2.57)$$

$$\Delta S^\ddagger = R \times (\text{intercept} - 23.76) \quad (2.58)$$

Fig. 2.4 illustrates the Eyring analysis. The key difference from the Arrhenius plot is the use of $\ln(k/T)$ rather than $\ln(k)$ on the y -axis. This seemingly small modification arises from the T -dependence in the pre-exponential factor of the Eyring equation and separates the enthalpic and entropic contributions more cleanly.

2.7.5 Physical Interpretation of Activation Parameters

The activation parameters from Eyring analysis provide rich mechanistic information:

Activation Enthalpy ($\Delta H^\ddagger$)

The activation enthalpy reflects the energy required to distort bonds in the reactants to reach the transition state. It is closely related to the Arrhenius activation energy:

$$E_a \approx \Delta H^\ddagger + RT$$

This relationship is derived formally in Section 2.7.6.

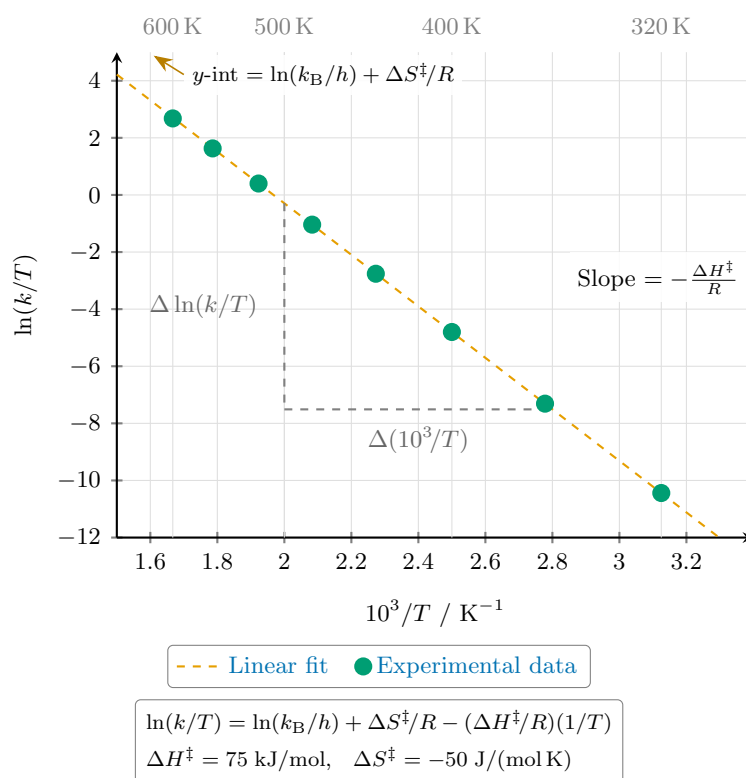


Figure 2.4: Eyring plot for determining activation parameters $\Delta H^\ddagger$ and $\Delta S^\ddagger$ from temperature-dependent rate data. Plotting $\ln(k/T)$ versus $1/T$ yields a straight line with slope $= -\Delta H^\ddagger/R$ and y -intercept $= \ln(k_B/h) + \Delta S^\ddagger/R$. The data correspond to $\Delta H^\ddagger = 75 \text{ kJ/mol}$ and $\Delta S^\ddagger = -50 \text{ J/(mol K)}$, values consistent with an associative surface reaction where the transition state is more ordered than the reactants.

For typical temperatures (300–500 K), the RT term is 2.5–4.2 kJ/mol, so E_a and $\Delta H^\ddagger$ are numerically similar. However, $\Delta H^\ddagger$ has a clearer thermodynamic interpretation as the enthalpy difference between reactants and the transition state.

Key points about $\Delta H^\ddagger$:

- **Lower $\Delta H^\ddagger$ means easier activation**—this is how catalysts work, by providing an alternative pathway with lower barrier
- **Involves bond distortion/breaking**: Reactions that break strong bonds have higher $\Delta H^\ddagger$
- **Temperature independent** (to first approximation): Unlike $\Delta G^\ddagger$, the enthalpy barrier does not change significantly with temperature

Activation Entropy ($\Delta S^\ddagger$)

The activation entropy reveals the “tightness” or “looseness” of the transition state relative to the reactants. This is often the most mechanistically informative parameter:

- $\Delta S^\ddagger < 0$ (**negative**): The transition state is more *ordered* than the reactants
 - Characteristic of **associative mechanisms**—two molecules combining
 - Loss of translational and/or rotational degrees of freedom
 - Typical values: -50 to -150 J/(mol·K)
- $\Delta S^\ddagger > 0$ (**positive**): The transition state is more *disordered* than the reactants
 - Characteristic of **dissociative mechanisms**—a molecule breaking apart
 - Gain of degrees of freedom
 - Typical values: $+20$ to $+80$ J/(mol·K)
- $\Delta S^\ddagger \approx 0$: Little change in structure/ordering between reactants and transition state
 - Unimolecular rearrangements
 - Intramolecular reactions

Remark 2.25 (Concept Check: Entropy and Rate). Two surface reactions have identical $\Delta H^\ddagger = 80$ kJ/mol but different entropies: Reaction A has $\Delta S^\ddagger = -80$ J/(mol·K); Reaction B has $\Delta S^\ddagger = +20$ J/(mol·K).

Without calculating, which reaction is faster at 500 K? Which has the higher pre-exponential factor?

Answer: Reaction B is faster. Both reactions have the same $\Delta H^\ddagger$, so the enthalpic contribution to $\Delta G^\ddagger$ is identical. However, Reaction B has the more positive $\Delta S^\ddagger$, making $\Delta G^\ddagger = \Delta H^\ddagger - T\Delta S^\ddagger$ lower (more favorable). Equivalently, Reaction B has the higher pre-exponential factor ($A \propto \exp(\Delta S^\ddagger/R)$), so its rate constant is larger at any temperature.

Remark 2.26 (Entropy in Surface Reactions). For adsorption and surface reactions, $\Delta S^\ddagger$ is often significantly negative because gas-phase molecules lose translational and rotational freedom when binding to the surface. A free gas molecule at 1 bar and 300 K has a translational entropy of approximately 120–150 J/(mol·K); much of this is lost upon adsorption.

This entropy penalty explains why adsorption is favored at low temperatures (where $T\Delta S$ is small) and desorption becomes thermodynamically favorable at high temperatures.

Remark 2.27 (Interpreting the Pre-Exponential). The pre-exponential factor A encodes the entropy of activation. From the Eyring relation, $A = e(k_B T/h) \exp(\Delta S^\ddagger/R)$, we can establish key benchmarks at typical catalytic temperatures:

- $A \sim 10^{13} \text{ s}^{-1}$: $\Delta S^\ddagger \approx 0$ — the transition state has similar entropy to the reactant state
- $A \gg 10^{13} \text{ s}^{-1}$: $\Delta S^\ddagger > 0$ — a “loose” transition state with more degrees of freedom than the reactants (e.g., dissociative desorption where fragments gain translational freedom)
- $A \ll 10^{13} \text{ s}^{-1}$: $\Delta S^\ddagger < 0$ — a “tight” transition state with fewer degrees of freedom (e.g., associative reactions requiring specific molecular orientation)

Thus, a reported pre-exponential factor immediately tells you something about the mechanism without performing a full Eyring analysis. Reference: Kolasinski, *Surface Science*, p. 241.

2.7.6 Comparing Arrhenius and Eyring Parameters

The relationships between Arrhenius and Eyring parameters are:

$$E_a = \Delta H^\ddagger + RT \quad (2.59)$$

$$A = \frac{e \cdot k_B T}{h} \exp\left(\frac{\Delta S^\ddagger}{R}\right) \quad (2.60)$$

where $e = 2.718\dots$ is Euler’s number (arising from the RT correction).

Eq. (2.60) shows that the pre-exponential factor contains entropic information:

- $A \approx 10^{13} \text{ s}^{-1}$ (at 300 K) corresponds to $\Delta S^\ddagger \approx 0$

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- $A < 10^{13} \text{ s}^{-1}$ indicates $\Delta S^\ddagger < 0$ (tight transition state)
- $A > 10^{13} \text{ s}^{-1}$ indicates $\Delta S^\ddagger > 0$ (loose transition state)

This interpretation of the pre-exponential factor is a key advantage of understanding the Eyring framework, even when using Arrhenius analysis for practical parameter extraction.

Remark 2.28 (Predict Before You Read: Enthalpy–Entropy Compensation). For a series of structurally similar catalysts (e.g., Pd supported on different oxide supports), one catalyst binds the substrate more strongly than the others.

Before reading further, predict:

- Will this catalyst have a *higher* or *lower* $\Delta H^\ddagger$?
- Will $\Delta S^\ddagger$ be *more positive* or *more negative*?
- Overall, will this catalyst be faster or slower than its weaker-binding counterpart?

Sketch your reasoning on paper, then read on to see how the enthalpy–entropy compensation effect resolves these questions.

2.7.7 Enthalpy–Entropy Compensation

A common observation in catalysis is **enthalpy–entropy compensation**: reactions with higher $\Delta H^\ddagger$ often also have more positive $\Delta S^\ddagger$. When comparing different catalysts or conditions, one sometimes finds:

$$\Delta H^\ddagger = \beta \Delta S^\ddagger + \gamma \quad (2.61)$$

where β has units of K (the isokinetic temperature) and γ has units of kJ/mol.

At the isokinetic temperature, all members of the series have the same rate constant because changes in $\Delta H^\ddagger$ are exactly compensated by changes in $T\Delta S^\ddagger$. The physical origin, as explained by Laidler, is as follows: across a family of catalysts, *weaker* catalyst–substrate binding raises $\Delta H^\ddagger$ (higher barrier to the transition state) but also makes $\Delta S^\ddagger$ more positive (a looser, less constrained transition state), partially compensating the effect on $\Delta G^\ddagger$. Conversely, stronger binding lowers the enthalpic barrier while creating a more ordered (lower entropy) transition state.

This phenomenon can arise from:

1. Changes in catalyst binding strength (weaker binding raises $\Delta H^\ddagger$ but also makes $\Delta S^\ddagger$ more positive, partially compensating the effect on $\Delta G^\ddagger$)

2. Solvent reorganization effects
3. Experimental artifacts from limited temperature ranges (apparent compensation)

Careful analysis is needed to distinguish genuine mechanistic compensation from statistical artifacts.

2.7.8 Python Visualization: Arrhenius vs. Eyring Comparison

The following Python script generates synthetic rate constant data from known Eyring parameters, then performs both Arrhenius and Eyring analyses side by side. This lets you see exactly how the two linearization methods extract different—but related—information from the same dataset.

```
1 # Eyring linearization: ln(k/T) vs 1/T
2 ln_kT = np.log(k_data / T)
3 eyr = linregress(1.0/T, ln_kT)
4 DH_fit = -eyr.slope * R
5 DS_fit = R * (eyr.intercept - np.log(kB / h))
```

Listing 2.5: Side-by-side Arrhenius and Eyring analysis (snippet)

Full implementation: see NB2.

Example 2.3: Eyring Parameter Extraction from Experimental Data

Problem: The rate constant for surface dehydrogenation of cyclohexane on Pt/Al₂O₃ was measured at four temperatures:

T (K)	k (s ⁻¹)
523	1.2×10^2
548	4.1×10^2
573	1.3×10^3
598	3.7×10^3

- (a) Prepare the Eyring linearization data: calculate $1/T$ and $\ln(k/T)$ for each point.
- (b) Determine $\Delta H^\ddagger$ and $\Delta S^\ddagger$ from the slope and intercept.
- (c) Interpret the sign of $\Delta S^\ddagger$ mechanistically.

Solution (Linearized Eyring Equation):

We use the linearized Eyring form (Eq. (2.56)):

$$\ln\left(\frac{k}{T}\right) = \underbrace{\ln\left(\frac{k_B}{h}\right) + \frac{\Delta S^\ddagger}{R}}_{\text{intercept}} - \frac{\Delta H^\ddagger}{R} \cdot \frac{1}{T}$$

(a) Linearization data:

T (K)	k (s^{-1})	$10^3/T$ (K^{-1})	$\ln(k/T)$
523	1.2×10^2	1.912	-1.472
548	4.1×10^2	1.825	-0.290
573	1.3×10^3	1.745	0.820
598	3.7×10^3	1.672	1.823

(b) Linear regression of $\ln(k/T)$ versus $1/T$ gives:

$$\begin{aligned}\text{slope} &= -1.374 \times 10^4 \text{ K} \\ \text{intercept} &= 24.79\end{aligned}$$

Extracting parameters using Eqs. (2.57) and (2.58):

$$\begin{aligned}\Delta H^\ddagger &= -\text{slope} \times R = (1.374 \times 10^4)(8.314) = 114 \text{ kJ/mol} \\ \Delta S^\ddagger &= R \times (\text{intercept} - 23.76) \\ &= (8.314)(24.79 - 23.76) = 8.6 \text{ J/(mol K)}\end{aligned}$$

Python verification:

```
import numpy as np
from scipy.stats import linregress
T = np.array([523, 548, 573, 598])
k = np.array([1.2e2, 4.1e2, 1.3e3, 3.7e3])
res = linregress(1/T, np.log(k/T))
DH = -res.slope * 8.314
DS = 8.314 * (res.intercept - 23.76)
print(f"DH_ddag = {DH/1e3:.1f} kJ/mol, DS_ddag = {DS:.1f} J/(mol*K)")
```

(c) The slightly positive $\Delta S^\ddagger \approx +9 \text{ J/(mol}\cdot\text{K)}$ indicates a “loose” transition state with slightly more disorder than the reactant state. This is consistent with a dehydrogenation mechanism where C–H bond cleavage generates a more mobile hydrogen atom at the transition state, gaining translational degrees of freedom on the surface.

Answer:

$$\Delta H^\ddagger = 114 \text{ kJ/mol}$$

$$\Delta S^\ddagger = +8.6 \text{ J/(mol K)}$$

$$\Delta S^\ddagger > 0: \text{ loose, dissociative TS}$$

Physical interpretation: The positive $\Delta S^\ddagger$ is characteristic of bond-breaking processes. By contrast, associative reactions (e.g., two adsorbed fragments combining) typically show $\Delta S^\ddagger < 0$, reflecting loss of freedom as fragments come together. The Eyring analysis thus provides mechanistic insight that the Arrhenius pre-exponential alone cannot cleanly separate.

2.8 Apparent vs. Intrinsic Activation Energy

When we study Langmuir–Hinshelwood kinetics, a subtlety emerges: the overall rate depends on both the rate constant k (for the surface reaction) *and* the adsorption equilibrium constant K (which determines surface coverage). Both quantities are temperature-dependent. The **apparent activation energy** E_{app} that we measure from an Arrhenius plot of the overall rate may differ substantially from the **intrinsic activation energy** E_a of the surface reaction step.

2.8.1 Definition of Apparent Activation Energy

The apparent activation energy is defined operationally from the temperature dependence of the observed rate:

$$E_{\text{app}} = -R \frac{d \ln(r)}{d(1/T)} \quad (2.62)$$

This is what you measure when you plot $\ln(r)$ versus $1/T$ and extract the slope. For a simple elementary reaction with rate $r = k \cdot [\text{reactants}]$, the apparent and intrinsic activation energies are identical. But for Langmuir–Hinshelwood kinetics, where coverage itself depends on temperature, the situation is more complex.

2.8.2 Coverage Effects on Apparent Activation Energy

Consider a unimolecular surface reaction with rate:

$$r = k\theta_A \quad (2.63)$$

2. Computational Perspectives on Surface Catalysis

where the coverage follows the Langmuir isotherm:

$$\theta_A = \frac{KP_A}{1 + KP_A} \quad (2.64)$$

The rate constant follows Arrhenius: $k = A \exp(-E_a/RT)$.

The adsorption equilibrium constant also depends on temperature via the van't Hoff equation:

$$K(T) = K_0 \exp\left(-\frac{\Delta H_{\text{ads}}}{RT}\right) \quad (2.65)$$

where ΔH_{ads} is the enthalpy of adsorption (negative for exothermic adsorption).

Low Coverage Limit ($KP_A \ll 1$)

When the surface is sparsely covered, $\theta_A \approx KP_A$, so:

$$r \approx k \cdot KP_A \quad (2.66)$$

Applying the definition of apparent activation energy:

$$E_{\text{app}} = -R \frac{d}{d(1/T)} \ln(k \cdot K) \quad (2.67)$$

$$= -R \frac{d}{d(1/T)} [\ln k + \ln K] \quad (2.68)$$

$$= E_a + \Delta H_{\text{ads}} \quad (2.69)$$

$$\boxed{E_{\text{app}} = E_a + \Delta H_{\text{ads}} \quad (\text{low coverage})} \quad (2.70)$$

Since adsorption is exothermic ($\Delta H_{\text{ads}} < 0$), the apparent activation energy is *lower* than the intrinsic value. Physically, this occurs because increasing temperature has two opposing effects:

1. It speeds up the surface reaction (via k)
2. It decreases coverage (via K), which slows the overall rate

The apparent activation energy reflects the net effect of these competing tendencies.

High Coverage Limit ($KP_A \gg 1$)

When the surface is nearly saturated, $\theta_A \approx 1$, so:

$$r \approx k \quad (2.71)$$

The rate depends only on the surface rate constant, and:

$$E_{\text{app}} = E_a \quad (\text{high coverage}) \quad (2.72)$$

The apparent activation energy equals the intrinsic activation energy because coverage no longer changes appreciably with temperature.

Remark 2.29 (Concept Check: Coverage Regime from E_{app}). An experimentalist measures $E_{\text{app}} = 85 \text{ kJ mol}^{-1}$ at $P = 1 \text{ bar}$ and $E_{\text{app}} = 40 \text{ kJ mol}^{-1}$ at $P = 0.01 \text{ bar}$ for the same reaction.

1. Is the surface coverage higher at 1 bar or 0.01 bar?
2. Can you estimate ΔH_{ads} ?

Answer: (1) Higher coverage at 1 bar (higher pressure $\rightarrow$ more adsorption). At 1 bar the high-coverage limit applies: $E_{\text{app}} = E_a = 85 \text{ kJ mol}^{-1}$ (Eq. (2.72)). At 0.01 bar, coverage is low: $E_{\text{app}} = E_a + \Delta H_{\text{ads}} = 40 \text{ kJ mol}^{-1}$ (Eq. (2.70)). (2) $\Delta H_{\text{ads}} \approx 40 - 85 = -45 \text{ kJ mol}^{-1}$ (exothermic, as expected for chemisorption).

Intermediate Coverage

At intermediate coverage, E_{app} takes values between E_a and $E_a + \Delta H_{\text{ads}}$. The exact value depends on the coverage and how it changes with temperature at the specific conditions of the experiment.

2.8.3 Python Visualization: Curved Arrhenius Plot

The following Python script generates the curved Arrhenius plot that arises from the interplay between intrinsic kinetics and temperature-dependent coverage. The two-panel figure reveals how the apparent activation energy transitions between limiting regimes.

```
1 K_ads = lambda T: K0 * np.exp(-DH_ads / (R*T))
2 theta = lambda T, P: K_ads(T)*P / (1 + K_ads(T)*P)
3 k      = lambda T: A_pre * np.exp(-Ea / (R*T))
4 r      = k(T) * theta(T, P_A)      # overall rate
```

Listing 2.6: Curved Arrhenius plot from coverage-dependent E_{app} (snippet)

Full implementation: see NB2.

Remark 2.30 (Key observations from Listing 2.6).

- **Curvature is the hallmark of LH kinetics:** A simple elementary reaction gives a perfectly straight Arrhenius plot. Curvature signals that the “effective concentration” (here, θ_A) is itself temperature-dependent.
- **Two limiting slopes are diagnostic:** At high temperature (low coverage), the slope reflects *both* the intrinsic barrier and the cost of adsorption: $E_{\text{app}} = E_a + \Delta H_{\text{ads}}$. At low temperature (saturated surface), coverage is pinned near unity and $E_{\text{app}} = E_a$.
- **The difference reveals $|\Delta H_{\text{ads}}|$:** Subtracting the two limiting slopes gives the adsorption enthalpy directly—a powerful experimental diagnostic.
- **Panel (b) connects to panel (a):** The coverage transition region in panel (b) corresponds exactly to the curvature region in the Arrhenius plot. Try changing ΔH_{ads} to -40 kJ/mol or -100 kJ/mol and observe how the transition shifts.

2.8.4 Diagnostic Value of Apparent Activation Energy

Changes in apparent activation energy provide valuable mechanistic information:

Remark 2.31 (Diagnostic Applications). The apparent activation energy can reveal:

- **Coverage regime:** If E_{app} changes with temperature or pressure, the surface is transitioning between coverage regimes
- **Adsorption enthalpy:** The difference between high-coverage and low-coverage E_{app} values equals $|\Delta H_{\text{ads}}|$
- **Mechanism changes:** Abrupt changes in E_{app} may indicate a change in rate-limiting step
- **Mass transfer limitations:** Onset of diffusion control typically shows $E_{\text{app}} \approx 10\text{--}20$ kJ/mol (activation energy for diffusion)

Remark 2.32 (The Narrow Temperature Range Trap). If the experimental temperature range lies entirely within one coverage regime—either all high-coverage or all low-coverage—the Arrhenius plot ap-

pears *perfectly straight*. The extracted slope yields E_{app} , which the experimentalist may mistakenly report as “the activation energy” without recognising that it is an apparent value conditioned on the coverage regime.

Only by spanning the coverage transition region (where θ passes through intermediate values) can one detect the curvature in the Arrhenius plot that distinguishes E_{a} from E_{app} and allows extraction of ΔH_{ads} . This is particularly insidious because the straight-line fit within either regime gives excellent R^2 values, offering no statistical warning that the extracted energy is regime-dependent.

Concrete example: For n -hexane cracking on H-ZSM-5, $E_{\text{app}} \approx 130$ kJ/mol in the low-coverage regime, reflecting $E_{\text{a}} + \Delta H_{\text{ads}} = 195 + (-65)$ kJ/mol. The intrinsic activation energy is $E_{\text{a}} \approx 195$ kJ/mol, accessible at high coverage (lower temperatures where n -hexane fills the zeolite pores). Measuring only in one temperature window would yield a perfectly linear Arrhenius plot yet miss the underlying adsorption thermodynamics entirely.

Best practice: Always report the temperature range alongside activation energy values. Where possible, collect data across both coverage regimes, or independently measure ΔH_{ads} (e.g., via calorimetry or van’t Hoff analysis of equilibrium data) to verify internal consistency.

2.8.5 Example: CO Oxidation

Consider CO oxidation on a Pt catalyst via a Langmuir–Hinshelwood mechanism. At low temperatures, the Pt surface is nearly saturated with CO (CO poisons the catalyst for O_2 adsorption). As temperature increases:

1. CO coverage decreases, opening sites for O_2 dissociation
2. The intrinsic surface reaction speeds up
3. The net effect depends on which factor dominates

Experimentally, one often observes:

- At low T : E_{app} is very high (or even the rate decreases with increasing T !) because CO desorption is required before reaction can occur
- At intermediate T : A maximum in rate may occur where coverage and temperature effects balance
- At high T : E_{app} approaches the intrinsic value as the surface becomes less inhibited

This behavior—captured by the observation of negative apparent orders and complex temperature dependence—is a hallmark of LH kinetics and motivates the need for careful kinetic analysis to extract intrinsic parameters.

2.8.6 Extracting Intrinsic Parameters

To determine the intrinsic activation energy E_a and the adsorption enthalpy ΔH_{ads} separately:

1. **Vary both temperature and pressure:** Measure rates over a range of T and P_A
2. **Fit the full LH rate expression:** Use nonlinear regression to fit:

$$r = \frac{k(T) \cdot K(T) \cdot P_A}{1 + K(T) \cdot P_A} \quad (2.73)$$

with $k(T) = A \exp(-E_a/RT)$ and $K(T) = K_0 \exp(-\Delta H_{\text{ads}}/RT)$

3. **Determine E_{app} per regime:** If possible, work in limiting regimes (low or high coverage) to simplify interpretation

The companion notebook (NB2) demonstrates these fitting procedures with synthetic and experimental data.

The following analysis extends to bimolecular surface reactions, where the interplay of two adsorbing species creates richer behaviour. Focus on the physical logic—which species dominates the surface—rather than memorising all the limiting cases.

2.8.7 Bimolecular LH Kinetics

For bimolecular LH reactions with rate:

$$r = k\theta_A\theta_B = k \frac{K_A P_A \cdot K_B P_B}{(1 + K_A P_A + K_B P_B)^2} \quad (2.74)$$

the apparent activation energy depends on the coverage regime in more complex ways:

- **Both low coverage** ($K_A P_A, K_B P_B \ll 1$): The denominator ≈ 1 , so $r \approx k K_A P_A K_B P_B$ and

$$E_{\text{app}} = E_a + \Delta H_{\text{ads,A}} + \Delta H_{\text{ads,B}}$$

- **A saturated, B low** ($K_A P_A \gg K_B P_B$ and $K_A P_A \gg 1$, but $K_B P_B \ll K_A P_A$): The denominator $\approx (K_A P_A)^2$, so $r \approx k K_B P_B / (K_A P_A)$ and

$$E_{\text{app}} = E_a - \Delta H_{\text{ads,A}} + \Delta H_{\text{ads,B}}$$

- **B saturated, A low** ($K_B P_B \gg K_A P_A$ and $K_B P_B \gg 1$): By symmetry, $r \approx k K_A P_A / (K_B P_B)$ and

$$E_{\text{app}} = E_{\text{a}} + \Delta H_{\text{ads,A}} - \Delta H_{\text{ads,B}}$$

Note that the case “both saturated” ($\theta_A \approx 1$ and $\theta_B \approx 1$ simultaneously) is *not* a physically meaningful regime in competitive adsorption, because both species compete for the same sites: $\theta_A + \theta_B + \theta_* = 1$, so both coverages cannot approach unity at the same time. When both $K_i P_i \gg 1$, the species with the larger $K_i P_i$ dominates the surface, reducing the other to a minority adsorbate. This is captured by the “A saturated” and “B saturated” cases above.

The negative signs in some terms arise because increasing temperature decreases coverage of the dominating species, opening sites for the minority co-reactant. These relationships provide powerful diagnostics but require careful experimental design to exploit fully.

Remark 2.33 (Brønsted–Evans–Polanyi: Activation Energy Meets Thermodynamics). For families of related reactions on different catalysts, a linear relationship often exists between the activation energy and the reaction energy:

$$E_{\text{a}} = \alpha \Delta E_{\text{rxn}} + \beta$$

where ΔE_{rxn} is the dissociation energy on the surface (the energy difference between the dissociated state and the gas-phase molecule) and α is a transfer coefficient ($0 < \alpha < 1$; note that this α is unrelated to the poisoning ratio defined in Eq. (2.30)). This **Brønsted–Evans–Polanyi (BEP)** relation implies that catalysts binding intermediates more strongly (more negative ΔE_{rxn}) tend to have lower activation barriers—but also greater difficulty releasing products. The optimal catalyst therefore has intermediate binding strength: strong enough to activate reactants, weak enough to release products. This is the **Sabatier principle**, which underlies the volcano plots used in catalyst design and screening. We will revisit this concept when discussing selectivity in Chapter 5 and catalyst optimization in later chapters.

Example 2.4: Identifying the Coverage Regime from E_{app}

Problem: The overall rate of a unimolecular LH reaction on a supported Rh catalyst was measured at three temperatures:

T (K)	r (mol·g ⁻¹ ·s ⁻¹)
450	3.5×10^{-4}
500	8.2×10^{-4}
550	1.6×10^{-3}

The intrinsic activation energy for the surface reaction step is known to be $E_a = 85$ kJ/mol, and the adsorption enthalpy is $\Delta H_{\text{ads}} = -50$ kJ/mol.

- Plot $\ln(r)$ versus $1/T$ and extract the apparent activation energy E_{app} .
- Compare E_{app} with the two limiting cases to identify the coverage regime.
- Is the catalyst surface predominantly empty or full under these conditions?

Solution (Arrhenius Analysis of Overall Rate):

- Prepare the linearization data:

T (K)	$10^3/T$ (K ⁻¹)	$\ln(r)$
450	2.222	-7.958
500	2.000	-7.106
550	1.818	-6.438

Linear regression of $\ln(r)$ versus $1/T$:

$$\text{slope} = -3.76 \times 10^3 \text{ K}$$

$$E_{\text{app}} = -\text{slope} \times R = (3.76 \times 10^3)(8.314) = 31 \text{ kJ/mol}$$

Python calculation:

```
import numpy as np
from scipy.stats import linregress
T = np.array([450, 500, 550])
r = np.array([3.5e-4, 8.2e-4, 1.6e-3])
res = linregress(1/T, np.log(r))
E_app = -res.slope * 8.314
print(f"E_app = {E_app/1e3:.1f} kJ/mol")
```

(b) Compare with the two limiting cases (Eqs. (2.70) and (2.72)):

Coverage regime	E_{app}
High coverage ($\theta_A \approx 1$)	$E_a = 85 \text{ kJ/mol}$
Low coverage ($\theta_A \ll 1$)	$E_a + \Delta H_{\text{ads}} = 85 + (-50) = 35 \text{ kJ/mol}$
Measured	31 kJ/mol

The measured $E_{\text{app}} = 31 \text{ kJ/mol}$ is very close to the low-coverage limit of 35 kJ/mol.

(c) Since $E_{\text{app}} \approx E_a + \Delta H_{\text{ads}}$, the surface is in the **low-coverage regime** ($\theta_A \ll 1$) over this temperature range. The rate expression is approximately $r \approx kK P_A$, and the temperature dependence reflects both the surface reaction barrier *and* the decreasing equilibrium coverage.

Answer:

$E_{\text{app}} = 31 \text{ kJ/mol}$; **low-coverage regime** ($\theta_A \ll 1$)

Physical interpretation: This diagnostic is powerful: without measuring coverage directly, we determined the adsorption regime from kinetic data alone. In practice, you would verify by checking whether E_{app} changes when you increase P_A (pushing coverage higher). If E_{app} increases toward E_a at higher pressures, the interpretation is confirmed.

2.9 Computational Exploration: Notebook NB2

The Python notebook NB2.ipynb accompanies this chapter. It provides hands-on computational exercises that reinforce and extend the analytical results developed in each section. The notebook is organized into six modules, mirroring the structure of the chapter.

1. Competitive Langmuir: Poisoning Sensitivity Sweeps (Section 2.3)

Sweep the poison binding constant K_I from 10 to 10^6 bar^{-1} and visualize how the critical poison pressure P_I^{crit} shifts on a logarithmic scale. Reproduce Fig. 2.1 and verify the analytical formula $P_I^{\text{crit}} = (1 + K_A P_A)/K_I$.

2. Dissociative vs. Molecular Isotherm Comparison (Section 2.4)

Plot molecular and dissociative Langmuir isotherms on the same axes for several K values. Identify the crossover pressure where both models predict the same coverage and explain why the dissociative isotherm rises more steeply at low pressures but saturates more gradually.

3. MVK Rate Surfaces: $r(P_A, P_{O_2})$ Contour Maps (Section 2.5)

Generate 2D contour plots of the MVK rate as a function of P_A and P_{O_2} for different $k_{\text{red}}/k_{\text{ox}}$ ratios. Identify the transition from reduction-limited to reoxidation-limited regimes and observe how the rate maximum shifts.

4. KIE Bar Chart Analysis (Section 2.6.1)

Given experimental k_H and k_D values for several candidate steps, compute $\text{KIE} = k_H/k_D$ and display as a bar chart. Compare KIE values against primary (>2) and secondary (≈ 1) thresholds to identify which step involves bond cleavage.

5. Side-by-Side Arrhenius vs. Eyring Fitting (Section 2.7)

Reproduce the analysis from Listing 2.5: generate noisy synthetic data from known Eyring parameters, perform both Arrhenius and Eyring regressions, and compare the extracted E_a against $\Delta H^\ddagger + RT$. Explore how noise level and temperature range affect parameter reliability.

6. Curved Arrhenius Plot: Coverage Regime Transitions (Section 2.8)

Reproduce Listing 2.6 and extend it: overlay curves for three different pressures ($P_A = 0.01, 1, 100$ bar) on the same Arrhenius plot. Observe how higher pressure shifts the transition to higher temperatures and extends the high-coverage ($E_{\text{app}} = E_a$) regime.

Remark 2.34 (Reflection questions for NB2). As you work through the notebook, consider:

- In the MVK contour map, what happens when $k_{\text{red}} \gg k_{\text{ox}}$? Which gas-phase pressure controls the rate in this limit, and why?
- The Eyring analysis extracts both $\Delta H^\ddagger$ and $\Delta S^\ddagger$, while Arrhenius gives only E_a and A . Under what experimental conditions (temperature range, noise level) does the additional information from Eyring analysis become unreliable?
- In the curved Arrhenius exercise, you overlay multiple pressures. Can you design an experiment—varying both T and P_A —that would let you extract E_a and ΔH_{ads} independently from a single

dataset?

2.10 Summary

- **Competitive adsorption and poisoning (Section 2.3):** The two-component Langmuir model shows that a strongly adsorbing inhibitor can suppress reactant coverage far below its gas-phase mole fraction. The dimensionless **poisoning ratio** $\alpha = K_I P_I / (K_A P_A)$ quantifies inhibition severity: when $\alpha \gg 1$, trace impurities dominate the surface. Python sensitivity plots reveal how α amplifies or dampens with changes in feed composition and temperature.
- **Dissociative adsorption (Section 2.4):** Molecules that dissociate upon adsorption (e.g., H_2 on Pd) follow $\theta = \sqrt{KP} / (1 + \sqrt{KP})$, producing a $\sqrt{P}$ dependence at low pressure—a diagnostic **fingerprint** distinguishing dissociative from molecular adsorption. Computational comparison of the two isotherms highlights the slower approach to saturation in the dissociative case.
- **Mars–van Krevelen mechanism (Section 2.5):** Lattice oxygen participates as a reactant, creating a redox cycle with two half-reactions: catalyst reduction by substrate (r_{red}) and reoxidation by gas-phase O_2 (r_{ox}). At steady state the overall rate behaves as two resistances in series: $1/r = 1/r_{\text{red}} + 1/r_{\text{ox}}$. Python rate maps over (P_A, P_{O_2}) space identify reduction-controlled vs. reoxidation-controlled regimes.
- **Experimental tools for mechanism elucidation (Section 2.6):** The **kinetic isotope effect** ($\text{KIE} = k_H/k_D$) indicates whether bond breaking is rate-limiting: $\text{KIE} \approx 2\text{--}7$ for primary effects, ≈ 1 when the isotopically substituted bond is not involved in the RDS. **Operando spectroscopy** (IR, Raman, XAS) identifies surface intermediates under working conditions, providing direct evidence for proposed mechanistic steps.
- **Eyring transition-state theory (Section 2.7):** The Eyring equation $k = (k_B T/h) \exp(\Delta S^\ddagger/R) \exp(-\Delta H^\ddagger/RT)$ separates enthalpic and entropic contributions to the rate constant. Plotting $\ln(k/T)$ vs. $1/T$ gives slope $= -\Delta H^\ddagger/R$ and intercept containing $\Delta S^\ddagger$. Side-by-side Arrhenius–Eyring regression extracts both E_a and $(\Delta H^\ddagger, \Delta S^\ddagger)$ from the same dataset.
- **Apparent vs. intrinsic activation energy (Section 2.8):** The **observed** (apparent) activation energy E_{app} depends on the coverage regime. At low coverage, $E_{\text{app}} = E_a + \Delta H_{\text{ads}}$; at high coverage,

$E_{\text{app}} \approx E_a$. This means the *same catalyst* can show different slopes on an Arrhenius plot at different temperatures. Curved Arrhenius plots signal a transition between coverage-limited and surface-reaction-limited kinetics.

2.11 Notation Introduced in This Chapter

Table 2.2: Notation introduced in Chapter 2

Symbol	Description	Typical Units
α	Poisoning ratio, $K_I P_I / (K_A P_A)$	dimensionless
θ_I	Coverage of inhibitor species I	dimensionless
r_{red}	MVK reduction half-reaction rate	$\text{mol s}^{-1} \text{g}_{\text{cat}}^{-1}$
r_{ox}	MVK reoxidation half-reaction rate	$\text{mol s}^{-1} \text{g}_{\text{cat}}^{-1}$
KIE	Kinetic isotope effect, k_H/k_D	dimensionless
$\Delta H^\ddagger$	Activation enthalpy (Eyring)	kJ mol^{-1}
$\Delta S^\ddagger$	Activation entropy (Eyring)	$\text{J mol}^{-1} \text{K}^{-1}$
h	Planck constant	J s
E_{app}	Apparent activation energy	kJ mol^{-1}

2.12 Problems

1. Competitive Adsorption: Poisoning Sensitivity Sweep

A Pt catalyst is used for propane dehydrogenation with $K_{\text{C}_3} = 3 \text{ bar}^{-1}$ and $P_{\text{C}_3} = 0.5 \text{ bar}$. The feed may contain trace sulfur impurities.

- Write a Python function `competitive_langmuir(P_A, K_A, P_I, K_I)` that returns θ_A , θ_I , and θ_* .
- Use your function to sweep P_S from 10^{-6} to 10^{-1} bar for $K_S = 10^3, 10^4$, and 10^5 bar^{-1} . Plot θ_{C_3} versus P_S on a semilog- x axis using the Wong 2011 palette.
- For each K_S , compute the critical poison pressure P_S^{crit} at which propane coverage drops to half its clean value. Verify numerically against $P_S^{\text{crit}} = (1 + K_A P_A) / K_S$.
- At what sulfur concentration (in ppm, assuming 1 bar total pressure) does the rate drop below 10% of its clean value for $K_S = 10^4 \text{ bar}^{-1}$?

Hint: NB2 provides `competitive_langmuir()` — use it directly or extend it to return θ_I and θ_ .*

2. Dissociative vs. Molecular Adsorption: Model Discrimination

Hydrogen uptake on a Ni/SiO₂ catalyst was measured at 400 K:

P_{H_2} (bar)	θ
0.005	0.26
0.02	0.45
0.05	0.58
0.10	0.67
0.50	0.85
1.00	0.90

- Write Python functions `molecular_langmuir(P, K)` and `dissociative_langmuir(P, K)`, each returning θ .
- Fit the data to both models using `scipy.optimize.curve_fit`. Report K and R^2 for each fit.
- Plot the data alongside both fitted curves on the same axes. Use the Wong 2011 palette.
- Create a diagnostic plot: plot θ vs P and θ vs $\sqrt{P}$ side by side. Which model gives a linear relationship at low pressure?
- Based on your analysis, is the adsorption molecular or dissociative? Is this consistent with the known chemistry of H₂ on Ni?
Hint: NB2 provides `langmuir_coverage()`. Write a dissociative variant and fit with `curve_fit`.

3. Mars–van Krevelen Rate Analysis

The oxidation of toluene over a V₂O₅/TiO₂ catalyst follows MVK kinetics. The following steady-state rate data were measured at 623 K:

P_{tol} (bar)	P_{O_2} (bar)	r ($\mu\text{mol g}^{-1} \text{s}^{-1}$)
0.01	0.20	2.8
0.02	0.20	5.1
0.05	0.20	9.7
0.02	0.05	3.2
0.02	0.10	4.2
0.02	0.40	5.8

The MVK rate expression (Eq. (2.47)), assuming $n = 1$ (molecular oxygen incorporation), is:

$$r = \frac{k_{\text{red}} k_{\text{ox}} P_A P_{\text{O}_2}}{k_{\text{red}} P_A + k_{\text{ox}} P_{\text{O}_2}}$$

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- (a) Linearize the MVK rate expression by taking $1/r$. Show that:

$$\frac{1}{r} = \frac{1}{k_{\text{red}} P_A} + \frac{1}{k_{\text{ox}} P_{O_2}}$$

- (b) Using the first three data points (constant P_{O_2}), plot $1/r$ versus $1/P_{\text{tol}}$ and extract k_{red} .
- (c) Using data points 2 and 4–6 (constant P_{tol}), plot $1/r$ versus $1/P_{O_2}$ and extract k_{ox} .
- (d) Write a Python script that performs both regressions and prints k_{red} and k_{ox} with units.
- (e) Is this catalyst reduction-limited or reoxidation-limited under typical conditions ($P_{\text{tol}} = 0.02$ bar, $P_{O_2} = 0.20$ bar)? Compare $k_{\text{red}} P_{\text{tol}}$ with $k_{\text{ox}} P_{O_2}$.
- Hint: Validate with `muk_rate()` from NB2. Note: `muk_rate(P_A, P_O2, k_ox, k_red)` takes `k_ox` before `k_red`.*

4. Kinetic Isotope Effect (KIE) Analysis

In ethylene hydrogenation over Pd/C, two candidate rate-determining steps are considered:

- (i) **H₂ dissociation:** $\text{H}_2 + 2* \rightarrow 2\text{H}^*$
(ii) **Ethylene insertion:** $\text{C}_2\text{H}_4 + \text{H}^* \rightarrow \text{C}_2\text{H}_5^*$

An experiment comparing H₂ and D₂ as the hydrogen source gives:

$$k_H = 4.2 \times 10^3 \text{ s}^{-1} \quad (\text{with H}_2)$$
$$k_D = 1.1 \times 10^3 \text{ s}^{-1} \quad (\text{with D}_2)$$

- (a) Calculate the kinetic isotope effect $\text{KIE} = k_H/k_D$.
- (b) Is this a primary or secondary KIE? Justify your classification using the thresholds from Eq. (2.51).
- (c) Which of the two candidate RDS is consistent with this KIE? Explain why.
- (d) Write a short Python script that creates a bar chart comparing k_H and k_D , annotated with the KIE value. Use `BLUE = '#0072B2'` and `ORANGE = '#E69F00'`.

Hint: The Wong 2011 palette constants are defined in NB2 as the `COLORS` dictionary.

5. Eyring Analysis: Activation Parameters for CO Oxidation

The rate constant for CO oxidation on a Pd/Al₂O₃ catalyst was measured over a range of temperatures:

T (K)	k (s^{-1})
400	2.5×10^1
425	9.8×10^1
450	3.4×10^2
475	1.1×10^3
500	3.0×10^3

- Prepare a table with columns: T , k , $1/T$, $\ln(k)$, $\ln(k/T)$.
- Perform both Arrhenius ($\ln k$ vs. $1/T$) and Eyring ($\ln(k/T)$ vs. $1/T$) linear regressions using Python. Report E_a , $\Delta H^\ddagger$, and $\Delta S^\ddagger$ with units.
- Verify the relationship $E_a \approx \Delta H^\ddagger + R\bar{T}$ where $\bar{T}$ is the mean temperature.
- Interpret the sign of $\Delta S^\ddagger$. Is the transition state “tight” or “loose”? What does this suggest about the mechanism?
- Using your fitted Eyring parameters, predict the rate constant at $T = 550$ K. Compare this with the Arrhenius prediction. How large is the discrepancy?

Hint: NB2 provides `full_temperature_analysis(T, k)` for both Arrhenius and Eyring regressions with diagnostic plots.

6. Apparent vs. Intrinsic Activation Energy from Curved Arrhenius Data

The overall rate of a unimolecular LH reaction was measured over a wide temperature range at $P_A = 0.5$ bar:

T (K)	r ($\text{mol}\cdot\text{g}^{-1}\cdot\text{s}^{-1}$)
350	6.0×10^{-6}
400	1.4×10^{-4}
450	1.8×10^{-3}
500	1.1×10^{-2}
550	3.0×10^{-2}
600	5.8×10^{-2}
650	9.2×10^{-2}

- Plot $\ln(r)$ versus $1/T$. Does the plot appear linear? Describe the curvature qualitatively.
- Fit a straight line to the three lowest-temperature points (350–450 K) and extract $E_{\text{app,low}}$.
- Fit a straight line to the three highest-temperature points (550–650 K) and extract $E_{\text{app,high}}$.

2. Computational Perspectives on Surface Catalysis

- (d) The difference $E_{\text{app,low}} - E_{\text{app,high}}$ should approximate $|\Delta H_{\text{ads}}|$. What is your estimate?
- (e) Write a Python script that reproduces the full analysis: plot the data with both tangent lines on a single Arrhenius plot, annotated with the two slopes. Use colors `'#0072B2'` (data), `'#D55E00'` (high- T tangent), and `'#009E73'` (low- T tangent).
Hint: NB2 provides `lh_rate_unimolecular_T()` for computing the temperature-dependent rate with combined Arrhenius $k(T)$ and van 't Hoff $K(T)$.

7. (Optional) Temperature Dependence of MVK Kinetics

Consider the toluene oxidation catalyst from Problem 3, now studied over a temperature range. Assume both rate constants follow Eyring kinetics:

$$k_{\text{red}}(T) = \frac{k_{\text{B}}T}{h} \exp\left(\frac{\Delta S_{\text{red}}^{\ddagger}}{R}\right) \exp\left(-\frac{\Delta H_{\text{red}}^{\ddagger}}{RT}\right)$$
$$k_{\text{ox}}(T) = \frac{k_{\text{B}}T}{h} \exp\left(\frac{\Delta S_{\text{ox}}^{\ddagger}}{R}\right) \exp\left(-\frac{\Delta H_{\text{ox}}^{\ddagger}}{RT}\right)$$

with parameters: $\Delta H_{\text{red}}^{\ddagger} = 60 \text{ kJ/mol}$, $\Delta S_{\text{red}}^{\ddagger} = -30 \text{ J/(mol K)}$, $\Delta H_{\text{ox}}^{\ddagger} = 45 \text{ kJ/mol}$, $\Delta S_{\text{ox}}^{\ddagger} = -80 \text{ J/(mol K)}$.

- (a) Write a Python function `mvk_rate(T, P_A, P_O2)` that computes the MVK rate using the temperature-dependent rate constants.
- (b) Plot r versus T for $T = 400\text{--}800 \text{ K}$ at fixed $P_{\text{tol}} = 0.02 \text{ bar}$, $P_{\text{O}_2} = 0.20 \text{ bar}$.
- (c) Construct an Arrhenius plot of $\ln(r)$ versus $1/T$. Is the plot linear, or does it show curvature? Explain why.
- (d) At what temperature does the rate-limiting step switch from reduction to reoxidation? Identify this as the crossover point where $k_{\text{red}}P_A = k_{\text{ox}}P_{\text{O}_2}$.
- (e) Generate a 2D contour plot of $r(T, P_A)$ at fixed $P_{\text{O}_2} = 0.20 \text{ bar}$. How does the optimal operating temperature depend on toluene concentration?

Hint: Combine `eyring()` from `utils.py` with `mvk_rate()` from NB2 to build the temperature-dependent MVK function.

Looking Ahead

This chapter developed computational tools for adsorption equilibria, surface kinetics, and temperature dependence. The rate expressions derived

here—Langmuir–Hinshelwood, Eley–Rideal, and Mars–van Krevelen—are the inputs to every reactor design and catalyst evaluation problem that follows.

Ideal Reactor Design (Chapter 3)

The LH and ER rate laws enter directly into the CSTR and PFR design equations. For a CSTR, $\tau = C_{A,0} X / r(C_{A,\text{exit}})$ becomes an algebraic equation with the Langmuir denominator from this chapter. For a PFR, the ODE $dX/d\tau = r(C_A)/C_{A,0}$ must be integrated numerically when the rate expression is nonlinear in concentration.

Catalyst Characterization (Chapter 3)

The adsorption isotherms reviewed here underpin key characterization techniques: BET surface area measurement extends the Langmuir model to multilayer physisorption, chemisorption titrations use molecular and dissociative isotherms to count active sites, and Temperature-Programmed Desorption connects peak temperatures to adsorption enthalpies via the Arrhenius framework.

Transport Limitations (Chapter 4)

Inside porous catalyst pellets, simultaneous diffusion and reaction create concentration gradients that reduce the effective rate. The Thiele modulus ϕ and effectiveness factor η quantify this penalty. Because the intrinsic rate depends on local coverage—governed by the Langmuir isotherm—the severity of transport limitations is coupled to the adsorption parameters K and k developed here.

Selectivity Engineering (Chapter 5)

When parallel or consecutive reactions compete on the same catalyst, the relative coverages of reactants and intermediates determine which products form preferentially. The competitive Langmuir framework and the poisoning ratio α from this chapter generalize to selectivity parameters: adjusting partial pressures or temperature shifts which species dominate the surface, steering the product distribution toward the desired outcome.

Chapter 3

Reactor Design and Catalyst Characterization

Learning Objectives

Every chemist who measures a reaction rate in a flask is performing batch reactor kinetics. This chapter formalises what you already do in the laboratory—extracting rate constants, determining reaction orders, and comparing catalyst performance—and extends these ideas to continuous-flow reactors used in industrial catalysis. The turnover frequency (TOF) bridges bench-scale chemistry and large-scale production by normalising activity to a per-site basis.

By the end of this chapter, you will be able to:

Part I: Reactor Design

- LO1. Derive** the batch reactor mass balance and apply it to extract kinetic parameters from experimental concentration-time data
- LO2. Apply** integrated rate laws (zeroth, first, and second order) to determine reaction order and rate constants from batch experiments
- LO3. Calculate** TOF and TON (turnover frequency and number) to compare catalysts
- LO4. Describe** the difference between differential and integral methods for kinetic analysis
- LO5. Apply** temperature-dependent rate constants from Chapter 2 to predict reactor performance across operating temperatures
- LO6. Explain** the concept of contact time in flow reactors and how differential reactor operation extracts kinetic data

Part II: Catalyst Characterization

- LO7. Apply** the BET equation to determine surface area from nitrogen adsorption isotherms and interpret the BET constant in terms of adsorption energetics
- LO8. Explain** the BJH method and classify pores as micropores (<2 nm), mesopores (2–50 nm), or macropores (>50 nm)
- LO9. Interpret** TPD profiles to extract desorption energies and surface site distributions using the Redhead equation
- LO10. Interpret** TPR and TPO profiles qualitatively to identify reducible/oxidizable species
- LO11. Use** X-ray diffraction (XRD) with the Scherrer equation to determine crystallite sizes and phase composition
- LO12. Interpret** XPS spectra to determine surface elemental composition, oxidation states, and chemical environments
- LO13. Evaluate** electron microscopy images (TEM, SEM) for particle size, morphology, and dispersion measurements
- LO14. Analyze** FTIR spectra of adsorbed probe molecules (especially CO) to characterize active site types and distributions

Chemistry connection: The integrated rate laws in this chapter are identical to those used in enzyme kinetics ($v = -d[S]/dt$) and in physical organic chemistry (pseudo-first-order conditions). If you have measured rate constants in a teaching laboratory, you have already practised batch reactor kinetics.

3.1 Introduction

In Chapter 2, we developed rate expressions for catalytic reactions—the Langmuir-Hinshelwood, Eley-Rideal, and Mars-van Krevelen mechanisms that describe how fast reactions occur on surfaces. These rate laws express reaction velocity as a function of concentration (or partial pressure) and temperature. But how do we *measure* these rates experimentally? How do we extract the parameters that define a rate law from laboratory data?

In Chapter 2, we explored how rate constants depend on temperature through the Arrhenius and Eyring frameworks. This chapter focuses on how to *measure* rate constants experimentally and design reactors that achieve target conversions.

We begin by focusing on **batch reactor kinetics**—the workhorse of the research laboratory. Batch experiments, where a fixed amount of reactant is

monitored over time, provide the direct connection between theory and experiment. From a simple concentration-versus-time curve, we can determine reaction order, rate constants, and activation energies. More importantly for catalysis, we can calculate the **turnover frequency (TOF)**—the number of reaction events per active site per unit time—which is the gold standard for comparing catalyst activity.

Understanding batch kinetics is essential for mechanistic studies. When you propose a mechanism in Chapter 2, it predicts a specific functional form for the rate law. Batch experiments let you test that prediction. Does the reaction follow first-order kinetics, suggesting a single rate-determining step? Does the rate saturate at high concentration, indicating Langmuir-Hinshelwood behavior? Does inhibition occur at high product concentration? The answers come from careful kinetic analysis of batch data.

We will also briefly examine flow systems, which are essential for industrial applications and increasingly important in modern catalysis research. However, our emphasis remains on the batch experiments that underpin mechanistic understanding and catalyst development.

In summary, this chapter develops two complementary skill sets. First, you will learn to extract kinetic parameters—rate constants, reaction orders, and activation energies—from batch reactor data, the bread and butter of laboratory kinetics. Second, you will meet the flow reactor concepts (contact time, differential reactor operation) needed to interpret modern catalytic data. The bridge between laboratory measurements and catalyst comparison is the **turnover frequency (TOF)**, which normalises catalytic activity to a per-site basis and enables meaningful comparison across different catalysts and reaction conditions.

3.2 Batch Reactor Kinetics

3.2.1 The Batch Reactor Mass Balance

A batch reactor is a closed vessel in which reactants are charged at time $t = 0$ and allowed to react. There is no flow in or out during the reaction. For a well-stirred batch reactor, concentration is uniform throughout the vessel at any instant.

The fundamental mass balance for reactant A is:

$$\boxed{\frac{dN_A}{dt} = V \cdot r_A} \quad (3.1)$$

where N_A is the number of moles of A, V is the reactor volume, and r_A is the rate of formation of A (negative for a reactant being consumed). For constant volume (typical for liquid-phase reactions), we can write:

$$\boxed{-\frac{dC_A}{dt} = r(C_A)} \quad (3.2)$$

where $r \equiv -r_A > 0$ is the rate of *consumption* of A. This sign convention ensures that r is positive when reactant is being consumed, which is the case for all forward reactions.

Remark 3.1 (Key Teaching Point). Eq. (3.2) is the master equation for batch kinetic analysis. It states that the rate at which concentration decreases equals the reaction rate. By measuring $C_A(t)$, we can determine r and thereby extract kinetic parameters.

3.2.2 Integrated Rate Laws

For simple power-law kinetics $r = kC_A^n$, we can integrate Eq. (3.2) analytically. The integrated forms provide the basis for determining reaction order from experimental data.

Zeroth-Order Kinetics ($n = 0$)

When the rate is independent of concentration:

$$r = k \quad (3.3)$$

Integration gives:

$$\boxed{C_A = C_{A0} - kt} \quad (3.4)$$

A plot of C_A versus t is linear with slope $-k$ and intercept C_{A0} .

When does zeroth-order apply? In catalysis, zeroth-order kinetics often arise when the surface is saturated with adsorbed reactant ($\theta \approx 1$). In this limit, the Langmuir-Hinshelwood rate becomes independent of gas-phase concentration because all sites are already occupied.

First-Order Kinetics ($n = 1$)

The most commonly encountered rate law:

$$r = kC_A \quad (3.5)$$

Separating variables and integrating:

$$\int_{C_{A0}}^{C_A} \frac{dC'_A}{C'_A} = -k \int_0^t dt' \quad (3.6)$$

yields:

$$\ln\left(\frac{C_A}{C_{A0}}\right) = -kt \quad \text{or equivalently} \quad C_A = C_{A0}e^{-kt} \quad (3.7)$$

A plot of $\ln(C_A)$ versus t is linear with slope $-k$.

When does first-order apply? First-order kinetics arise in catalysis when coverage is proportional to pressure (low coverage regime of the Langmuir isotherm) or when a single unimolecular surface step is rate-determining.

Second-Order Kinetics ($n = 2$)

For a reaction second-order in a single reactant:

$$r = kC_A^2 \quad (3.8)$$

Integration gives:

$$\frac{1}{C_A} - \frac{1}{C_{A0}} = kt \quad \text{or equivalently} \quad \frac{1}{C_A} = \frac{1}{C_{A0}} + kt \quad (3.9)$$

A plot of $1/C_A$ versus t is linear with slope k and intercept $1/C_{A0}$.

Remark 3.2 (Linearization Strategy for Kinetic Analysis). The linearization strategy is the same in all cases: identify a function of concentration that varies linearly with time. This is the basis of the **integral method** for determining reaction order.

Remark 3.3 (From Rate Laws to Surface Mechanisms). The observed kinetic order encodes the surface coverage regime (Section 2.2):

- **Zeroth order** ($r = k$): surface saturated ($\theta_A \approx 1$); increasing gas-phase concentration cannot increase the rate.
- **First order** ($r = kC_A$): low coverage ($\theta_A \ll 1$, $\theta_A \approx K_A P_A$); rate proportional to pressure.
- **Fractional order** (e.g., $n = 0.5$): often diagnostic of dissociative adsorption ($\theta \propto P^{1/2}$ at low coverage; Section 2.4).

By varying C_{A0} across experiments and observing how the apparent order changes, one can map the transition from low-coverage to saturation—a direct window into the Langmuir isotherm.

3.2.3 Half-Life Analysis

The **half-life** $t_{1/2}$ is the time required for the concentration to decrease to half its initial value. Half-life analysis provides a simple diagnostic for reaction order.

Half-Life Expressions

Setting $C_A = C_{A0}/2$ in each integrated rate law:

Zeroth-order:

$$t_{1/2} = \frac{C_{A0}}{2k} \quad (3.10)$$

The half-life is proportional to initial concentration.

First-order:

$$t_{1/2} = \frac{\ln 2}{k} \approx \frac{0.693}{k} \quad (3.11)$$

The half-life is *independent* of initial concentration—a unique signature of first-order kinetics.

Second-order:

$$t_{1/2} = \frac{1}{kC_{A0}} \quad (3.12)$$

The half-life is inversely proportional to initial concentration.

Using Half-Life to Determine Reaction Order

The diagnostic power of half-life analysis is clear: run experiments at different initial concentrations and observe how $t_{1/2}$ changes.

- If $t_{1/2}$ increases with C_{A0} : zeroth-order
- If $t_{1/2}$ is independent of C_{A0} : first-order
- If $t_{1/2}$ decreases with C_{A0} : second-order (or higher)

More generally, for $r = kC_A^n$, the half-life scales as:

$$t_{1/2} \propto C_{A0}^{1-n} \quad (3.13)$$

Taking logarithms: $\ln(t_{1/2}) = \text{const} + (1 - n) \ln(C_{A0})$. A plot of $\ln(t_{1/2})$ versus $\ln(C_{A0})$ has slope $(1 - n)$, allowing determination of non-integer orders.

3.2.4 Differential vs. Integral Methods

Two complementary approaches exist for extracting kinetic parameters from batch data:

The Integral Method

We have already seen this approach: assume a rate law, integrate it, and test whether the data fit the predicted linear form.

Procedure:

1. Hypothesize a reaction order n
2. Plot the appropriate function: C_A (if $n = 0$), $\ln C_A$ (if $n = 1$), or $1/C_A$ (if $n = 2$) versus t
3. If the plot is linear, the hypothesis is confirmed; extract k from the slope
4. If not linear, try a different order

Advantages: Uses all data points; less sensitive to experimental noise; directly gives k .

Disadvantages: Requires guessing the correct order; may not distinguish between similar orders.

The Differential Method

This approach determines the rate directly at each data point, then fits the rate-concentration relationship.

Procedure:

1. From $C_A(t)$ data, calculate the rate $r = -dC_A/dt$ at each point (e.g., by numerical differentiation or graphical tangent)
2. Plot $\ln(r)$ versus $\ln(C_A)$
3. For $r = kC_A^n$, this plot is linear with slope n and intercept $\ln(k)$

Advantages: Does not require assuming an order; directly determines n ; works for non-integer orders.

Disadvantages: Numerical differentiation amplifies noise and requires smooth or pre-smoothed data.

Remark 3.4 (Practical Recommendation). For high-quality data, use the differential method to establish reaction order, then use the integral method to obtain a precise value of k . When data are noisy, rely on the integral method with several trial orders.

Remark 3.5 (The Method of Initial Rates). A powerful alternative to following a single experiment over time is to run *multiple* batch experiments, each at a different initial concentration C_{A0} , and measure only the initial rate $r_0 = -dC_A/dt|_{t=0}$ from each run.

For a power-law rate expression $r = kC_A^n$, we have $r_0 = kC_{A0}^n$, so that

$$\ln(r_0) = \ln(k) + n \ln(C_{A0}) \quad (3.14)$$

A plot of $\ln(r_0)$ versus $\ln(C_{A0})$ yields a straight line with slope n and intercept $\ln(k)$.

This method has several practical advantages:

- **No product inhibition**—at $t \approx 0$, product concentrations are negligible.
- **No catalyst deactivation**—measurements are taken before significant activity loss.
- **No reverse reaction**—the system is far from equilibrium initially.
- **No need to assume integer order**— n emerges directly from the slope.

The method of initial rates is widely used in heterogeneous catalysis for determining rate law form and is also the standard approach for measuring turnover frequencies (see Section 3.3.4).

3.3 Turnover Frequency and Turnover Number

Rate constants like k depend on how we define the rate—per unit volume, per gram of catalyst, or per unit surface area. To compare catalysts fairly, we need a metric that accounts for the number of active sites. This is where **turnover frequency** and **turnover number** become essential.

3.3.1 Definitions

Definition 3.1 (Turnover Frequency (TOF)). The turnover frequency is the number of times a catalytic cycle is completed per active site per unit time:

$$\text{TOF} = \frac{\text{moles of product formed}}{\text{moles of active sites} \times \text{time}} = \frac{r_{\text{mol}}}{N_{\text{sites}}} \quad (3.15)$$

TOF has units of s^{-1} (or min^{-1} , h^{-1}).

Definition 3.2 (Turnover Number (TON)). The turnover number is the total number of catalytic cycles completed per active site over the lifetime of the catalyst (or during a specified reaction period):

$$\text{TON} = \frac{\text{total moles of product formed}}{\text{moles of active sites}} = \text{TOF} \times t \quad (3.16)$$

TON is dimensionless (mol/mol).

Remark 3.6 (TOF vs. TON: Activity vs. Productivity). The distinction is important: TOF measures activity (how fast), while TON measures productivity (how much). A catalyst with high TOF but low stability may have lower TON than a slower but more durable catalyst.

Remark 3.7 (Historical Context: From Ostwald to Volcano Plots). The TOF concept traces back to Ostwald’s Nobel Prize-winning definition of catalysis (1909). Sabatier (1912 Nobel) demonstrated that optimal catalytic activity requires intermediate binding strength—the *Sabatier principle*—which maps directly onto TOF: too-weak binding gives low coverage and low TOF; too-strong binding poisons the surface. The resulting *volcano plot* (TOF vs. binding energy) remains the central tool for computational catalyst screening, as recognised by Ertl’s Nobel Prize (2007) for molecular-level surface science.

3.3.2 Counting Active Sites

The challenge in calculating TOF lies in determining the number of active sites, N_{sites} . Several experimental techniques are used:

- **Chemisorption:** Adsorb a probe molecule (H_2 , CO , O_2) that binds specifically to active sites. The uptake, corrected for stoichiometry, gives the number of surface metal atoms. For example, CO adsorbs 1:1 on many transition metals, so CO uptake directly measures surface metal sites.
- **Titration methods:** For acid catalysts, titrate surface acid sites with a base (e.g., pyridine). For redox catalysts, use selective oxidation/reduction cycles.
- **Spectroscopy:** IR of adsorbed probe molecules can distinguish site types (e.g., terrace vs. edge sites).

- **For homogeneous catalysts:** The number of metal centers is typically known from the amount of catalyst added.

Remark 3.8 (Site Counting Uncertainty). Site counting is often the largest source of uncertainty in TOF values. Different probe molecules may give different site counts if they access different populations of sites. Always report how sites were measured when quoting TOF values.

3.3.3 Site-Specific Activity

TOF allows meaningful comparison of fundamentally different catalysts:

- A bulk metal powder versus a supported nanoparticle catalyst
- Catalysts with different metal loadings
- Homogeneous versus heterogeneous catalysts

Example comparison:

Catalyst	Rate (mol/g _{cat} /h)	TOF (h ⁻¹)
Pd/C (5%, large particles)	2.0	150
Pd/C (1%, small particles)	1.5	400

The 5% catalyst has higher rate per gram, but the 1% catalyst has higher intrinsic activity per site. The smaller particles expose more surface atoms, so each gram of Pd contributes more active sites, but each site is also more active (possibly due to different surface structure at small particle sizes).

Remark 3.9 (Structure Sensitivity (Boudart, 1969)). Catalytic reactions fall into two classes: **structure-insensitive** reactions where TOF is roughly constant regardless of particle size (e.g., CO oxidation on Pd), and **structure-sensitive** reactions where TOF depends strongly on particle size and crystal face (e.g., NH₃ synthesis on Fe, where the (111) face is ~100× more active than (110)). For structure-sensitive reactions, optimizing particle size and exposed facets—not just metal loading—is critical.

Remark 3.10 (TOF Across Catalysis: How Fast Is Fast?). To develop physical intuition for TOF magnitudes, consider typical ranges across different catalyst classes:

Catalyst type	Typical TOF range
Enzymes	$1\text{--}10^4 \text{ s}^{-1}$ (extremes: $\sim 10^6 \text{ s}^{-1}$)
Homogeneous organometallic	$10^{-2}\text{--}10^2 \text{ s}^{-1}$
Heterogeneous metals	$10^{-2}\text{--}10^2 \text{ s}^{-1}$
Heterogeneous oxide/acid	$10^{-4}\text{--}10^{-1} \text{ s}^{-1}$

Direct comparison across categories can be misleading because operating conditions (temperature, pressure, solvent) differ enormously. However, the overlap between homogeneous and heterogeneous metal catalysts is noteworthy: intrinsic per-site activity is often comparable. The practical advantage of heterogeneous catalysis lies in separation, thermal stability, and recyclability rather than superior per-site kinetics.

Remark 3.11 (Catalyst Deactivation: Why TON Is Finite). Real catalysts deactivate through three mechanisms: (1) **Poisoning**—strong chemisorption of impurities (e.g., S, thiols) blocks sites irreversibly; (2) **Sintering**—nanoparticles merge at high temperatures, reducing dispersion; (3) **Coking**—carbonaceous deposits block pores and active sites. TON is the key economic metric: pharmaceutical catalysts typically require $\text{TON} > 10^4$, while bulk chemicals need $\text{TON} > 10^6$. Note that catalyst lifetime = TON/TOF , so a fast but unstable catalyst may be less useful than a slower but durable one.

3.3.4 Calculating TOF from Batch Data

The standard procedure for determining TOF from batch experiments:

1. **Measure the reaction rate:** From batch $C_A(t)$ data, determine the rate (usually the initial rate r_0 to avoid complications from product inhibition or catalyst deactivation).

2. **Calculate moles converted per unit time:**

$$r_{\text{mol}} = r_0 \times V = -V \left. \frac{dC_A}{dt} \right|_{t=0} \quad (3.17)$$

3. **Determine the number of active sites:** From chemisorption or other characterization:

$$N_{\text{sites}} = (\text{mass of metal}) \times (\text{dispersion}) \times \frac{1}{M_{\text{metal}}} \quad (3.18)$$

where dispersion is the fraction of metal atoms on the surface.

4. Calculate TOF:

$$\text{TOF} = \frac{r_{\text{mol}}}{N_{\text{sites}}} \quad (3.19)$$

3.4 Flow Reactor Concepts

While batch reactors dominate the research laboratory, continuous flow systems are essential for industrial catalysis and increasingly common in modern synthetic chemistry (flow microreactors). This section introduces the key concepts needed to interpret flow reactor data; detailed reactor sizing is covered in chemical engineering courses.

3.4.1 Contact Time

In flow systems, the key time parameter is the **contact time**—the average time a fluid element spends in contact with the catalyst.

Definition 3.3 (Contact Time). For a flow reactor with catalyst volume (or mass) V_{cat} and volumetric flow rate $\dot{V}$:

$$t_{\text{contact}} = \frac{V_{\text{cat}}}{\dot{V}} \quad (3.20)$$

Contact time has units of time (s, min) and is the flow analog of batch reaction time: longer contact time gives higher conversion.

3.4.2 Extracting Kinetic Data from Flow Reactors

At very low conversion ($X < 5\%$), a flow reactor operates as a **differential reactor**: concentration changes negligibly across the catalyst bed, and the rate can be calculated directly:

$$r = \frac{F_{A0}X}{W} = \frac{C_{A0}\dot{V}X}{W} \quad (3.21)$$

This is the flow analog of the initial rate method in batch systems. By varying C_{A0} or $\dot{V}$ and measuring X , one can determine rate-concentration relationships without following a full time course.

3.4.3 CSTR and PFR: Two Idealized Flow Reactors

The **CSTR** (continuous stirred tank reactor) assumes perfect mixing throughout the vessel, so the exit concentration equals the concentration everywhere inside. The **PFR** (plug flow reactor) assumes no axial mixing, so concentration decreases continuously along the flow direction. For first-order kinetics

3. Reactor Design and Catalyst Characterization

($r = kC_A$), the analytical solutions are:

$$\text{CSTR: } X = \frac{k\tau}{1 + k\tau} \quad (3.22)$$

$$\text{PFR: } X = 1 - \exp(-k\tau) \quad (3.23)$$

where $\tau = V/\dot{V}$ is the space time. The PFR always achieves higher conversion than the CSTR at the same τ for positive-order kinetics, because back-mixing in the CSTR forces the entire reactor to operate at the low exit concentration.

Remark 3.12 (The Three-Way Catalytic Converter). The automotive catalytic converter is one of the most successful applications of heterogeneous catalysis. The monolith substrate contains thousands of parallel channels coated with Pt/Pd/Rh catalyst, each behaving as a small PFR. Exhaust gases (CO, NO_x, hydrocarbons) undergo simultaneous oxidation and reduction with a contact time of only 10–50 ms at highway speeds, yet >95% conversion is required to meet emission standards. This demands extremely active catalysts and careful management of transport limitations (Chapter 4).

3.5 Worked Examples

Example 3.1: Determining Reaction Order from Batch Data

Problem: A catalytic hydrogenation reaction is studied in a batch reactor. The following concentration data are obtained:

Time (min)	C_A (mol/L)
0	1.000
10	0.607
20	0.368
30	0.223
40	0.135
50	0.082

Determine the reaction order and rate constant.

Solution (Integral Method with Linearization):

We test the integrated rate laws by constructing linearization plots.

Test for first-order: Calculate $\ln(C_A)$ at each time:

t (min)	C_A (mol/L)	$\ln(C_A)$
0	1.000	0.000
10	0.607	-0.500
20	0.368	-1.000
30	0.223	-1.500
40	0.135	-2.000
50	0.082	-2.501

The $\ln(C_A)$ values decrease linearly with time: $\ln(C_A) = -0.050 t$. This confirms **first-order kinetics**.

From Eq. (3.7): slope = $-k$

$$k = 0.050 \text{ min}^{-1} = 8.3 \times 10^{-4} \text{ s}^{-1}$$

Answer:

First-order; $k = 0.050 \text{ min}^{-1}$; $t_{1/2} = 13.9 \text{ min}$

Physical interpretation: The reaction is first-order with rate constant $k = 0.050 \text{ min}^{-1}$. The half-life is $t_{1/2} = \ln(2)/k = 13.9 \text{ min}$, which can be verified from the data (at $t = 14 \text{ min}$, C_A would be approximately 0.5 mol/L).

Example 3.2: Turnover Frequency Calculation

Problem: A Pd/Al₂O₃ catalyst (2.0% Pd by mass) is used for acetylene hydrogenation in a batch reactor. The initial rate of acetylene consumption is $4.2 \times 10^{-4} \text{ mol/(L}\cdot\text{s)}$ in a 0.500 L reactor containing 0.250 g of catalyst. CO chemisorption indicates a Pd dispersion of 35% (35% of Pd atoms are on the surface). Calculate the TOF.

Solution (Site-Normalized Rate Calculation):

Step 1: Calculate moles of product formed per unit time.

$$r_{\text{mol}} = r_0 \times V = (4.2 \times 10^{-4} \text{ mol}/(\text{L}\cdot\text{s})) \times (0.500 \text{ L}) = 2.1 \times 10^{-4} \text{ mol/s}$$

Step 2: Calculate moles of surface Pd atoms.

Mass of Pd in catalyst:

$$m_{\text{Pd}} = 0.250 \text{ g} \times 0.020 = 5.0 \times 10^{-3} \text{ g}$$

Total moles of Pd ($M_{\text{Pd}} = 106.4 \text{ g/mol}$):

$$n_{\text{Pd,total}} = \frac{5.0 \times 10^{-3} \text{ g}}{106.4 \text{ g/mol}} = 4.70 \times 10^{-5} \text{ mol}$$

Moles of surface Pd (active sites):

$$N_{\text{sites}} = 4.70 \times 10^{-5} \text{ mol} \times 0.35 = 1.65 \times 10^{-5} \text{ mol}$$

Step 3: Calculate TOF.

$$\text{TOF} = \frac{r_{\text{mol}}}{N_{\text{sites}}} = \frac{2.1 \times 10^{-4} \text{ mol/s}}{1.65 \times 10^{-5} \text{ mol}} = 12.8 \text{ s}^{-1}$$

Answer:

$$\text{TOF} = 12.8 \text{ s}^{-1} (\approx 770 \text{ min}^{-1})$$

Physical interpretation: Each surface Pd atom completes about 13 catalytic cycles per second. For comparison, enzyme catalysis typically shows TOF values of 10^2 – 10^6 s^{-1} , while heterogeneous catalysts typically range from 10^{-2} – 10^2 s^{-1} .

Example 3.3: Arrhenius Parameter Extraction

Problem: A first-order catalytic reaction is studied at five temperatures. The rate constants are:

T (K)	k (s^{-1})
350	1.80×10^{-3}
370	5.83×10^{-3}
390	1.68×10^{-2}
410	4.35×10^{-2}
430	1.03×10^{-1}

Determine the activation energy and pre-exponential factor.

Solution (Arrhenius Linearization):

This example applies the Arrhenius analysis technique developed in Chapter 2 to batch reactor data, demonstrating how kinetic parameters extracted from laboratory measurements feed into reactor performance prediction.

Prepare data for Arrhenius plot:

T (K)	$1/T$ (K^{-1})	k (s^{-1})	$\ln(k)$
350	2.857×10^{-3}	1.80×10^{-3}	-6.32
370	2.703×10^{-3}	5.83×10^{-3}	-5.14
390	2.564×10^{-3}	1.68×10^{-2}	-4.09
410	2.439×10^{-3}	4.35×10^{-2}	-3.14
430	2.326×10^{-3}	1.03×10^{-1}	-2.27

Linear regression of $\ln(k)$ vs. $1/T$:

$$\ln(k) = 15.45 - 7620 \times (1/T)$$

Activation energy:

$$\text{slope} = -\frac{E_a}{R} = -7620 \text{ K}$$

$$E_a = 7620 \times 8.314 \text{ J}/(\text{mol}\cdot\text{K}) = 63.4 \text{ kJ/mol}$$

Pre-exponential factor:

$$\text{intercept} = \ln(A) = 15.45$$

$$A = e^{15.45} = 5.1 \times 10^6 \text{ s}^{-1}$$

Answer:

$$E_a = 63.4 \text{ kJ/mol}; A = 5.1 \times 10^6 \text{ s}^{-1}$$

Physical interpretation: The activation energy is typical for a surface reaction. The pre-exponential factor is in the range expected for a first-order unimolecular surface step (typically 10^6 – 10^{13} s^{-1}).

Example 3.4: Half-Life Analysis

Problem: To determine the reaction order for a catalytic decomposition, experiments are run at three different initial concentrations. The half-lives are measured:

C_{A0} (mol/L)	$t_{1/2}$ (min)
0.50	42
1.00	21
2.00	10.5

Determine the reaction order and rate constant.

Solution (Half-Life Scaling Method):

First, examine how $t_{1/2}$ changes with C_{A0} :

- When C_{A0} doubles ($0.50 \rightarrow 1.00$), $t_{1/2}$ halves ($42 \rightarrow 21$)
- When C_{A0} doubles again ($1.00 \rightarrow 2.00$), $t_{1/2}$ halves again ($21 \rightarrow 10.5$)

This inverse dependence ($t_{1/2} \propto 1/C_{A0}$) is the hallmark of **second-order** kinetics.

To confirm, use the general relationship $t_{1/2} \propto C_{A0}^{1-n}$:

Taking the ratio of half-lives:

$$\frac{t_{1/2,1}}{t_{1/2,2}} = \left(\frac{C_{A0,1}}{C_{A0,2}} \right)^{1-n}$$

Using the first two data points:

$$\frac{42}{21} = \left(\frac{0.50}{1.00} \right)^{1-n} = (0.5)^{1-n}$$

$$2 = (0.5)^{1-n} \Rightarrow 1 - n = -1 \Rightarrow n = 2$$

For second-order kinetics, from Eq. (3.12):

$$k = \frac{1}{t_{1/2} C_{A0}}$$

Using the data at $C_{A0} = 1.00$ mol/L:

$$k = \frac{1}{(21 \text{ min})(1.00 \text{ mol/L})} = 0.0476 \text{ L}/(\text{mol}\cdot\text{min})$$

Verify with other data points:

- At $C_{A0} = 0.50$: $k = 1/(42 \times 0.50) = 0.0476 \text{ L}/(\text{mol}\cdot\text{min})$ ✓
- At $C_{A0} = 2.00$: $k = 1/(10.5 \times 2.00) = 0.0476 \text{ L}/(\text{mol}\cdot\text{min})$ ✓

Answer:

Second-order; $k = 0.0476 \text{ L}/(\text{mol}\cdot\text{min})$

Physical interpretation: The inverse relationship between half-life and initial concentration is a definitive signature of second-order kinetics. In SI units, $k = 7.94 \times 10^{-4} \text{ L}/(\text{mol}\cdot\text{s})$. This diagnostic is useful when only half-life data are available, without full concentration–time profiles.

Remark 3.13 (Self-Assessment: From Data to Design). Before attempting the problems, check that you can connect the chapter's key concepts:

1. You measure the half-life of a catalytic reaction at three different initial concentrations and find $t_{1/2}$ is constant. What is the reaction order? If you double the catalyst loading (same reactor volume), how does $t_{1/2}$ change?

2. A Pd/Al₂O₃ catalyst (3% Pd, 40% dispersion, $M_{\text{Pd}} = 106.4 \text{ g/mol}$) converts 0.05 mol per gram of catalyst per hour. Estimate the TOF.
3. You want to measure the initial rate of a catalytic reaction using a flow reactor. At what conversion level should the reactor operate, and why? How would you vary conditions to map out the full rate law?

Brief answers: (1) Constant $t_{1/2}$ implies first-order kinetics. Doubling catalyst loading doubles k_{eff} (more active sites per unit volume), so $t_{1/2} = \ln 2/k_{\text{eff}}$ halves. (2) Moles of Pd per gram of catalyst = $0.03/106.4 = 2.82 \times 10^{-4} \text{ mol g}^{-1}$; surface Pd = $0.40 \times 2.82 \times 10^{-4} = 1.13 \times 10^{-4} \text{ mol g}^{-1}$; TOF = $0.05/(1.13 \times 10^{-4}) = 443 \text{ h}^{-1} \approx 0.12 \text{ s}^{-1}$. (3) Operate below $\sim 5\%$ conversion (differential reactor mode) so that the concentration is approximately constant across the bed. Vary inlet concentration C_{A0} across runs to determine reaction order, just as in the method of initial rates for batch experiments.

Part I developed tools for extracting kinetic parameters and introduced TOF as the standard metric for comparing catalysts. But to calculate TOF we need the number of active sites, and to understand *why* one catalyst outperforms another we must characterize its physical and chemical properties. In short: *kinetics tells you how fast; characterisation tells you why*. The following sections introduce the experimental techniques that measure surface area, active site populations, crystallite size, and electronic environment—the quantities that connect macroscopic rates to atomic-level catalyst properties.

3.6 Introduction: Why Characterization Matters

In Chapter 2, we developed the theoretical framework for understanding how molecules adsorb on catalyst surfaces. We introduced surface coverage θ , the equilibrium constant K , and the Langmuir isotherm. To apply these concepts to real catalysts, however, we need to know fundamental properties of the material: *How much surface area is available? What types of active sites are present? What is the metal oxidation state? How large are the catalyst particles?*

Catalyst characterization provides the experimental answers to these questions. The techniques covered in this chapter form the essential toolkit

that every catalysis researcher uses to understand structure–activity relationships. Without proper characterization, we cannot:

- Compare catalysts synthesized by different methods
- Understand why one catalyst outperforms another
- Diagnose deactivation mechanisms
- Rationally design improved catalysts
- Reproduce literature results reliably

This part of the chapter surveys seven complementary techniques, each revealing a different facet of the catalyst. Rather than memorising every equation, focus on *what question each technique answers*:

- **How much surface is available?** → BET (N_2 physisorption)
- **What pore structure does the catalyst have?** → BJH, t-plot
- **How strongly do molecules bind?** → TPD (Redhead equation)
- **What crystalline phases and sizes are present?** → XRD (Bragg, Scherrer)
- **What oxidation states exist at the surface?** → XPS
- **What do the particles look like?** → TEM/SEM
- **What types of active sites are present?** → IR/Raman of probe molecules

Each technique answers one question well; together they build a complete picture of the working catalyst.

3.7 Surface Area and Porosity

The catalytic activity of a heterogeneous catalyst depends critically on its surface area. More surface area means more sites available for adsorption and reaction. Industrial catalysts are designed with high surface areas, typically achieved through porous structures with complex internal geometries.

3.7.1 The BET Method: Physical Basis

The Brunauer–Emmett–Teller (BET) method is the standard technique for measuring catalyst surface area. It extends the Langmuir monolayer adsorption model to account for **multilayer adsorption**, which occurs with physically adsorbed (physisorbed) gases like nitrogen at low temperatures.

Remark 3.14 (Physisorption vs. Chemisorption). The BET method relies on physisorption, where gas molecules are held on the surface by weak van der Waals forces (5–40 kJ/mol). This contrasts with chemisorption, where chemical bonds form between adsorbate and surface (40–400 kJ/mol). Physisorption is reversible and allows multi-layer formation; chemisorption typically stops at monolayer coverage.

The key assumptions of the BET model are:

- (i) Gas molecules adsorb in infinite layers on the surface
- (ii) There is no interaction between adsorption layers
- (iii) The Langmuir theory applies to each layer
- (iv) The heat of adsorption for the first layer (E_1) differs from subsequent layers
- (v) For layers 2 and above, the heat of adsorption equals the heat of liquefaction (E_L)

3.7.2 Derivation of the BET Equation

The BET equation relates the volume of gas adsorbed to the relative pressure P/P_0 (where P_0 is the saturation vapor pressure of the adsorbate):

The BET equation was derived by Brunauer, Emmett, and Teller (1938) by writing a kinetic balance for each adsorption layer, assuming that all layers beyond the first behave like the bulk liquid. The linearized form of the resulting equation (Eq. (3.24)) is the basis for standard surface area analysis.

The result is the **BET equation**:

$$\boxed{\frac{P/P_0}{V(1 - P/P_0)} = \frac{1}{V_m c} + \frac{(c - 1)}{V_m c} \left(\frac{P}{P_0} \right)} \quad (3.24)$$

where:

- V = volume of gas adsorbed at pressure P (at STP)
- V_m = volume of gas required for monolayer coverage (at STP)
- P_0 = saturation vapor pressure of the adsorbate
- c = BET constant, related to the energy of adsorption

The BET constant c is defined as:

$$c = \exp\left(\frac{E_1 - E_L}{RT}\right) \quad (3.25)$$

where E_1 is the heat of adsorption for the first layer and E_L is the heat of liquefaction of the adsorbate.

Remark 3.15 (Physical Interpretation of c). The BET constant reflects the relative affinity of the surface for the adsorbate compared to adsorbate–adsorbate interactions:

- Large c (>100): Strong surface–adsorbate interaction; first layer fills before multilayer forms; isotherm shows sharp knee
- Small c (<20): Weak surface–adsorbate interaction; multilayer forms before monolayer complete; gradual isotherm shape
- Typical oxide supports: $c \approx 50$ –200
- Carbons and weakly interacting surfaces: $c < 20$

3.7.3 BET Linearization and Data Analysis

Eq. (3.24) is already in linear form. Plotting $\frac{P/P_0}{V(1-P/P_0)}$ versus P/P_0 yields a straight line with:

$$\text{Slope} = \frac{c-1}{V_m c} \quad (3.26)$$

$$\text{Intercept} = \frac{1}{V_m c} \quad (3.27)$$

From these, we can solve for the monolayer volume and BET constant:

$$V_m = \frac{1}{\text{slope} + \text{intercept}}, \quad c = 1 + \frac{\text{slope}}{\text{intercept}} \quad (3.28)$$

Fig. 3.1 illustrates the graphical analysis central to the BET method. The main plot shows experimental data transformed according to Eq. (3.24): plotting $(P/P_0)/[V(1-P/P_0)]$ on the y -axis against P/P_0 on the x -axis yields a straight line within the valid range ($0.05 < P/P_0 < 0.30$, indicated by the shaded region). The slope and intercept of this line provide the two unknowns, V_m and c , through Eq. (3.28). For the example shown, the slope of 0.0198 g/cm^3 and intercept of 0.0002 g/cm^3 yield $V_m = 50 \text{ cm}^3/\text{g}$ and $c = 100$. The inset displays the original isotherm (volume adsorbed versus

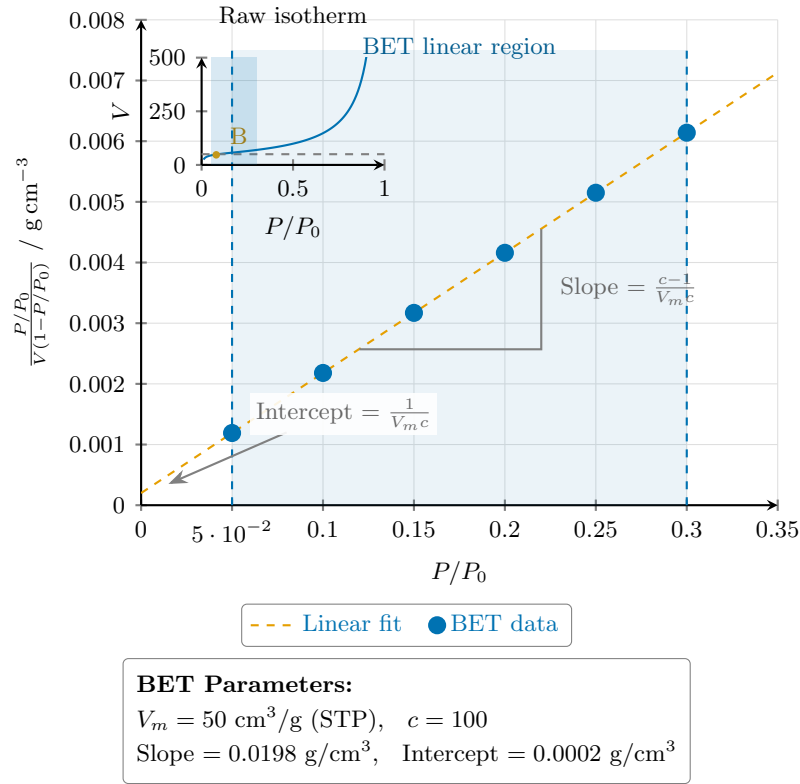


Figure 3.1: Linearized BET plot showing the transformation used to determine monolayer volume V_m and BET constant c . Data points in the linear region ($P/P_0 = 0.05\text{--}0.30$) are fitted to extract slope and intercept. The inset shows the raw Type II adsorption isotherm with Point B (the “knee”) corresponding approximately to monolayer completion.

relative pressure), which exhibits the characteristic Type II shape expected for multilayer adsorption on nonporous or macroporous materials. Point B, located at the “knee” of the isotherm, corresponds roughly to monolayer completion and occurs near V_m . The large value of c in this example produces a sharp knee, indicating strong adsorbate–surface interactions relative to adsorbate–adsorbate interactions.

Remark 3.16 (Practical Tips for BET Analysis).

1. The BET equation is only valid in the relative pressure range $0.05 < P/P_0 < 0.30$. Outside this range, the linear approximation breaks down. (Some references extend to 0.35, but the Rouquerol consistency criteria should always be checked.)
2. Always check that the correlation coefficient $R^2 > 0.999$ for the

linear fit.

3. The intercept should be positive. A negative intercept indicates invalid data or inappropriate pressure range.
4. The BET constant c should be positive (typically $c > 2$). Values $c < 2$ suggest the model assumptions are violated.

Remark 3.17 (IUPAC Classification of Adsorption Isotherms). IUPAC recognizes six standard physisorption isotherm types, each reflecting distinct adsorbent–adsorbate characteristics:

- **Type I** — Concave to the P/P_0 axis, reaching a plateau at moderate P/P_0 . Characteristic of *microporous* solids (e.g. zeolites, activated carbons) where pore filling, rather than multilayer formation, dominates. Langmuir-like behavior.
- **Type II** — S-shaped (sigmoid), with unrestricted multilayer adsorption. Typical of *nonporous or macroporous* materials. The BET equation applies here; Point B marks approximate monolayer completion.
- **Type III** — Convex to the P/P_0 axis throughout. Arises when the adsorbate–surface interaction is weaker than the adsorbate–adsorbate interaction ($c < 2$). Rare in practice (e.g. H_2O on graphite).
- **Type IV** — Resembles Type II at low P/P_0 but exhibits a *hysteresis loop* at higher pressures due to capillary condensation. Characteristic of *mesoporous* materials. BET analysis applies to the initial (pre-hysteresis) portion; the BJH method (Section 3.7.6) extracts the pore size distribution from the hysteresis region.
- **Type V** — Similar to Type III (weak adsorbate–surface interaction) but with a hysteresis loop. Indicates mesoporous structure combined with weak interaction. Uncommon.
- **Type VI** — Stepwise multilayer adsorption on a highly uniform, nonporous surface (e.g. Ar on graphitized carbon black at 77 K). Each step corresponds to completion of one adsorbed layer.

Note: Standard BET analysis is reliable only for Type II isotherms and the initial portion of Type IV isotherms. Applying the BET equation to Type I or Type III data yields physically meaningless V_m and c values.

3.7.4 Calculating Surface Area

Once V_m is known, the specific surface area S_{BET} (m^2/g) is calculated from:

$$S_{\text{BET}} = \frac{V_m \cdot N_A \cdot \sigma}{V_{\text{mol}} \cdot m} \quad (3.29)$$

where:

- $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$ (Avogadro's number)
- σ = cross-sectional area of the adsorbate molecule
- $V_{\text{mol}} = 22414 \text{ cm}^3/\text{mol}$ (molar volume at STP)
- m = sample mass (g)

For nitrogen at 77 K (the standard BET adsorbate), $\sigma_{\text{N}_2} = 0.162 \text{ nm}^2 = 16.2 \times 10^{-20} \text{ m}^2$.

This simplifies to a convenient working formula:

$$S_{\text{BET}} (\text{m}^2/\text{g}) = 4.35 \times V_m (\text{cm}^3/\text{g at STP}) \quad (3.30)$$

Remark 3.18 (Concept Check: Interpreting a BET Result). A researcher reports $S_{\text{BET}} = 250 \text{ m}^2 \text{ g}^{-1}$ with $c = 15$ for a mesoporous silica.

1. Is the BET constant c large or small? What does this tell you about the strength of the N_2 -surface interaction?
2. The researcher also reports $S_{\text{BET}} = 850 \text{ m}^2 \text{ g}^{-1}$ for an activated carbon. Without any other information, would you expect c to be larger or smaller than 15?
3. A third sample gives a negative intercept in the BET plot. Should the result be trusted?

Answers: (1) $c = 15$ is small (typical range 50–300 for oxides). Since $c = \exp[(E_1 - E_L)/RT]$, small c means the first-layer adsorption energy E_1 is only slightly above the liquefaction energy E_L —the surface interacts weakly with N_2 , consistent with silica. (2) Activated carbons typically have $c > 100$ due to stronger dispersion interactions in micropores. However, for highly microporous materials the BET method may overestimate surface area. (3) A negative intercept gives a negative or meaningless c ; the BET model is invalid in this pressure range. Re-measure using a narrower P/P_0 window (0.05–0.15).

3.7.5 Single-Point BET Method

For routine measurements, a simplified single-point BET method is often used. When c is large (strong surface–adsorbate interaction), the intercept in Eq. (3.24) becomes negligible compared to the slope, and:

$$V_m \approx V \left(1 - \frac{P}{P_0} \right) \quad (3.31)$$

measured at a single relative pressure (typically $P/P_0 = 0.3$).

Remark 3.19 (When to Use Multi-Point vs. Single-Point BET).

- Use **multi-point BET** for: accurate measurements, unknown samples, samples with low surface area, publication-quality data
- Use **single-point BET** for: quality control, screening, samples with well-characterized supports where c is known to be large

3.7.6 Pore Size Distribution: The BJH Method

Beyond total surface area, the distribution of pore sizes profoundly affects catalyst performance. The Barrett–Joyner–Halenda (BJH) method extracts pore size distributions from nitrogen adsorption–desorption isotherms.

Pores are classified by IUPAC as:

- **Micropores:** diameter < 2 nm
- **Mesopores:** diameter 2–50 nm
- **Macropores:** diameter > 50 nm

The BJH method applies to mesopores and is based on the **Kelvin equation**, which relates the vapor pressure over a curved meniscus to the pore radius:

$$\ln \left(\frac{P}{P_0} \right) = - \frac{2\gamma V_L}{RT \cdot r_k} \quad (3.32)$$

where γ is the surface tension of liquid nitrogen, V_L is the molar volume of liquid nitrogen, and r_k is the Kelvin radius (radius of the meniscus). This form assumes complete wetting ($\cos \theta_c = 1$), which is valid for N_2 on most oxide surfaces. The general form includes a $\cos \theta_c$ factor in the numerator.

The actual pore radius r_p includes a correction for the thickness t of the adsorbed layer already present on the pore walls:

$$r_p = r_k + t \quad (3.33)$$

The BJH method works by:

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1. Starting at the highest P/P_0 (all pores filled)
2. Decreasing pressure stepwise, calculating the pore radius that empties at each step using Eq. (3.32)
3. Correcting for the change in t -layer thickness
4. Computing incremental pore volume at each radius

Remark 3.20 (Hysteresis Loops and Pore Geometry). The shape of the adsorption–desorption hysteresis loop provides information about pore geometry:

- **H1 (parallel branches):** Cylindrical pores, uniform size
- **H2 (triangular):** Ink-bottle pores (narrow neck, wide body)
- **H3 (no limiting uptake):** Slit-shaped pores, plate-like particles
- **H4 (nearly parallel):** Narrow slit pores

Remark 3.21 (The BJH Cavitation Artifact). For materials with ink-bottle pores (narrow necks, wide bodies), the desorption branch of the isotherm shows a forced closure of the hysteresis loop near $P/P_0 \approx 0.42$ (for N_2 at 77 K). This produces a spurious sharp peak near $d \approx 3.8$ nm in the BJH pore size distribution that does *not* correspond to a real pore size. Always compare adsorption-branch and desorption-branch BJH analyses: if the desorption branch shows a sharp peak at ~ 3.8 nm but the adsorption branch does not, the peak is a cavitation artifact.

3.7.7 The t-Plot Method for Micropore Analysis

The t-plot method separates micropore volume from external surface area. It compares the experimental adsorption isotherm to a standard isotherm measured on a nonporous reference material.

The statistical thickness t of the adsorbed layer on a flat surface is calculated from:

$$t = \left(\frac{V}{V_m} \right) \times t_m \quad (3.34)$$

where t_m is the thickness of a monolayer (0.354 nm for N_2).

For the Harkins–Jura equation, commonly used for oxides:

$$t \text{ (nm)} = \left[\frac{13.99}{0.034 - \log_{10}(P/P_0)} \right]^{0.5} \quad (3.35)$$

Interpreting t-plots:

- **Linear through origin:** No micropores; all surface is external
- **Positive intercept:** Micropores present; intercept = micropore volume
- **Upward deviation at high t :** Capillary condensation in mesopores
- **Slope of linear region:** External (non-micropore) surface area

3.8 Temperature-Programmed Techniques

Temperature-programmed methods monitor the interaction of gases with catalyst surfaces as temperature is systematically varied. These techniques provide information about binding energies, site distributions, and reducibility/oxidizability of catalyst components.

3.8.1 Temperature-Programmed Desorption (TPD)

In TPD, a probe molecule is first adsorbed on the catalyst surface at low temperature, then the temperature is increased at a constant rate β (K/min) while monitoring the desorbed species with a mass spectrometer or thermal conductivity detector.

The desorption rate for first-order kinetics is:

$$-\frac{d\theta}{dt} = \nu \cdot \theta \cdot \exp\left(-\frac{E_d}{RT}\right) \quad (3.36)$$

where ν is the pre-exponential factor (typically 10^{12} – 10^{13} s $^{-1}$) and E_d is the desorption activation energy, defined as a positive quantity. For non-activated adsorption, E_d equals the magnitude of the adsorption enthalpy, $|\Delta H_{\text{ads}}|$.

The Redhead Equation

For first-order desorption with a linear temperature ramp, the peak maximum temperature T_p is related to the desorption energy by the **Redhead equation**:

$$\frac{E_d}{RT_p^2} = \frac{\nu}{\beta} \exp\left(-\frac{E_d}{RT_p}\right) \quad (3.37)$$

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This transcendental equation (i.e., it cannot be solved algebraically for E_d) can be approximated for typical conditions ($E_d/RT_p = 20$ –50) as:

$$E_d = RT_p \left[\ln \left(\frac{\nu T_p}{\beta} \right) - 3.64 \right] \quad (3.38)$$

where β must be in K/s when ν is in s^{-1} (convert from K/min by dividing by 60).

Remark 3.22 (Using the Redhead Equation). With typical values $\nu = 10^{13} \text{ s}^{-1}$ and $\beta = 10 \text{ K/min}$, the Redhead equation simplifies to:

$$E_d \text{ (kJ/mol)} \approx 0.28 \times T_p \text{ (K)} \quad (3.39)$$

This rule of thumb provides quick estimates: a desorption peak at 400 K corresponds to $E_d \approx 112 \text{ kJ/mol}$. The coefficient 0.28 is valid to within $\pm 3\%$ over the range $T_p = 300$ –700 K for these standard conditions.

Remark 3.23 (Limitations of the Redhead Equation). The Redhead equation assumes: (1) first-order desorption—verify desorption order before applying; (2) a known pre-exponential factor ν —a factor-of-10 uncertainty in ν changes E_d by $RT_p \ln 10 \approx 8$ –12 kJ/mol at typical TPD temperatures (400–600 K); (3) coverage-independent desorption energy (no lateral interactions). For more accurate analysis, leading-edge methods or complete Polanyi–Wigner fitting should be used. Always report the assumed ν value alongside the calculated E_d .

Remark 3.24 (Concept Check: Applying the Redhead Equation). CO desorbs from three different metal surfaces with the following TPD peak temperatures (all measured at $\beta = 10 \text{ K/min}$):

Surface	T_p (K)
Cu(111)	200
Ni(111)	400
Pt(111)	480

1. Using the rule of thumb, estimate E_d for each surface.
2. Rank the metals by CO binding strength. Does this ranking match what you would expect from the d-band model?

3. If you change the heating rate from $\beta = 5$ K/min to $\beta = 20$ K/min, does T_p shift to higher or lower temperature? Why?

Answers: (1) Cu: $0.28 \times 200 = 56$ kJ mol⁻¹; Ni: $0.28 \times 400 = 112$ kJ mol⁻¹; Pt: $0.28 \times 480 = 134$ kJ mol⁻¹. (2) Binding strength: Pt > Ni > Cu, consistent with the d-band model—Pt and Ni have d-band centres closer to the Fermi level than Cu, strengthening back-donation into CO $2\pi^*$. (3) Higher β shifts T_p upward: faster heating gives less time for desorption at each T , so a higher temperature is needed to achieve the same desorption rate.

When Redhead assumptions break down, model-free methods—such as leading-edge analysis at varying initial coverages—can extract coverage-dependent desorption energies without assuming a specific desorption order or pre-exponential.

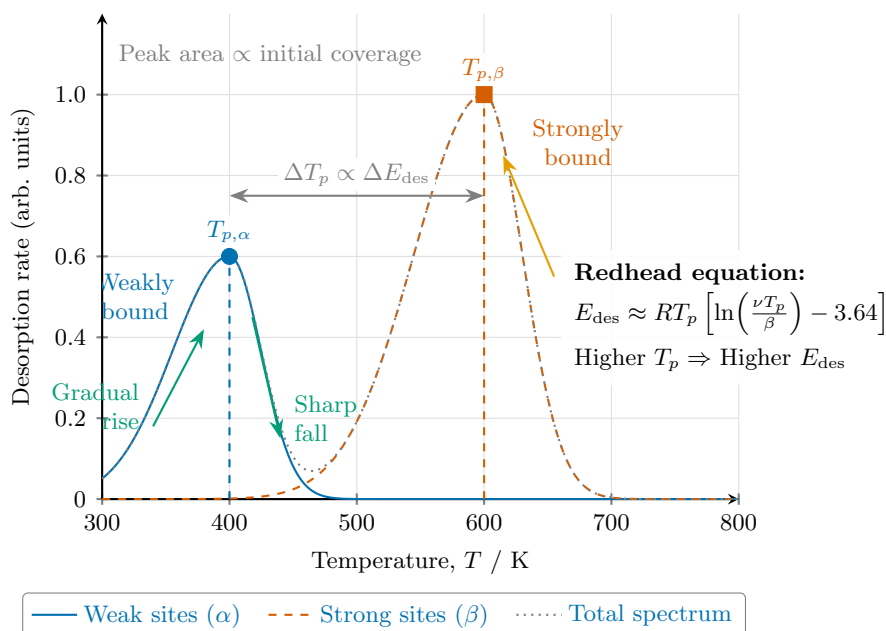


Figure 3.2: TPD spectrum showing desorption peaks from weakly bound (α , 400 K) and strongly bound (β , 600 K) adsorbate states. Higher peak temperatures indicate stronger binding according to the Redhead equation (Eq. (3.38)). The asymmetric peak shape—gradual rise followed by sharp decay—is characteristic of first-order desorption kinetics.

Fig. 3.2 displays a typical TPD spectrum exhibiting two distinct desorption peaks. The low-temperature α peak at $T_{p,\alpha} = 400$ K arises from weakly bound adsorbates (such as molecules on terrace sites or physisorbed species),

while the high-temperature β peak at $T_{p,\beta} = 600$ K corresponds to strongly bound adsorbates (for example, molecules on step edges or chemisorbed species). Using the Redhead approximation (Eq. (3.38)) with typical values $\nu = 10^{13} \text{ s}^{-1}$ and $\beta = 10 \text{ K/min}$, we estimate desorption energies of approximately $E_{d,\alpha} \approx 112 \text{ kJ/mol}$ and $E_{d,\beta} \approx 168 \text{ kJ/mol}$ (consistent with the rule of thumb $E_d \approx 0.28 \times T_p$). Notice the characteristic asymmetric peak shape: the gradual leading edge reflects the exponential increase in desorption rate as temperature rises (Arrhenius acceleration), while the sharp trailing edge results from rapid depletion of adsorbate coverage past the peak. The peak area is proportional to the initial coverage of each species, allowing quantitative determination of site populations. In practice, peak deconvolution (shown as separate curves) is often necessary when multiple binding states produce overlapping features.

Peak Shape Analysis

The shape of TPD peaks contains additional information:

- **Asymmetric peaks** (gradual rise, sharp fall): First-order desorption kinetics
- **Symmetric peaks**: Second-order desorption (e.g., associative desorption of H_2 from 2H^*)
- **Multiple peaks**: Different binding sites or desorption states
- **Peak area**: Proportional to amount adsorbed (surface coverage)
- **Coverage dependence**: For first-order desorption, T_p is independent of initial coverage θ_0 . For second-order, T_p decreases with increasing θ_0 .

Remark 3.25 (Coverage Dependence as Desorption Order Diagnostic). One of the most reliable tests for desorption order: for first-order desorption, T_p is independent of initial coverage θ_0 . For second-order desorption, T_p shifts to lower temperatures as θ_0 increases (because the rate depends on θ^2 , so higher coverage accelerates desorption). Experimentally: run TPD at several initial coverages and plot T_p vs. θ_0 .

Remark 3.26 (NH_3 -TPD for Acid Site Characterization). Ammonia TPD is widely used to characterize acid sites on zeolites and oxide catalysts. NH_3 adsorbs on both Brønsted and Lewis acid sites.

A typical NH_3 -TPD profile shows:

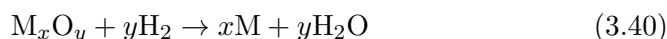
- **Low-temperature peak** (150–250°C): Weakly bound NH_3 , physisorbed or on weak Lewis sites
- **High-temperature peak** (350–500°C): Strongly bound NH_3 on strong Brønsted or Lewis acid sites

The total area under the TPD curve (after baseline correction) gives the total number of acid sites. Peak deconvolution separates contributions from different site types.

3.8.2 Temperature-Programmed Reduction (TPR)

TPR characterizes the reducibility of oxide catalysts by monitoring hydrogen consumption as temperature increases. A dilute H_2 stream (typically 5–10% H_2 in Ar or N_2) flows over the sample while temperature increases linearly.

The reduction reaction for a metal oxide:



Interpreting TPR Profiles

Definition 3.4 (Reduction Temperature). The **reduction temperature** T_{red} (peak maximum) indicates the ease of reduction. Lower T_{red} means easier reduction (more weakly bound oxygen or more reducible metal species).

Key factors affecting T_{red} :

- **Metal identity:** Cu^{2+} reduces at lower T than Ni^{2+} , which reduces lower than Co^{2+}
- **Metal–support interaction:** Strong interaction increases T_{red}
- **Particle size:** Smaller particles may reduce at higher T due to stronger oxide–support bonding
- **Oxidation state:** Higher oxidation states generally reduce at higher T

Remark 3.27 (TPR Peak Assignment). Multiple TPR peaks indicate different reducible species. For supported metal catalysts:

- Lowest T peak: Bulk oxide not interacting with support
- Intermediate T peak: Oxide species at support interface
- Highest T peak: Highly dispersed species or ions incorporated into support lattice

Quantitative Analysis

The hydrogen consumption can be quantified by integrating the TPR peak area and comparing to a calibration standard:

$$\text{H}_2 \text{ consumed (mol/g)} = \frac{\text{Peak area} \times \text{Calibration factor}}{\text{Sample mass}} \quad (3.41)$$

From stoichiometry, this determines:

- Total amount of reducible species
- Average oxidation state (if multiple states present)
- Degree of reduction achieved

3.8.3 Temperature-Programmed Oxidation (TPO)

TPO is the oxidative counterpart to TPR. A dilute O₂ stream flows over a pre-reduced sample (or coked catalyst), and oxygen consumption or product evolution (CO₂, H₂O) is monitored.

Applications include:

- Characterizing coke deposits on deactivated catalysts
- Studying oxidation behavior of reduced metals
- Determining carbon content and reactivity

For coke analysis, TPO with mass spectrometry detection distinguishes:

- **Soft coke** (200–400°C): Aliphatic, hydrogen-rich deposits
- **Hard coke** (400–600°C): Aromatic, graphitic deposits

3.9 X-ray Techniques

X-ray-based methods provide fundamental structural and compositional information about catalysts. We focus on two complementary techniques: X-ray diffraction (XRD) for bulk crystalline structure and X-ray photoelectron spectroscopy (XPS) for surface composition.

3.9.1 X-ray Diffraction (XRD)

XRD identifies crystalline phases and determines crystallite sizes in polycrystalline catalysts. When X-rays interact with a crystalline material, constructive interference occurs when **Bragg's law** is satisfied:

$$\boxed{n\lambda = 2d \sin \theta} \quad (3.42)$$

where:

- n = diffraction order (integer)
- λ = X-ray wavelength (Cu $K\alpha$ = 0.15406 nm)
- d = interplanar spacing (lattice plane distance)
- θ = diffraction angle (Bragg angle)

XRD Pattern Interpretation

An XRD pattern (diffractogram) plots intensity versus diffraction angle 2θ . Each crystalline phase produces a characteristic set of peaks at specific 2θ values.

Phase identification proceeds by:

1. Measuring peak positions (2θ values)
2. Calculating d -spacings from Bragg's law
3. Comparing to reference patterns (ICDD/JCPDS database)

The techniques in this chapter—XRD (Bragg, 1915 Nobel Prize), XPS (Siegbahn, 1981 Nobel Prize), and electron microscopy (Ruska, 1986 Nobel Prize)—were developed by physicists but are now indispensable in every catalysis and materials chemistry laboratory.

The Scherrer Equation: Crystallite Size

Small crystallites produce broadened XRD peaks. The **Scherrer equation** relates peak width to crystallite size:

$$D = \frac{K\lambda}{\beta \cos \theta} \quad (3.43)$$

where:

- D = crystallite size (nm)
- K = shape factor (typically 0.89–0.94 for spheres)
- λ = X-ray wavelength (nm)
- β = peak full width at half maximum (FWHM) in radians
- θ = Bragg angle (radians)

Remark 3.28 (Instrumental Broadening Correction). The measured peak width β_{obs} includes contributions from both crystallite

size and instrumental effects:

$$\beta^2 = \beta_{\text{obs}}^2 - \beta_{\text{inst}}^2 \quad (3.44)$$

The instrumental broadening β_{inst} is determined by measuring a standard with large crystallites (e.g., LaB_6 , Si).

Remark 3.29 (Limitations of the Scherrer Equation).

- Valid for crystallite sizes 3–100 nm
- Smaller crystallites: peaks become too broad to measure accurately
- Larger crystallites: broadening approaches instrumental limit
- Assumes uniform crystallite size; polydisperse samples give a volume-weighted average size
- Other broadening sources (strain, defects) can affect results

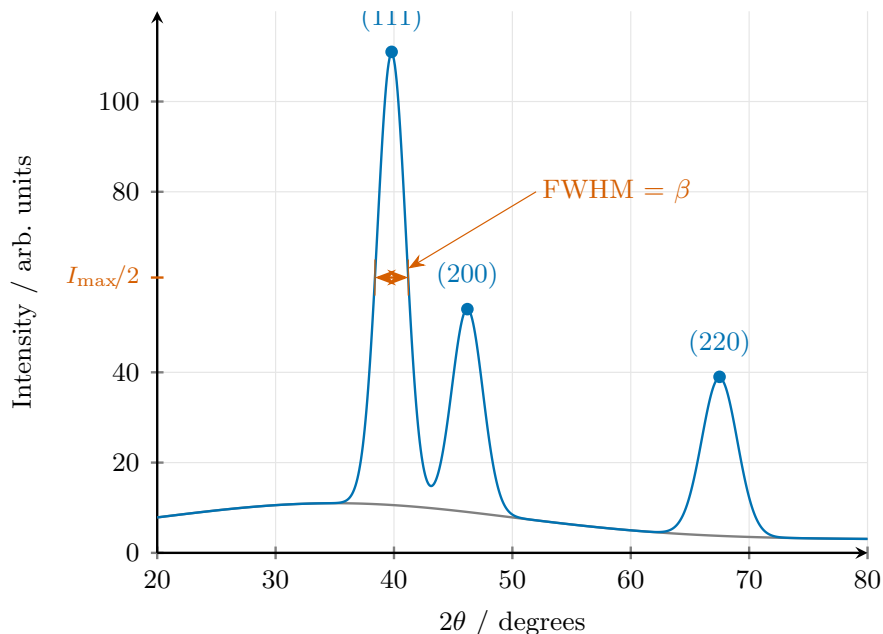
Remark 3.30 (Concept Check: Scherrer Equation). Two $\text{Pt}/\text{Al}_2\text{O}_3$ catalysts give Pt(111) XRD peaks at the same 2θ angle but with FWHM values of 0.5° and 2.0° (after instrumental broadening correction).

1. Which catalyst has larger crystallites?
2. By what factor do the crystallite sizes differ?
3. Assuming spherical particles, which catalyst has higher metal dispersion?

Answers: (1) The catalyst with the *narrower* peak (0.5°) has larger crystallites—broader peaks mean smaller crystallites. (2) Since $D \propto 1/\beta$, the sizes differ by a factor of $2.0/0.5 = 4$. (3) The catalyst with the broader peak (smaller crystallites) has higher dispersion, as a larger fraction of atoms reside on the surface of smaller particles.

Note: The Scherrer equation gives the *crystallite* size, which may differ from the *particle* size measured by TEM. A single particle can contain multiple crystallites (polycrystalline), so the Scherrer size is often smaller than the TEM particle size. Always compare both measurements when both are available.

When microstrain from defects or dislocations is suspected, the Williamson–Hall method separates size and strain contributions to peak broadening by exploiting their different θ -dependence; otherwise, the Scherrer equation alone suffices for size estimation.



Bragg: $n\lambda = 2d \sin \theta$ **Scherrer:** $D = K\lambda/(\beta \cos \theta)$, $K \approx 0.9$, β = FWHM in rad

Figure 3.3: XRD pattern of Pt nanoparticles supported on γ -Al₂O₃. Sharp peaks correspond to fcc Pt crystallites with Miller indices labeled. The FWHM (β) of the (111) peak at $2\theta = 39.8^\circ$ is used with the Scherrer equation to estimate crystallite size.

Fig. 3.3 shows a characteristic XRD pattern for a supported Pt catalyst. Three diffraction peaks are observed at $2\theta = 39.8^\circ$, 46.2° , and 67.5° , corresponding to the (111), (200), and (220) crystallographic planes of face-centered cubic (fcc) platinum metal, respectively. The peak positions, determined by Bragg’s law (Eq. (3.42)), confirm the presence of metallic Pt rather than oxidized forms such as PtO or PtO₂. The peak intensities reflect both the multiplicity of equivalent planes and the crystallite orientation distribution. For quantitative crystallite size analysis, we focus on the (111) peak because it is typically the most intense and well-resolved. The full width at half maximum (FWHM, denoted β in the Scherrer equation) is measured as shown—in this example, $\beta \approx 2.8^\circ$ or 0.049 rad after instrumental broadening correction. Applying Eq. (3.43) with $K = 0.9$ and Cu K α radiation ($\lambda = 0.15406$ nm) yields an estimated crystallite size of approximately 3 nm. The broad baseline hump arises from the amorphous alumina

support, which lacks long-range crystalline order.

3.9.2 X-ray Photoelectron Spectroscopy (XPS)

XPS (also called ESCA—Electron Spectroscopy for Chemical Analysis) is the premier technique for determining surface elemental composition and chemical states. It is inherently surface-sensitive, probing only the top 1–10 nm of the sample. The signal intensity decays exponentially with depth: approximately 63% of the signal originates from the top λ (inelastic mean free path, typically 1–3 nm), and 95% from the top 3λ . XPS is therefore not a monolayer technique but averages over several atomic layers.

Physical Basis

XPS is based on the photoelectric effect: X-ray photons eject core-level electrons from the sample surface, and the kinetic energy of each photoelectron is measured. The binding energy is then obtained from $E_B = h\nu - E_K - \phi$, where $h\nu$ is the photon energy and ϕ is the spectrometer work function, providing element-specific and oxidation-state-specific identification.

Elemental Identification

Each element has characteristic core-level binding energies. For example:

- C 1s: ~ 285 eV
- O 1s: ~ 530 eV
- N 1s: ~ 400 eV
- Pt 4f_{7/2}: ~ 71 eV (metal), ~ 74 eV (oxide)

Chemical State Information: Binding Energy Shifts

The power of XPS lies in its ability to distinguish different chemical environments through **binding energy shifts**. Higher oxidation states generally exhibit higher binding energies because:

- Fewer valence electrons means less shielding of core electrons
- Core electrons are held more tightly, requiring more energy for ejection

Table 3.1: Example XPS binding energies for different oxidation states

Species	Binding Energy (eV)	Assignment
Pt 4f _{7/2}	71.0–71.5	Pt ⁰ (metallic)
	72.5–73.5	Pt ²⁺ (PtO, Pt(OH) ₂)
	74.5–75.5	Pt ⁴⁺ (PtO ₂)
Cu 2p _{3/2}	932.5	Cu ⁰ or Cu ⁺
	933.5–934.5	Cu ²⁺ (CuO)
Fe 2p _{3/2}	706.5–707.5	Fe ⁰ (metallic)
	709.5–710.8	Fe ²⁺ (FeO)
	710.7–711.6	Fe ³⁺ (Fe ₂ O ₃) [†]

[†]Fe²⁺ and Fe³⁺ binding energies overlap significantly; satellite peaks and spectral fitting are required for reliable assignment.

Quantitative Analysis

Surface composition is determined from peak areas corrected by sensitivity factors:

$$x_i = \frac{A_i/S_i}{\sum_j A_j/S_j} \quad (3.45)$$

where x_i is the atomic fraction of element i , A_i is the peak area, and S_i is the relative sensitivity factor (RSF).

Remark 3.31 (Surface vs. Bulk Composition). XPS measures *surface* composition, which often differs from bulk composition. For supported metal catalysts:

- Surface enrichment of active metal indicates good dispersion
- Low surface metal content may indicate encapsulation by support
- Comparison of XPS composition with bulk analysis (ICP, XRF) reveals segregation effects

3.10 Electron Microscopy

Electron microscopy provides direct visualization of catalyst morphology, particle size, and in favorable cases, atomic structure. The two primary techniques are transmission electron microscopy (TEM) and scanning electron microscopy (SEM).

3.10.1 Transmission Electron Microscopy (TEM)

TEM transmits a high-energy electron beam (100–300 keV) through an ultrathin sample (<100 nm), forming a magnified image on a detector. Resolution can reach sub-nanometer, allowing direct imaging of metal nanoparticles on supports.

Particle Size Measurement

For supported metal catalysts, TEM provides:

- Direct measurement of individual particle diameters
- Particle size distributions from counting many particles
- Shape information (spherical, cubic, rod-like, etc.)
- Particle location relative to support

Statistical analysis of TEM particle sizes yields:

$$\text{Number-average diameter: } \bar{d}_n = \frac{\sum n_i d_i}{\sum n_i} \quad (3.46)$$

$$\text{Surface-average diameter: } \bar{d}_s = \frac{\sum n_i d_i^3}{\sum n_i d_i^2} \quad (3.47)$$

$$\text{Volume-average diameter: } \bar{d}_v = \frac{\sum n_i d_i^4}{\sum n_i d_i^3} \quad (3.48)$$

The surface-average diameter $\bar{d}_s$ is most relevant for catalysis because reaction occurs on the surface.

Remark 3.32 (Concept Check: Number vs Surface Average). A TEM analysis of a Pt/C catalyst finds 200 particles: 150 with $d = 2$ nm and 50 with $d = 6$ nm.

1. Calculate $\bar{d}_n$ and $\bar{d}_s$.
2. Why is $\bar{d}_s$ always $\geq \bar{d}_n$, and which is more relevant for catalysis?

Answers: (1) $\bar{d}_n = (150 \times 2 + 50 \times 6)/200 = 3.0$ nm. $\bar{d}_s = (150 \times 2^3 + 50 \times 6^3)/(150 \times 2^2 + 50 \times 6^2) = (1200 + 10800)/(600 + 1800) = 12000/2400 = 5.0$ nm. (2) $\bar{d}_s \geq \bar{d}_n$ because the surface-average weights larger particles more heavily (they contribute disproportionately more surface area). Since catalysis occurs on the surface, $\bar{d}_s$ better represents the “catalytically active” average particle size.

Remark 3.33 (How Many Particles Should You Count?). TEM particle size analysis is only as reliable as the sample size. As a rule of thumb, count at least **200 particles** for $\pm 10\%$ accuracy on $\bar{d}_s$, and **300–500 particles** for $\pm 5\%$ accuracy. Broad or bimodal distributions require even more (>500) to capture the tails reliably. Always report the number of particles counted (N) alongside $\bar{d}_n$, $\bar{d}_s$, and the standard deviation—a histogram without N is incomplete. To quantify distribution width, compute the **polydispersity index** $\text{PDI} = \bar{d}_v/\bar{d}_n$: a value of 1.0 indicates a perfectly monodisperse sample, while $\text{PDI} > 1.5$ signals a broadly distributed population. Software tools such as ImageJ or Digital Micrograph can semi-automate particle identification and measurement, but automated selections should always be verified manually—overlapping particles, low-contrast edges, and support artifacts commonly introduce errors that inflate or deflate the reported size.

High-Resolution TEM (HRTEM)

HRTEM can resolve individual crystal lattice planes (lattice fringes), enabling direct measurement of interplanar spacings for phase identification and detection of surface defects in catalyst nanoparticles.

Metal Dispersion from TEM

Metal dispersion D (fraction of metal atoms at the surface) can be estimated from particle size assuming spherical particles:

$$D = \frac{6V_m}{a \cdot \bar{d}_s} \quad (3.49)$$

where V_m is the volume per metal atom, a is the surface area per metal atom, and $\bar{d}_s$ is the surface-average diameter.

For Pt (fcc, $a = 0.392$ nm):

$$D_{\text{Pt}} \approx \frac{1.13}{\bar{d}_s \text{ (nm)}} \quad (3.50)$$

Remark 3.34 (Chemisorption-Based Dispersion Measurement).

The most common method for measuring metal dispersion uses selective chemisorption of probe molecules (CO , H_2 , O_2) on surface metal atoms. The stoichiometry assumption is critical: $\text{CO}:\text{Pt} = 1:1$ (linear/atop binding) or $1:2$ (bridge binding); $\text{H}_2:\text{Pt} = 1:2$ (dissociative adsorption). Standard practice assumes $\text{CO}:\text{Pt} = 1:1$ for Pt and

Pd, giving $D = n_{\text{CO}}/n_{\text{metal, total}}$. Dispersion values from chemisorption and TEM may differ because chemisorption counts chemically accessible sites while TEM measures geometric size.

3.10.2 Scanning Electron Microscopy (SEM)

SEM scans a focused electron beam across the sample surface, detecting secondary electrons (SE) or backscattered electrons (BSE) to form an image. SEM provides:

- 3D-like surface topography images
- Lower magnification/larger field of view than TEM
- Minimal sample preparation
- Elemental mapping when combined with EDS

Remark 3.35 (Choosing Between TEM and SEM).

- Use **TEM** when you need: high resolution (<1 nm), internal structure, particle size of small nanoparticles (<20 nm), lattice imaging
- Use **SEM** when you need: surface morphology, particle/crystal shape, large area imaging, minimal sample preparation, thick samples

3.11 Spectroscopic Methods

Vibrational spectroscopy (IR, Raman) provides molecular-level information about adsorbed species, functional groups, and catalyst structure. These techniques are particularly powerful when combined with probe molecules.

3.11.1 Infrared Spectroscopy of Adsorbed Species

Infrared spectroscopy detects molecular vibrations by measuring absorption of IR radiation. For catalyst characterization, we focus on species adsorbed on the catalyst surface.

Transmission vs. Diffuse Reflectance Modes

- **Transmission IR:** Sample pressed into a thin pellet (with KBr for dilution); IR beam passes through the sample

- **DRIFTS** (Diffuse Reflectance Infrared Fourier Transform Spectroscopy): Powder sample in a cup; diffusely reflected IR collected. Better for strongly absorbing or opaque samples

CO as a Probe Molecule

Carbon monoxide is the most widely used probe molecule for characterizing metal catalysts because:

- Strong IR absorption ($\nu_{\text{CO}} \approx 2000\text{--}2200\text{ cm}^{-1}$)
- Sensitive to metal identity, oxidation state, and coordination
- Forms well-defined surface species
- Commercially available, easy to handle

Definition 3.5 (CO Adsorption Modes). CO adsorbs on metals in several configurations with characteristic frequencies:

- **Linear (on-top)**: CO bonded to one metal atom; $\nu_{\text{CO}} = 2000\text{--}2130\text{ cm}^{-1}$
- **Bridging**: CO bonded to two metal atoms; $\nu_{\text{CO}} = 1850\text{--}2000\text{ cm}^{-1}$
- **Three-fold hollow**: CO bonded to three metal atoms; $\nu_{\text{CO}} = 1700\text{--}1850\text{ cm}^{-1}$

Fig. 3.4 illustrates the three principal CO adsorption geometries and their characteristic IR frequency ranges. The key to understanding this frequency–structure correlation lies in the electronic nature of the metal–CO bond. When CO adsorbs on a metal surface, two simultaneous interactions occur: (i) donation of electron density from the CO 5σ orbital (located primarily on carbon) into empty metal orbitals, and (ii) back-donation from filled metal d -orbitals into the CO $2\pi^*$ antibonding orbital. The second interaction is crucial—it weakens the C–O bond by populating an antibonding orbital, thereby lowering the stretching frequency. In the linear (atop) geometry, CO interacts with only one metal atom, limiting back-donation and preserving much of the C–O triple bond character ($\nu_{\text{CO}} = 2000\text{--}2130\text{ cm}^{-1}$). In bridging and hollow sites, CO interacts with two or three metal atoms, respectively, allowing greater back-donation and progressively weakening the C–O bond ($\nu_{\text{CO}} = 1850\text{--}2000\text{ cm}^{-1}$ for bridge; $1700\text{--}1850\text{ cm}^{-1}$ for hollow). This systematic frequency shift makes CO-FTIR spectroscopy a powerful tool for characterizing the local coordination environment on catalyst surfaces.

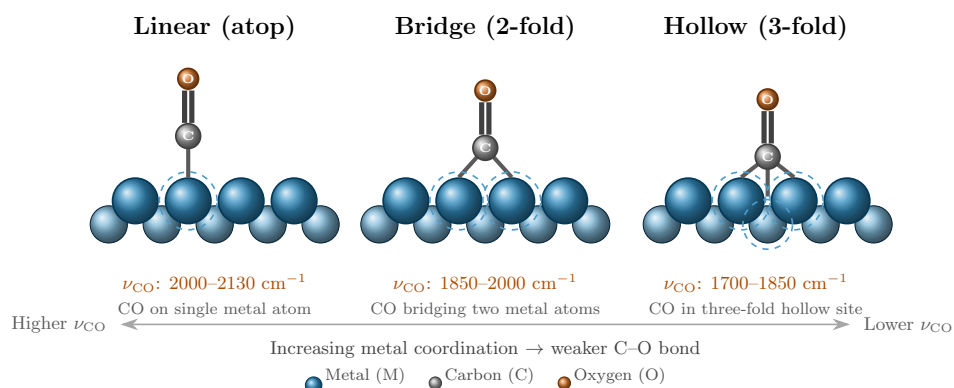


Figure 3.4: CO adsorption modes on metal surfaces showing the relationship between coordination number and vibrational frequency. Linear (atop) coordination to a single metal atom preserves the C–O bond order and gives the highest ν_{CO} . As the carbon atom coordinates to additional metal atoms (bridging, then hollow sites), electron density flows from the metal d -orbitals into the CO π^* antibonding orbital, weakening the C–O bond and lowering the stretching frequency.

Remark 3.36 (Safety: Carbon Monoxide in the Laboratory). CO is the most widely used probe molecule in catalyst characterisation, but it is an odourless, colourless, and highly toxic gas. The occupational exposure limit is TLV-TWA = 25 ppm; the IDLH (Immediately Dangerous to Life or Health) concentration is 1200 ppm. Always perform CO chemisorption and CO-FTIR experiments in a well-ventilated fume hood with a continuous CO monitor. Ensure that exhaust lines from chemisorption instruments vent to fume hoods or building exhaust, not into the laboratory. For H_2 used in TPR experiments, see also the safety notes on hydrogen handling in Section 2.6.4.

On metallic surfaces, only vibrations with a dynamic dipole moment perpendicular to the surface are IR-active in reflection mode (the **metal surface selection rule**); the parallel component is cancelled by its image dipole in the metal, which explains why some expected modes may be absent or weak in spectra of adsorbed species.

The frequency–structure correlation for adsorbed CO is explained by the **Dewar–Chatt–Duncanson (DCD) model**: the CO 5σ lone pair donates electron density into empty metal orbitals (σ -donation), while filled metal d -orbitals donate electron density back into the CO $2\pi^*$ antibonding orbital (π -back-donation). Back-donation is the dominant effect—it weakens the

C–O bond and lowers ν_{CO} , so bridging and hollow CO (receiving back-donation from two or three metal atoms, respectively) show progressively lower stretching frequencies than linear (atop) CO.

Remark 3.37 (Concept Check: CO Stretching Frequency). A CO-FTIR spectrum of a Pd catalyst shows bands at 2075 cm^{-1} and 1950 cm^{-1} .

1. Assign each band to a CO adsorption geometry.
2. If the catalyst is partially oxidised (some Pd^{2+} present), would you expect the linear CO band to shift to higher or lower frequency?
3. Free CO absorbs at 2143 cm^{-1} . What does the red-shift for adsorbed CO tell you about the metal–CO interaction?

Answers: (1) 2075 cm^{-1} = linear (atop) CO; 1950 cm^{-1} = bridging CO. (2) Higher frequency—oxidation reduces metal d -electron density, decreasing π -back-donation into the CO $2\pi^*$ orbital, so the C–O bond is less weakened. (3) The red-shift from 2143 cm^{-1} directly measures the extent of π -back-donation: larger red-shift = more back-donation = more electron-rich metal surface.

Interpreting CO-FTIR Spectra

Key interpretations from CO-FTIR:

- **Peak position:** Indicates metal identity and oxidation state
 - Pt^0 : $\nu_{\text{CO}} \approx 2050\text{--}2090\text{ cm}^{-1}$
 - Pd^0 : $\nu_{\text{CO}} \approx 2050\text{--}2090\text{ cm}^{-1}$ (linear), $\sim 1970\text{ cm}^{-1}$ (bridge)
 - Oxidized metal: Higher frequencies due to less back-donation
- **Peak width:** Related to site heterogeneity; broader peaks indicate more diverse sites
- **Multiple peaks:** Different adsorption sites (terraces, edges, corners)
- **Coverage-dependent shifts:** Dipole–dipole interactions shift peaks to higher frequency at high coverage

Remark 3.38 (Singleton Frequency). The “singleton frequency” (CO frequency extrapolated to zero coverage) eliminates dipole–dipole coupling effects and provides the truest measure of metal–CO inter-

action strength.

Other Probe Molecules

- **Pyridine:** Probes acid sites
 - Brønsted acid: protonated pyridinium ion, $\sim 1545\text{ cm}^{-1}$
 - Lewis acid: coordinated pyridine, $\sim 1450\text{ cm}^{-1}$
- **NO:** Similar to CO but can form bent species; useful for distinguishing metal oxidation states
- **CO₂:** Probes basic sites via carbonate/bicarbonate formation

Remark 3.39 (Pyridine IR: Distinguishing Brønsted from Lewis Acid Sites). Pyridine is the gold-standard probe molecule for characterising acid sites in zeolites and oxide catalysts. When pyridine adsorbs on a solid acid, the nature of the acid site determines the product:

- **Brønsted acid sites** (bridging hydroxyls, e.g., Si–OH–Al in zeolites) protonate pyridine to form the pyridinium ion (PyH⁺). The characteristic ring vibration appears at **1545 cm^{−1}** (ν_{19b} mode).
- **Lewis acid sites** (coordinatively unsaturated metal cations, e.g., Al³⁺, Ti⁴⁺) coordinate pyridine through the nitrogen lone pair without proton transfer. The characteristic ring vibration appears at **1450 cm^{−1}** (ν_{19b} mode).
- **Hydrogen-bonded pyridine** (on weak silanol groups) absorbs near 1440 cm^{-1} —easily distinguished from the Lewis acid band by its lower frequency and lower thermal stability.

Quantification. Using extinction coefficients (ε_B for Brønsted and ε_L for Lewis bands), the concentration of each site type can be calculated from the integrated peak areas, giving a complete acid-site census of the catalyst. Published ε values are material-dependent, so literature values specific to the zeolite framework should be used.

Temperature-programmed desorption of pyridine. By recording IR spectra after stepwise heating under vacuum, one obtains the acid-strength distribution. Sites that retain pyridine at $\geq 450^\circ\text{C}$ are strong Brønsted acids; those that lose pyridine below 200°C are weak. This thermal stability parallels the desorption energy concepts from TPD analysis (Section 3.8.1).

Connection to zeolites (Chapter 6): Pyridine-FTIR is routinely used to characterise H-ZSM-5 (MFI), H-Beta (*BEA), and H-Y (FAU) zeolites, directly measuring how the Si/Al ratio controls Brønsted acid site density. One practical caveat: pyridine (kinetic diameter ≈ 5.7 Å) cannot enter small-pore zeolites such as CHA (≈ 3.8 Å pore opening), so smaller probe molecules like ammonia or acetonitrile are used instead.

3.11.2 Raman Spectroscopy

Raman spectroscopy complements IR by probing vibrations through inelastic light scattering. Key advantages:

- Water does not interfere strongly (unlike IR)
- Can probe M=O, M–O–M vibrations strongly (often IR-weak)
- Spatial resolution possible with Raman microscopy

Applications in catalysis:

- Metal oxide structure (V_2O_5 , MoO_3 , WO_3 supported catalysts)
- Carbon materials (D and G bands for graphitic carbon)
- Zeolite framework vibrations
- In situ/operando studies of working catalysts

Most characterization techniques in this chapter are applied *ex situ*, but the catalyst's structure under reaction conditions can differ substantially from its resting state. *In situ* (controlled atmosphere) and *operando* (simultaneous spectroscopy and activity measurement) variants of IR, Raman, XPS, and XRD now exist and represent the gold standard for correlating catalyst structure with performance.

3.12 Worked Examples: Characterization

Example 3.5: BET Surface Area Calculation

Problem: A nitrogen adsorption experiment on a γ -Al₂O₃ support (sample mass 0.2534 g) yields the following data in the BET region:

P/P_0	Volume adsorbed (cm ³ STP/g)
0.05	48.2
0.10	53.1
0.15	57.3
0.20	61.2
0.25	65.0
0.30	69.1

Calculate (a) the BET surface area and (b) the BET constant c .

Solution (BET Linear Transform):

Step 1: Calculate the BET transform $y = \frac{P/P_0}{V(1-P/P_0)}$ for each data point:

P/P_0	V (cm ³ /g)	$1 - P/P_0$	$\frac{P/P_0}{V(1-P/P_0)}$
0.05	48.2	0.95	0.00109
0.10	53.1	0.90	0.00209
0.15	57.3	0.85	0.00308
0.20	61.2	0.80	0.00408
0.25	65.0	0.75	0.00513
0.30	69.1	0.70	0.00620

Step 2: Linear regression of y versus P/P_0 :

$$\text{Slope} = 0.02038 \text{ g/cm}^3$$

$$\text{Intercept} = 0.000047 \text{ g/cm}^3$$

$$R^2 = 0.9997$$

Step 3: Calculate V_m and c using Eq. (3.28):

$$V_m = \frac{1}{\text{slope} + \text{intercept}} = \frac{1}{0.02038 + 0.000047} = 49.0 \text{ cm}^3/\text{g (STP)}$$

$$c = 1 + \frac{\text{slope}}{\text{intercept}} = 1 + \frac{0.02038}{0.000047} = 435$$

Step 4: Calculate surface area using Eq. (3.30):

$$S_{\text{BET}} = 4.35 \times V_m = 4.35 \times 49.0 = 213 \text{ m}^2/\text{g}$$

Answer:

$$S_{\text{BET}} = 213 \text{ m}^2/\text{g}; c = 435$$

Physical interpretation: The high c value ($\gg 100$) indicates strong interaction between N_2 and the alumina surface, consistent with hydroxylated mesoporous $\gamma\text{-Al}_2\text{O}_3$. The surface area of $\sim 210 \text{ m}^2/\text{g}$ is characteristic of high-surface-area $\gamma\text{-Al}_2\text{O}_3$ supports used in industrial catalysis.

Example 3.6: Crystallite Size from XRD

Problem: An XRD pattern of a Pt/C catalyst shows a Pt(111) peak at $2\theta = 39.76^\circ$ with $\text{FWHM} = 1.85^\circ$. The instrumental broadening determined from a LaB_6 standard is 0.10° . Calculate the Pt crystallite size. (Cu $K\alpha$ radiation: $\lambda = 0.15406 \text{ nm}$)

Solution (Scherrer Equation):

Step 1: Correct for instrumental broadening:

$$\beta = \sqrt{\beta_{\text{obs}}^2 - \beta_{\text{inst}}^2} = \sqrt{1.85^2 - 0.10^2} = 1.847^\circ$$

Step 2: Convert to radians:

$$\beta = 1.847^\circ \times \frac{\pi}{180^\circ} = 0.0322 \text{ rad}$$

Step 3: Calculate θ (half of 2θ):

$$\theta = \frac{39.76^\circ}{2} = 19.88^\circ = 0.347 \text{ rad}$$

Step 4: Apply the Scherrer equation (Eq. (3.43)) with $K = 0.89$:

$$D = \frac{K\lambda}{\beta \cos \theta} = \frac{0.89 \times 0.15406}{0.0322 \times \cos(19.88^\circ)} = \frac{0.1371}{0.0303} = 4.5 \text{ nm}$$

Answer:

$$D = 4.5 \text{ nm (Pt crystallite size)}$$

Physical interpretation: This particle size corresponds to a dispersion of $D = 1.13/4.5 \approx 0.25 = 25\%$, meaning approximately 25% of Pt atoms are at the surface. This is a typical dispersion for fuel cell catalysts.

Remark 3.40 (Self-Assessment: Choosing the Right Technique).

For each scenario below, identify which characterisation technique(s) you would use and what specific information you would extract:

1. A fresh Pd/Al₂O₃ catalyst has lower activity than expected. You suspect the metal particles are too large. Which technique gives the particle size most directly?
2. After running a zeolite catalyst for 100 h, the conversion drops by 50%. You suspect either coking or dealumination. How would you distinguish between these two causes?
3. You have synthesised a Cu/ZnO/Al₂O₃ catalyst for methanol synthesis and need to verify that the copper is in the metallic state (Cu⁰) rather than oxidised (Cu²⁺). Which technique is most surface-sensitive?

Brief answers: (1) **TEM** gives particle size directly from images; alternatively, **XRD** (Scherrer equation) gives crystallite size from peak broadening, and **CO chemisorption** gives dispersion. Use TEM + chemisorption for confirmation. (2) Coking: **TPO** (exotherm at 400–600 °C from coke burn-off) and **TGA** (mass loss). Dealumination: **XRD** (unit cell contraction), ²⁷**Al MAS-NMR** (octahedral Al signal), or **NH₃-TPD** (reduced acid site count). (3) **XPS**: Cu 2p_{3/2} binding energy distinguishes Cu⁰ (932.6 eV) from Cu²⁺ (933.6 eV + shake-up satellite). XRD can confirm crystalline Cu⁰ phase but misses amorphous oxides.

3.13 Summary

Reactor Design

- **Batch reactor mass balance:** $-\frac{dC_A}{dt} = r(C_A)$ is the fundamental equation relating measurable concentration changes to reaction rate
- **Integrated rate laws:**
 - Zeroth-order: $C_A = C_{A0} - kt$ (linear C_A vs. t)
 - First-order: $\ln(C_A/C_{A0}) = -kt$ (linear $\ln C_A$ vs. t)
 - Second-order: $1/C_A - 1/C_{A0} = kt$ (linear $1/C_A$ vs. t)
- **Half-life behavior:** Zeroth-order: $t_{1/2} \propto C_{A0}$; First-order: $t_{1/2} = \ln(2)/k$ (independent of C_{A0}); Second-order: $t_{1/2} \propto 1/C_{A0}$
- **Turnover frequency (TOF):** $\text{TOF} = \frac{\text{moles product}}{\text{moles active sites} \times \text{time}}$ — the standard metric for comparing catalyst activity
- **Turnover number (TON):** Total catalytic cycles per active site over the catalyst lifetime; $\text{TON} = \text{TOF} \times \text{time}$
- **Contact time:** $t_{\text{contact}} = V_{\text{cat}}/\dot{V}$ characterizes residence time in flow systems
- **Temperature-dependent kinetics:** Arrhenius and Eyring analysis for extracting activation parameters was developed in Chapter 2; the rate constant $k(T)$ enters all reactor design equations through its temperature dependence

Catalyst Characterization

- **BET method:** Determines total surface area from N_2 adsorption isotherms via multilayer theory. The linearized BET equation yields V_m and c , giving $S_{\text{BET}} = 4.35 \times V_m$ (m^2/g).
- **BJH method:** Determines mesopore size distribution from the desorption branch of the isotherm using the Kelvin equation. Pores classified as micro- (< 2 nm), meso- (2–50 nm), and macropores (> 50 nm).
- **t-plot:** Separates micropore volume from external surface area by comparing to a standard isotherm.
- **TPD:** Reveals binding energies and site distributions. The Redhead equation relates peak temperature to desorption energy: $E_d \approx 0.28 \times T_p$ (kJ/mol).

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- **TPR/TPO:** Characterize reducibility and oxidation behavior. Peak temperature indicates ease of reduction/oxidation; peak area quantifies reducible/oxidizable species.
- **XRD:** Identifies crystalline phases via Bragg's law ($n\lambda = 2d \sin \theta$). The Scherrer equation ($D = K\lambda/\beta \cos \theta$) gives crystallite size from peak broadening.
- **XPS:** Determines surface composition and oxidation states. Binding energy shifts reveal chemical environment. Surface-sensitive (top 1–10 nm).
- **TEM:** Provides direct particle imaging and size measurement. HRTEM resolves lattice fringes; the surface-average diameter is most relevant for catalytic applications.
- **CO-FTIR:** Characterizes metal active sites through CO stretching frequencies. Linear CO (2000–2130 cm^{-1}) and bridging CO (1850–2000 cm^{-1}) reveal site geometry.
- **Integration:** Complete catalyst characterization requires multiple complementary techniques. Surface area (BET) + structure (XRD) + surface composition (XPS) + particle size (TEM) + site characterization (IR, TPD) provide a comprehensive picture.

Remark 3.41 (Safety Note: Thermal Runaway). Many catalytic reactions are highly exothermic. Heat generation scales with volume ($\propto R^3$) but heat removal scales with surface area ($\propto R^2$), so a reaction that is safely isothermal at laboratory scale may exhibit **thermal runaway** at larger scale. Always perform a thermal hazard assessment before scaling up exothermic catalytic reactions.

3.14 Notation Introduced in This Chapter

3.15 Problems

1. **Integral method analysis.** The following concentration-time data are obtained for a catalytic reaction in a batch reactor:

t (min)	0	5	10	15	20
C_A (mol/L)	2.00	1.60	1.20	0.80	0.40

- (a) Determine the reaction order using the integral method.

Table 3.2: Reactor design notation

Symbol	Description	Typical Units
N_A	Number of moles of species A	mol
C_A	Concentration of species A	mol/L
C_{A0}	Initial concentration of species A	mol/L
r	Rate of consumption (positive)	mol/(L·s)
k	Rate constant	depends on order
n	Reaction order	dimensionless
$t_{1/2}$	Half-life	s, min
TOF	Turnover frequency	s^{-1} , h^{-1}
TON	Turnover number	dimensionless
N_{sites}	Number of moles of active sites	mol
t_{contact}	Contact time (flow systems)	s
$\dot{V}$	Volumetric flow rate	L/s, mL/min
W/F_{A0}	Catalyst weight over molar flow rate	$g_{\text{cat}} \cdot s / \text{mol}$

Note: For activation parameters (E_a , A , $\Delta H^\ddagger$, $\Delta S^\ddagger$), see the notation table in Chapter 2.

- (b) Calculate the rate constant with appropriate units.
- (c) Predict the time required for 95% conversion.

NB3 tools: Use `scipy.stats.linregress` for linear fits. Compare zeroth-, first-, and second-order linearizations.

2. **TOF calculation.** A Pt/SiO₂ catalyst (3.5% Pt by mass) is tested for propane dehydrogenation. In a batch reactor with 1.00 g catalyst and 2.0 L volume, the initial rate of propane consumption is 2.8×10^{-5} mol/(L·s). Hydrogen chemisorption indicates a Pt dispersion of 45%.

- (a) Calculate the TOF in units of s^{-1} .
- (b) If the reaction runs for 4 hours before the catalyst deactivates completely, what is the TON?
- (c) A new catalyst preparation gives TOF = $0.8 s^{-1}$. Which catalyst is more active per site?

NB3 tools: Use `calculate_tof()` from NB3 to verify your calculation.

3. **Arrhenius analysis.** A surface reaction has the following temperature dependence:

T (K)	400	420	440	460	480
k (s^{-1})	0.012	0.031	0.075	0.17	0.36

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Table 3.3: Characterization notation

Symbol	Description	Typical Units
V	Volume of gas adsorbed	cm^3 (STP)/g
V_m	Monolayer volume	cm^3 (STP)/g
P_0	Saturation vapor pressure	Pa, bar
c	BET constant	dimensionless
S_{BET}	BET surface area	m^2/g
E_1	Heat of adsorption (first layer)	kJ/mol
E_L	Heat of liquefaction	kJ/mol
σ	Molecular cross-sectional area	nm^2
r_p	Pore radius	nm
t	Statistical thickness of adsorbed layer	nm
E_d	Desorption activation energy	kJ/mol
T_p	TPD peak temperature	K
ν	Pre-exponential factor (desorption)	s^{-1}
$\beta^\dagger$	Heating rate (TPD)	K/min, K/s
T_{red}	Reduction temperature	K, $^\circ\text{C}$
λ	X-ray wavelength	nm
d	Interplanar spacing	nm
D	Crystallite size (Scherrer)	nm
$\beta^\dagger$	Peak FWHM (XRD)	rad
E_B	Binding energy (XPS)	eV
E_K	Kinetic energy (XPS)	eV
D (dispersion)	Metal dispersion	dimensionless
$\bar{d}_s$	Surface-average particle diameter	nm
ν_{CO}	CO stretching frequency	cm^{-1}
K	Scherrer shape factor (~ 0.9)	dimensionless
γ	Surface tension of liquid N_2	N/m
V_L	Molar volume of liquid N_2	cm^3/mol
r_k	Kelvin radius	nm

$^\dagger\beta$ denotes heating rate in TPD context and peak FWHM in XRD context.

D denotes crystallite size in the Scherrer equation and metal dispersion in the context of chemisorption/TEM analysis; context disambiguates.

- (a) Construct an Arrhenius plot and determine E_a and A .
- (b) Predict the rate constant at 500 K.
- (c) Perform an Eyring analysis to determine $\Delta H^\ddagger$ and $\Delta S^\ddagger$.
- (d) Interpret the sign of $\Delta S^\ddagger$ in terms of the transition state structure.

NB3 tools: Use `scipy.stats.linregress` for both the Arrhenius ($\ln k$ vs $1/T$) and Eyring ($\ln(k/T)$ vs $1/T$) plots.

4. **Comparing catalysts.** Two catalysts are tested for the same reaction under identical conditions:

Catalyst	Rate (mol/g _{cat} /h)	Loading (%)	Dispersion (%)
Catalyst A	0.45	5.0	20
Catalyst B	0.28	1.0	60

Both catalysts use Pt ($M_{\text{Pt}} = 195$ g/mol).

- (a) Calculate the TOF for each catalyst.
- (b) Which catalyst is more active on a per-site basis?
- (c) Suggest a reason why the two catalysts might have different intrinsic activities.

NB3 tools: Use `calculate_tof()` and `dispersion_from_size()` from NB3.

5. BET Surface Area Analysis

The following nitrogen adsorption data were obtained for a silica support at 77 K (sample mass = 0.1823 g):

P/P_0	V_{ads} (cm ³ STP/g)
0.05	85.3
0.10	94.2
0.15	101.8
0.20	108.7
0.25	115.4
0.30	122.5

- (a) Construct the BET plot and determine the slope and intercept.
- (b) Calculate the monolayer volume V_m and BET constant c .
- (c) Calculate the specific surface area in m²/g.

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- (d) Interpret the BET constant c for this silica. Compare it with values typically observed for alumina ($c > 200$). What does the value of c indicate about the strength of the N_2 -surface interaction?

NB3 tools: Use `bet_analysis()` from NB3 to verify your BET linearization.

6. XRD Crystallite Size

A Pd/C catalyst shows the following XRD peaks (Cu $K\alpha$ radiation):

Peak	2θ ($^\circ$)	FWHM ($^\circ$)
Pd(111)	40.12	2.34
Pd(200)	46.66	2.68

The instrumental broadening is 0.12° .

- (a) Calculate the crystallite size from each peak using the Scherrer equation ($K = 0.89$).
- (b) Should the two values be identical? Explain any differences.
- (c) Estimate the Pd dispersion from the average crystallite size.
- (d) If the catalyst contains 5 wt% Pd, estimate the number of surface Pd atoms per gram of catalyst.

NB3 tools: Use `scherrer_size()` and `dispersion_from_size()` from NB3. Correct for instrumental broadening first.

7. NH_3 -TPD of Zeolite Acid Sites

An NH_3 -TPD experiment on an H-ZSM-5 zeolite (heating rate 10 K/min) shows two peaks:

- Peak 1: $T_p = 210^\circ\text{C}$, Area = 0.35 (arbitrary units)
 - Peak 2: $T_p = 420^\circ\text{C}$, Area = 0.45 (arbitrary units)
- (a) Using the Redhead equation (assume $\nu = 10^{13} \text{ s}^{-1}$), calculate the desorption energies for both peaks. *Note: ensure consistent units for the heating rate (β) when using the Redhead equation.*
- (b) Which peak corresponds to strong acid sites and which to weak acid sites?
- (c) Calculate the ratio of strong to weak acid sites based on peak areas.

- (d) If the total acid site concentration is 0.8 mmol/g, what is the concentration of strong acid sites?

NB3 tools: Use `redhead.equation()` from NB3. Convert heating rate to K/s: `redhead.equation(T-peak, beta=10/60)`.

8. XPS Analysis of Supported Cu Catalyst

XPS analysis of a Cu/ZnO catalyst after different treatments gives the following Cu 2p_{3/2} binding energies:

Treatment	E_B (eV)	Area (%)
As-prepared (calcined)	933.8	100
Reduced at 250°C	932.4 (75%), 933.6 (25%)	100
Reduced at 400°C	932.4	100

- (a) Assign oxidation states to each binding energy observed. *Note: Cu(0) and Cu(I) have very similar 2p_{3/2} binding energies. In practice, the Cu LMM Auger parameter is used to distinguish them.*
- (b) What percentage of Cu is reduced after treatment at 250°C?
- (c) Why might some Cu remain oxidized even after reduction at 250°C?
- (d) If the Cu 2p/Zn 2p area ratio increases after reduction, what does this suggest about the catalyst structure?

9. Integrated Characterization Study

A Pt/TiO₂ catalyst is characterized by multiple techniques with the following results:

- BET: $S = 45 \text{ m}^2/\text{g}$
 - XRD: Pt(111) peak FWHM = 1.2° at $2\theta = 39.8^\circ$; TiO₂ shows anatase phase
 - TEM: Average particle size $\bar{d}_s = 3.2 \text{ nm}$ (from 200 particles counted)
 - XPS: Pt 4f_{7/2} at 71.2 eV (100%), surface Pt/Ti atomic ratio = 0.08
 - CO chemisorption: $15.2 \text{ } \mu\text{mol CO/g catalyst}$
- (a) Calculate the Pt crystallite size from XRD ($K = 0.9$, instrumental broadening = 0.08°). How does it compare to TEM?
- (b) Calculate the Pt dispersion from TEM particle size.

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- (c) From CO chemisorption data, calculate the Pt dispersion assuming 1:1 CO:Pt_{surface} stoichiometry. The catalyst contains 1.5 wt% Pt.
- (d) Are the dispersion values from (b) and (c) consistent? Discuss possible reasons for any discrepancies.
- (e) What does the XPS Pt 4f binding energy tell you about the Pt oxidation state?

NB3 tools: `scherrer_size()`, `dispersion_from_size(metal='Pt')`, `active_surface_area()`.

10. CO-FTIR Site Assignment

CO-FTIR spectra of a Pt/Al₂O₃ catalyst at different CO pressures show the following bands:

P_{CO} (mbar)	Band 1 (cm ⁻¹)	Band 2 (cm ⁻¹)
0.1	2068	2048
1.0	2073	2055
10	2078	2062
100	2084	2068

- (a) Assign the two bands to different types of Pt surface sites.
- (b) Explain why both bands shift to higher frequency as CO pressure increases.
- (c) Extrapolate to estimate the singleton frequencies (zero-coverage limit) for each band.
- (d) After exposure to 1% O₂ at 100°C, a new band appears at 2118 cm⁻¹. What species does this represent?
- (e) How would you use CO-FTIR to distinguish between a bimetallic PtRu catalyst and a physical mixture of Pt/C and Ru/C?

11. TEM Particle Size Distribution Analysis

A TEM study of a Pd/C catalyst counts 180 particles with the following size distribution:

Particle diameter (nm)	Number of particles
1.0	12
2.0	38
3.0	55
4.0	40
5.0	20
6.0	10
8.0	5

- Calculate the number-average diameter $\bar{d}_n$ and the surface-average diameter $\bar{d}_s$.
- Calculate the polydispersity index $\text{PDI} = \bar{d}_w/\bar{d}_n$ and comment on the breadth of the distribution.
- Estimate the metal dispersion using $D_{\text{Pd}} \approx 1.12/\bar{d}_s$ (nm).
- A colleague reports $\bar{d}_n = 2.8$ nm from counting only 25 particles in a single TEM image. Explain why this value may be unreliable and how many particles should be counted for $\pm 10\%$ accuracy.
- XRD gives a Scherrer crystallite size of 2.5 nm for the same catalyst. Is it consistent with the TEM data? Explain any discrepancy.

NB3 tools: Use `dispersion_from_size(metal='Pd')` from NB3 to verify your dispersion estimate.

Looking Ahead

In this chapter, we developed two complementary toolkits: reactor design equations for extracting kinetic parameters from batch and flow experiments, and characterization techniques that reveal the physical and chemical properties of the catalyst itself. Together, these tools let us connect measured rates to intrinsic surface chemistry on a per-site basis. We assumed, however, that measured rates reflect the true activity of the catalyst surface—that every active site experiences the same concentration of reactants, and that a single reaction dominates. The remaining chapters relax these assumptions and add the layers of complexity encountered in real catalytic systems.

Chapter 4 addresses the most important departure from ideal behavior: **transport limitations**. Real catalysts are porous, and reactants must diffuse through narrow pores to reach interior active sites. When diffusion is slow relative to reaction, concentration gradients develop within the particle, and measured rates fall below their intrinsic values. The **Thiele modulus** ϕ and **effectiveness factor** η quantify this effect, allowing us to diagnose whether laboratory data reflect true surface kinetics or are masked by diffusion. The BET surface area and BJH pore size distribution measured here enter directly into the calculation of effective diffusivity D_{eff} , since D_{eff} depends on pore diameter, tortuosity, and porosity—all quantities extracted from the nitrogen adsorption isotherm.

Chapter 5 extends our analysis from a single reaction to **competing reaction networks**—parallel and consecutive pathways that produce both desired and undesired products. The choice of reactor type (CSTR vs. PFR) and operating conditions (temperature, contact time) profoundly affects product distribution, and optimizing selectivity often conflicts with

3. Reactor Design and Catalyst Characterization

maximizing conversion. Different active site types may catalyze parallel reaction pathways with different selectivities; the site-type distributions revealed by NH_3 -TPD (weak vs. strong acid sites) and CO-FTIR (terrace vs. edge vs. corner metal sites) inform selectivity models. Understanding *which* sites are responsible for the desired product—and which for unwanted side reactions—is the foundation of selectivity engineering.

Chapter 6 explores **microporous zeolite catalysts and carbon nanotube supports**. For zeolites, shape selectivity—the ability to discriminate between reactants, products, or transition states based on molecular geometry—adds a steric dimension to the kinetic and transport frameworks developed here. The same chapter examines CNT supports where anomalously fast diffusion inside nanotube channels ($D_{\text{CNT}} = F \cdot D_{\text{K}}$ with $F = 10$ – $10,000$) modifies the effectiveness factor. The characterisation toolkit from this chapter applies directly: XRD for framework identification, NH_3 -TPD for acid site quantification, TEM for particle imaging, and Raman spectroscopy for CNT quality assessment.

Finally, Appendix G brings all concepts together in the context of **industrial catalytic synthesis**—Pd-catalyzed cross-coupling, asymmetric catalysis, and green chemistry—where reactor design, transport analysis, selectivity engineering, and catalyst characterization must be integrated to develop practical processes. The ability to select, perform, and interpret the right combination of characterization techniques alongside appropriate reactor models is what distinguishes a catalyst scientist from someone who merely runs reactions.

Beyond the chapters ahead, the tools you have learned here reach into domains that may seem far removed from heterogeneous catalysis but are, in fact, built on the same principles.

The enzyme kinetics parallel. If you take a biochemistry course, you will encounter the Michaelis–Menten equation,

$$v = \frac{V_{\text{max}}[\text{S}]}{K_M + [\text{S}]},$$

which describes enzyme-catalyzed reactions. Compare this to the unimolecular Langmuir–Hinshelwood rate expression $r = kK_A P_A / (1 + K_A P_A)$: the two are mathematically identical, with $V_{\text{max}} \leftrightarrow k$ and $K_M \leftrightarrow 1/K_A$. Biochemists extract k_{cat} (the enzymatic TOF) and K_M using progress-curve analysis and Lineweaver–Burk plots—the same integrated rate-law methods you practised in this chapter. The kinetic frameworks of heterogeneous catalysis and enzymology are two dialects of the same language. Recognising these connections will serve you well whenever you work at the boundary between chemistry, biology, and engineering.

Chapter 4

Transport Limitations in Porous Catalysts

Learning Objectives

By the end of this chapter, you will be able to:

- LO1.** Distinguish **intrinsic kinetics** from **transport-limited observed rates** and explain when each regime dominates.
- LO2.** Explain the difference between **external film mass transfer** and **internal pore diffusion**, identifying which resistance controls in a given situation.
- LO3.** Define and calculate the **Thiele modulus** (ϕ) for first-order kinetics in spherical and slab geometries.
- LO4.** Compute the **effectiveness factor** (η) and interpret its limiting behavior for kinetic control ($\phi \ll 1$) and diffusion control ($\phi \gg 1$).
- LO5.** Apply the **Weisz–Prater criterion** to diagnose whether internal diffusion limitations are significant from experimental data.
- LO6.** Connect diffusion behavior to practical catalyst systems including **zeolites** (micropore diffusion) and **nanotubes** (Knudsen diffusion).
- LO7.** Integrate the effectiveness factor into reactor design equations:
$$r_{\text{obs}} = \eta \cdot r_{\text{intrinsic}}.$$

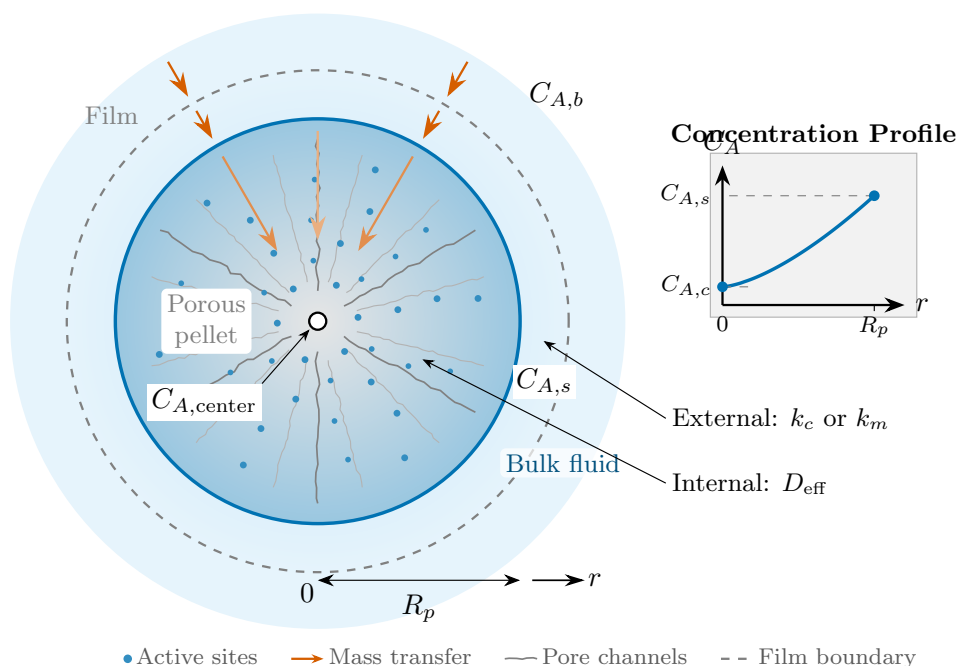


Figure 4.1: Cross-section of a spherical catalyst pellet showing mass transfer resistances. Reactant A diffuses from the bulk fluid ($C_{A,b}$) through the external boundary layer to the pellet surface ($C_{A,s}$), then through the porous interior where it reacts at active sites. The concentration decreases from surface to center due to simultaneous diffusion and reaction. The inset shows the radial concentration profile. Two key transport parameters govern this process: the external mass transfer coefficient k_c (or k_m) and the effective diffusivity D_{eff} .

4.1 Why Transport Limitations Matter

In Chapter 2, we developed rate expressions for surface reactions assuming that reactant concentrations at active sites equal their bulk values. In Chapter 3, we used these intrinsic rate laws to design CSTRs and PFRs. Now we confront a critical reality check: *porous catalysts may not operate at their intrinsic rates because molecules cannot diffuse fast enough to reach all active sites.*

A porous catalyst pellet has an external surface exposed to the bulk fluid and an extensive network of internal pores where most active sites reside. For a reactant A , the concentration can drop from $C_{A,b}$ in the bulk fluid, to $C_{A,s}$ at the pellet surface, to even lower values $C_A(r)$ inside the pores—often reaching a minimum at the pellet center. Even with fast intrinsic kinetics, the catalyst interior may be “starved” of reactant.

Understanding transport limitations is essential for:

- **Zeolites:** Micropores create strong diffusion resistance; shape selectivity depends on relative diffusion rates of different molecules. This is exploited in pharmaceutical synthesis for chiral separations and selective transformations.
- **Supported metal catalysts:** Palladium on carbon (Pd/C) and other heterogeneous catalysts for cross-coupling reactions can be diffusion-limited when large organic substrates must access active sites in porous supports.
- **Nanotube catalysts:** Small pore diameters lead to Knudsen diffusion, where molecules collide more frequently with pore walls than with each other.

This chapter develops the theoretical framework to quantify these effects and provides practical tools for diagnosis and design. These concepts are directly relevant to catalytic organic synthesis (see the appendix on catalytic organic synthesis, appendix G), where supported metal catalysts and zeolites are widely used.

Remark 4.1 (Safety Note: Handling Porous Catalyst Pellets).

Porous catalyst pellets present two practical hazards in the laboratory:

- **Inhalable dust.** Crushing catalysts to diagnose diffusion limitations (Section 4.6.2) generates fine particulates. Catalyst dust (e.g., Ni, Co, Cr oxides) can be toxic or carcinogenic. Always crush pellets in a fume hood and wear appropriate respiratory protection.
- **Thermal runaway.** For exothermic reactions, poor heat removal inside large pellets can lead to hot spots (Section 4.9). In extreme cases ($\beta\gamma > 4$), multiple steady states and ignition phenomena can cause sudden temperature excursions. Monitor reactor temperature closely when scaling up pellet size.

Fig. 4.1 illustrates the essential physics of mass transfer in a porous catalyst. A reactant molecule in the bulk fluid must overcome two sequential resistances before reaching an active site in the pellet interior. First, it must diffuse through the *external boundary layer*—a thin stagnant film of fluid surrounding the pellet—where the concentration drops from $C_{A,b}$ to $C_{A,s}$. This external resistance depends on fluid velocity and pellet geometry. Second, the molecule must navigate the tortuous pore network inside the

pellet, where it simultaneously diffuses and reacts. As molecules penetrate deeper, they are progressively consumed by reaction, creating a concentration gradient that decreases toward the pellet center. In the limiting case of severe diffusion limitation, only a thin shell near the surface participates in reaction—the interior is effectively “starved” of reactant. Understanding and quantifying these effects is the central goal of this chapter.

4.2 Two Types of Mass Transfer Limitations

Before diving into mathematics, we must clearly distinguish two fundamentally different transport resistances that can limit catalyst performance.

4.2.1 External Film Mass Transfer

Reactant molecules must cross a boundary layer (the “film”) surrounding each catalyst pellet before reaching the external surface. The flux through this film is proportional to the concentration difference:

$$\text{Molar flux} = k_c (C_{A,b} - C_{A,s}) \quad (4.1)$$

where k_c is the external mass transfer coefficient (units: m/s).

Key diagnostic: If external mass transfer limits the rate, *increasing fluid velocity or stirring rate will increase the observed rate* because higher flow reduces the boundary layer thickness and increases k_c .

4.2.2 Internal Pore Diffusion

Even when external film resistance is negligible (so $C_{A,s} \approx C_{A,b}$), reactant must still diffuse through the tortuous pore network to reach active sites deep inside the pellet. Internal diffusion becomes limiting when:

- Pellets are large (long diffusion path)
- Pores are small or tortuous (low effective diffusivity)
- Intrinsic reaction is fast (reactant consumed before reaching center)
- Adsorption is strong (common in zeolites, slows molecular motion)

Key diagnostic: If internal diffusion limits the rate, *crushing the catalyst into smaller particles will increase the observed rate* because molecules have shorter distances to diffuse.

Remark 4.2 (Chapter Focus: Internal Diffusion). This chapter focuses primarily on internal diffusion and the effectiveness factor η . External mass transfer effects are addressed briefly in Section 4.8 and

combined with internal resistance in Section 4.9.

Remark 4.3 (Concept Check: External vs. Internal Resistance).

A supported Pd catalyst is tested for a hydrogenation reaction in a stirred batch reactor. Consider two separate experiments:

1. Doubling the stirring speed increases the observed rate by 30%.
2. Crushing the pellets to half their diameter increases the observed rate by 25%.

For each experiment, state which transport resistance (external, internal, or neither) the observation implicates, and explain your reasoning.

Hint: Review the key diagnostics for external (Section 4.2.1) and internal (Section 4.2.2) limitations above.

4.3 Reaction–Diffusion Inside a Porous Pellet

4.3.1 The Physical Picture

Inside a porous catalyst pellet, two processes occur simultaneously:

1. **Diffusion:** Reactant molecules move down the concentration gradient toward the pellet center.
2. **Reaction:** At every point inside the pellet, reactant is consumed by the catalytic reaction.

At steady state, these processes balance: diffusion supplies exactly as much reactant as reaction consumes at each location.

4.3.2 The Effective Diffusivity

Diffusion inside a porous material is slower than in free space because:

- Only a fraction of the pellet volume is pore space (porosity ε)
- Pores are not straight paths (tortuosity τ_p)

The **effective diffusivity** accounts for these geometric factors:

$$D_{\text{eff}} = \frac{\varepsilon}{\tau_p} D_{\text{pore}} \quad (4.2)$$

where D_{pore} is the diffusivity inside a single pore. Typical values: $D_{\text{eff}} \approx 10^{-6}$ to 10^{-7} m²/s for gases in mesoporous catalysts.

Remark 4.4 (Estimating Tortuosity). Tortuosity τ_p accounts for the longer path length through tortuous pores compared to a straight line. Typical values:

- $\tau_p \approx 3$: Random cylindrical pores with isotropic orientation (theoretical minimum for random media)
- $\tau_p = 3\text{--}5$: Well-connected pore networks (most commercial catalysts)
- $\tau_p = 5\text{--}10+$: Poorly connected or partially blocked pore networks

When no experimental data are available, $\tau_p = 3\text{--}4$ is a reasonable first estimate for pelletized oxide catalysts.

4.3.3 The Governing Equation for a Spherical Pellet

For a spherical pellet of radius R_p with effective diffusivity D_{eff} , a steady-state mass balance yields the governing equation. The derivation relies on **Fick's first law**: the molar flux of species A is proportional to the concentration gradient,

$$J_A = -D_{\text{eff}} \frac{dC_A}{dr} \quad (4.3)$$

where the negative sign indicates that diffusion proceeds from high to low concentration.

Remark 4.5 (Shell-Balance Derivation of the Reaction–Diffusion Equation).

We derive Eq. (4.22) from first principles using a steady-state mass balance on a thin spherical shell.

Step 1: Define the control volume. Consider a thin shell between radii r and $r + dr$ inside a spherical pellet of radius R_p . The shell has volume $dV = 4\pi r^2 dr$ and the diffusive flux passes through surfaces of area $4\pi r^2$.

Step 2: Write the molar balance at steady state. At steady state the net rate of diffusion *into* the shell equals the rate of consumption by reaction *within* the shell:

$$\left[-D_{\text{eff}} 4\pi r^2 \frac{dC_A}{dr} \right]_r - \left[-D_{\text{eff}} 4\pi r^2 \frac{dC_A}{dr} \right]_{r+dr} = r_{\text{intrinsic}}(C_A) 4\pi r^2 dr \quad (4.4)$$

The left-hand side is the molar inflow at r minus the molar outflow at $r + dr$ (Fick's first law, with the negative sign giving positive flux toward the center where concentration is lower).

Step 3: Take the limit $dr \rightarrow 0$. Dividing both sides by $4\pi dr$ and recognising the definition of a derivative, we obtain

$$\boxed{D_{\text{eff}} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dC_A}{dr} \right) = r_{\text{intrinsic}}(C_A)} \quad (4.5)$$

This is the general steady-state reaction–diffusion equation for a spherical pellet with any intrinsic kinetics $r_{\text{intrinsic}}(C_A)$.

Step 4: First-order kinetics. Substituting $r_{\text{intrinsic}} = k C_A$ gives Eq. (4.22) directly.

Step 5: Non-dimensionalise to reveal the Thiele modulus. Define the dimensionless variables $\xi = r/R_p$ and $\psi = C_A/C_{A,s}$. Substituting into Eq. (4.22) and simplifying yields

$$\frac{d^2(\xi \psi)}{d\xi^2} = \phi^2 (\xi \psi), \quad \phi = R_p \sqrt{\frac{k}{D_{\text{eff}}}} \quad (4.6)$$

The Thiele modulus ϕ thus emerges naturally as the *only* parameter governing the concentration profile. This is the power of non-dimensionalisation: regardless of the individual values of R_p , k , and D_{eff} , the shape of the internal concentration profile depends solely on ϕ .

For the important case of **first-order intrinsic kinetics** ($r_{\text{intrinsic}} = kC_A$), the equation becomes:

$$\boxed{\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dC_A}{dr} \right) = \frac{k}{D_{\text{eff}}} C_A} \quad (4.7)$$

This second-order ODE requires two boundary conditions:

$$\text{Symmetry at center: } \left. \frac{dC_A}{dr} \right|_{r=0} = 0 \quad (4.8)$$

$$\text{Surface concentration: } C_A(R_p) = C_{A,s} \quad (4.9)$$

The surface boundary condition assumes external film resistance is negligible. We will relax this assumption in Section 4.9.

Remark 4.6 (Concept Check: Shell Balance Reasoning). In the shell-balance derivation (Section 4.3.3), consider what happens at steady state when the intrinsic rate constant k increases (e.g., due to a temperature rise):

1. Does the concentration gradient inside the pellet become steeper or shallower? Why?
2. Would the concentration at the pellet centre increase, decrease, or stay the same?

Hint: Think about the balance between diffusive supply and reactive consumption at each point inside the pellet.

4.4 The Thiele Modulus

4.4.1 Definition and Physical Meaning

To understand when diffusion matters, we introduce the **Thiele modulus**, the key dimensionless group for porous catalyst analysis.

Definition 4.1 (Thiele Modulus for First-Order Reaction in a Sphere).

$$\phi = R_p \sqrt{\frac{k}{D_{\text{eff}}}} \quad (4.10)$$

where R_p is the pellet radius, k is the first-order rate constant, and D_{eff} is the effective diffusivity.

The Thiele modulus compares two characteristic rates:

- **Numerator** ($R_p \sqrt{k}$): Related to how fast reaction consumes reactant
- **Denominator** ($\sqrt{D_{\text{eff}}}$): Related to how fast diffusion supplies reactant

Physical interpretation:

- $\phi \ll 1$: Diffusion is fast compared to reaction. Concentration is nearly uniform throughout the pellet.
- $\phi \gg 1$: Reaction is fast compared to diffusion. Reactant is consumed near the surface; the pellet center is “starved.”

4.4.2 Thiele Modulus for a Flat Slab

For a flat slab of half-thickness L (reaction occurs on both faces):

$$\phi_{\text{slab}} = L \sqrt{\frac{k}{D_{\text{eff}}}} \quad (4.11)$$

The slab geometry is mathematically simpler and provides analytical solutions that build intuition for the more complex spherical case.

Remark 4.7 (Generalized Thiele Modulus for Non-First-Order Kinetics).

So far we have defined ϕ only for first-order kinetics. For an n th-order intrinsic rate law $r = k C_A^n$, the **generalized Thiele modulus** is

$$\phi_n = L \sqrt{\frac{n+1}{2} \frac{k C_{A,s}^{n-1}}{D_{\text{eff}}}} \quad (4.12)$$

where L is the characteristic length (half-thickness for a slab, $R_p/3$ for a sphere—see Section 4.4.2) and $C_{A,s}$ is the surface concentration.

Verification for $n = 1$. Substituting $n = 1$ into Eq. (4.12):

$$\phi_1 = L \sqrt{\frac{1+1}{2} \frac{k C_{A,s}^0}{D_{\text{eff}}}} = L \sqrt{\frac{k}{D_{\text{eff}}}}$$

which is exactly the standard first-order result (Eq. (4.11)).

Zero-order kinetics ($n = 0$). For $r = k$ (rate independent of concentration):

$$\phi_0 = L \sqrt{\frac{k}{2 D_{\text{eff}} C_{A,s}}}$$

Here $\eta = 1$ as long as $\phi_0 < 1$, because the entire pellet has sufficient reactant. Only when $\phi_0 > 1$ does a “dead core” appear where the concentration drops to zero before reaching the center.

Langmuir–Hinshelwood kinetics. For rate expressions of the form $r = k C_A / (1 + K C_A)$, no single reaction order applies. The generalized modulus provides a useful approximation by evaluating an effective order at the surface concentration, but computing the exact η requires numerical integration of the reaction–diffusion equation.

The key insight: When the η – ϕ data for *any* reaction order are plotted against ϕ_n (rather than the first-order ϕ), the curves collapse nearly onto the first-order result. This universality is what makes the generalized modulus so powerful in practice—you can read η from a single master curve regardless of kinetic order.

Remark 4.8 (The Aris Characteristic Length: Unifying Pellet Geometries).

Aris (1957) showed that defining the characteristic length as the

ratio of pellet volume to external surface area,

$$L_c = \frac{V_p}{S_{\text{ext}}} \quad (4.13)$$

collapses the $\eta(\phi)$ curves for *all* pellet geometries to within approximately 10% of each other.

Geometry	V_p/S_{ext}	L_c
Infinite slab (sealed one side, thickness L)	$L \cdot A/A$	L
Infinite cylinder (radius R)	$\pi R^2 \ell / 2\pi R \ell$	$R/2$
Sphere (radius R_p)	$\frac{4}{3}\pi R_p^3 / 4\pi R_p^2$	$R_p/3$

Using this characteristic length, we define

$$\phi_c = L_c \sqrt{\frac{k}{D_{\text{eff}}}} \quad (4.14)$$

Note that $\phi_c = \phi_{\text{sphere}}/3$ for a sphere, which explains why the sphere effectiveness factor formula, when plotted against ϕ_c instead of ϕ , overlaps closely with the slab result.

Practical value for irregular shapes. Commercial catalyst pellets—rings, trilobes, wagon wheels, extrudates—are rarely perfect spheres or slabs. For any such pellet you can measure or calculate V_p/S_{ext} , and then use the simple *slab* effectiveness factor as a universal approximation:

$$\eta \approx \frac{\tanh(\phi_c)}{\phi_c} \quad (4.15)$$

This single formula replaces the need for geometry-specific solutions and works remarkably well for first-order kinetics in any pellet shape.

Remark 4.9 (Concept Check: Thiele Modulus Estimation). A first-order reaction with $k = 10 \text{ s}^{-1}$ and $D_{\text{eff}} = 2 \times 10^{-6} \text{ m}^2/\text{s}$ is carried out in spherical pellets of radius $R_p = 2 \text{ mm}$.

1. Calculate ϕ and predict the operating regime (kinetic, transition, or diffusion-limited).
2. Without recalculating, state how ϕ would change if: (a) the

pellet radius is halved; (b) the temperature is raised so that k quadruples.

3. What pellet radius would place this system at the boundary of kinetic control ($\phi \approx 0.3$)?

4.5 The Effectiveness Factor

4.5.1 Definition

The **effectiveness factor** quantifies how much the actual reaction rate is reduced compared to what we would observe if the entire pellet operated at surface conditions.

Definition 4.2 (Effectiveness Factor).

$$\eta \equiv \frac{\text{Actual rate in porous pellet}}{\text{Rate if entire pellet were at } C_{A,s}} = \frac{\int_V r(C_A(r)) dV}{r(C_{A,s}) \cdot V} \quad (4.16)$$

The effectiveness factor allows us to write:

$$r_{\text{obs}} = \eta \cdot r_{\text{intrinsic}}(C_{A,s}) \quad (4.17)$$

This is the practical payoff: once we know η , we can use the intrinsic rate law with a simple correction factor.

4.5.2 Effectiveness Factor for a Flat Slab

For first-order kinetics in a flat slab, solving the reaction–diffusion equation yields:

Remark 4.10 (Deriving the Slab Effectiveness Factor). We start from the steady-state reaction–diffusion equation for a first-order reaction inside a flat slab of half-thickness L , with the coordinate x measured from the centre plane:

$$D_{\text{eff}} \frac{d^2 C_A}{dx^2} = k C_A \quad (4.18)$$

The boundary conditions are:

1. **Symmetry at the centre:** $\left. \frac{dC_A}{dx} \right|_{x=0} = 0$
2. **Surface concentration:** $C_A(L) = C_{A,s}$

Defining $\phi = L\sqrt{k/D_{\text{eff}}}$, the general solution of Eq. (4.18) is a linear combination of cosh and sinh. The symmetry boundary condition eliminates sinh, and evaluating at $x = L$ fixes the remaining constant, giving the **concentration profile**:

$$\boxed{\frac{C_A(x)}{C_{A,s}} = \frac{\cosh(\phi x/L)}{\cosh \phi}} \quad (4.19)$$

This profile is flat ($C_A \approx C_{A,s}$) when $\phi \ll 1$ and steep when $\phi \gg 1$, consistent with the physical picture in Fig. 4.3.

To obtain the effectiveness factor, we compute the volume-averaged concentration (which equals the volume-averaged rate for first-order kinetics):

$$\begin{aligned} \eta &= \frac{1}{L} \int_0^L \frac{C_A(x)}{C_{A,s}} dx = \frac{1}{L} \int_0^L \frac{\cosh(\phi x/L)}{\cosh \phi} dx \\ &= \frac{1}{L} \cdot \frac{L}{\phi \cosh \phi} \left[\sinh(\phi x/L) \right]_0^L = \frac{\sinh \phi}{\phi \cosh \phi} \end{aligned} \quad (4.20)$$

Recognising $\sinh \phi / \cosh \phi = \tanh \phi$, we arrive at $\eta_{\text{slab}} = \tanh \phi / \phi$ as stated in Eq. (4.21).

$$\boxed{\eta_{\text{slab}} = \frac{\tanh \phi}{\phi}} \quad (4.21)$$

This elegant result shows that η depends only on ϕ —all the physical parameters (L, k, D_{eff}) enter through the single dimensionless group.

4.5.3 Effectiveness Factor for a Sphere

For first-order kinetics in a spherical pellet:

Remark 4.11 (Deriving the Sphere Effectiveness Factor). For a spherical pellet of radius R_p , the steady-state reaction–diffusion equation in radial coordinates is:

$$D_{\text{eff}} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dC_A}{dr} \right) = k C_A \quad (4.22)$$

The key trick is the substitution $u = r C_A$, which transforms this into the simpler equation:

$$\frac{d^2 u}{dr^2} = \frac{k}{D_{\text{eff}}} u$$

whose general solution is $u = A \sinh(r\sqrt{k/D_{\text{eff}}}) + B \cosh(r\sqrt{k/D_{\text{eff}}})$.

We apply two boundary conditions:

1. **Finite concentration at the centre** ($r = 0$): Since $C_A = u/r$ and $u(0) = B$ (because $\sinh(0) = 0$, $\cosh(0) = 1$), we would have $C_A(0) = B/0 \rightarrow \infty$ unless $B = 0$. With $B = 0$, L'Hôpital's rule gives the finite value $C_A(0) = A\sqrt{k/D_{\text{eff}}}$.
2. **Surface concentration:** $u(R_p) = R_p C_{A,s}$, which gives $A = R_p C_{A,s} / \sinh \phi$ where $\phi = R_p \sqrt{k/D_{\text{eff}}}$.

Substituting back, we obtain the **concentration profile**:

$$\boxed{\frac{C_A(r)}{C_{A,s}} = \frac{R_p}{r} \cdot \frac{\sinh(\phi r/R_p)}{\sinh \phi}} \quad (4.23)$$

Notice that this profile satisfies $C_A(R_p) = C_{A,s}$ and remains finite at $r = 0$ (use L'Hôpital's rule: $\lim_{r \rightarrow 0} \sinh(\phi r/R_p)/r = \phi/R_p$).

The effectiveness factor is the volume-averaged rate divided by the surface rate. For a sphere:

$$\begin{aligned} \eta &= \frac{3}{R_p^3} \int_0^{R_p} \frac{C_A(r)}{C_{A,s}} r^2 dr = \frac{3}{R_p^3} \int_0^{R_p} \frac{R_p \sinh(\phi r/R_p)}{\sinh \phi} r dr \\ &= \frac{3}{R_p^2 \sinh \phi} \int_0^{R_p} r \sinh\left(\frac{\phi r}{R_p}\right) dr \end{aligned} \quad (4.24)$$

Evaluating the integral by parts and simplifying:

$$\eta = \frac{3}{\phi^2} (\phi \coth \phi - 1)$$

which is the sphere result in Eq. (4.25). You can verify that $\eta \rightarrow 1$ when $\phi \rightarrow 0$ by expanding $\coth \phi \approx 1/\phi + \phi/3 - \dots$.

$$\boxed{\eta_{\text{sphere}} = \frac{3}{\phi^2} (\phi \coth \phi - 1)} \quad (4.25)$$

where $\coth \phi = \cosh \phi / \sinh \phi$ is the hyperbolic cotangent.

4.5.4 Limiting Behavior

The limiting cases of Eq. (4.25) provide essential physical insight:

4. Transport Limitations in Porous Catalysts

Regime	Condition	Effectiveness Factor
Kinetic control	$\phi \ll 1$	$\eta \approx 1$
Strong diffusion limitation	$\phi \gg 1$	$\eta \approx \frac{3}{\phi}$ (sphere)

Physical interpretation of the limits:

- **Kinetic control** ($\phi \ll 1$): Diffusion is fast enough to maintain nearly uniform concentration. The entire pellet volume participates in reaction at full capacity. The intrinsic kinetics control the rate.
- **Diffusion control** ($\phi \gg 1$): Only a thin shell near the pellet surface has significant reactant concentration. The effective reaction volume shrinks as ϕ increases. Note that $\eta \propto 1/\phi$, so even if intrinsic k is huge, the observed rate increases only as $\sqrt{k}$ (since $\phi \propto \sqrt{k}$ and $r_{\text{obs}} = \eta \cdot k C_{A,s} \propto k/\sqrt{k} = \sqrt{k}$).

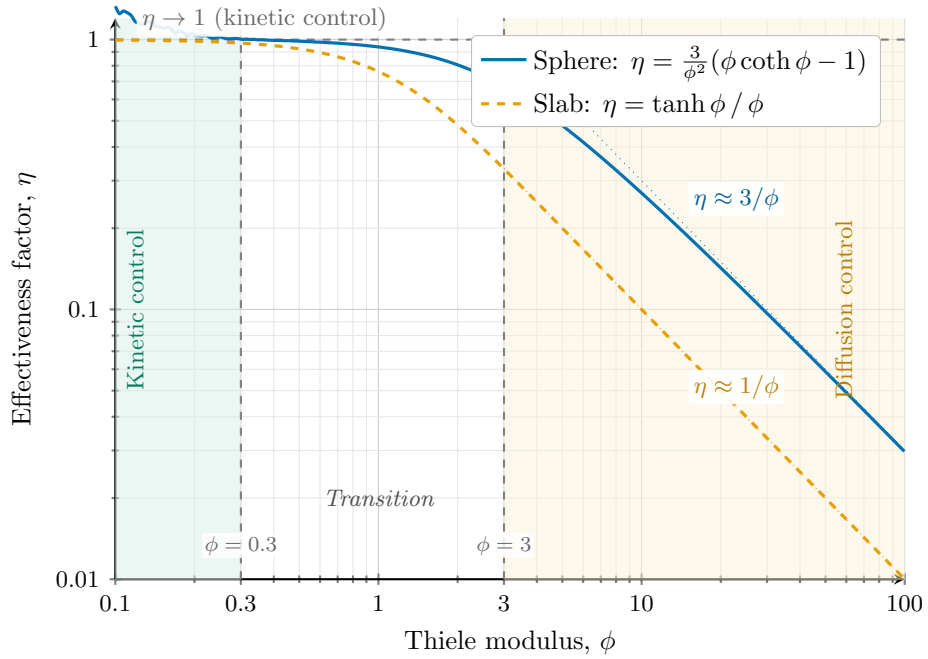


Figure 4.2: Effectiveness factor η as a function of Thiele modulus ϕ for spherical (Eq. (4.25)) and slab (Eq. (4.21)) catalyst geometries on a log-log scale. The kinetic control regime ($\phi < 0.3$) shows $\eta \approx 1$, while diffusion control ($\phi > 3$) exhibits the characteristic $\eta \propto 1/\phi$ behavior. Dotted lines show the asymptotic approximations $\eta \approx 3/\phi$ (sphere) and $\eta \approx 1/\phi$ (slab).

Fig. 4.2 presents the complete picture of how η varies with ϕ . Three distinct regions emerge. In the *kinetic control* regime ($\phi < 0.3$, green shaded

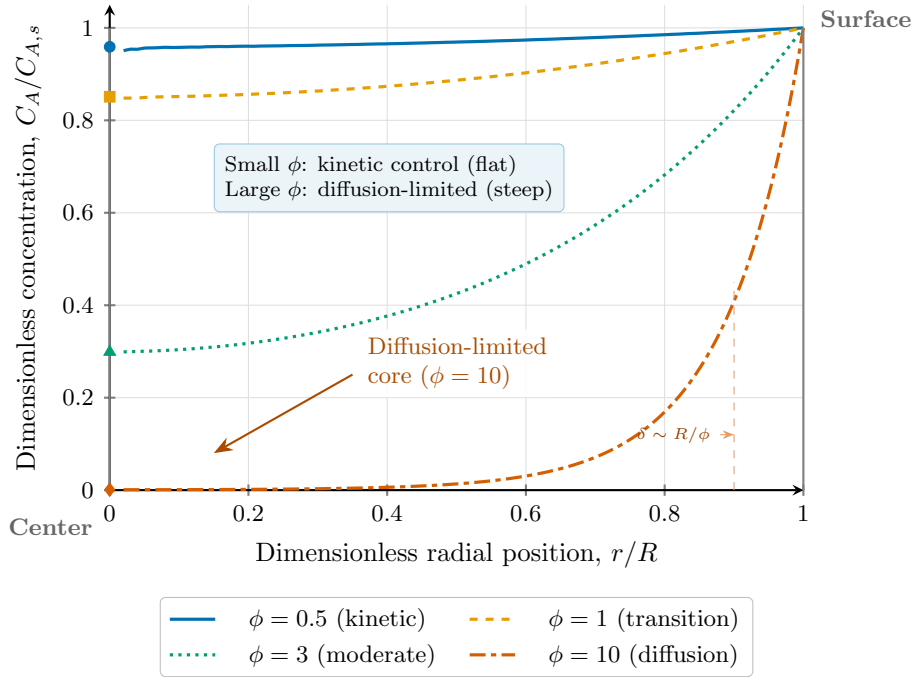


Figure 4.3: Dimensionless concentration profiles $C_A/C_{A,s}$ inside a spherical catalyst pellet for first-order reaction at different Thiele modulus values. Small ϕ indicates kinetic control with uniform concentration; large ϕ indicates diffusion limitation with reactant depletion in the pellet interior. For $\phi = 10$, significant reaction occurs only within a thin shell of thickness $\delta \sim R_p/\phi$ near the surface.

region), diffusion is fast enough to maintain nearly uniform concentration throughout the pellet, and $\eta \approx 1$ —the catalyst operates at its full intrinsic capacity. In the *diffusion control* regime ($\phi > 3$, orange shaded region), reaction at the surface consumes reactant faster than diffusion can supply it, and the effectiveness factor falls as $\eta \propto 1/\phi$. Between these extremes lies a *transition* region where both effects are comparable. The log-log plot reveals that the curves for sphere and slab geometries are nearly parallel at large ϕ , differing only by a geometric factor (3 versus 1 in the numerator). This universality is a powerful result: regardless of pellet shape, the same dimensionless group ϕ captures the competition between reaction and diffusion.

Fig. 4.3 shows why the effectiveness factor decreases with increasing ϕ . For $\phi = 0.5$ (kinetic control), the concentration is nearly uniform from surface to center—the entire pellet volume participates in reaction at essentially full capacity, and $\eta \approx 1$. As ϕ increases, the concentration profile steepens progressively. For $\phi = 10$ (severe diffusion limitation), the concentration plunges to nearly zero within a thin shell near the surface. The “diffusion-

limited core” at $r/R < 0.8$ contributes almost nothing to the overall reaction rate because reactant cannot penetrate that far. This visual connection between concentration profile steepness and effectiveness factor is essential for physical intuition: *the steeper the gradient, the lower the effectiveness factor*. The characteristic penetration depth scales as $\delta \sim R_p/\phi$, explaining why only the outer $\sim 10\%$ of the pellet is active when $\phi = 10$.

Remark 4.12 (Apparent Activation Energy in the Diffusion-Limited Regime). When internal diffusion dominates ($\phi \gg 1$), the observed rate becomes:

$$r_{\text{obs}} = \eta \cdot k C_{A,s} \approx \frac{3}{\phi} \cdot k C_{A,s} = \frac{3}{R_p} \sqrt{D_{\text{eff}} \cdot k} C_{A,s}$$

Since $k = A e^{-E_a/RT}$ and D_{eff} has a weak temperature dependence ($E_D \ll E_a$), the observed rate scales as $e^{-E_a/(2RT)}$, giving:

$$E_{\text{app}} \approx \frac{E_a}{2}$$

This halving of the apparent activation energy is a classic diagnostic for internal mass transfer limitations (Section 2.8). If crushing the catalyst pellet restores the full E_a , diffusion was limiting.

Remark 4.13 (Concept Check: Effectiveness Factor Limits). Consider a spherical pellet with $\phi = 20$.

1. Without using the exact formula, estimate η using the asymptotic approximation. What fraction of the pellet volume is effectively “wasted”?
2. If the pellet radius is reduced by a factor of 4, what is the new ϕ ? Is the system still in the diffusion-limited regime?
3. An exothermic reaction has $\beta = 0.2$ (Prater number). Could $\eta > 1$? Briefly explain the physics.

4.6 Diagnosing Diffusion Limitations: The Weisz–Prater Criterion

In practice, we often have experimental data (observed rates) but do not know the intrinsic kinetics. How can we determine if internal diffusion is important?

4.6.1 The Weisz–Prater Number

The **Weisz–Prater criterion** provides a diagnostic based on observable quantities:

Definition 4.3 (Weisz–Prater Number).

$$N_{WP} = \frac{r_{\text{obs}} \cdot R_p^2}{C_{A,s} \cdot D_{\text{eff}}} \quad (4.26)$$

Interpretation:

- $N_{WP} \ll 1$: Internal diffusion is *negligible*. The pellet operates at nearly uniform concentration.
- $N_{WP} \gtrsim 1$: Internal diffusion is *significant*. The observed rate is lower than the intrinsic rate.

4.6.2 Connection to Thiele Modulus

For first-order kinetics, the Weisz–Prater number is related to the Thiele modulus and effectiveness factor:

$$N_{WP} = \eta \cdot \phi^2 \quad (4.27)$$

This relationship shows why N_{WP} is useful: it can be calculated from measurable quantities (r_{obs} , R_p , $C_{A,s}$, D_{eff}) without knowing the intrinsic rate constant k .

Remark 4.14 (Practical Diagnostics: WP Thresholds and the Particle Size Test).

Part 1 — Refined WP thresholds by reaction order. The simple criterion “ $N_{WP} \ll 1$ ” can be made quantitative. For an n th-order reaction the condition $\eta > 0.95$ (less than 5% rate reduction due to diffusion) requires:

Order n	Criterion ($\eta > 0.95$)	Physical reason
0 (zero)	$N_{WP} < 6$	Rate independent of C_A ; diffusion matters only when $C_A \rightarrow 0$
1 (first)	$N_{WP} < 0.3$	Benchmark case
2 (second)	$N_{WP} < 0.15$	Rate drops quadratically with C_A ; more sensitive

These thresholds follow from the generalised Thiele modulus analysis for n th-order kinetics. For first-order kinetics the criterion $N_{WP} < 0.3$ is the most widely used benchmark. Higher-order reactions are more

sensitive to internal gradients because the rate drops more steeply as concentration decreases, hence the lower threshold for $n = 2$. Conversely, zero-order kinetics is insensitive to concentration until the reactant is completely consumed (dead core), giving the much larger threshold of 6.

Part 2 — The particle size test (crushing test). The most direct experimental diagnostic for internal diffusion limitations requires *no knowledge of D_{eff}* :

1. Measure the observed rate r_{obs} at two or more pellet sizes (e.g., the original pellet and a crushed fraction).
2. **If r_{obs} is independent of $R_p \rightarrow$ kinetic control ($\phi < 0.3$).** The entire pellet volume is fully utilised.
3. **If $r_{\text{obs}} \propto 1/R_p \rightarrow$ strong diffusion limitation ($\phi > 3$).** This follows because $\eta \approx 3/\phi \propto 1/R_p$, so $r_{\text{obs}} = \eta \cdot k C_{A,s} \propto 1/R_p$.
4. **If r_{obs} changes but not inversely with $R_p \rightarrow$ transition regime ($0.3 < \phi < 3$).**

This particle size test is complementary to the Weisz–Prater calculation: the WP criterion checks whether diffusion *could* matter given the observed rate and pellet properties, while the crushing test directly *demonstrates* whether diffusion is limiting by changing the characteristic diffusion length. Together, they form the standard toolkit for diagnosing internal mass transfer effects in heterogeneous catalysis.

Remark 4.15 (Concept Check: Weisz–Prater Diagnosis). You measure $r_{\text{obs}} = 0.05 \text{ mol}/(\text{L}\cdot\text{s})$ for a first-order reaction over 2 mm radius pellets with $C_{A,s} = 0.5 \text{ mol/L}$ and $D_{\text{eff}} = 1 \times 10^{-6} \text{ m}^2/\text{s}$.

1. Calculate N_{WP} . Is internal diffusion significant?
2. Your colleague suggests switching to 0.5 mm pellets. Predict the new N_{WP} without recalculating from scratch.

Hint: Note that $N_{WP} \propto R_p^2$.

4.7 Applications to Microporous Materials

4.7.1 Zeolites: Micropore Diffusion and Shape Selectivity

Zeolites present three “transport multipliers” that make diffusion limitations particularly important:

1. **Very small pore sizes** (typically 0.3–1.2 nm): Molecular motion inside zeolite crystals is orders of magnitude slower than in mesoporous supports.
2. **Strong adsorption**: High coverage reduces the fraction of molecules free to diffuse; mobility can be further hindered by site-hopping mechanisms.
3. **Shape selectivity**: Molecules of different sizes and shapes diffuse at very different rates, leading to selectivity that depends on ϕ .

Practical consequence: Zeolites often have small D_{eff} values (sometimes 10^{-10} to 10^{-14} m²/s), so ϕ can be large even for moderate pellet sizes. This reduces η and can alter apparent reaction orders and activation energies. We return to zeolite transport in detail in Chapter 6.

4.7.2 Nanotubes and Knudsen Diffusion

When pore diameters are comparable to or smaller than the mean free path of gas molecules, **Knudsen diffusion** dominates. In this regime, molecules collide more frequently with pore walls than with each other.

The Knudsen diffusivity scales as:

$$D_K = \frac{2}{3} r_p \sqrt{\frac{8RT}{\pi M}} \propto r_p \sqrt{\frac{T}{M}} \quad (4.28)$$

where:

- r_p = pore radius
- T = absolute temperature
- M = molar mass of the diffusing species
- R = gas constant

Remark 4.16 (Diffusivity Ranges by Pore Size). The dominant diffusion mechanism depends on pore diameter relative to the mean free path:

4. Transport Limitations in Porous Catalysts

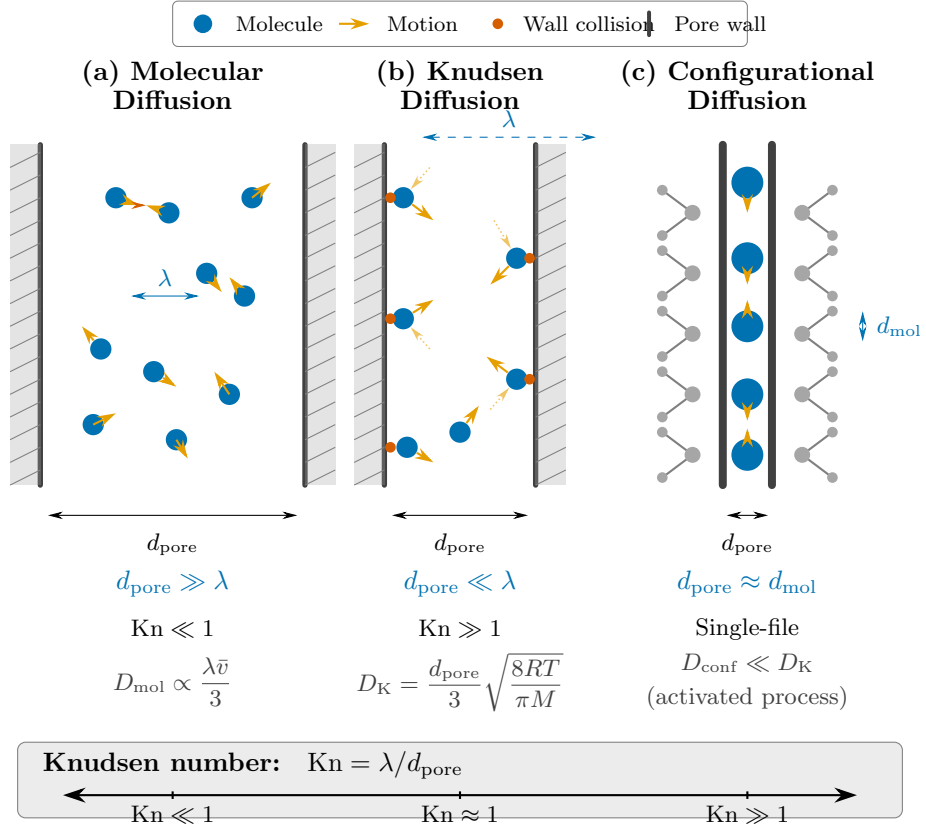


Figure 4.4: Three diffusion regimes in porous materials characterized by the Knudsen number $\text{Kn} = \lambda/d_{\text{pore}}$, where λ is the mean free path: (a) bulk/molecular diffusion in wide pores ($\text{Kn} \ll 1$), where molecule–molecule collisions dominate; (b) Knudsen diffusion in narrow pores ($\text{Kn} \gg 1$), where molecule–wall collisions dominate (Eq. (4.28)); (c) configurational diffusion in zeolite micropores, where pore diameter approaches molecular size and motion becomes an activated process.

- **Macropores** ($d_{\text{pore}} > 50 \text{ nm}$): Molecular diffusion dominates; $D \sim 10^{-5}\text{--}10^{-4} \text{ m}^2/\text{s}$ (gases)
- **Mesopores** (2–50 nm): Knudsen diffusion becomes significant; $D_K \sim 10^{-7}\text{--}10^{-5} \text{ m}^2/\text{s}$
- **Micropores** ($d_{\text{pore}} < 2 \text{ nm}$): Configurational diffusion; $D \sim 10^{-14}\text{--}10^{-8} \text{ m}^2/\text{s}$

The transition between regimes is governed by the Knudsen number $\text{Kn} = \lambda/d_{\text{pore}}$, where λ is the mean free path.

Remark 4.17 (Activated Diffusion in Micropores). Unlike molecular or Knudsen diffusion, configurational diffusion through micropores is an *activated process* that follows an Arrhenius-type temperature dependence:

$$D_{\text{conf}} = D_0 \exp\left(-\frac{E_D}{RT}\right) \quad (4.29)$$

where E_D is the activation energy for diffusion. This temperature dependence arises because molecules must overcome energy barriers to squeeze through narrow constrictions between sorption sites.

Typical values of E_D depend strongly on the ratio of molecular size to pore aperture:

- Small molecules in wide zeolite channels (e.g., methane in FAU): $E_D \approx 5\text{--}15$ kJ/mol
- Branched hydrocarbons in medium-pore zeolites (e.g., isobutane in MFI): $E_D \approx 30\text{--}60$ kJ/mol
- Bulky molecules near the pore aperture limit (e.g., xylenes in MFI): $E_D \approx 40\text{--}100$ kJ/mol

Important consequence for apparent activation energy.

Recall from Section 4.5.4 that when internal diffusion controls ($\phi \gg 1$), the apparent activation energy for a conventional (non-activated) diffusivity is $E_a^{\text{app}} \approx E_a/2$. However, when the diffusivity itself is activated, the halving rule is modified:

$$E_a^{\text{app}} \approx \frac{E_a + E_D}{2} \quad (4.30)$$

For zeolites with large E_D , the apparent activation energy may be much closer to E_a than to $E_a/2$, potentially *masking* diffusion limitations. This makes the Weisz–Prater criterion (Eq. (4.26)) a more reliable diagnostic than the activation energy test.

This connects directly to the shape selectivity discussion in Chapter 6, where molecules of different sizes experience dramatically different E_D values within the same framework—providing a molecular-level basis for reactant and product selectivity.

Key insight for nanotube catalysts: Smaller tube radius $\rightarrow$ smaller $D_K \rightarrow$ larger $\phi \rightarrow$ smaller η . While nanostructured materials increase surface area, diffusion can become the bottleneck that limits catalyst utilization. However, carbon nanotubes can also exhibit *enhanced* transport relative to conventional Knudsen predictions, as we explore in Chapter 6.

Remark 4.18 (Concept Check: Diffusion Regime Identification).

For each scenario below, identify the dominant diffusion mechanism (molecular, Knudsen, or configurational) and estimate the order of magnitude of D_{eff} :

1. N_2 ($M = 28$ g/mol) at 500 K in a γ -alumina support with 8 nm pores, $\varepsilon = 0.5$, $\tau_p = 4$. The mean free path is $\lambda \approx 100$ nm.
2. Benzene in ZSM-5 zeolite (pore aperture ≈ 0.55 nm) at 400 K.
3. O_2 at 300 K in a macroporous ceramic monolith ($d_{\text{pore}} = 5$ μm).

Hint: Compare λ to d_{pore} for the first and third cases; consider molecule-to-pore size ratio for the second.

Fig. 4.4 illustrates the three distinct diffusion regimes encountered in porous catalysts, each characterized by a different dominant transport mechanism. In *molecular (bulk) diffusion* (panel a), pores are wide compared to the mean free path λ , and molecules collide primarily with each other—the same mechanism as in homogeneous gas mixtures. The molecular diffusivity D_{mol} depends on temperature and pressure but not on pore geometry. In *Knudsen diffusion* (panel b), pores are narrow enough that molecules collide more frequently with the walls than with each other. Each wall collision randomizes the molecule's direction, creating a “random walk” along the pore axis. The Knudsen diffusivity D_K (Eq. (4.28)) scales with pore diameter and $\sqrt{T/M}$ —larger molecules diffuse more slowly. In *configurational diffusion* (panel c), pore diameters approach molecular dimensions. Molecules must “squeeze” through constrictions in single file, making diffusion an activated process with strong temperature dependence. Zeolites typically operate in this regime, with D_{conf} values orders of magnitude smaller than D_K . The Knudsen number $\text{Kn} = \lambda/d_{\text{pore}}$ determines which regime applies: $\text{Kn} \ll 1$ for molecular diffusion, $\text{Kn} \gg 1$ for Knudsen diffusion, and $d_{\text{pore}} \approx d_{\text{mol}}$ for configurational diffusion.

4.8 External Mass Transfer

While internal diffusion is often the dominant resistance in industrial catalysis, external film mass transfer can also be important, especially in liquid-phase reactions or at high reaction rates.

4.8.1 The External Mass Transfer Coefficient

At steady state, the rate of mass transfer through the film equals the reaction rate:

$$k_c a_p (C_{A,b} - C_{A,s}) = r_{\text{obs}} \quad (4.31)$$

where a_p is the external surface area per unit volume.

Correlations for k_c typically involve the Sherwood number:

$$Sh = \frac{k_c d_p}{D_{AB}} = f(Re, Sc) \quad (4.32)$$

4.8.2 Diagnostic: The Carberry Number

A practical diagnostic for external mass transfer limitation:

$$Ca = \frac{r_{\text{obs}}}{k_c a_p C_{A,b}} \quad (4.33)$$

If $Ca \ll 1$, external mass transfer is negligible and $C_{A,s} \approx C_{A,b}$.

Remark 4.19 (The Mears Criterion). An alternative diagnostic for external mass transfer limitations is the **Mears criterion**:

$$C_M = \frac{r_{\text{obs}} \rho_b R_p n}{k_c C_{A,b}} < 0.15$$

where ρ_b is the bulk catalyst density and n is the reaction order. When $C_M < 0.15$, the external concentration drop is less than 5% of the bulk value. The Mears criterion complements the Carberry number by explicitly accounting for reaction order.

Remark 4.20 (Estimating k_c : Sherwood Number Correlations).

To apply the Carberry number (Eq. (4.33)) or the Mears criterion (Section 4.8.2), we need a value for the external mass transfer coefficient k_c . This is obtained from the Sherwood number $Sh = k_c d_p / D_{AB}$ (Eq. (4.32)) using empirical correlations.

Single sphere in unbounded fluid (Frössling/Ranz–Marshall):

$$Sh = 2 + 0.6 Re^{1/2} Sc^{1/3} \quad (4.34)$$

Packed beds (Wakao–Funazkri, 1978):

$$Sh = 2 + 1.1 Re^{0.6} Sc^{1/3} \quad (3 < Re < 10^4) \quad (4.35)$$

Here $Re = \rho u d_p / \mu$ is the particle Reynolds number (based on superficial velocity u) and $Sc = \mu / (\rho D_{AB})$ is the Schmidt number.

The lower limit $Sh = 2$ corresponds to pure diffusion from a sphere into a stagnant medium—the minimum possible mass transfer rate.

The Schmidt number reveals why *liquid-phase* reactions are far more susceptible to external mass transfer limitation than gas-phase reactions:

- **Gas phase:** $D_{AB} \sim 10^{-5} \text{ m}^2/\text{s} \Rightarrow Sc \approx 0.5\text{--}2$. The high gas-phase diffusivity keeps k_c large, so external limitation rarely controls.
- **Liquid phase:** $D_{AB} \sim 10^{-9} \text{ m}^2/\text{s} \Rightarrow Sc \approx 100\text{--}10\,000$. The low liquid-phase diffusivity makes k_c much smaller, and external limitation is far more likely.

This is one reason why liquid-phase reactions over supported catalysts (e.g., Pd/C in Suzuki cross-coupling) frequently operate in the externally limited regime, requiring vigorous stirring or small catalyst particles to alleviate mass transfer resistance.

4.9 Combined Internal and External Resistances

When both resistances are significant, we must solve the coupled problem. For first-order kinetics, this introduces the **Biot number for mass transfer**:

$$Bi_m = \frac{k_c R_p}{D_{\text{eff}}} \quad (4.36)$$

The Biot number compares external mass transfer rate to internal diffusion rate:

- $Bi_m \gg 1$: External transfer is fast; internal diffusion controls
- $Bi_m \ll 1$: External transfer is slow; film resistance controls

The overall effectiveness factor accounting for both resistances is:

$$\boxed{\eta_{\text{overall}} = \frac{\eta}{1 + \eta \phi^2 / (3 Bi_m)}} \quad (\text{sphere}) \quad (4.37)$$

The factor of 3 in the denominator arises from the volume-to-surface-area ratio of a sphere ($V_p/S_{\text{ext}} = R_p/3$). For a slab of half-thickness L , replace $\phi^2/(3 Bi_m)$ with $\phi_{\text{slab}}^2/Bi_{m,\text{slab}}$ (no factor of 3).

This expression shows that:

- When $Bi_m \gg \eta \phi^2$: external resistance is negligible, $\eta_{\text{overall}} \rightarrow \eta$ (internal control)

- When $Bi_m \ll \eta\phi^2$: external resistance dominates, $\eta_{\text{overall}} \rightarrow 3 Bi_m / \phi^2$ (external control)
- Both resistances matter when $Bi_m \sim \eta\phi^2$

Remark 4.21 (Non-Isothermal Pellets: When $\eta > 1$). All of the analysis presented so far assumes *isothermal* pellets—that is, the temperature inside the pellet equals the surface temperature T_s everywhere. For exothermic reactions, however, the heat released by reaction inside the pellet can raise the internal temperature above T_s , increasing the local rate constant and potentially making the pellet *more* effective than the isothermal prediction.

The **Prater number** quantifies the maximum dimensionless temperature rise inside the pellet:

$$\beta = \frac{(-\Delta H_{\text{rxn}}) D_{\text{eff}} C_{A,s}}{\lambda_{\text{eff}} T_s} \quad (4.38)$$

where λ_{eff} is the effective thermal conductivity of the pellet. The maximum possible temperature rise for a fully diffusion-limited pellet is $\Delta T_{\text{max}} = \beta T_s$.

Consequences for the effectiveness factor:

- $\beta > 0$ (exothermic): Internal temperature rises $\Rightarrow$ local k increases $\Rightarrow \eta$ can exceed 1.
- $\beta < 0$ (endothermic): Internal temperature drops $\Rightarrow$ local k decreases $\Rightarrow \eta$ is further reduced beyond the isothermal prediction.

Typical values of β are 0.01–0.3 for gas-phase oxidations and less than 0.01 for liquid-phase reactions. The effectiveness factor exceeding unity has been experimentally confirmed for highly exothermic gas-phase reactions (e.g., oxidation of H_2 on Pt catalysts).

For a quantitative treatment, the **Weisz–Hicks analysis** shows that *multiple steady states* can exist when $\beta\gamma > 4$, where $\gamma = E_a/(RT_s)$ is the Arrhenius number. In such cases, ignition and extinction phenomena occur: a pellet may jump between a low-conversion (kinetically controlled) and a high-conversion (diffusion-controlled, hot) steady state.

In practice, the isothermal assumption ($\beta \approx 0$) is adequate for the majority of systems treated in this course, but you should be aware that $\eta > 1$ is not a mathematical artifact—it is a real physical phenomenon arising from the coupling of heat and mass transport.

4.10 Integrating Effectiveness Factor into Reactor Design

The practical value of the effectiveness factor is that it allows us to use reactor design equations from Chapter 3 with a simple modification.

4.10.1 Observed Rate Law

For any reactor calculation, replace the intrinsic rate with the observed rate:

$$r_{\text{obs}}(C_A, T) = \eta(\phi) \cdot r_{\text{intrinsic}}(C_A, T) \quad (4.39)$$

where the Thiele modulus itself depends on temperature through $k(T)$ and possibly $D_{\text{eff}}(T)$:

$$\phi(T) = R_p \sqrt{\frac{k(T)}{D_{\text{eff}}(T)}} \quad (4.40)$$

4.10.2 PFR Design with Effectiveness Factor

From Chapter 3, the PFR design equation for constant density is:

$$\frac{dX}{d\tau} = \frac{r(C_A)}{C_{A0}} \quad (4.41)$$

Including the effectiveness factor:

$$\frac{dX}{d\tau} = \frac{\eta(\phi) \cdot r_{\text{intrinsic}}(C_A)}{C_{A0}} \quad (4.42)$$

For first-order kinetics with constant η :

$$X = 1 - \exp(-\eta \cdot k \cdot \tau) = 1 - \exp(-k_{\text{eff}} \cdot \tau) \quad (4.43)$$

where $k_{\text{eff}} = \eta \cdot k$ is the “effective” rate constant.

Remark 4.22 (Temperature Dependence of Effectiveness Factor).

In non-isothermal reactions, η varies with temperature because both $k(T)$ and $D_{\text{eff}}(T)$ change (Chapter 2). This temperature dependence affects catalyst performance in both batch and flow systems and is relevant to supported catalyst design for organic synthesis (see the appendix, appendix G).

4.11 Worked Examples

Example 4.1: Effect of Pellet Size on Effectiveness Factor

Problem: A first-order catalytic reaction has $k = 1.0 \text{ s}^{-1}$ and $D_{\text{eff}} = 1.0 \times 10^{-6} \text{ m}^2/\text{s}$. Calculate the effectiveness factor for spherical pellets with radii of:

- (a) $R_p = 0.1 \text{ mm}$
- (b) $R_p = 1.0 \text{ mm}$
- (c) $R_p = 5.0 \text{ mm}$

Solution (Thiele Modulus for Spheres):

The Thiele modulus for a sphere is $\phi = R_p \sqrt{k/D_{\text{eff}}}$.

First, calculate $\sqrt{k/D_{\text{eff}}}$:

$$\sqrt{\frac{k}{D_{\text{eff}}}} = \sqrt{\frac{1.0 \text{ s}^{-1}}{1.0 \times 10^{-6} \text{ m}^2/\text{s}}} = \sqrt{10^6 \text{ m}^{-2}} = 10^3 \text{ m}^{-1}$$

(a) $R_p = 0.1 \text{ mm} = 1.0 \times 10^{-4} \text{ m}$:

$$\phi = (1.0 \times 10^{-4})(10^3) = 0.1$$

Since $\phi \ll 1$, we expect $\eta \approx 1$. Using the exact formula:

$$\begin{aligned} \eta &= \frac{3}{0.1^2} (0.1 \coth(0.1) - 1) = \frac{3}{0.01} (1.0033 - 1) \\ &= 300 \times 0.0033 \approx 0.999 \approx 1 \end{aligned}$$

(b) $R_p = 1.0 \text{ mm} = 1.0 \times 10^{-3} \text{ m}$:

$$\phi = (1.0 \times 10^{-3})(10^3) = 1.0$$

$$\eta = \frac{3}{1^2} (1.0 \coth(1.0) - 1) = 3(1.313 - 1) = 0.94$$

(c) $R_p = 5.0 \text{ mm} = 5.0 \times 10^{-3} \text{ m}$:

$$\phi = (5.0 \times 10^{-3})(10^3) = 5.0$$

Since $\phi \gg 1$, we can try the asymptotic approximation $\eta \approx 3/\phi = 3/5 = 0.60$.

However, the exact calculation gives:

$$\eta = \frac{3}{5^2} (5 \coth(5) - 1) = \frac{3}{25} (5.0005 - 1) = 0.12 \times 4.0005 = 0.48$$

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The asymptotic approximation $3/\phi = 0.60$ overestimates η by about 25% at $\phi = 5$. The asymptotic formula becomes accurate for $\phi \gtrsim 10$; at moderate ϕ , the exact expression must be used.

R_p (mm)	ϕ	η	Regime
0.1	0.1	0.999	Kinetic control
1.0	1.0	0.94	Transition
5.0	5.0	0.48	Diffusion limited

Answer:

$\eta = 0.999, 0.94, 0.48$ for $R_p = 0.1, 1.0, 5.0$ mm

Physical interpretation: A 50-fold increase in pellet radius (0.1 to 5 mm) reduces catalyst utilization from nearly 100% to only 48%. This demonstrates why industrial catalysts often use small pellets or thin washcoats on monolith supports.

Example 4.2: Impact of Diffusion Limitation on Reactor Conversion

Problem: A PFR operates with first-order intrinsic kinetics ($k = 0.5 \text{ s}^{-1}$) and space time $\tau = 4 \text{ s}$. If internal diffusion reduces the effectiveness factor to $\eta = 0.4$, compare the conversion with and without diffusion limitation.

Solution (First-Order PFR Design Equation):

For first-order kinetics in a PFR:

$$X = 1 - \exp(-k_{\text{eff}} \cdot \tau)$$

Without diffusion limitation ($\eta = 1$):

$$k_{\text{eff}} = k = 0.5 \text{ s}^{-1}$$

$$X = 1 - \exp(-0.5 \times 4) = 1 - \exp(-2) = 1 - 0.135 = 0.865 = 86.5\%$$

With diffusion limitation ($\eta = 0.4$):

$$k_{\text{eff}} = \eta \cdot k = 0.4 \times 0.5 = 0.2 \text{ s}^{-1}$$

$$X = 1 - \exp(-0.2 \times 4) = 1 - \exp(-0.8) = 1 - 0.449 = 0.551 = 55.1\%$$

Answer:

$$X = 86.5\% \text{ (no limitation) vs. } 55.1\% \text{ (diffusion-limited)}$$

Physical interpretation: Conversion drops by 31 percentage points due to internal diffusion limitation. To achieve 86.5% conversion with the diffusion-limited catalyst would require increasing τ from 4 s to 10 s (a factor of 2.5 increase in reactor volume).

Example 4.3: Weisz–Prater Diagnostic

Problem: In a laboratory reactor, you measure an observed rate of $r_{\text{obs}} = 0.05 \text{ mol}/(\text{L}\cdot\text{s})$ for a first-order reaction using spherical catalyst pellets with $R_p = 2 \text{ mm}$. The surface concentration is $C_{A,s} = 0.5 \text{ mol/L}$ and you estimate $D_{\text{eff}} = 5 \times 10^{-7} \text{ m}^2/\text{s}$. Is internal diffusion significant?

Solution (Weisz–Prater Criterion):

Calculate the Weisz–Prater number:

$$N_{WP} = \frac{r_{\text{obs}} \cdot R_p^2}{C_{A,s} \cdot D_{\text{eff}}}$$

Convert units carefully:

- $r_{\text{obs}} = 0.05 \text{ mol}/(\text{L}\cdot\text{s}) = 50 \text{ mol}/(\text{m}^3\cdot\text{s})$
- $R_p = 2 \text{ mm} = 2 \times 10^{-3} \text{ m}$
- $C_{A,s} = 0.5 \text{ mol/L} = 500 \text{ mol}/\text{m}^3$

$$N_{WP} = \frac{50 \times (2 \times 10^{-3})^2}{500 \times 5 \times 10^{-7}} = \frac{50 \times 4 \times 10^{-6}}{2.5 \times 10^{-4}} = \frac{2 \times 10^{-4}}{2.5 \times 10^{-4}} = 0.8$$

Answer:

$N_{WP} = 0.8$; internal diffusion is moderately significant

Physical interpretation: We are in the transition region where the observed rate is somewhat lower than the intrinsic rate. To eliminate diffusion effects, consider reducing pellet size (e.g., crush to $R_p = 0.5$ mm would give $N_{WP} = 0.05 \ll 1$).

Example 4.4: Knudsen Diffusion in Nanotube Catalyst

Problem: A carbon nanotube catalyst has pore radius $r_p = 5$ nm. Estimate the Knudsen diffusivity for CO_2 ($M = 44$ g/mol) at 500 K and compare to the molecular diffusivity in a mesoporous support ($D_{AB} \approx 10^{-5}$ m²/s).

Solution (Knudsen Diffusivity Formula):

The Knudsen diffusivity is:

$$D_K = \frac{2}{3} r_p \sqrt{\frac{8RT}{\pi M}}$$

Substituting values (with $R = 8.314$ J/(mol·K) and $M = 0.044$ kg/mol):

$$\begin{aligned} D_K &= \frac{2}{3} (5 \times 10^{-9}) \sqrt{\frac{8 \times 8.314 \times 500}{\pi \times 0.044}} \\ D_K &= 3.33 \times 10^{-9} \sqrt{\frac{33256}{0.138}} = 3.33 \times 10^{-9} \sqrt{2.41 \times 10^5} \\ D_K &= 3.33 \times 10^{-9} \times 491 = 1.6 \times 10^{-6} \text{ m}^2/\text{s} \end{aligned}$$

Answer:

$D_K \approx 1.6 \times 10^{-6}$ m²/s, about 6 times smaller than D_{AB}

Physical interpretation: Nanoscale pores significantly reduce effective diffusivity compared to mesopores. This can lead to larger ϕ and smaller η despite the high surface area of nanotube catalysts.

4.12 Computational Exploration: Notebook NB4

The companion Jupyter notebook `NB4.ipynb` provides hands-on exploration of the concepts in this chapter. In the notebook, you will:

1. Implement the effectiveness factor function $\eta(\phi)$ for spherical pellets.
2. Generate plots of η versus ϕ and verify the limiting behaviors.
3. Explore how η changes with pellet radius R_p , rate constant k , and effective diffusivity D_{eff} .
4. Integrate diffusion-limited kinetics ($r_{\text{obs}} = \eta \cdot r_{\text{intrinsic}}$) into a PFR model and compare conversion profiles for different pellet sizes.
5. Calculate the Weisz–Prater number for various operating conditions.

Reflection questions for the notebook:

- At what value of ϕ does η drop below 0.5?
- Which design variable most effectively restores η to near unity: reducing R_p , increasing D_{eff} , or decreasing k ?
- Does diffusion limitation have a larger effect on conversion at high τ (high conversion) or low τ (low conversion)? Explain.

4.13 Summary

- **Two transport resistances** affect observed rates in porous catalysts: external film mass transfer and internal pore diffusion.
- The **Thiele modulus** $\phi = R_p \sqrt{k/D_{\text{eff}}}$ is the key dimensionless group comparing reaction rate to diffusion rate.
- The **effectiveness factor** η quantifies catalyst utilization:
 - Sphere: $\eta = (3/\phi^2)(\phi \coth \phi - 1)$
 - Slab: $\eta = \tanh \phi / \phi$
- **Limiting behavior:**
 - Kinetic control ($\phi \ll 1$): $\eta \approx 1$
 - Diffusion control ($\phi \gg 1$): $\eta \approx 3/\phi$ (sphere)
- The **Weisz–Prater criterion** $N_{WP} = r_{\text{obs}} R_p^2 / (C_{A,s} D_{\text{eff}})$ diagnoses diffusion limitations from experimental data.

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- **Microporous materials** (zeolites, nanotubes) have low D_{eff} due to small pores and possibly Knudsen diffusion, making $\eta < 1$ even for small pellets.
- **Reactor design** uses the observed rate: $r_{\text{obs}} = \eta \cdot r_{\text{intrinsic}}$.

4.14 Notation Introduced in This Chapter

Symbol	Description	Units
<i>Dimensionless groups</i>		
ϕ	Thiele modulus	—
η	Effectiveness factor	—
N_{WP}	Weisz–Prater number	—
Bi_m	Mass transfer Biot number	—
Sh	Sherwood number	—
Ca	Carberry number	—
<i>Transport properties</i>		
D_{eff}	Effective diffusivity	m^2/s
D_K	Knudsen diffusivity	m^2/s
D_{AB}	Molecular (bulk) diffusivity	m^2/s
k_c	External mass transfer coefficient	m/s
<i>Pellet geometry</i>		
R_p	Pellet radius (sphere)	m
L	Half-thickness (slab)	m
r_p	Pore radius	m
d_p	Pellet diameter	m or mm
ε	Pellet porosity	—
τ_p	Pore tortuosity	—
a_p	External surface area per unit volume	m^{-1}
<i>Concentrations and rates</i>		
$C_{A,b}$	Bulk concentration	mol/m^3
$C_{A,s}$	Surface concentration	mol/m^3
r_{obs}	Observed (apparent) reaction rate	$\text{mol}/(\text{m}^3 \cdot \text{s})$
$r_{\text{intrinsic}}$	Intrinsic reaction rate	$\text{mol}/(\text{m}^3 \cdot \text{s})$

Remark 4.23 (Self-Assessment Checklist). Before attempting the problems, verify that you can:

1. Explain, in one sentence, why $\eta < 1$ for a porous pellet with fast kinetics.
2. Write the Thiele modulus for a sphere from memory and state

its physical meaning.

3. Sketch a log-log plot of η vs. ϕ showing both asymptotic limits.
4. State the Weisz–Prater criterion and explain why it is preferable to the Thiele modulus for experimental diagnosis.
5. Identify which diffusion regime (molecular, Knudsen, configurational) dominates for a given pore size.
6. Predict what happens to the apparent activation energy when a reaction transitions from kinetic to diffusion control.

If any of these feel uncertain, revisit the corresponding section before proceeding.

4.15 Problems

1. **Effective diffusivity estimation (warm-up).** A γ -alumina catalyst pellet has porosity $\varepsilon = 0.45$ and tortuosity $\tau_p = 3.5$. The gas-phase molecular diffusivity of reactant A at 500 K is $D_{AB} = 2.8 \times 10^{-5} \text{ m}^2/\text{s}$.
 - (a) Calculate D_{eff} using Eq. (4.2).
 - (b) If the pellet has average pore radius $r_p = 5 \text{ nm}$, calculate the Knudsen diffusivity D_K for a molecule with molar mass $M = 44 \text{ g/mol}$ at 500 K (Eq. (4.28)). Is molecular or Knudsen diffusion dominant?
 - (c) Compute the combined pore diffusivity using $1/D_{\text{pore}} = 1/D_{AB} + 1/D_K$ and the resulting D_{eff} .
2. **Thiele modulus calculation and interpretation.** A first-order reaction occurs in spherical catalyst pellets with intrinsic rate constant $k = 2.5 \text{ s}^{-1}$ and effective diffusivity $D_{\text{eff}} = 8.0 \times 10^{-7} \text{ m}^2/\text{s}$.
 - (a) Calculate the Thiele modulus ϕ for pellet radii $R_p = 0.5, 2.0$, and 5.0 mm .
 - (b) For each case, predict the operating regime (kinetic control, transition, or diffusion-limited) without calculating η .
 - (c) Which pellet size would you recommend for laboratory kinetic studies where you want to measure intrinsic kinetics? Justify your answer.
 - (d) If the reaction rate constant doubles due to a temperature increase, how does ϕ change for each pellet size?

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3. **Effectiveness factor for different geometries.** Consider a first-order catalytic reaction with $k = 1.0 \text{ s}^{-1}$ and $D_{\text{eff}} = 2.0 \times 10^{-6} \text{ m}^2/\text{s}$.

- (a) For a flat slab catalyst of half-thickness $L = 3.0 \text{ mm}$, calculate ϕ_{slab} and η_{slab} using $\eta_{\text{slab}} = \tanh \phi / \phi$.
- (b) For a spherical pellet of radius $R_p = 3.0 \text{ mm}$, calculate ϕ_{sphere} and η_{sphere} using $\eta_{\text{sphere}} = (3/\phi^2)(\phi \coth \phi - 1)$.
- (c) Compare the effectiveness factors. Which geometry has better catalyst utilization for the same characteristic length? Explain physically why this is the case.
- (d) For the spherical pellet, verify that the asymptotic approximation $\eta \approx 3/\phi$ (valid for $\phi \gg 1$) gives a reasonable estimate.

4. **Weisz–Prater criterion application.** In a laboratory packed-bed reactor, you measure the following data for a first-order gas-phase reaction over spherical catalyst pellets ($R_p = 1.5 \text{ mm}$):

- Observed reaction rate: $r_{\text{obs}} = 0.12 \text{ mol}/(\text{L}\cdot\text{s})$
- Surface concentration of reactant: $C_{A,s} = 0.8 \text{ mol/L}$
- Estimated effective diffusivity: $D_{\text{eff}} = 3.0 \times 10^{-7} \text{ m}^2/\text{s}$

- (a) Calculate the Weisz–Prater number N_{WP} for this system.
- (b) Is internal diffusion limitation significant? Justify your conclusion.
- (c) The experiment is repeated with crushed catalyst ($R_p = 0.3 \text{ mm}$). Predict the new N_{WP} value.
- (d) If crushing the catalyst increases the observed rate by 40%, what does this tell you about the original operating regime?

5. **Effect of particle size on observed rate.** A catalytic cracking reaction follows first-order kinetics with intrinsic rate constant $k = 4.0 \text{ s}^{-1}$ at 500°C . The effective diffusivity in the catalyst pores is $D_{\text{eff}} = 5.0 \times 10^{-7} \text{ m}^2/\text{s}$. The feed concentration is $C_{A0} = 1.0 \text{ mol/L}$.

- (a) Calculate η for spherical pellets with $R_p = 0.5, 1.0, 2.0$, and 4.0 mm .
- (b) Calculate the observed rate $r_{\text{obs}} = \eta \cdot k \cdot C_A$ for each particle size (assuming $C_A = C_{A0}$).
- (c) Plot r_{obs} versus R_p on a log-log scale. What is the apparent slope in the diffusion-limited regime?

- (d) A process engineer proposes using 6.0 mm pellets to reduce pressure drop. Estimate the penalty in catalyst utilization compared to 1.0 mm pellets. *Hint: Use `thiele_modulus_sphere()` and `effectiveness_sphere()` from NB4 to compute η for each pellet size, then plot.*
6. **Temperature effects on effectiveness factor.** A first-order reaction in a spherical catalyst pellet ($R_p = 2.0$ mm) has the following temperature-dependent parameters:
- Rate constant: $k(T) = 5.0 \times 10^6 \exp(-50000/RT) \text{ s}^{-1}$, where $R = 8.314 \text{ J/(mol}\cdot\text{K)}$
 - Effective diffusivity: $D_{\text{eff}}(T) = 1.0 \times 10^{-6} (T/300)^{1.75} \text{ m}^2/\text{s}$ (molecular diffusion regime)
- (a) Calculate ϕ and η at $T = 400, 500$, and 600 K.
- (b) Calculate the observed rate constant $k_{\text{eff}} = \eta \cdot k$ at each temperature.
- (c) Make an Arrhenius plot of $\ln(k_{\text{eff}})$ versus $1/T$. What is the apparent activation energy in the diffusion-limited regime?
- (d) Compare the apparent activation energy to the true activation energy. Explain the difference using the relationship between η and ϕ at high ϕ . *Hint: Adapt the Arrhenius plot code from NB4 (Part 7: Apparent Activation Energy) to automate the temperature sweep and regression.*
7. **Python analysis: effectiveness factor curves and design optimization.** Using Python (NB4), explore the effectiveness factor behavior and apply it to reactor design.
- (a) Write a function `eta_sphere(phi)` that computes $\eta = \frac{3}{\phi^2}(\phi \coth \phi - 1)$ for spheres. Handle the limit $\phi \rightarrow 0$ numerically.
- (b) Generate a log-log plot of η vs. ϕ for $\phi \in [0.01, 100]$. Add the asymptotic limits: $\eta = 1$ for $\phi \ll 1$ and $\eta = 3/\phi$ for $\phi \gg 1$.
- (c) For a PFR with first-order kinetics and space time $\tau = 5$ s, compute and plot the conversion $X = 1 - \exp(-\eta \cdot k \cdot \tau)$ as a function of pellet radius R_p (from 0.1 to 10 mm), using $k = 1.0 \text{ s}^{-1}$ and $D_{\text{eff}} = 1.0 \times 10^{-6} \text{ m}^2/\text{s}$.
- (d) An economic analysis shows that catalyst cost scales as $1/R_p^2$ (smaller pellets are more expensive) while reactor volume scales as $1/\eta$. Define a cost function $\text{Cost} \propto 1/R_p^2 + \alpha/\eta$ where α is a weighting factor. For $\alpha = 0.5, 1.0$, and 2.0 , find the optimal pellet radius that minimizes total cost.

8. **Zeolite vs. nanotube catalyst: Thiele modulus comparison.** A first-order isomerization reaction with intrinsic rate constant $k = 50 \text{ s}^{-1}$ is studied on two different catalyst supports.

- **Zeolite ZSM-5:** Crystal radius $R = 1.0 \text{ }\mu\text{m}$; $D_{\text{eff}} = 5.0 \times 10^{-12} \text{ m}^2/\text{s}$ (configurational diffusion).
 - **Carbon nanotube composite:** Effective pellet radius $R = 0.5 \text{ mm}$; $D_{\text{eff}} = 2.0 \times 10^{-5} \text{ m}^2/\text{s}$ (enhanced Knudsen transport, Chapter 6).
- (a) Calculate ϕ and η for each catalyst. Which system is more diffusion-limited, despite having a much larger pellet?
 - (b) Explain why ZSM-5 can be severely diffusion-limited even at micrometre crystal sizes, while the nanotube composite with a millimetre-scale pellet operates efficiently.
 - (c) For the zeolite, estimate the apparent activation energy given $E_a = 80 \text{ kJ/mol}$ and $E_D = 40 \text{ kJ/mol}$ (Eq. (4.30)). How does this differ from the nanotube case where diffusion is non-activated?
Hint: Compute ϕ and η using NB4.

Looking Ahead

In this chapter, we quantified how internal diffusion reduces catalyst utilization through the effectiveness factor η . The concepts developed here—the Thiele modulus, the effectiveness factor, and diagnostic criteria such as the Weisz–Prater number—form the foundation for several important extensions in the chapters ahead.

Chapter 5 examines how transport limitations affect not just overall rates but also **product selectivity**. In parallel reaction networks, diffusion can shift selectivity toward the reaction with lower activation energy; in consecutive networks, diffusion can reduce the yield of a desired intermediate by allowing further reaction before the product escapes the pellet. The interplay between ϕ , η , and selectivity is a central consideration in catalyst design.

Chapter 6 introduces **zeolites**, where the microporous framework creates exceptionally strong diffusion limitations. Configurational diffusion ($D \sim 10^{-10}$ – $10^{-14} \text{ m}^2/\text{s}$) leads to large Thiele moduli even for sub-millimeter crystals. Shape selectivity—the ability of zeolites to discriminate between molecules based on size—is fundamentally a transport phenomenon governed by the relative diffusion rates of different species through micropores.

Chapter 6 explores **carbon nanotubes** as catalyst supports, where an intriguing reversal occurs: the smooth, atomically uniform nanotube interior can *enhance* Knudsen diffusion by factors of 10–10,000 compared to

conventional mesoporous supports. This enhancement modifies the effective Thiele modulus and can dramatically improve catalyst utilisation in nanoscale systems.

Finally, the appendix on catalytic organic synthesis (appendix [G](#)) integrates transport considerations into practical catalytic systems, where supported metal catalysts (e.g., Pd/C) and zeolites are widely used. Choosing the right pellet size, pore structure, and support material requires balancing intrinsic activity, diffusion limitations, and selectivity trade-offs developed throughout these chapters.

Chapter 5

Selectivity and Reaction Networks

Learning Objectives

By the end of this chapter, you will be able to:

- LO1. Define** conversion, yield, and selectivity (both overall and instantaneous) using consistent notation
- LO2. Model** parallel and consecutive reaction networks in PFR and CSTR configurations
- LO3. Predict** when an intermediate product exhibits a “selectivity window” with an optimal residence time
- LO4. Apply** the optimal residence time formula τ^* to maximize intermediate yield for consecutive reactions
- LO5. Explain** how temperature affects selectivity through the Arrhenius dependence of competing rate constants
- LO6. Apply** Python (NB5) to compute and plot $X_A(\tau)$, $Y_B(\tau)$, $S_B(\tau)$, and optimal operating conditions

5.1 Introduction

In Chapter 3, we developed design equations for batch and flow reactors and learned to calculate conversion for single reactions. However, most catalytic processes involve multiple reactions occurring simultaneously. The goal is rarely to simply “maximize conversion”—instead, fine chemical and pharmaceutical processes aim to **maximize the desired product while minimizing byproducts**. This is the domain of *selectivity engineering*.

Consider asymmetric hydrogenation in pharmaceutical synthesis: the desired product is a single enantiomer, but the opposite enantiomer may also form. Success depends on choosing the right chiral catalyst, reaction conditions, and residence time to favor the desired stereochemical outcome. This chapter provides the quantitative framework for making such decisions.

We examine two fundamental reaction network topologies: **parallel reactions** (where a reactant can form multiple products directly) and **consecutive reactions** (where the desired product can react further to form unwanted byproducts). You will discover that parallel reactions are controlled primarily by temperature (through the ratio of rate constants), while consecutive reactions require careful optimization of residence time to “catch” the intermediate at its maximum concentration.

Remark 5.1 (Safety: Hazardous Reagents). Several industrially important selectivity problems involve hazardous reagents:

- **Ethylene oxide production** ($\text{Ag}/\alpha\text{-Al}_2\text{O}_3$ catalyst): Ethylene oxide is a **flammable, toxic, and potentially explosive** intermediate. Industrial reactors operate at low per-pass conversion (8–12%) partly for selectivity and partly because high ethylene oxide concentrations create explosion risks. Strict temperature control prevents thermal runaway.
- **Chlorination reactions**: Chlorine gas (Cl_2) is **highly toxic** (occupational exposure limit 0.5 ppm). Consecutive chlorination (e.g., $\text{CH}_4 \rightarrow \text{CH}_3\text{Cl} \rightarrow \text{CH}_2\text{Cl}_2 \rightarrow \dots$) requires careful Cl_2 /substrate ratio control to stop at the desired substitution level. All work with Cl_2 must be conducted in a well-ventilated fume hood with appropriate scrubbing of exhaust gases.

5.2 Conversion, Yield, and Selectivity

Before analyzing reaction networks, we establish precise definitions. Consistent notation prevents confusion when comparing results across different references and industrial contexts.

5.2.1 Conversion of Reactant A

Conversion measures the fraction of reactant consumed:

$$X_A \equiv \frac{F_{A0} - F_A}{F_{A0}} \quad (5.1)$$

5. Selectivity and Reaction Networks

For constant-density (liquid-phase or dilute gas-phase) systems where volumetric flow rate $\dot{V}$ is constant:

$$X_A = \frac{C_{A0} - C_A}{C_{A0}} \quad (5.2)$$

Rearranging gives $C_A = C_{A0}(1 - X_A)$, as used throughout Chapter 3.

5.2.2 Yield of Product B

The **overall yield** of product B (relative to the initial amount of reactant A) is:

$$Y_B \equiv \frac{F_B - F_{B0}}{F_{A0}} \quad (5.3)$$

When the feed contains no B ($F_{B0} = 0$) and volumetric flow is constant:

$$Y_B = \frac{C_B}{C_{A0}} \quad (5.4)$$

Yield answers the question: “Of all the A we fed, how much ended up as B?”

Remark 5.2 (Yield Bounds). Yield is bounded: $0 \leq Y_B \leq 1$ for reactions with 1:1 stoichiometry. For reactions where multiple moles of product form per mole of reactant, adjust the definition accordingly.

5.2.3 Selectivity: Overall and Instantaneous

Overall selectivity of B relative to A consumed measures what fraction of the converted reactant became the desired product:

$$S_{B/A} \equiv \frac{\text{moles of B formed}}{\text{moles of A consumed}} = \frac{F_B - F_{B0}}{F_{A0} - F_A} \quad (5.5)$$

When $F_{B0} = 0$, this simplifies to the important relationship:

$$S_{B/A} = \frac{Y_B}{X_A} \quad (5.6)$$

Remark 5.3 (Concept Check: Selectivity–Yield–Conversion Triangle). A reactor converts 60% of A. The yield of desired product B is $Y_B = 0.42$.

1. What is the selectivity $S_{B/A}$?

2. If the only other product is C, what is Y_C ?
3. If operating conditions change so that $S_{B/A}$ drops to 0.50 while X_A stays at 0.60, what is the new Y_B ?

Hint: Use $S_{B/A} = Y_B/X_A$ and $Y_B + Y_C = X_A$ (for $A \rightarrow B$, $A \rightarrow C$ with 1:1 stoichiometry).

For comparing two products B and C, define the **product selectivity ratio**:

$$S_{B/C} \equiv \frac{F_B - F_{B0}}{F_C - F_{C0}} \quad (5.7)$$

When both products start at zero, $S_{B/C} = C_B/C_C$.

Instantaneous selectivity compares reaction rates directly:

$$s_{B/C} \equiv \frac{r_B}{r_C} \quad (5.8)$$

This is powerful because it connects directly to mechanism and rate constants, independent of reactor configuration. For simple kinetics, instantaneous selectivity equals the ratio of rate constants.

5.2.4 The Selectivity-Conversion Tradeoff

A key insight for process design: **high conversion and high selectivity often conflict**. At low conversion, selectivity may be high because undesired secondary reactions have not yet consumed the product. As conversion increases, these secondary reactions accumulate, reducing selectivity.

The process chemist's challenge is to find operating conditions that achieve acceptable conversion while maintaining high selectivity—or to recycle unreacted feed to achieve high overall conversion at low per-pass conversion.

Remark 5.4 (Resolving the Selectivity–Conversion Tradeoff: Recycle). The selectivity–conversion conflict is resolved in industrial practice by operating at **low per-pass conversion** (maintaining high selectivity) and **recycling** unconverted reactant. With complete recycle of unreacted A and perfect separation, the overall conversion approaches 100% even though each pass converts only a small fraction.

The effective overall yield equals the per-pass selectivity times the

overall conversion:

$$Y_{B,\text{overall}} \approx S_{B/A}(\text{low } X) \times X_{\text{overall}} \quad (5.9)$$

This approximation is exact for parallel reactions (where selectivity is independent of conversion) and a good estimate for consecutive reactions when per-pass conversion is low enough that secondary reactions are negligible.

A classic industrial example is **ethylene oxide production** on an $\text{Ag}/\alpha\text{-Al}_2\text{O}_3$ catalyst. The per-pass conversion is kept at only 8–12% to maintain selectivity $S_{\text{EO}} > 80\%$; unreacted ethylene is recycled after separation, and the overall ethylene conversion exceeds 95% with recycle. Similarly, in **methanol synthesis** ($\text{CO}_2 + 3\text{H}_2 \rightarrow \text{CH}_3\text{OH} + \text{H}_2\text{O}$ over $\text{Cu}/\text{ZnO}/\text{Al}_2\text{O}_3$), per-pass conversion is limited by equilibrium to about 25%, yet recycle achieves >99% overall conversion.

The cost of recycle—separation, compression, and piping—must be weighed against the gain in selectivity. This economic optimization is a core challenge in process design and connects directly to Problem 6, which asks about recycle as a design modification.

5.3 Reactor Modeling for Multiple Reactions

Before analyzing specific networks, we establish the general framework for multi-component reactor modeling.

5.3.1 PFR: System of ODEs in Residence Time

For constant-density systems, the PFR design equation for each species becomes:

$$\boxed{\frac{dC_i}{d\tau} = R_i(\mathbf{C}, T)} \quad (5.10)$$

where R_i is the **net rate of formation** of species i (mol/L/s), determined by summing contributions from all reactions weighted by stoichiometry. The vector $\mathbf{C} = [C_A, C_B, C_C, \dots]$ represents all species concentrations.

This is a system of coupled ordinary differential equations integrated from $\tau = 0$ (inlet) to the desired outlet residence time.

5.3.2 CSTR: System of Algebraic Equations

At steady state, the CSTR mole balance for each species gives:

$$0 = C_{i,\text{in}} - C_{i,\text{out}} + \tau \cdot R_i(\mathbf{C}_{\text{out}}, T) \quad (5.11)$$

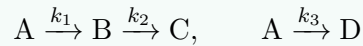
Rearranging:

$$C_{i,\text{out}} = C_{i,\text{in}} + \tau \cdot R_i(\mathbf{C}_{\text{out}}, T) \quad (5.12)$$

This is a set of **coupled nonlinear algebraic equations** that must be solved simultaneously for all outlet concentrations. In Python, we use `scipy.optimize.root` or `fsolve`.

Remark 5.5 (Key Conceptual Difference). PFR = integrate along τ (composition changes continuously).
CSTR = solve steady-state algebra at exit conditions (entire reactor at exit composition).

Remark 5.6 (Mixed Parallel–Consecutive Networks). Real reaction systems rarely consist of pure parallel or pure consecutive networks. A common industrial pattern combines both:



where B is desired, C is the overproduct, and D is a parallel byproduct.

The design principles from individual networks still apply, but must be combined:

- **Residence time** (from consecutive): operate near τ^* to maximize B before it converts to C
- **Temperature** (from parallel): choose T to maximize k_1 relative to both k_2 and k_3
- **Concentration** (from non-equal-order parallel, see Section 5.4): choose reactor type based on reaction orders of the competing pathways

A classic example is the **chlorination of toluene**: monochlorotoluene (desired) forms by electrophilic substitution (k_1), but can overchlorinate to dichlorotoluene (k_2), while a parallel side reaction produces benzyl chloride (k_3). Controlling both temperature and τ simultaneously is essential.

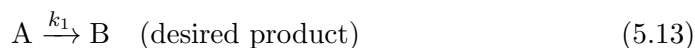
For complex networks with three or more reactions, analytical solutions are rarely available. The ODE/algebraic framework (Eqs. (5.10) and (5.12)) must be solved numerically, as practised in NB5.

5.4 Network 1: Parallel Reactions

Parallel reactions occur when a single reactant can form multiple products through independent pathways. This is common in organic synthesis where a substrate can react at different functional groups or through different mechanisms (e.g., substitution vs. elimination, or attack at different aromatic positions).

5.4.1 Reaction Scheme

Consider the simplest parallel network:



Assuming first-order kinetics in A for both reactions:

$$r_1 = k_1 C_A, \quad r_2 = k_2 C_A \quad (5.15)$$

The net rates of formation are:

$$R_A = -(k_1 + k_2)C_A \quad (5.16)$$

$$R_B = k_1 C_A \quad (5.17)$$

$$R_C = k_2 C_A \quad (5.18)$$

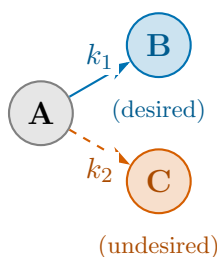


Figure 5.1: Parallel reaction network: reactant A can form either desired product B (rate constant k_1 , solid arrow) or undesired byproduct C (rate constant k_2 , dashed arrow). The competing pathways share the same reactant, so their rates are given by Eq. (5.15). The ratio of products formed depends only on the ratio k_1/k_2 , regardless of conversion.

Fig. 5.1 illustrates the essential feature of parallel reactions: a single reactant partitions between two independent pathways. Because both reactions consume A at rates proportional to C_A , the competition between them is determined entirely by the rate constants k_1 and k_2 . At any instant, the

fraction of A molecules that react to form B versus C is fixed by the ratio $k_1/(k_1 + k_2)$. This fundamental insight—that parallel first-order reactions yield products in a constant ratio—has profound implications for reactor design, as we shall see.

5.4.2 PFR Solution for Parallel Reactions

Remark 5.7 (Deriving the Parallel PFR Solution). We solve the system of ODEs that govern the PFR in three steps.

Step 1: Solve for $C_A(\tau)$. The total consumption rate of A is $-(k_1 + k_2)C_A$, giving the ODE

$$\frac{dC_A}{d\tau} = -(k_1 + k_2)C_A.$$

Separating variables: $dC_A/C_A = -(k_1 + k_2) d\tau$. Integrating with the initial condition $C_A(0) = C_{A0}$:

$$C_A(\tau) = C_{A0} e^{-(k_1 + k_2)\tau}.$$

Step 2: Solve for $C_B(\tau)$. Substituting the result for C_A into the ODE for B:

$$\frac{dC_B}{d\tau} = k_1 C_A = k_1 C_{A0} e^{-(k_1 + k_2)\tau}.$$

This is a known function of τ . Integrating with $C_B(0) = 0$:

$$C_B(\tau) = \frac{k_1 C_{A0}}{k_1 + k_2} \left[1 - e^{-(k_1 + k_2)\tau} \right].$$

Step 3: Solve for $C_C(\tau)$. By symmetry (replacing k_1 with k_2):

$$C_C(\tau) = \frac{k_2 C_{A0}}{k_1 + k_2} \left[1 - e^{-(k_1 + k_2)\tau} \right].$$

Verification (mass balance): Adding all three concentrations:

$$C_A + C_B + C_C = C_{A0} e^{-(k_1 + k_2)\tau} + C_{A0} \left(1 - e^{-(k_1 + k_2)\tau} \right) = C_{A0} \quad \checkmark$$

The mass balance is satisfied at all τ , confirming the solution. Note the key feature: $C_B/C_C = k_1/k_2$ at all times, independent of τ .

For the PFR, the concentration of A decays as:

$$\frac{dC_A}{d\tau} = -(k_1 + k_2)C_A \quad \Rightarrow \quad C_A(\tau) = C_{A0} e^{-(k_1 + k_2)\tau} \quad (5.19)$$

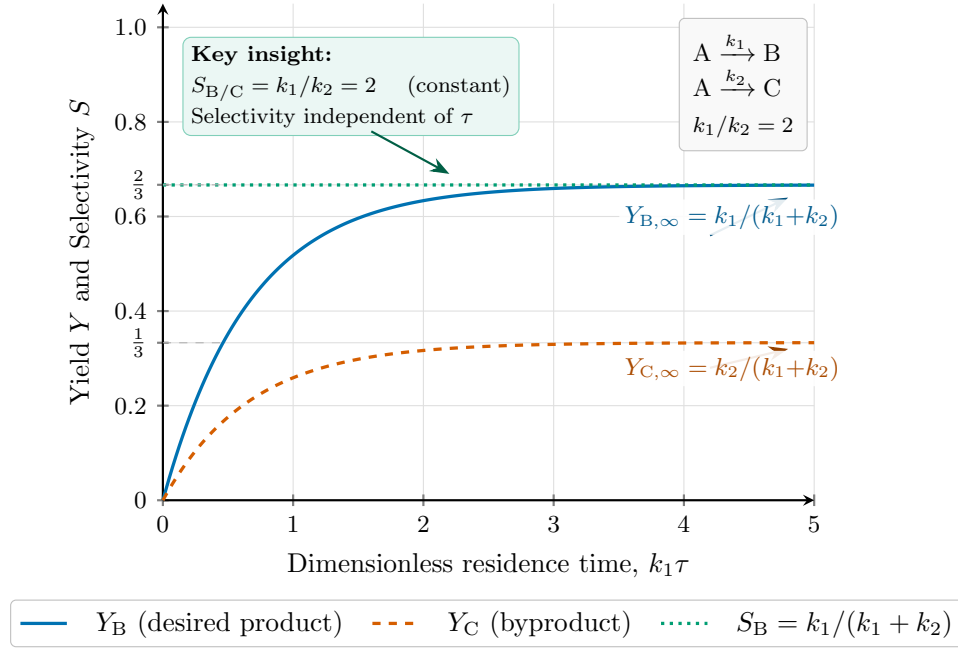


Figure 5.2: Yields and selectivity versus dimensionless residence time for parallel first-order reactions with $k_1/k_2 = 2$. Both Y_B and Y_C increase monotonically with τ , approaching their asymptotic values $k_1/(k_1 + k_2) = 2/3$ and $k_2/(k_1 + k_2) = 1/3$ respectively (Eqs. (5.20) and (5.21)). The selectivity $S_{B/A} = k_1/(k_1 + k_2)$ remains constant at all residence times (Eq. (5.22))—a hallmark of parallel reaction networks.

Substituting into the equations for B and C and integrating:

$$C_B(\tau) = C_{A0} \frac{k_1}{k_1 + k_2} \left(1 - e^{-(k_1 + k_2)\tau} \right) \quad (5.20)$$

$$C_C(\tau) = C_{A0} \frac{k_2}{k_1 + k_2} \left(1 - e^{-(k_1 + k_2)\tau} \right) \quad (5.21)$$

5.4.3 The Fundamental Result: Selectivity is Independent of Residence Time

Taking the ratio of product concentrations:

$$\frac{C_B}{C_C} = \frac{k_1}{k_2} \Rightarrow S_{B/C} = \frac{k_1}{k_2} \quad (5.22)$$

This is a **crucial result**: for parallel first-order reactions with the same reactant, the product selectivity ratio is **constant and independent of residence time**. It depends only on the **ratio of rate constants**.

Fig. 5.2 demonstrates the key result graphically: as residence time increases, both product yields rise toward their asymptotic limits, but the *ratio* of Y_B to Y_C never changes. The horizontal selectivity line emphasizes this point—no matter how long the reactants spend in the reactor, the product distribution is locked in by the rate constant ratio. This behavior contrasts sharply with consecutive reactions (discussed in Section 5.5), where residence time profoundly affects selectivity. For the practicing engineer, this result is liberating: when dealing with parallel reactions, conversion and selectivity are decoupled, allowing independent optimization of each objective.

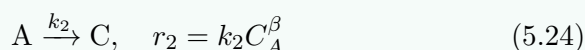
Remark 5.8 (Practical Implication). Since residence time does not affect selectivity in parallel first-order reactions, the chemist has freedom to choose τ based solely on conversion requirements. Selectivity must be controlled through other means—primarily temperature.

Remark 5.9 (Concept Check: Parallel Selectivity). For parallel first-order reactions $A \rightarrow B$ ($k_1 = 0.3 \text{ s}^{-1}$) and $A \rightarrow C$ ($k_2 = 0.1 \text{ s}^{-1}$):

1. What is the selectivity $S_{B/C}$?
2. If the PFR residence time is doubled, how does $S_{B/C}$ change?
3. What happens to $S_{B/C}$ if $E_{a1} > E_{a2}$ and the temperature is raised by 50 K?

Answers: (1) $S_{B/C} = k_1/k_2 = 3$; (2) unchanged—selectivity is independent of τ for equal-order parallel reactions; (3) $S_{B/C}$ increases because the reaction with higher E_a responds more steeply to temperature.

Remark 5.10 (When Selectivity Depends on Concentration: Non-Equal-Order Reactions). The result $S_{B/C} = k_1/k_2$ (constant, independent of conversion) holds *only* when both reactions have the same order in A. For competing reactions of **different orders**:



the instantaneous selectivity becomes:

$$s_{B/C} = \frac{r_1}{r_2} = \frac{k_1}{k_2} C_A^{\alpha-\beta} \quad (5.25)$$

Now selectivity depends on concentration! This gives a powerful reactor design rule:

Condition	Favored by	Preferred reactor
$\alpha > \beta$	High C_A	PFR or batch
$\alpha < \beta$	Low C_A	CSTR (low exit C_A)
$\alpha = \beta$	Independent of C_A	No preference

Example: if $A \rightarrow B$ is second-order ($\alpha = 2$) and $A \rightarrow C$ is first-order ($\beta = 1$), then $s_{B/C} = (k_1/k_2)C_A$. A PFR keeps C_A high (especially near the inlet), favoring B production.

This concentration-based selectivity principle **combines** with the temperature principle from Section 5.4.5. Optimal reactor design considers both levers simultaneously.

5.4.4 CSTR Solution for Parallel Reactions

For the CSTR, steady-state balances give:

$$C_A = \frac{C_{A0}}{1 + (k_1 + k_2)\tau} \quad (5.26)$$

$$C_B = \tau \cdot k_1 C_A = \frac{k_1 \tau C_{A0}}{1 + (k_1 + k_2)\tau} \quad (5.27)$$

$$C_C = \tau \cdot k_2 C_A = \frac{k_2 \tau C_{A0}}{1 + (k_1 + k_2)\tau} \quad (5.28)$$

Once again:

$$\frac{C_B}{C_C} = \frac{k_1}{k_2} \quad (5.29)$$

The selectivity result holds for both reactor types. For simple parallel first-order kinetics, reactor type does not significantly affect selectivity.

5.4.5 Temperature Control of Parallel Selectivity

Since selectivity depends on k_1/k_2 , and rate constants follow the Arrhenius equation:

$$\frac{k_1}{k_2} = \frac{A_1}{A_2} \exp\left(-\frac{E_{a1} - E_{a2}}{RT}\right) \quad (5.30)$$

This gives us the **primary control lever** for parallel reactions:

$$\boxed{\frac{k_1}{k_2} \propto \exp\left(-\frac{\Delta E_a}{RT}\right), \quad \text{where } \Delta E_a = E_{a1} - E_{a2}} \quad (5.31)$$

- If $E_{a1} > E_{a2}$ (desired reaction has higher activation energy): **Increasing T improves selectivity to B**
- If $E_{a1} < E_{a2}$ (undesired reaction has higher activation energy): **Decreasing T improves selectivity to B**

Remark 5.11 (The Selectivity Crossover Temperature). Setting $k_1(T) = k_2(T)$ in the Arrhenius expressions (Eq. (5.30)) gives the crossover temperature where neither reaction is favored:

$$T_{\text{cross}} = \frac{E_{a1} - E_{a2}}{R \ln(A_1/A_2)} \quad (5.32)$$

This expression requires $E_{a1} \neq E_{a2}$ (otherwise the Arrhenius lines are parallel and never intersect) and the argument of the logarithm must be positive ($A_1/A_2 > 0$, which is always true since pre-exponential factors are positive).

- For $T > T_{\text{cross}}$: the reaction with **higher** E_a dominates (its rate constant responds more steeply to temperature).
- For $T < T_{\text{cross}}$: the reaction with **lower** E_a dominates.

Design implication: operate well away from T_{cross} in the direction that favors the desired product. Near T_{cross} , selectivity is insensitive to temperature changes, and other control strategies (concentration, catalyst modification) become more important.

Note that T_{cross} can be negative (i.e., nonphysical) if the signs of numerator and denominator do not produce a positive result—meaning one reaction is always faster than the other at all achievable temperatures.

Remark 5.12 (Practical Constraints). In practice, temperature cannot be adjusted arbitrarily. Catalyst stability, equilibrium limitations, and mass transfer effects all constrain the operating window. The engineer must balance selectivity against these constraints.

5.5 Network 2: Consecutive Reactions

Consecutive (or series) reactions occur when the desired product can react further to form secondary products. This is extremely common—the intermediate we want is often more reactive than the starting material.

5.5.1 Reaction Scheme

The prototype consecutive network is:



where B is the desired intermediate product and C is the undesired final product.

Assuming first-order kinetics in each step:

$$r_1 = k_1 C_A, \quad r_2 = k_2 C_B \quad (5.34)$$

The net rates of formation are:

$$R_A = -k_1 C_A \quad (5.35)$$

$$R_B = k_1 C_A - k_2 C_B \quad (5.36)$$

$$R_C = k_2 C_B \quad (5.37)$$

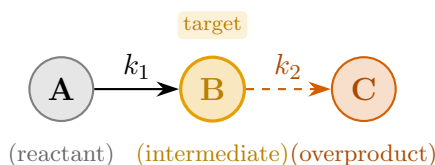


Figure 5.3: Consecutive reaction network: reactant A converts to intermediate B (rate constant k_1), which then reacts further to form overproduct C (rate constant k_2). The intermediate B is the target product, highlighted with a “target” label. The challenge is to harvest B before it reacts onward to C, as described by Eq. (5.34).

Fig. 5.3 highlights the central challenge of consecutive reactions: the desired product B is inherently unstable in the reaction environment—it is consumed by the very process that creates it. Unlike parallel reactions where selectivity is fixed, here the yield of B depends critically on *when* we stop the reaction. Stop too early, and little A has converted to B; stop too late, and B has been lost to C. This creates a “selectivity window” that we must identify and exploit.

5.5.2 PFR Solution: The Classic A to B to C Problem

Remark 5.13 (Deriving the Consecutive PFR Solution by Integrating Factor). The ODE for species B (Eq. (5.42)) is a **first-order linear ODE** in standard form:

$$\frac{dC_B}{d\tau} + k_2 C_B = k_1 C_{A0} e^{-k_1 \tau} \quad (5.38)$$

This has the structure $dy/dx + P(x)y = Q(x)$ with $P = k_2$ (constant) and $Q = k_1 C_{A0} e^{-k_1 \tau}$.

Step 1: Integrating factor. The integrating factor is $\mu(\tau) = e^{\int k_2 d\tau} = e^{k_2 \tau}$. Multiplying both sides by μ :

$$\frac{d}{d\tau}[e^{k_2 \tau} C_B] = k_1 C_{A0} e^{(k_2 - k_1)\tau} \quad (5.39)$$

Step 2: Integrate both sides.

$$e^{k_2 \tau} C_B = \frac{k_1 C_{A0}}{k_2 - k_1} e^{(k_2 - k_1)\tau} + \mathcal{C} \quad (k_1 \neq k_2) \quad (5.40)$$

Step 3: Apply initial condition $C_B(0) = 0$:

$$0 = \frac{k_1 C_{A0}}{k_2 - k_1} + \mathcal{C} \implies \mathcal{C} = -\frac{k_1 C_{A0}}{k_2 - k_1}$$

Step 4: Solve for C_B . Dividing by $e^{k_2 \tau}$:

$$C_B(\tau) = \frac{k_1 C_{A0}}{k_2 - k_1} (e^{-k_1 \tau} - e^{-k_2 \tau})$$

which is exactly Eq. (5.43). Note the restriction $k_1 \neq k_2$; the equal-rate-constant case requires separate treatment (see Section 5.5.4).

The concentration of A decays as a simple first-order process:

$$\boxed{C_A(\tau) = C_{A0} e^{-k_1 \tau}} \quad (5.41)$$

The intermediate B satisfies:

$$\frac{dC_B}{d\tau} = k_1 C_A - k_2 C_B = k_1 C_{A0} e^{-k_1 \tau} - k_2 C_B \quad (5.42)$$

This is a first-order linear ODE. With initial condition $C_B(0) = 0$, the solution is:

$$\boxed{C_B(\tau) = C_{A0} \frac{k_1}{k_2 - k_1} (e^{-k_1 \tau} - e^{-k_2 \tau}) \quad (k_2 \neq k_1)} \quad (5.43)$$

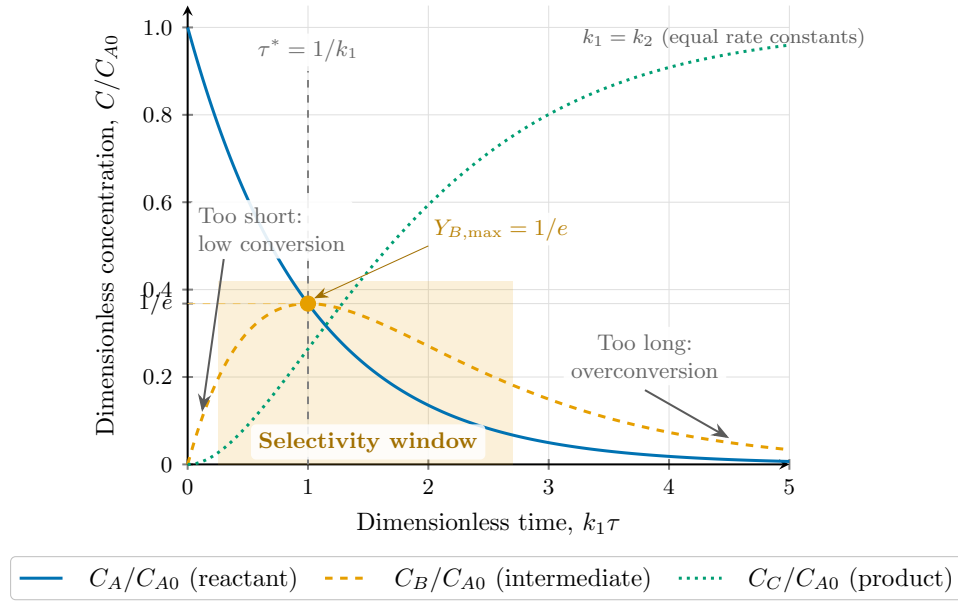


Figure 5.4: Concentration profiles for consecutive first-order reactions with equal rate constants ($k_1 = k_2$). Reactant A decays exponentially (Eq. (5.41)), while intermediate B rises then falls through a maximum at $\tau^* = 1/k_1$ (Eq. (5.50)). The overproduct C accumulates sigmoidally. The shaded region indicates the “selectivity window” where Y_B is appreciable. Operating within this window is essential for maximizing intermediate yield.

By mass balance, $C_A + C_B + C_C = C_{A0}$:

$$C_C(\tau) = C_{A0} - C_A(\tau) - C_B(\tau) \quad (5.44)$$

5.5.3 The Selectivity Window: B Has a Maximum

The concentration profile of B rises from zero (no B initially), passes through a maximum, and then decreases as B is consumed to form C. To find the optimal residence time, set $dC_B/d\tau = 0$:

Remark 5.14 (Deriving τ^* and $Y_{B,\max}$).

Part 1 — Finding the optimal residence time τ^* .

Differentiating Eq. (5.43) with respect to τ :

$$\frac{dC_B}{d\tau} = C_{A0} \frac{k_1}{k_2 - k_1} (-k_1 e^{-k_1\tau} + k_2 e^{-k_2\tau}) \quad (5.45)$$

Setting this equal to zero (the pre-factor $C_{A0} k_1 / (k_2 - k_1) \neq 0$):

$$k_1 e^{-k_1\tau^*} = k_2 e^{-k_2\tau^*}$$

Rearranging:

$$e^{(k_2-k_1)\tau^*} = \frac{k_2}{k_1}$$

Taking the natural logarithm of both sides:

$$\tau^* = \frac{\ln(k_2/k_1)}{k_2 - k_1}$$

which is precisely Eq. (5.50).

Part 2 — Finding $Y_{B,\max}$.

To substitute τ^* back into C_B , we establish two identities. From $e^{(k_2-k_1)\tau^*} = k_2/k_1$, raise both sides to the powers $k_1/(k_2 - k_1)$ and $k_2/(k_2 - k_1)$:

$$e^{-k_1\tau^*} = \left(\frac{k_1}{k_2}\right)^{k_1/(k_2-k_1)} \quad (5.46)$$

$$e^{-k_2\tau^*} = \left(\frac{k_1}{k_2}\right)^{k_2/(k_2-k_1)} \quad (5.47)$$

Substituting into $Y_B = C_B/C_{A0}$ from Eq. (5.43):

$$Y_{B,\max} = \frac{k_1}{k_2 - k_1} \left[\left(\frac{k_1}{k_2}\right)^{k_1/(k_2-k_1)} - \left(\frac{k_1}{k_2}\right)^{k_2/(k_2-k_1)} \right]$$

Factoring out $(k_1/k_2)^{k_2/(k_2-k_1)}$ from the bracket, one can show after algebraic simplification that:

$$Y_{B,\max} = \left(\frac{k_1}{k_2}\right)^{k_2/(k_2-k_1)} \quad (5.48)$$

which is the compact result given in Eq. (5.51).

$$k_1 C_{A0} e^{-k_1\tau^*} - k_2 C_B(\tau^*) = 0 \quad (5.49)$$

Solving (after some algebra) gives the **optimal residence time**:

$$\tau^* = \frac{1}{k_2 - k_1} \ln \left(\frac{k_2}{k_1} \right) \quad (k_1 \neq k_2) \quad (5.50)$$

For the case $k_1 > k_2$, the formula still applies with the understanding that the argument of the logarithm is less than 1 (giving a negative value that makes τ^* positive after division by the negative denominator).

Remark 5.15 (Physical Interpretation). The existence of an optimal τ^* is the hallmark of consecutive reactions:

- **Too short** residence time: Not enough A has converted to B
- **Too long** residence time: B has been over-reacted to C
- **At τ^* :** The rate of B formation (from A) exactly balances the rate of B consumption (to C)

This is the **selectivity window**—the operating region where B yield is maximized.

Fig. 5.4 vividly illustrates the selectivity window concept. The concentration of B traces a characteristic arc: starting from zero, rising as A converts to B, reaching a maximum at τ^* , and then declining as B is consumed faster than it is produced. The shaded region marks the operating window where significant amounts of B can be harvested. Notice that even at the optimal point, the maximum yield is only $Y_{B,\max} = 1/e \approx 0.368$ for this equal-rate-constant case—a substantial fraction of reactant is inevitably lost to overproduction. The annotations “too short” and “too long” remind us that deviation from the optimal residence time in either direction reduces yield, though for different reasons: insufficient conversion versus excessive overconversion.

Remark 5.16 (Concept Check: Selectivity Window). For consecutive reactions $A \xrightarrow{k_1} B \xrightarrow{k_2} C$ with $k_1 = 0.1 \text{ s}^{-1}$ and $k_2 = 0.4 \text{ s}^{-1}$:

1. Calculate the optimal residence time τ^* using Eq. (5.50).
2. Without calculating $Y_{B,\max}$, predict whether it will be closer to 0.2 or 0.8. Justify by considering the ratio k_1/k_2 .
3. If you could redesign the catalyst to cut k_2 in half (to 0.2 s^{-1}), what would happen to $Y_{B,\max}$ qualitatively?

Hint: When $k_2 \gg k_1$, intermediate B is consumed almost as fast as it forms, limiting achievable yield.

5.5.4 Maximum Yield of Intermediate

Substituting τ^* back into the expression for C_B gives the maximum yield. After simplification:

$$Y_{B,\max} = \frac{C_B(\tau^*)}{C_{A0}} = \left(\frac{k_1}{k_2}\right)^{k_2/(k_2-k_1)} \quad (5.51)$$

This result shows that maximum yield depends only on the ratio k_1/k_2 :

- When $k_1 \gg k_2$ (slow second reaction): $Y_{B,\max} \rightarrow 1$ (nearly all A converts to B before any B converts to C)
- When $k_1 \ll k_2$ (fast second reaction): $Y_{B,\max} \rightarrow 0$ (B is consumed almost as fast as it forms)
- When $k_1 = k_2$: $Y_{B,\max} = 1/e \approx 0.368$

Remark 5.17 (Special Case: Equal Rate Constants ($k_1 = k_2$)).

When $k_1 = k_2 = k$, the general formula Eq. (5.43) becomes the indeterminate form $0/0$. We resolve this by applying L'Hôpital's rule, differentiating the numerator and denominator with respect to k_2 and then taking the limit $k_2 \rightarrow k_1$:

$$\lim_{k_2 \rightarrow k_1} \frac{k_1(e^{-k_1\tau} - e^{-k_2\tau})}{k_2 - k_1} = k_1 \tau e^{-k_1\tau} \quad (5.52)$$

Therefore the concentration of intermediate B simplifies to:

$$C_B(\tau) = C_{A0} k \tau e^{-k\tau} \quad (k_1 = k_2 = k) \quad (5.53)$$

Optimal residence time. Setting $dC_B/d\tau = 0$:

$$k C_{A0} e^{-k\tau} (1 - k\tau) = 0 \implies \tau^* = \frac{1}{k}$$

Maximum yield. Substituting $\tau^* = 1/k$ into Eq. (5.53):

$$Y_{B,\max} = k \cdot \frac{1}{k} \cdot e^{-1} = \frac{1}{e} \approx 0.368$$

This is the case plotted in Fig. 5.4.

Physical insight: even when both steps proceed at exactly the same rate, the maximum achievable yield of intermediate is only 36.8%—a fundamental kinetic constraint. Achieving higher yields requires making $k_1 \gg k_2$, which is the goal of catalyst design and temperature optimization.

5.5.5 CSTR Solution: Why Mixing Hurts Intermediate Yield

For a CSTR with consecutive reactions, the steady-state balances give:

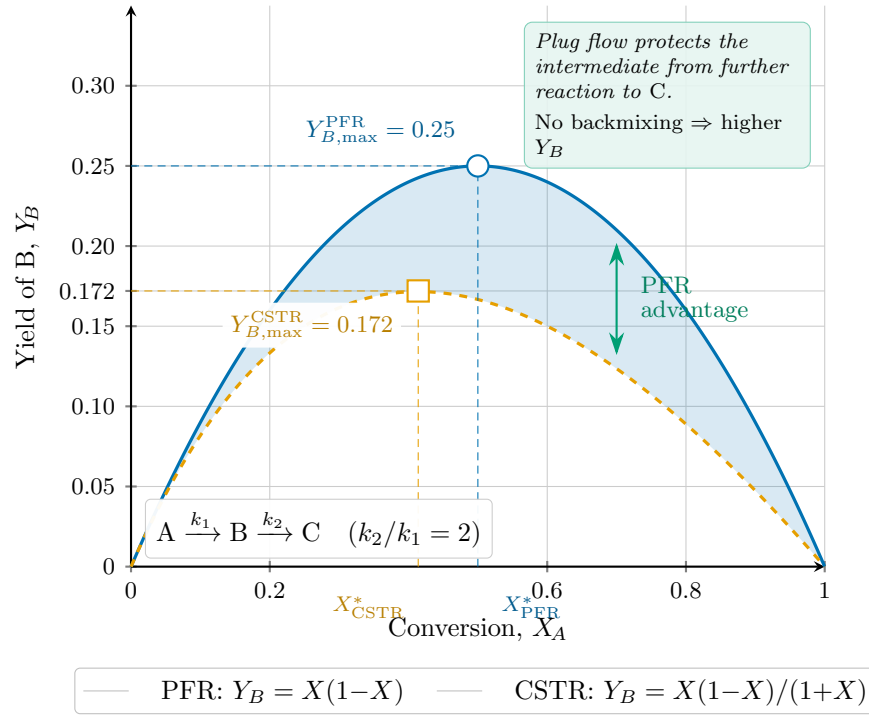


Figure 5.5: Yield of intermediate B versus conversion for consecutive first-order reactions with $k_2/k_1 = 2$, comparing PFR (solid blue) and CSTR (dashed orange). The PFR achieves a maximum yield of $Y_{B,\max}^{\text{PFR}} = 0.25$ at $X = 0.5$, while the CSTR reaches only $Y_{B,\max}^{\text{CSTR}} = 0.172$ at $X = 0.414$. The shaded region represents the “PFR advantage”—the additional yield obtained by avoiding backmixing. This advantage arises because plug flow protects freshly-formed B from immediate exposure to the full residence time (Eqs. (5.43) and (5.55)).

$$C_A = \frac{C_{A0}}{1 + k_1\tau} \quad (5.54)$$

For B, balancing formation from A and consumption to C:

$$C_B = \frac{k_1\tau C_A}{1 + k_2\tau} = \frac{k_1\tau C_{A0}}{(1 + k_1\tau)(1 + k_2\tau)} \quad (5.55)$$

The CSTR also exhibits a maximum in C_B , but this maximum is **lower** than for the PFR at equivalent conditions. Setting $dY_B/d\tau = 0$ and solving gives the **CSTR optimal residence time**:

$$\tau_{\text{CSTR}}^* = \frac{1}{\sqrt{k_1 k_2}} \quad (5.56)$$

Substituting back yields the **CSTR maximum yield**:

$$Y_{B,\max}^{\text{CSTR}} = \frac{1}{\left(1 + \sqrt{k_2/k_1}\right)^2} \quad (5.57)$$

Comparing with the PFR result (Eq. (5.51)), the CSTR maximum yield is always lower than the PFR value. For equal rate constants ($k_1 = k_2$), the CSTR gives $Y_{B,\max}^{\text{CSTR}} = 1/4 = 0.25$, compared to $Y_{B,\max}^{\text{PFR}} = 1/e \approx 0.368$.

Physical explanation: In a CSTR, freshly-formed B is immediately mixed into the entire reactor volume, where it experiences the full residence time and continues reacting to C. In a PFR, B formed near the inlet has not yet had much time to react further—the concentration profile protects early-formed B from over-reaction.

Fig. 5.5 quantifies the penalty of backmixing for consecutive reactions. At any given conversion, the PFR achieves higher intermediate yield than the CSTR—the blue curve lies entirely above the orange curve. The physical explanation lies in the concentration profiles within each reactor. In a PFR, molecules of B formed near the inlet have not yet accumulated the full residence time; they are “protected” by the plug flow displacement until they approach the outlet. In a CSTR, by contrast, freshly-formed B is instantly mixed into the bulk, where it experiences the average (long) residence time and reacts onward to C. This mixing-induced yield loss is intrinsic to the CSTR design and cannot be overcome by adjusting operating conditions. For intermediate products in consecutive reactions, plug flow is nearly always superior.

Remark 5.18 (Practical Implication). For intermediate products in consecutive reactions, **plug flow behavior** is almost always better than a fully mixed reactor. In laboratory synthesis, this principle explains why slow addition (dropwise) of reagents often gives better selectivity than adding all reagent at once—the concentration gradient mimics plug flow behavior.

Remark 5.19 (Concept Check: PFR vs CSTR for Intermediates).

For consecutive first-order reactions with $k_1 = k_2 = k$:

1. Calculate $Y_{B,\max}$ for the PFR and CSTR using Eqs. (5.51) and (5.57).
2. By what percentage does the PFR outperform the CSTR?
3. Explain in one sentence *why* backmixing hurts intermediate yield.

Answers: PFR gives $1/e \approx 0.368$; CSTR gives $1/4 = 0.250$. The PFR advantage is $(0.368 - 0.250)/0.250 = 47\%$. Backmixing exposes freshly-formed B to the full average residence time, accelerating its consumption to C.

5.5.6 Multiple CSTRs in Series

A practical approximation to PFR behavior is achieved by placing N CSTRs in series, each with residence time τ/N . As $N \rightarrow \infty$, the cascade approaches PFR performance.

For intermediate selectivity:

- 1 CSTR: Lowest $Y_{B,\max}$
- 3 CSTRs in series: Significantly improved
- 10+ CSTRs: Approaches PFR
- PFR: Highest $Y_{B,\max}$

This cascade concept is used in practice when CSTR-type equipment (stirred tanks) is preferred for mechanical or heat transfer reasons, but PFR-like selectivity is desired.

Remark 5.20 (Diffusion Limitations and Selectivity). Internal diffusion limitations (Chapter 4) can significantly affect selectivity in consecutive reaction networks. In the diffusion-limited regime, reactant concentration drops steeply near the pellet center, effectively creating a local environment of high conversion. For consecutive reactions $A \rightarrow B \rightarrow C$, this means the desired intermediate B is exposed to further reaction before it can diffuse out of the pellet. The result is **lower selectivity to B than predicted by intrinsic kinetics alone**. Reducing pellet size (lowering ϕ) or increasing pore diameter improves both effectiveness factor and selectivity—a rare case where rate and selectivity benefit from the same design change.

5.6 Temperature Effects on Selectivity

Temperature affects consecutive reaction selectivity through two mechanisms: changing the ratio k_1/k_2 and shifting the optimal residence time $\tau^*(T)$.

5.6.1 Arrhenius Dependence of Rate Constants

Recall from Chapter 2 that both rate constants follow the Arrhenius equation:

$$k_1(T) = A_1 \exp\left(-\frac{E_{a1}}{RT}\right), \quad k_2(T) = A_2 \exp\left(-\frac{E_{a2}}{RT}\right) \quad (5.58)$$

The ratio becomes:

$$\frac{k_1(T)}{k_2(T)} = \frac{A_1}{A_2} \exp\left(-\frac{E_{a1} - E_{a2}}{RT}\right) \quad (5.59)$$

5.6.2 Effect on Maximum Yield

Since $Y_{B,\max}$ depends on k_1/k_2 (Eq. (5.51)):

- If $E_{a1} > E_{a2}$: Higher T increases k_1/k_2 , **improving** $Y_{B,\max}$
- If $E_{a1} < E_{a2}$: Higher T decreases k_1/k_2 , **reducing** $Y_{B,\max}$

5.6.3 Effect on Optimal Residence Time

The optimal residence time also varies with temperature:

$$\tau^*(T) = \frac{1}{k_2(T) - k_1(T)} \ln\left(\frac{k_2(T)}{k_1(T)}\right) \quad (5.60)$$

As temperature increases, both rate constants increase, generally making τ^* shorter. However, the sensitivity depends on the difference in activation energies.

Remark 5.21 (Practical Synthesis Message). There is often a “sweet spot” temperature that balances:

1. Selectivity (ratio k_1/k_2)
2. Productivity (higher T means faster overall rates)
3. Catalyst stability (excessive T may cause deactivation)
4. Equilibrium (for reversible reactions, high T may be unfavorable)

Finding this optimum is a core challenge in synthetic chemistry and process development.

Remark 5.22 (Beyond Isothermal: Temperature Programming for Consecutive Reactions). All analysis so far assumes isothermal operation, but for consecutive reactions $A \rightarrow B \rightarrow C$ where $E_{a1} > E_{a2}$, a **decreasing temperature profile** can exceed the isothermal $Y_{B,\max}$.

The strategy is: start at **high temperature** (maximizes k_1/k_2 , rapidly converts A to B), then **cool** as B accumulates (minimizes k_2 , slowing B $\rightarrow$ C). In a **batch reactor**, this is achieved by a programmed temperature ramp downward during the reaction. In a **PFR**, a temperature gradient along the reactor length (hot inlet $\rightarrow$ cool outlet) can be implemented with segmented heat exchangers or countercurrent cooling. In a **CSTR**, this approach is not applicable because the entire reactor operates at a single uniform temperature by definition.

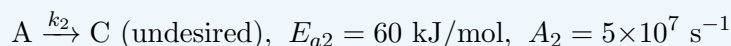
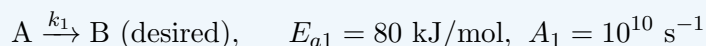
The improvement over isothermal operation is most significant when the difference $E_{a1} - E_{a2}$ is large (strong temperature sensitivity of selectivity) and when the isothermal $Y_{B,\max}$ is moderate (0.3–0.7). In practice, thermal inertia of industrial reactors limits the achievable cooling rate, so this technique finds its greatest use in laboratory and pilot-scale batch synthesis.

This is a powerful concept for the catalytic organic synthesis applications discussed in appendix G.

5.7 Worked Examples

Example 5.1: Parallel Reactions: Temperature Selection

Problem: An organic substrate A undergoes parallel reactions:



Calculate $S_{B/C}$ at $T = 400 \text{ K}$ and $T = 500 \text{ K}$. Which temperature favors the desired product?

Solution (Arrhenius Rate Constant Comparison):

First, calculate the rate constants at each temperature using the Arrhenius equation with $R = 8.314 \text{ J/(mol}\cdot\text{K)}$:

At $T = 400$ K:

$$k_1 = (1.0 \times 10^{10}) e^{-80000/(8.314 \times 400)} = (1.0 \times 10^{10}) e^{-24.06} = 0.36 \text{ s}^{-1}$$

$$k_2 = (5.0 \times 10^7) e^{-60000/(8.314 \times 400)} = (5.0 \times 10^7) e^{-18.04} = 0.730 \text{ s}^{-1}$$

At $T = 500$ K:

$$k_1 = (1.0 \times 10^{10}) e^{-80000/(8.314 \times 500)} = (1.0 \times 10^{10}) e^{-19.24} = 43.9 \text{ s}^{-1}$$

$$k_2 = (5.0 \times 10^7) e^{-60000/(8.314 \times 500)} = (5.0 \times 10^7) e^{-14.43} = 27.0 \text{ s}^{-1}$$

The selectivity ratios are:

$$S_{B/C}(400 \text{ K}) = \frac{k_1}{k_2} = \frac{0.36}{0.730} = 0.49$$

$$S_{B/C}(500 \text{ K}) = \frac{k_1}{k_2} = \frac{43.9}{27.0} = 1.63$$

Answer:

$$S_{B/C} = 0.49 \text{ at } 400 \text{ K}; S_{B/C} = 1.63 \text{ at } 500 \text{ K}$$

Physical interpretation: Since $E_{a1} > E_{a2}$, increasing temperature favors the desired reaction. At 400 K, the undesired product dominates; at 500 K, the desired product B is favored 1.6:1 over C. We should operate at higher temperature to maximize selectivity to B.

Example 5.2: Consecutive Reactions: Optimal Residence Time in PFR

Problem: Consider consecutive reactions $A \rightarrow B \rightarrow C$ in a PFR with:

- $C_{A0} = 1.0 \text{ mol/L}$
- $k_1 = 0.4 \text{ s}^{-1}$
- $k_2 = 1.0 \text{ s}^{-1}$

Calculate: (a) the optimal residence time τ^* , (b) the maximum yield $Y_{B,\max}$, and (c) the conversion X_A at τ^* .

Solution (Optimal Residence Time):

(a) Using Eq. (5.50):

$$\tau^* = \frac{1}{k_2 - k_1} \ln \left(\frac{k_2}{k_1} \right) = \frac{1}{1.0 - 0.4} \ln \left(\frac{1.0}{0.4} \right) = \frac{1}{0.6} \ln(2.5)$$

$$\tau^* = \frac{0.916}{0.6} = 1.53 \text{ s}$$

(b) The maximum concentration of B using Eq. (5.43):

$$\begin{aligned} C_B(\tau^*) &= C_{A0} \frac{k_1}{k_2 - k_1} \left(e^{-k_1 \tau^*} - e^{-k_2 \tau^*} \right) \\ &= (1.0) \frac{0.4}{0.6} \left(e^{-0.4 \times 1.53} - e^{-1.0 \times 1.53} \right) \\ &= 0.667 \left(e^{-0.612} - e^{-1.53} \right) \\ &= 0.667(0.542 - 0.217) \\ &= 0.667 \times 0.325 = 0.217 \text{ mol/L} \end{aligned}$$

Therefore:

$$Y_{B,\max} = \frac{0.217}{1.0} = 0.217 \text{ (21.7\%)}$$

(c) Conversion at τ^* :

$$\begin{aligned} C_A(\tau^*) &= C_{A0} e^{-k_1 \tau^*} = 1.0 \times e^{-0.612} = 0.542 \text{ mol/L} \\ X_A &= 1 - \frac{C_A}{C_{A0}} = 1 - 0.542 = 0.458 \text{ (45.8\%)} \end{aligned}$$

Answer:

$$\tau^* = 1.53 \text{ s}; Y_{B,\max} = 21.7\%; X_A = 45.8\%$$

Physical interpretation: The relatively low maximum yield (despite optimizing τ) reflects the unfavorable kinetics: $k_2 > k_1$ means B reacts faster than it forms. To improve yield, we would need to find conditions (catalyst, temperature) that increase k_1/k_2 .

Example 5.3: PFR vs CSTR Comparison for Intermediate Yield

Problem: Using the same kinetics as Section 5.7 ($k_1 = 0.4 \text{ s}^{-1}$, $k_2 = 1.0 \text{ s}^{-1}$, $C_{A0} = 1.0 \text{ mol/L}$), compare the maximum yield of B achievable in a PFR versus a single CSTR.

Solution (Reactor Comparison):

PFR (from previous example): $Y_{B,\max}^{\text{PFR}} = 0.217$ at $\tau^* = 1.53 \text{ s}$.

CSTR: Using Eq. (5.55), the yield is:

$$Y_B = \frac{C_B}{C_{A0}} = \frac{k_1 \tau}{(1 + k_1 \tau)(1 + k_2 \tau)}$$

To find the maximum, differentiate with respect to τ and set equal to zero. The result is:

$$\tau_{\text{CSTR}}^* = \frac{1}{\sqrt{k_1 k_2}} = \frac{1}{\sqrt{0.4 \times 1.0}} = \frac{1}{0.632} = 1.58 \text{ s}$$

Substituting back:

$$\begin{aligned} Y_{B,\max}^{\text{CSTR}} &= \frac{0.4 \times 1.58}{(1 + 0.4 \times 1.58)(1 + 1.0 \times 1.58)} \\ &= \frac{0.632}{(1.632)(2.58)} \\ &= \frac{0.632}{4.21} = 0.150 \text{ (15.0\%)} \end{aligned}$$

Answer:

PFR: $Y_{B,\max} = 21.7\%$; **CSTR:** $Y_{B,\max} = 15.0\%$

Physical interpretation: The PFR achieves 45% higher yield than the CSTR. Both reactors have similar optimal residence times (≈ 1.5 – 1.6 s), but the PFR's plug flow behavior protects the intermediate from over-reaction, giving significantly better selectivity.

Example 5.4: Temperature Effect on Consecutive Reaction Selectivity

Problem: The consecutive reaction $A \rightarrow B \rightarrow C$ has Arrhenius parameters:

$$k_1 : E_{a1} = 50 \text{ kJ/mol}, \quad A_1 = 1.0 \times 10^6 \text{ s}^{-1}$$

$$k_2 : E_{a2} = 70 \text{ kJ/mol}, \quad A_2 = 1.0 \times 10^8 \text{ s}^{-1}$$

Compare τ^* and $Y_{B,\max}$ at $T = 400 \text{ K}$ and $T = 450 \text{ K}$ in a PFR.

Solution (Temperature Dependence of Rate Constant Ratio):

At $T = 400 \text{ K}$:

$$k_1 = (1.0 \times 10^6) e^{-50000/(8.314 \times 400)} = (1.0 \times 10^6) e^{-15.03} = 0.295 \text{ s}^{-1}$$

$$k_2 = (1.0 \times 10^8) e^{-70000/(8.314 \times 400)} = (1.0 \times 10^8) e^{-21.05} = 0.0722 \text{ s}^{-1}$$

Since $k_1 > k_2$:

$$\tau^*(400 \text{ K}) = \frac{1}{k_2 - k_1} \ln \left(\frac{k_2}{k_1} \right) = \frac{\ln(0.0722/0.295)}{0.0722 - 0.295} = \frac{-1.41}{-0.223} = 6.31 \text{ s}$$

At $T = 450 \text{ K}$:

$$k_1 = (1.0 \times 10^6) e^{-50000/(8.314 \times 450)} = (1.0 \times 10^6) e^{-13.36} = 1.57 \text{ s}^{-1}$$

$$k_2 = (1.0 \times 10^8) e^{-70000/(8.314 \times 450)} = (1.0 \times 10^8) e^{-18.71} = 0.749 \text{ s}^{-1}$$

$$\tau^*(450 \text{ K}) = \frac{\ln(0.749/1.57)}{0.749 - 1.57} = \frac{-0.741}{-0.821} = 0.90 \text{ s}$$

For maximum yield, calculate k_1/k_2 and apply Eq. (5.51):

$$\text{At } 400 \text{ K: } k_1/k_2 = 0.295/0.0722 = 4.09$$

$$Y_{B,\max} = (4.09)^{-0.324} = 0.634$$

$$\text{At } 450 \text{ K: } k_1/k_2 = 1.57/0.749 = 2.10$$

$$Y_{B,\max} = (2.10)^{-0.912} = 0.509$$

Since $E_{a2} > E_{a1}$, increasing temperature reduces the ratio k_1/k_2 , which **reduces** the maximum achievable yield from 63.4% to 50.9%.

Answer:

At 400 K: $\tau^* = 6.31$ s, $k_1/k_2 = 4.09$, $Y_{B,\max} = 63.4\%$

At 450 K: $\tau^* = 0.90$ s, $k_1/k_2 = 2.10$, $Y_{B,\max} = 50.9\%$

Physical interpretation: Lower temperature is better for selectivity because the undesired second reaction has higher activation energy. At 400 K the maximum yield is 63.4% versus only 50.9% at 450 K. The tradeoff is that the overall reaction is slower at 400 K, requiring a residence time of 6.31 s versus 0.90 s—a factor of seven longer—and thus a correspondingly larger reactor volume.

5.8 Computational Tools: NB5

The Python notebook NB5 (“Selectivity Windows + Reactor Choice + Temperature Sweep”) provides computational tools for:

1. Simulating parallel and consecutive reaction networks in PFR (ODE integration using `solve_ivp`)
2. Computing CSTR outputs by solving algebraic balances using `scipy.optimize.root`
3. Calculating X_A , Y_B , $S_{B/A}$, and $S_{B/C}$ as functions of τ
4. Visualizing the selectivity window for consecutive reactions
5. Comparing PFR vs CSTR performance for intermediate yield
6. Adding Arrhenius temperature dependence to explore selectivity vs temperature tradeoffs
7. (Optional) Comparing performance of N CSTRs in series approaching PFR behavior

Remark 5.23 (Expected Outputs from NB5). Your notebook analysis should produce:

- Plot 1: $Y_B(\tau)$ and $Y_C(\tau)$ for parallel reactions, with $S_{B/C}$ shown as constant
- Plot 2: $Y_B(\tau)$ for consecutive reactions showing the clear maximum (PFR vs CSTR)

- Plot 3: Temperature sweep showing maximum Y_B vs T or $\tau^*(T)$ vs T
- Printed summary: Optimal τ for B in both PFR and CSTR

5.9 Summary

- **Conversion:** $X_A = (F_{A0} - F_A)/F_{A0} = (C_{A0} - C_A)/C_{A0}$
- **Yield:** $Y_B = F_B/F_{A0} = C_B/C_{A0}$ (fraction of fed A that became B)
- **Selectivity:** $S_{B/A} = Y_B/X_A$ (fraction of reacted A that became B)
- **Parallel reactions** ($A \rightarrow B, A \rightarrow C$):
 - Selectivity $S_{B/C} = k_1/k_2$ is **independent of τ**
 - Temperature is the primary control: $k_1/k_2 \propto \exp(-\Delta E_a/RT)$
 - If $E_{a1} > E_{a2}$: higher T improves selectivity to B
- **Consecutive reactions** ($A \rightarrow B \rightarrow C$):
 - Intermediate B has a maximum at optimal residence time
 - PFR: $\tau^* = \frac{1}{k_2 - k_1} \ln \left(\frac{k_2}{k_1} \right)$
 - PFR achieves higher $Y_{B,\max}$ than CSTR (plug flow protects intermediate)
 - Multiple CSTRs in series approach PFR performance
- **Design strategy:**
 - For parallel reactions: choose T to maximize k_1/k_2 ; τ controls conversion only
 - For consecutive reactions: use PFR, operate at τ^* ; adjust T to improve k_1/k_2 ratio

5.10 Notation Introduced in This Chapter

Remark 5.24 (Self-Assessment Checklist). Before attempting the problems, verify that you can:

1. Write $S_{B/A} = Y_B/X_A$ from memory and explain what each quantity measures.
2. State why selectivity is independent of τ for parallel first-order

Table 5.1: Notation introduced in Chapter 5

Symbol	Description	Typical Units
<i>Performance metrics</i>		
X_A	Fractional conversion of reactant A	dimensionless
Y_B	Overall yield of product B	dimensionless
$S_{B/A}$	Overall selectivity of B relative to A consumed	dimensionless
$S_{B/C}$	Product selectivity ratio (B to C)	dimensionless
$s_{B/C}$	Instantaneous selectivity (rate ratio)	dimensionless
$Y_{B,\max}$	Maximum achievable yield of intermediate B	dimensionless
<i>Kinetic parameters</i>		
k_1, k_2	Rate constants for reactions 1 and 2	s^{-1} (first-order)
ΔE_a	Difference in activation energies ($E_{a1} - E_{a2}$)	kJ/mol
T_{cross}	Crossover temperature where $k_1 = k_2$	K
α, β	Reaction orders for non-equal-order parallel reactions	dimensionless
<i>Design variables</i>		
τ^*	Optimal residence time for max. intermediate yield	s
R_i	Net rate of formation of species i	$\text{mol}/(\text{L}\cdot\text{s})$
F_i	Molar flow rate of species i	mol/s

reactions.

3. Sketch $C_B(\tau)$ for consecutive reactions and mark τ^* .
4. Explain in one sentence why the PFR gives higher $Y_{B,\max}$ than the CSTR for consecutive reactions.
5. Given E_{a1} and E_{a2} , predict whether raising temperature improves or hurts selectivity.
6. Distinguish between overall selectivity $S_{B/A}$ and instantaneous selectivity $s_{B/C}$.

If any of these feel uncertain, revisit the corresponding section before proceeding.

5.11 Problems

1. **Definitions.** A reactor processes 100 mol/h of A. At the outlet, 30 mol/h of A remains, 50 mol/h of B has formed, and 20 mol/h of C has formed.
 - (a) Calculate the conversion X_A .
 - (b) Calculate the yields Y_B and Y_C .
 - (c) Calculate the selectivities $S_{B/A}$, $S_{C/A}$, and $S_{B/C}$.
 - (d) Verify that $Y_B + Y_C = X_A$ (why must this be true for $A \rightarrow B$, $A \rightarrow C$?).
2. **Parallel reactions—temperature selection.** For parallel reactions with:

$$k_1 : E_{a1} = 100 \text{ kJ/mol}, \quad A_1 = 10^{12} \text{ s}^{-1}$$

$$k_2 : E_{a2} = 60 \text{ kJ/mol}, \quad A_2 = 10^8 \text{ s}^{-1}$$

- (a) Calculate $S_{B/C}$ at 350 K, 400 K, and 450 K.
 - (b) At what temperature does $S_{B/C} = 1$ (equal selectivity)?
 - (c) Recommend an operating temperature if $S_{B/C} > 5$ is required.
Hint: Compare your hand calculations with the temperature sweep in NB5 (Part 4: Temperature Effects on Selectivity).
3. **Consecutive reactions—PFR design.** For $A \xrightarrow{k_1} B \xrightarrow{k_2} C$ with $k_1 = 0.2 \text{ min}^{-1}$ and $k_2 = 0.5 \text{ min}^{-1}$:
 - (a) Calculate the optimal residence time τ^* for maximum B yield.

- (b) Calculate $Y_{B,\max}$ and X_A at this optimum.
- (c) If operating at $\tau = 2\tau^*$, what are Y_B and X_A ?
- (d) Discuss the tradeoff between selectivity and conversion.
Hint: Verify using `tau_opt_pfr()` and `consecutive_pfr_analytical()` in NB5.
4. **PFR vs CSTR comparison (NB5).** Using NB5 with $k_1 = 0.3 \text{ s}^{-1}$ and $k_2 = 0.6 \text{ s}^{-1}$:
- (a) Plot $Y_B(\tau)$ for both PFR and CSTR from $\tau = 0$ to 15 s.
- (b) Determine τ^* and $Y_{B,\max}$ for each reactor type.
- (c) By what percentage does the PFR outperform the CSTR in maximum yield?
- (d) How many CSTRs in series would approximate 90% of the PFR yield improvement?
5. **Temperature optimization (NB5).** For consecutive reactions with Arrhenius parameters:
- $$k_1 : E_{a1} = 40 \text{ kJ/mol}, \quad A_1 = 10^6 \text{ s}^{-1}$$
- $$k_2 : E_{a2} = 80 \text{ kJ/mol}, \quad A_2 = 10^{10} \text{ s}^{-1}$$
- (a) At what temperature does $k_1 = k_2$?
- (b) Use NB5 to plot $Y_{B,\max}(T)$ and $\tau^*(T)$ from 300 K to 600 K.
- (c) Recommend an operating temperature for maximum yield of B.
- (d) Discuss the productivity tradeoff (reaction rate vs selectivity).
6. **Design problem: Synthesis optimization.** You are developing a flow synthesis for producing intermediate B from A, with subsequent reaction to byproduct C. The process requires:
- Feed rate: 1000 mol/h of pure A at $C_{A0} = 2.0 \text{ mol/L}$
 - Target conversion: $X_A \geq 0.60$
 - Target selectivity: $S_{B/A} \geq 0.70$

Given $k_1 = 0.5 \text{ min}^{-1}$ and $k_2 = 0.8 \text{ min}^{-1}$ at 400 K:

- (a) Is it possible to meet both targets simultaneously? (Calculate $Y_{B,\max}$ and compare)
- (b) What is the maximum achievable $S_{B/A}$ at $X_A = 0.60$?
- (c) If targets cannot be met, propose a design modification (recycle, temperature change, etc.).
Hint: Use `consecutive_pfr_analytical()` in NB5 to plot Y_B and $S_{B/A}$ vs. τ and identify the feasible region.

Looking Ahead

In this chapter, we developed the framework for analyzing selectivity in reaction networks with competing pathways. The key insight is that reactor configuration and residence time significantly impact intermediate yield in consecutive reactions, while temperature is the primary lever for parallel reactions. The remaining chapters apply and extend these selectivity principles in increasingly specialized contexts.

Chapter 6 introduces **zeolites**—microporous crystalline aluminosilicates whose channel dimensions are comparable to molecular sizes. This geometric confinement enables three forms of *shape selectivity*: reactant selectivity (only molecules small enough to enter the pores can react), product selectivity (only products small enough to diffuse out of the pores are observed), and transition-state selectivity (only reactions whose transition states fit within the pore geometry can proceed). Shape selectivity provides a fundamentally different control lever for selectivity that complements the kinetic and thermal approaches developed here.

Chapter 6 examines **carbon nanotubes** as catalyst supports, where confinement within the nanotube interior produces dramatically enhanced diffusion coefficients ($D_{\text{CNT}} = F \cdot D_K$ with enhancement factors $F = 10\text{--}10,000$). These modified transport properties alter the Thiele modulus and effectiveness factor, which in turn affects selectivity in consecutive reaction networks—the intermediate B can escape more rapidly from the catalyst interior, reducing over-reaction to C.

Finally, Appendix G brings all selectivity concepts together in the context of **catalytic organic synthesis**, including Pd-catalyzed cross-coupling (Suzuki, Heck) and asymmetric catalysis. These systems present classic selectivity challenges—maximizing yield of the desired coupling product while controlling homo-coupling side reactions, and achieving high enantioselectivity through chiral catalyst design. You will see how catalyst selection, reaction conditions, and mechanistic understanding combine to achieve practical synthetic goals.

Chapter 6

Zeolites and Carbon Nanotubes in Catalysis

Learning Objectives

By the end of this chapter, you will be able to:

1. Describe zeolite framework structures using building units (TO_4 tetrahedra, SBUs, CBUs) and identify common framework types (FAU, MFI, CHA, MOR) and their pore systems.
2. Explain how the Si/Al ratio controls Brønsted acidity and predict how framework composition affects catalytic activity and selectivity.
3. Apply shape selectivity concepts (reactant, product, and transition-state selectivity) to predict and interpret zeolite catalysis outcomes.
4. Classify carbon nanotubes (SWCNTs vs. MWCNTs), relate chiral indices (n, m) to diameter, and describe how surface functionalization enables metal nanoparticle anchoring.
5. Compare **configurational diffusion** in zeolite micropores with **enhanced Knudsen diffusion** in CNT channels, and apply the Thiele modulus framework to both materials.
6. Evaluate the advantages and limitations of CNTs as catalyst supports compared to conventional materials and zeolites.
7. Select and justify appropriate characterization techniques for each material: XRD, t-plot, NH_3 -TPD, and pyridine-FTIR for zeolites; Raman (I_D/I_G), TGA, XPS, and TEM for CNTs.
8. Describe key industrial applications of zeolites (FCC, xylene isomerization, MTO, SCR) and CNTs (selective hydrogenation, electrocatal-

ysis) and connect catalyst design choices to the underlying structure–property relationships.

6.1 Introduction

The previous chapters developed general tools for analyzing heterogeneous catalysis: adsorption isotherms, surface kinetic mechanisms, reactor design equations, and the Thiele modulus framework for transport limitations. In this chapter, we apply those tools to two classes of nanostructured materials—**zeolites** and **carbon nanotubes (CNTs)**—that represent opposite extremes of nanoconfinement.

Zeolites are crystalline aluminosilicates with uniform micropores of 3–12 Å, comparable in size to small organic molecules. This molecular-scale confinement produces **shape selectivity**—the ability to discriminate between molecules based on size and geometry—but it also imposes severe diffusion limitations. Configurational diffusion coefficients in zeolite micropores can be 10^{10} – 10^{13} times smaller than in mesoporous materials, making transport through even a micrometer-sized crystal the dominant resistance.

Carbon nanotubes, by contrast, are hollow cylinders of rolled graphene with inner diameters of 1–50 nm and atomically smooth walls. This smoothness produces the opposite transport behavior: molecules undergo near-specular reflection at the graphitic surface, preserving axial momentum and yielding diffusion coefficients 10–10,000 times *higher* than classical Knudsen theory predicts. Where zeolite micropores hinder transport, CNT channels enhance it.

The unifying theme is **nanoconfinement**: both materials confine reactant molecules to channels with sub-100 nm dimensions, but the wall character—rough and ionic in zeolites, smooth and graphitic in CNTs—determines whether confinement helps or hinders catalysis. The Thiele modulus framework from Chapter 4 applies to both; only the magnitude and physical origin of D_{eff} differ.

We begin with the crystallographic foundations of zeolite structure, then progress through acidity, shape selectivity, modification, and industrial applications, before examining CNT structure, synthesis, surface chemistry, and catalytic uses. The chapter concludes with unified sections on diffusion and characterization that bring both materials together.

6.2 Zeolite Structure and Framework Types

Understanding the catalytic behavior of zeolites requires a firm grasp of their crystal structure, because it is the framework topology—the specific arrangement of atoms in three-dimensional space—that determines the pore geometry, and hence the shape-selective properties of the catalyst.

6.2.1 Primary and Secondary Building Units

All zeolite structures are built from a single fundamental unit: the **TO₄ tetrahedron**, in which a central T-atom (Si⁴⁺ or Al³⁺) is coordinated to four oxygen atoms. Each oxygen bridges exactly two T-atoms (**corner-sharing connectivity**), creating a fully connected three-dimensional network with stoichiometry TO₂ and the general formula:

$$M_{x/n}[Al_xSi_{1-x}O_2] \cdot mH_2O \quad (6.1)$$

where M is a charge-balancing cation of valence n . Each Al³⁺ substitution introduces one negative framework charge, compensated by an extra-framework cation; when that cation is H⁺, the result is a Brønsted acid site.

The T–O–T bond angles (typically 140–160°) provide enough geometric flexibility to construct a remarkable variety of open frameworks, each with a distinct channel system. The **framework density** (T-atoms per 1000 Å³) quantifies openness: dense frameworks like quartz exceed 20, while zeolites fall in the range 12–19, reflecting 20–50% void volume.

Secondary building units (SBUs) are small assemblies of tetrahedra—single 4-rings, 6-rings, double 4-rings (d4r), and double 6-rings (d6r). **Composite building units (CBUs)** are larger polyhedral cages; the most famous is the **sodalite cage** (β -cage), a truncated octahedron of 24 tetrahedra. In LTA, sodalite cages connect through d4r; in FAU, through d6r; in SOD, they share 4-ring faces directly—the same building block, connected differently, produces completely different pore systems. Fig. 6.1 shows common CBUs.

Remark 6.1 (Löwenstein’s Rule). Al–O–Al linkages are forbidden in zeolite frameworks: the electrostatic repulsion between adjacent [AlO₄][−] tetrahedra would destabilize the structure. Consequently Si/Al ≥ 1, setting an upper limit on acid site density. At Si/Al = 1 the acid sites are numerous but weak (neighboring negative charges partially cancel); at high Si/Al the sites are isolated and individually stronger—a central design trade-off.

6.2.2 Common Framework Types

The IZA recognizes over 250 framework topologies; a handful dominate industrial catalysis.

FAU (Faujasite/Zeolite Y)

The FAU framework (Fig. 6.2) is built from sodalite cages connected through d6r units in a diamond-like arrangement, creating large **supercages** (~13 Å

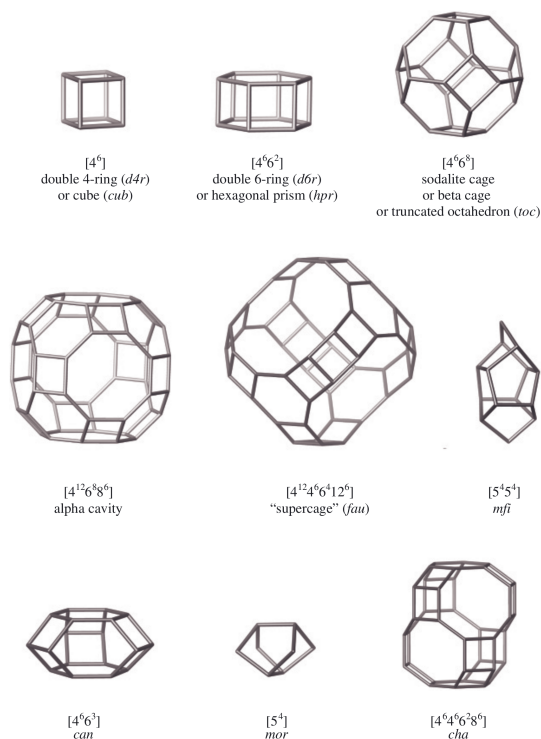


Figure 6.1: Common polyhedral composite building units (CBUs) with their face symbols and names. Top row: double 4-ring (*d4r*, $[4^6]$), double 6-ring (*d6r*, $[4^66^2]$), and sodalite cage ($[4^66^8]$). Middle row: alpha cavity, supercage (*fau*), and *mfi* cage. Adapted from Xu *et al.*, *Chemistry of Zeolites and Related Porous Materials*, Wiley, 2007.

diameter) accessible through four 12-MR windows (7.4 Å). The 3D channel system gives molecules multiple pathways, resisting deactivation by pore blockage. FAU is the single most commercially important topology, central to fluid catalytic cracking (FCC). The USY variant (Si/Al raised from 2.5 to 5–30 by steaming) provides hydrothermal stability at 700–750 °C regeneration temperatures.

MFI (ZSM-5)

The MFI framework (Fig. 6.3), discovered at Mobil Oil in 1972, is the most versatile medium-pore zeolite. It is built from **pentasil units** ($[5^8]$) that form chains and corrugated sheets. The pore system comprises straight channels (5.3 × 5.6 Å along $[010]$) and sinusoidal channels (5.1 × 5.5 Å along $[100]$), intersecting at expanded cavities (~9 Å). The three-dimensional connectivity provides resistance to coke deactivation, while the 10-MR apertures impose strong steric constraints. The *para*-xylene selectivity of ZSM-

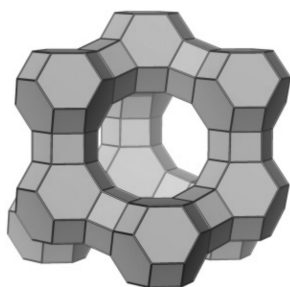


Figure 6.2: Framework structure of FAU (faujasite), formed by linking sodalite cages through double 6-rings. The large central void is the supercage (~ 13 Å diameter), accessible through four 12-MR windows (7.4 Å). Adapted from Xu *et al.*, Wiley, 2007.

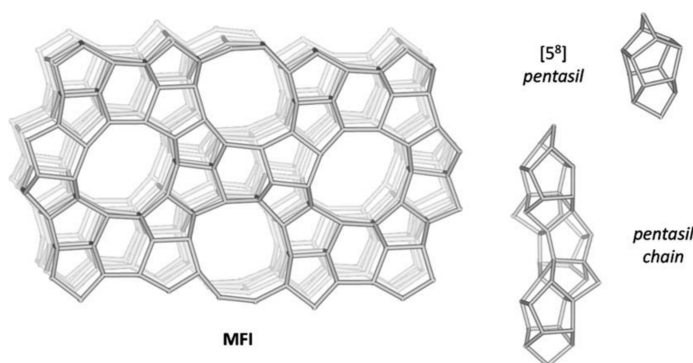


Figure 6.3: MFI framework: straight channels (5.3×5.6 Å) along [010] and sinusoidal channels (5.1×5.5 Å) along [100], with intersection cavities (~ 9 Å). Adapted from Čejka *et al.*, RSC, 2017.

5—where the linear isomer diffuses $\sim 200\times$ faster than *ortho/meta*—is the textbook illustration of product shape selectivity (Section 6.4).

CHA (Chabazite/SSZ-13)

The CHA framework features d6r units in AABBC stacking, creating ellipsoidal cages ($\sim 7.3 \times 12$ Å) connected through 8-MR windows (3.8 Å). This “cage effect”—spacious interior, restricted exit—is exploited in SAPO-34 for methanol-to-olefins (MTO), where only ethylene and propylene escape, and in Cu-SSZ-13 for diesel SCR, where isolated Cu^{2+} ions in d6r cages resist sintering far better than in Cu-ZSM-5.

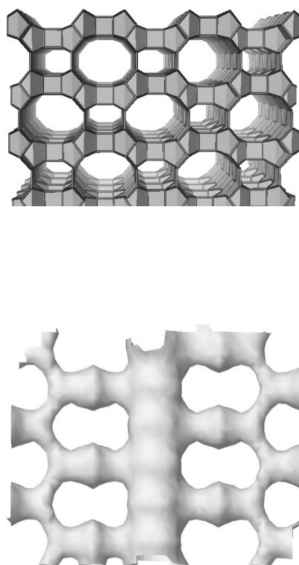


Figure 6.4: MOR (mordenite): one-dimensional 12-MR channels (6.5×7.0 Å) with 8-ring side pockets. Adapted from Xu *et al.*, Wiley, 2007.

MOR (Mordenite)

Mordenite (Fig. 6.4) has **one-dimensional** 12-MR channels (6.5×7.0 Å) connected only by small 8-MR side pockets. Any channel obstruction blocks all sites behind it, making MOR more susceptible to deactivation than 3D frameworks. BEA (zeolite beta) is another large-pore framework with three-dimensional 12-MR channels (~ 6.7 Å) used in aromatic alkylation.

6.2.3 Pore Size Classification

Table 6.1: Classification of zeolite pore sizes

Category	Ring Size	Pore Diameter	Examples
Small pore	8-MR	3–4 Å	CHA, LTA
Medium pore	10-MR	5–6 Å	MFI, FER
Large pore	12-MR	6–8 Å	FAU, BEA, MOR
Extra-large pore	>12-MR	>8 Å	UTD-1, ITQ-33

By choosing the appropriate ring size, one selects which molecules can access the internal acid sites. As pore size decreases, the diffusion mechanism transitions from Knudsen-like behavior to **configurational diffusion**, where diffusivities depend exponentially on the difference between molecular and pore diameters—the physical basis for zeolites’ extraordinary molecular

discrimination.

Remark 6.2 (Framework Density and Practical Stability). **Framework density** (FD), defined as the number of T-atoms per 1000 Å³, quantifies the trade-off between void volume and structural robustness. Dense frameworks (FD > 18, e.g., MFI at 18.4) offer excellent hydrothermal stability and moderate pore volume—ideal for harsh conditions. Open frameworks (FD < 14, e.g., FAU at 12.7) provide high pore volume but are inherently less stable and require high Si/Al ratios for industrial use.

As a rule of thumb, **steaming stability correlates with both FD and Si/Al**: higher values of both improve resistance to framework collapse. This explains why USY (dealuminated FAU, Si/Al = 5–30) survives FCC regeneration at 700–750 °C, and why MFI-based catalysts dominate xylene isomerization at ~400 °C. When selecting a zeolite for a high-temperature application, consider both framework topology and Si/Al ratio—neither alone guarantees stability.

Remark 6.3 (Concept Check: Framework Identification).

1. A zeolite has 12-membered ring windows and three-dimensional channels with supercages of ~13 Å diameter. Which framework type is this? *Answer: FAU (faujasite).*
2. A colleague proposes synthesizing a zeolite with Si/Al = 0.5. Why is this impossible? *Answer: Löwenstein's rule forbids Al–O–Al linkages, so Si/Al ≥ 1.*
3. Rank FAU, MFI, and CHA from largest to smallest pore aperture. *Answer: FAU (12-MR, 7.4 Å) > MFI (10-MR, 5.5 Å) > CHA (8-MR, 3.8 Å).*

6.3 Acidity and the Si/Al Ratio

The catalytic power of zeolites originates from their acid sites. Understanding the nature, density, and strength of these sites is essential for catalyst design.

6.3.1 Origin of Brønsted Acidity

When Al³⁺ substitutes for Si⁴⁺ at a tetrahedral site, the framework acquires one unit of negative charge per aluminum atom. In catalytic zeolites, this charge is compensated by a proton bonded to a bridging oxygen, creating a

where $M_{\text{framework}}$ is the average molar mass per T-atom. Since $x = 1/(1+R)$ with $R = \text{Si}/\text{Al}$, a zeolite with $\text{Si}/\text{Al} = 1$ has maximum acid density ($\sim 8\text{--}10$ mmol/g), while $\text{Si}/\text{Al} = 50$ gives a theoretical maximum of ~ 0.33 mmol/g (experimental NH_3 -TPD values are often lower due to incomplete site accessibility).

The Si/Al ratio affects not only the *number* of acid sites but also their *individual strength*. At low Si/Al, neighboring Al atoms partially cancel each other's effect, producing weaker sites. At high Si/Al, isolated Al atoms produce fewer but stronger Brønsted sites—explaining the seemingly paradoxical result that zeolites with *fewer* acid sites can exhibit *higher* activity per site.

Remark 6.4 (The Si/Al Trade-off). Decreasing Si/Al increases acid site number but decreases individual strength (more Al–O–Si–O–Al linkages, lower hydrothermal stability). The optimum depends on the reaction: FCC uses low Si/Al for high throughput; xylene isomerization uses high Si/Al for strong, isolated sites and selectivity.

6.3.4 Acid Strength Distribution

Even a single zeolite crystal contains acid sites with a distribution of strengths, arising from (i) multiple crystallographically distinct T-sites with different Si–O–Al bond angles, (ii) variation in the next-nearest-neighbour environment (isolated vs. partially clustered Al), and (iii) interactions with nearby EFAL. The acid strength distribution is measured by NH_3 -TPD for total acidity (Section 6.13.1) and pyridine-FTIR for Brønsted/Lewis speciation (Section 6.13.1).

Remark 6.5 (Concept Check: Si/Al Ratio and Acidity).

1. Zeolite A has $\text{Si}/\text{Al} = 5$ and Zeolite B has $\text{Si}/\text{Al} = 50$. Which has more acid sites per gram? Which has stronger individual acid sites? *Answer: A has more sites; B has stronger individual sites (isolated Al atoms).*
2. A zeolite is steamed at 700°C . What happens to the Si/Al ratio, and what new type of acid site appears? *Answer: Si/Al increases (dealumination); Lewis acid sites from EFAL species.*
3. Why does NH_3 -TPD alone not tell you the Brønsted/Lewis ratio? *Answer: NH_3 adsorbs on both site types indiscriminately. Pyridine-FTIR is needed for speciation.*

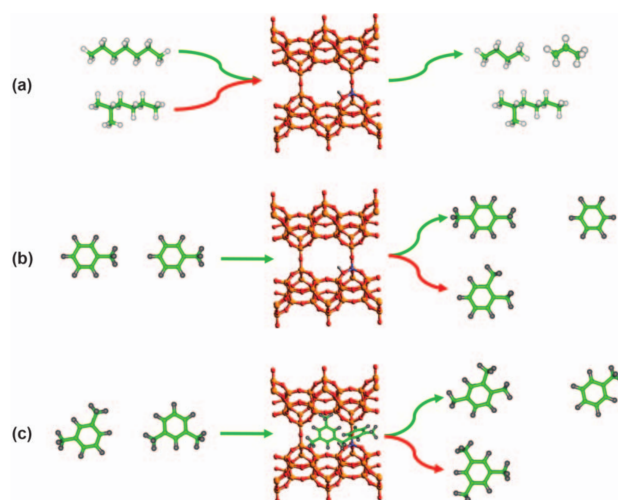


Figure 6.6: (a) **Reactant selectivity**: only molecules small enough to enter reach active sites. (b) **Product selectivity**: all isomers form inside, but the most compact diffuses out fastest. (c) **Transition-state selectivity**: bulky transition states cannot form in confined space. Adapted from Čejka et al., RSC, 2017.

6.4 Shape Selectivity

Shape selectivity is the defining feature that distinguishes zeolite catalysis from catalysis on open surfaces. Because zeolite pore dimensions (3–8 Å) are comparable to the kinetic diameters of small organic molecules, the catalyst discriminates between molecules—and transition states—based on size and shape. Three classical types are recognized (Fig. 6.6).

6.4.1 Reactant Shape Selectivity

Molecules whose kinetic diameter exceeds the pore aperture cannot enter the zeolite and therefore cannot reach the internal acid sites. A classic example is selective cracking of *n*-paraffins over erionite (ERI, 8-MR, 3.6×5.1 Å): *n*-hexane (4.3 Å) enters and cracks, while 2,3-dimethylbutane (6.3 Å) is excluded, giving selectivity ratios exceeding 100:

$$\text{Selectivity} = \frac{r_{\text{small}}}{r_{\text{large}}} \gg 1 \quad \text{when } d_{\text{large}} > d_{\text{pore}} \quad (6.4)$$

Reactant selectivity is strongest in small-pore (8-MR) zeolites and weakest in large-pore (12-MR) systems.

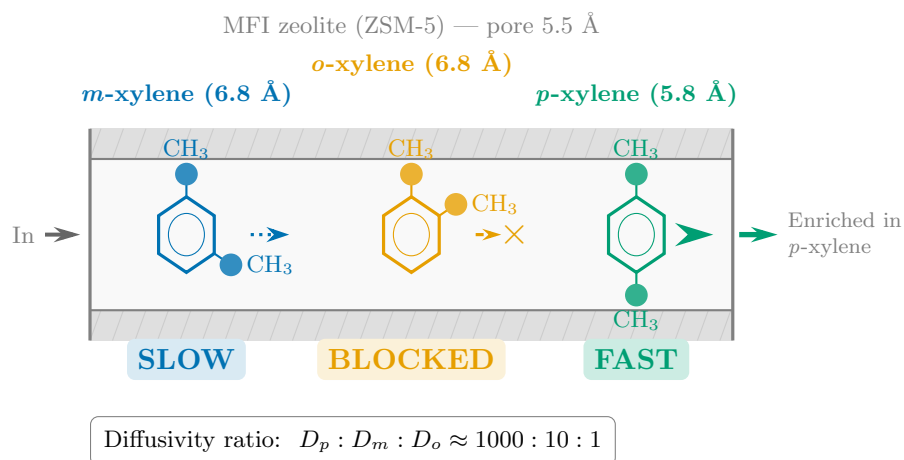


Figure 6.7: Product shape selectivity in ZSM-5: only *para*-xylene (~ 5.8 Å) diffuses rapidly through 10-ring channels (~ 5.5 Å). Bulkier *ortho*- and *meta*-xylenes (~ 6.8 Å) experience restricted diffusion and re-isomerize before escaping.

6.4.2 Product Shape Selectivity

When multiple product isomers form inside the pores but diffuse out at different rates, the most compact isomer escapes preferentially. By Le Chatelier's principle, the internal equilibrium shifts to replenish the fast-diffusing isomer.

The most celebrated example is ***para*-xylene production** over ZSM-5 (Fig. 6.7). Thermodynamic equilibrium at $\sim 400^\circ\text{C}$ gives $\sim 24\%$ *para*-, 53% *meta*-, 23% *ortho*-xylene. Over H-ZSM-5, *para*-selectivity exceeds 90% : the linear *para*-isomer (5.8 Å) diffuses $\sim 200\times$ faster through the 10-MR channels than the bulkier *ortho/meta*-isomers (6.8 Å). Trapped isomers re-equilibrate, continuously regenerating *para*-xylene. Global *p*-xylene production exceeds 50 Mt/yr, essentially all relying on this shape-selective mechanism.

6.4.3 Transition-State Shape Selectivity

Here, both reactants and products fit freely in the pores; the constraint acts on the *transition state* itself. If a reaction pathway requires a transition state too bulky to form in the micropore, that pathway is suppressed. For example, in alkane cracking over ZSM-5, the bulky bimolecular transition state (two hydrocarbon molecules in close proximity) cannot form in the 5.5 Å channels, so only compact unimolecular (protolytic) cracking proceeds—producing predominantly light olefins rather than branched heavy products.

The key diagnostic: transition-state selectivity is *independent of crystal size*, whereas reactant and product selectivity are not. Haag and coworkers confirmed this for *n*-hexane cracking over ZSM-5: crystals of 0.05 and 2.7 μm gave identical product selectivity.

Remark 6.6 (Concept Check: Shape Selectivity).

1. In *para*-xylene production over ZSM-5, reducing the crystal size from 2 μm to 0.1 μm would likely *decrease* the *para*-selectivity. Why? *Answer: Smaller crystals reduce diffusion path length, giving less time for slower ortho/meta isomers to re-equilibrate before escaping. Product shape selectivity requires sufficient diffusion resistance.*
2. A reaction shows identical product selectivity on 0.05 μm and 2.7 μm crystals of the same zeolite. Which type of shape selectivity is operating? *Answer: Transition-state selectivity—it is independent of crystal size.*
3. Would you expect strong reactant selectivity in a 12-MR zeolite like FAU? *Answer: No. The large 7.4 Å windows admit most organic molecules. Reactant selectivity requires small-pore (8-MR) zeolites.*

Remark 6.7 (Distinguishing the Three Types).

- **Reactant selectivity:** Test different-sized reactants; only the smaller one reacts.
- **Product selectivity:** Selectivity deviates from equilibrium and depends on crystal size.
- **Transition-state selectivity:** Rates depend on TS geometry; selectivity is independent of crystal size.

Remark 6.8 (Quantifying Shape Selectivity: The Selectivity Index).

Weisz (1980) defined the **shape selectivity index** as the ratio of effective diffusivities: $S_{A/B} = D_{\text{eff},A}/D_{\text{eff},B}$. However, the *observed* selectivity depends on which regime the catalyst operates in. In kinetic control ($\phi \ll 1$), selectivity is governed entirely by intrinsic rate constants—diffusivity differences are irrelevant. In the diffusion-limited regime ($\phi \gg 1$), the observed selectivity amplifies as $\sqrt{D_{\text{desired}}/D_{\text{undesired}}}$, not the full diffusivity ratio.

For the *para*-xylene example (Section 6.14), $S_{p/o} = D_p/D_o = 200$, but the effective selectivity enhancement in the diffusion-limited regime scales as $\sqrt{200} \approx 14$. This is still enormous—and it means

that **maximizing shape selectivity requires deliberately operating in the diffusion-limited regime**, an unusual situation where transport limitations are not a problem to be solved but a feature to be exploited.

6.4.4 Molecular Traffic Control

Zeolites with two or more distinct intersecting channel systems (e.g., MFI) may exhibit a fourth mechanism: **molecular traffic control**. Reactant and product molecules preferentially diffuse through *different* channels, reducing counter-diffusion and enhancing both transport rates and selectivity. The concept, first proposed by Derouane and Gabelica, explains why zeolites with intersecting multi-dimensional pore systems often outperform one-dimensional zeolites of similar pore size. Evidence comes primarily from molecular dynamics simulations showing anisotropic transport in MFI crystals.

Remark 6.9 (The Window Effect). Goring (1973) observed a striking **non-monotonic** dependence of diffusivity on chain length for *n*-alkanes in erionite (ERI): *n*-C₈ diffuses *faster* than either *n*-C₆ or *n*-C₁₀. The physical origin is a resonance between molecular length and the cage-to-cage distance. When a molecule's length is **commensurate** with the spacing between 8-MR windows, it threads through without needing to coil or deform inside the cage. Non-commensurate molecules must compress to fit through the window, dramatically reducing D_{eff} .

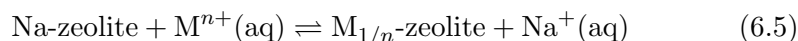
The practical lesson is that shape selectivity is *not* always a simple monotonic function of molecular size. Designing a shape-selective process requires measuring diffusivities for the specific molecule/zeolite pair rather than relying on kinetic diameter arguments alone. The window effect has been confirmed by molecular dynamics simulations and is most pronounced in cage-type zeolites with narrow connecting windows (8-MR systems such as ERI and CHA).

6.5 Ion Exchange and Post-Synthetic Modification

The as-synthesized zeolite is often just the starting point. **Post-synthetic modifications** fine-tune acidity, hydrothermal stability, and pore architecture for specific applications.

6.5.1 Ion Exchange

The charge-compensating cations (typically Na^+) in as-synthesized zeolites are mobile and can be replaced:



The most important application is preparing the **acid form** (H-zeolite) via a two-step indirect route: (1) ammonium exchange (NH_4Cl at $60\text{--}80^\circ\text{C}$ replaces Na^+ under mild conditions), followed by (2) calcination at $400\text{--}550^\circ\text{C}$ ($\text{NH}_4^+ \rightarrow \text{H}^+ + \text{NH}_3 \uparrow$), creating the Brønsted acid site. Direct acid exchange would dealuminate and destroy the framework, especially for Al-rich zeolites.

6.5.2 Dealumination

Dealumination removes framework aluminum, increasing Si/Al and conferring enhanced hydrothermal stability and hydrophobicity. The most important example is **ultra-stable Y (USY)** for FCC: steaming at $500\text{--}800^\circ\text{C}$ hydrolyzes Al from the framework, raising Si/Al from ~ 2.5 to $5\text{--}40$. The extracted aluminum forms **extra-framework aluminum (EFAL)** species that act as Lewis acid sites (Section 6.3.2). Acid leaching can complement steaming by dissolving EFAL and restoring micropore access. Fig. 6.8 illustrates the process.

6.5.3 Desilication

Desilication uses alkaline solutions ($0.1\text{--}0.5\text{ M NaOH}$, $60\text{--}80^\circ\text{C}$) to selectively dissolve silicon, creating intracrystalline mesopores ($5\text{--}20\text{ nm}$). The selectivity arises because framework Al^- repels OH^- , protecting Si–O–Al bonds while Si–O–Si bonds are preferentially hydrolyzed. These mesopores serve as transport highways, dramatically shortening the effective diffusion path to microporous active sites (Fig. 6.9). The optimal Si/Al range for desilication is $25\text{--}50$; lower values provide insufficient dissolution, higher values risk framework collapse.

Remark 6.10 (Desilication vs. Dealumination).

Treatment	Si/Al Effect	Porosity Effect
Dealumination	Increases	Creates small mesopores
Desilication	Decreases	Creates larger mesopores

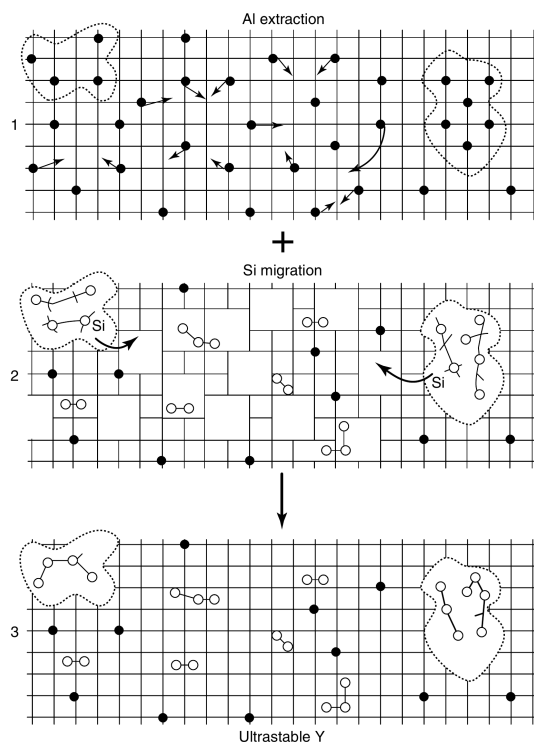


Figure 6.8: Dealumination by steaming: Al atoms are extracted, mobile Si species partially heal vacancies, and the final USY retains crystallinity with intracrystalline mesopores. Adapted from Groen *et al.*, Springer, 2007.

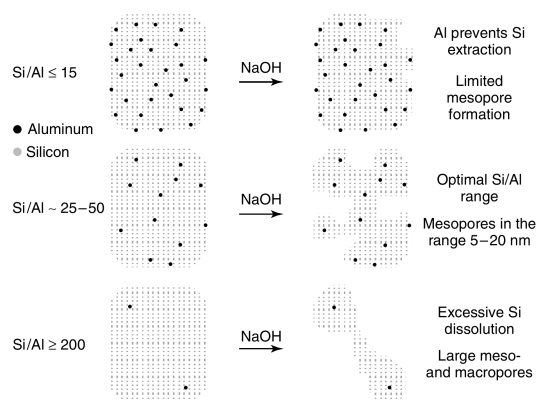


Figure 6.9: Mesopore formation during NaOH desilication depends on Si/Al: too low (≤ 15) gives limited mesoporosity; optimal ($\approx 25\text{--}50$) produces 5–20 nm mesopores; too high (≥ 200) causes uncontrolled dissolution. Adapted from Groen *et al.*, Springer, 2007.

Remark 6.11 (Designing Hierarchical Zeolites: Balancing Transport and Selectivity).

The design principle is straightforward: introduce **mesopores** (2–50 nm) as transport highways while retaining micropores for shape-selective catalysis. Quantitatively, reducing the effective diffusion path from 1 μm to 100 nm reduces ϕ by 10 $\times$, potentially shifting from $\eta \approx 0.3$ to $\eta > 0.9$. However, mesopores also create **non-selective bypass routes**—reactants can reach acid sites without passing through shape-selective micropore windows, eroding the selectivity advantage. Design guidelines for the optimal trade-off: (1) a mesopore volume fraction of 10–30% gives the best transport/selectivity balance; (2) maintain framework crystallinity above 80%; (3) mesopore diameter should exceed the reactant molecular diameter by $>5\times$ to avoid secondary confinement effects. Methods ranked by simplicity: desilication (NaOH, cheapest) $>$ hard templating (carbon black) $>$ soft templating (surfactant micelles) $>$ bottom-up seed assembly (most control). Industrial impact: hierarchical USY for FCC increases gasoline yield by 2–5% relative to conventional USY at the same conversion.

6.5.4 Metal Incorporation

Bifunctional catalysis—combining metal and acid sites—is essential for processes like hydroisomerization, where a metal site dehydrogenates an alkane, the olefin migrates to a Brønsted site for skeletal rearrangement, and then returns for rehydrogenation. Metals are introduced by **ion exchange** (e.g., Cu^{2+} in SSZ-13 for SCR), **impregnation** (e.g., Pt on mordenite for hydroisomerization), or **isomorphous substitution** during synthesis. The most celebrated example of the latter is **TS-1** (titanium silicalite-1): isolated Ti^{4+} sites in the MFI framework activate H_2O_2 for selective epoxidation of propylene, replacing chlorohydrin routes and eliminating chlorinated waste.

6.6 Industrial Applications

Zeolites account for the majority of solid acid catalysts used worldwide. We survey four major applications, connecting each to the structural and transport principles developed above.

6.6.1 Fluid Catalytic Cracking (FCC)

FCC is the largest application of zeolite catalysts, converting **vacuum gas oil** (VGO, bp $>340^\circ\text{C}$) into gasoline, light olefins, and light cycle oil via

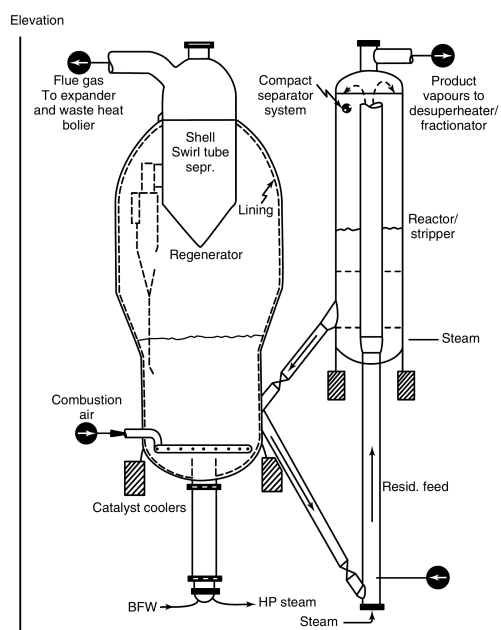


Figure 6.10: FCC unit: feed contacts hot regenerated catalyst in the riser (500–540°C, 2–4 s), spent catalyst is regenerated by coke combustion ($\sim 700^\circ\text{C}$). Adapted from Weitkamp and Puppe, Springer, 1999.

carbenium ion intermediates at Brønsted acid sites. The catalyst is **ultra-stable Y** (USY, FAU framework) embedded in an amorphous silica-alumina matrix that pre-cracks molecules too bulky for the 7.4 Å supercage windows. The FCC unit (Fig. 6.10) operates at 500–540°C with 2–4 s contact time; spent catalyst circulates to the regenerator where coke is burned at $\sim 700^\circ\text{C}$, providing the heat for endothermic cracking. Fig. 6.11 shows riser profiles. Many modern FCC catalysts include ZSM-5 as an additive to boost propylene yield through selective cracking in the 10-MR channels.

Remark 6.12 (Carbenium Ion Mechanism in FCC). A **carbenium ion** (R_3C^+) is a trivalent carbocation formed when a Brønsted acid site protonates an olefin or abstracts a hydride from a paraffin. In FCC, these intermediates undergo three key reactions:

- (i) **β -scission**—C–C bond cleavage at the β -position, producing a smaller olefin and a new carbenium ion;
- (ii) **isomerization**—hydride and methyl shifts that rearrange the carbon skeleton;
- (iii) **hydride transfer**—propagation by abstracting H^- from a neutral feed molecule.

Tertiary carbenium ions are more stable than secondary or primary

6. Zeolites and Carbon Nanotubes in Catalysis

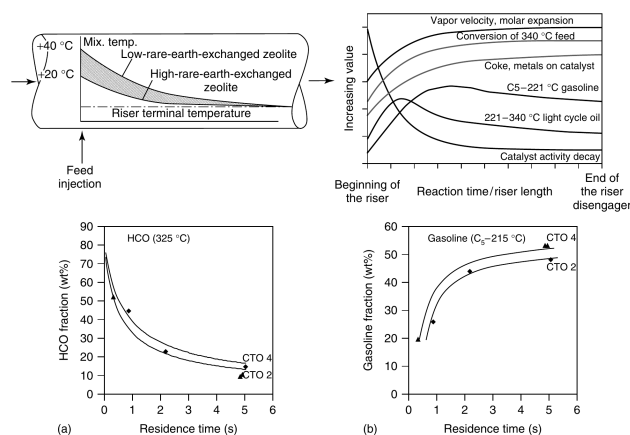


Figure 6.11: Temperature and product yield profiles along the FCC riser. Bottom panels: conversion and gasoline yield vs. residence time at different catalyst-to-oil ratios. Adapted from Weitkamp and Puppe, Springer, 1999.

(recall hyperconjugation from organic chemistry), so branched products are kinetically favoured.

6.6.2 Xylene Isomerization

para-Xylene (>50 Mt/yr, precursor to PET) is produced by exploiting **product shape selectivity** (Section 6.4.2) over H-ZSM-5. Inside the pore system, acid sites equilibrate all three xylene isomers, but *p*-xylene (5.8 Å) escapes the 10-MR channels $\sim 200\times$ faster than *o*- and *m*-xylene (6.8 Å), yielding >90% *para*-selectivity despite a 24% thermodynamic equilibrium value. The industrial **xylene loop** integrates a Parex adsorptive separation unit (BaX zeolite, simulated moving bed) with a Pt/H-ZSM-5 isomerization reactor at 350–450 °C, enabling near-complete conversion of the C₈ aromatic pool to *p*-xylene.

6.6.3 Methanol-to-Olefins (MTO)

MTO converts methanol (from natural gas or coal) to ethylene and propylene via the **hydrocarbon pool mechanism** (Fig. 6.12): methylated aromatic intermediates trapped in zeolite cages undergo repeated methylation–cracking cycles. SAPO-34 (CHA, 8-MR) achieves >80% C₂+C₃ selectivity because only light olefins can exit the 3.8 Å windows; ZSM-5 (MFI, 10-MR) gives a broader product slate but longer catalyst life. China has deployed MTO at >10 Mt/yr capacity.

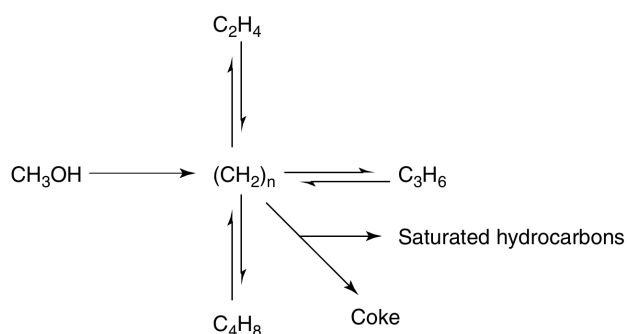
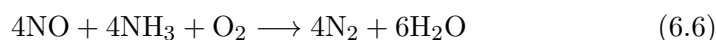


Figure 6.12: Hydrocarbon pool mechanism: methanol feeds into methylated aromatic intermediates inside zeolite cages, producing light olefins as primary products. Adapted from Weitkamp and Puppe, Springer, 1999.

6.6.4 Environmental Catalysis: Cu-Zeolite SCR

Cu-SSZ-13 (CHA) is the dominant catalyst for selective catalytic reduction (SCR) of NO_x with NH_3 in diesel exhaust:



The small CHA cages confine Cu^{2+} ions at specific sites, preventing migration and sintering that plagued earlier Cu-ZSM-5 catalysts. The mechanism involves a Cu^{2+}/Cu^+ redox cycle; at low temperatures ($<250^\circ C$), re-oxidation of Cu^+ requires transient mobile Cu^+-NH_3 complexes, explaining the second-order dependence of activity on Cu loading.

Remark 6.13 (Zeolite vs. Vanadia SCR). Cu-zeolite catalysts offer higher low-temperature activity, no vanadium volatilization, and better compatibility with particulate filters than V_2O_5/TiO_2 , but are more sensitive to hydrothermal aging.

Part II: Carbon Nanotubes

We now turn to a second class of nanoconfined catalytic material. While zeolites exploit *micropore constriction* for shape selectivity, carbon nanotubes exploit *atomically smooth channels* for enhanced transport and electronic modification of supported metals. The same Thiele modulus framework applies to both, but with opposite consequences for diffusion.

6.7 Structure and Classification

A carbon nanotube is a graphene sheet rolled into a seamless cylinder. The rolling direction, specified by integer *chiral indices* (n, m) , determines both the tube diameter and its electronic character. The diameter follows from the graphene lattice constant $a = 0.246$ nm:

$$d = \frac{a}{\pi} \sqrt{n^2 + nm + m^2} \quad (6.7)$$

Three chirality types exist: **armchair** (n, n) , **zigzag** $(n, 0)$, and **chiral** $(n \neq m)$. A nanotube is metallic when $(n - m) \bmod 3 = 0$ and semiconducting otherwise; statistically one-third of random (n, m) pairs are metallic. Metallic CNTs are preferred for electrocatalysis due to superior electron transport.

6.7.1 Single-Walled vs Multi-Walled Nanotubes

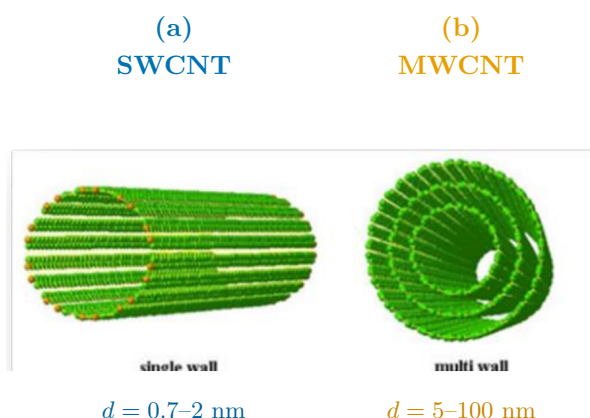


Figure 6.13: (a) Single-walled and (b) multi-walled carbon nanotube with cutaway view. Inter-wall spacing ≈ 0.34 nm. Adapted from Rashad et al., 2016.

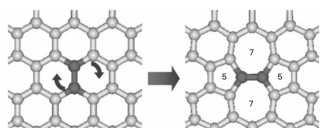
Property	SWCNT	MWCNT
Diameter	0.4–2 nm	5–100 nm
Surface area	300–900 m ² /g	200–400 m ² /g
Synthesis control	More difficult	Easier
Cost	Higher	Lower

6.7.2 Defects and Their Catalytic Implications

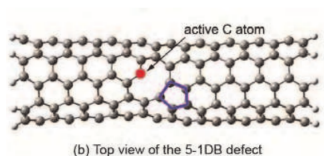
Real CNTs contain structural defects that serve as functional sites for metal anchoring: **Stone–Wales defects** (pentagon-heptagon pairs), **vacancies**

(dangling bonds), **edge sites** (open tube ends), and **dopant atoms** (substitutional N or B). The Raman I_D/I_G ratio (Section 6.13) provides a non-destructive measure of defect density.

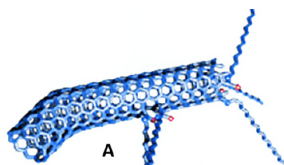
(a) Stone–Wales Defect



(b) Vacancy Defect



(c) Edge Site



(d) N-Doping

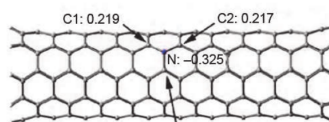


Figure 6.14: (a) Stone–Wales 5-7-7-5 rearrangement, (b) monovacancy with dangling bonds, (c) open tube end with functionalization sites, (d) substitutional N atom with charge redistribution. These defects serve as primary anchoring sites for metal nanoparticles. Adapted from Harris (2009) and Liu et al. (2006).

Remark 6.14 (Defect Density as a Design Parameter). More defects improve metal dispersion but disrupt the sp^2 network, reducing conductivity and thermal stability. A moderate defect density ($I_D/I_G \approx 0.3\text{--}0.8$) usually provides the best balance. Pristine CNTs ($I_D/I_G < 0.1$) give poor metal dispersion; heavily functionalized tubes ($I_D/I_G > 1.5$) resemble amorphous carbon.

Remark 6.15 (Concept Check: CNT Classification).

1. A nanotube has chiral indices (6,6). What is its chirality type? Is it metallic or semiconducting? *Answer: Armchair ($n = m$); metallic (always for armchair tubes).*
2. Calculate the diameter of a (10,0) zigzag nanotube. *Answer: $d = (0.246/\pi)\sqrt{100 + 0 + 0} = 0.78 \text{ nm}$.*
3. For catalyst support, would you choose pristine CNTs ($I_D/I_G <$

0.1) or moderately functionalized CNTs ($I_D/I_G \approx 0.5$)? *Answer: Moderately functionalized—pristine CNTs give poor metal dispersion due to lack of anchoring sites.*

6.8 Synthesis Methods

The synthesis method strongly influences CNT purity, defect density, and suitability as a catalyst support.

Earlier methods—arc discharge and laser ablation—produce high-crystallinity CNTs but with limited scalability and poor diameter control, restricting their use to research-scale quantities.

6.8.1 Chemical Vapor Deposition (CVD)

CVD is the most important method for catalytic applications due to its scalability and tunability. A carbon precursor (CH_4 , C_2H_2 , CO , or alcohols) decomposes over Fe, Co, or Ni nanoparticles at 600–1000°C.

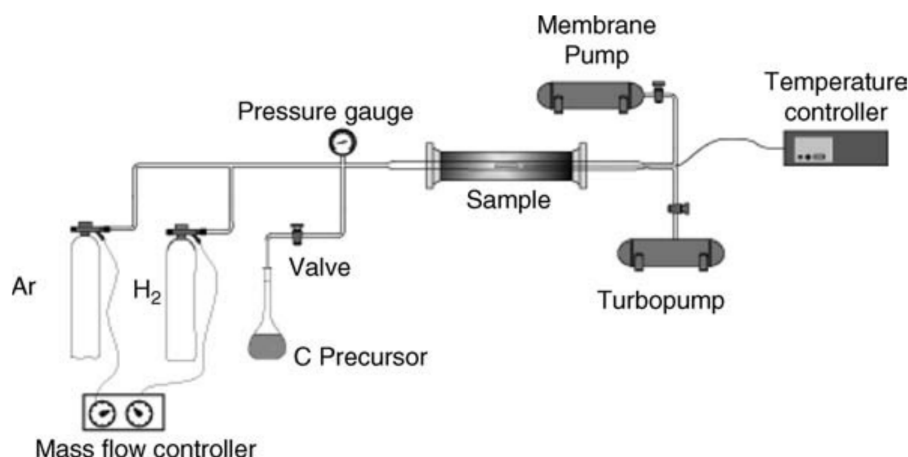


Figure 6.15: CVD system for carbon nanotube synthesis. Hydrocarbon precursors decompose at 600–1000°C over metal catalyst nanoparticles, offering direct control over diameter, length, and quality. Adapted from Guldi and Martín (Eds.), Wiley-VCH, 2010.

Remark 6.16 (CVD Catalyst Particle Size Controls CNT Diameter).

In CVD growth, the nanotube diameter closely matches the catalyst particle diameter, linking directly to Eq. (6.7). Fe, Co, and Ni are standard CVD catalysts because they catalyze precursor decomposition, have moderate carbon solubility, and form nanoparticles

in the 0.5–5 nm range on oxide supports.

CVD offers the best combination of scalability, low cost, and diameter control compared with arc discharge and laser ablation, which produce higher crystallinity but at lower throughput. CNT quality from CVD is tunable through precursor choice and growth temperature.

As-synthesized CNTs require purification to remove amorphous carbon (thermal oxidation at 350–500°C), residual metal catalyst (acid washing with HCl or HNO₃), and graphitic particles (sonication and centrifugation). Purification conditions must be balanced against unintended surface functionalization and structural damage.

Remark 6.17 (Purification vs. Functionalization Trade-off). Aggressive purification introduces oxygen-containing functional groups and structural defects. For catalyst support applications, this can be advantageous (better metal dispersion) or detrimental (reduced thermal stability) depending on the target reaction.

6.9 Surface Chemistry

Pristine CNTs present chemically inert sp^2 surfaces. Surface modification is therefore essential for most catalytic applications.

6.9.1 Pristine CNT Surface Properties

Defect-free CNT sidewalls interact with molecules primarily via weak van der Waals forces (physisorption, <20 kJ/mol; cf. Chapter 2):

- Low chemical reactivity—no dangling bonds or unsaturated sites
- Hydrophobic character with poor wetting by aqueous solutions
- Weak metal adhesion leading to nanoparticle agglomeration

6.9.2 Oxidative Functionalization

Treatment with oxidizing agents (conc. HNO₃, HNO₃/H₂SO₄, KMnO₄, or ozone) introduces oxygen-containing functional groups that convert the hydrophobic surface into a hydrophilic one and serve as metal anchoring sites:

Group	Location	Thermal Stability
Carboxylic (–COOH)	Edges, defects	Decomposes > 200°C
Hydroxyl (–OH)	Sidewalls, edges	Decomposes > 300°C
Carbonyl (C=O)	Defect sites	Stable to 500°C
Epoxide (C–O–C)	Sidewalls	Decomposes > 200°C

Remark 6.18 (Safety: CNT Functionalization Reagents). The oxidizing agents used for CNT functionalization pose serious hazards: concentrated HNO_3 releases toxic NO_2 fumes; piranha solution ($\text{H}_2\text{SO}_4/\text{H}_2\text{O}_2$) is violently exothermic and can detonate with organics; KMnO_4 is a strong oxidizer. Perform all functionalization in a well-ventilated fume hood with appropriate PPE.

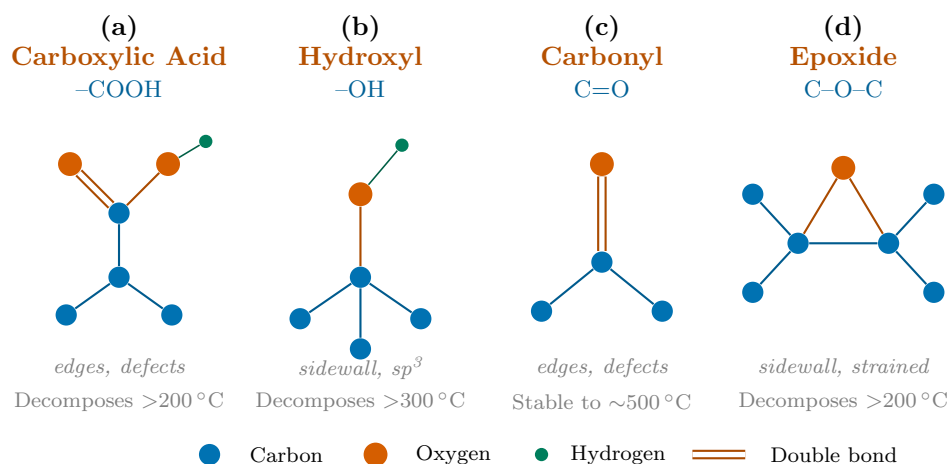


Figure 6.16: Principal oxygen-containing functional groups on oxidized CNTs: (a) carboxylic acid, (b) hydroxyl, (c) carbonyl, (d) epoxide. These groups serve as anchoring sites for metal nanoparticle precursors during catalyst preparation.

6.9.3 Heteroatom Doping

Substitutional doping with nitrogen or boron creates new active sites. Four nitrogen configurations are distinguishable by XPS binding energies (Chapter 3):

- **Pyridinic-N** ($E_b = 398.5$ eV): edge N in six-membered ring, basic character
- **Pyrrolic-N** ($E_b = 400.1$ eV): N in five-membered ring
- **Graphitic-N** ($E_b = 401.3$ eV): substitutional N, electron donor
- **Oxidized-N** ($E_b = 402\text{--}405$ eV): N-O species, surface acidity

N-doping enhances metal dispersion and anchoring, modifies electronic properties of supported particles, provides intrinsic ORR activity, and improves nanoparticle stability under reaction conditions. Covalent functionalization (oxidation, doping) is preferred for heterogeneous catalysis requiring strong anchoring and thermal stability, while non-covalent approaches (π - π

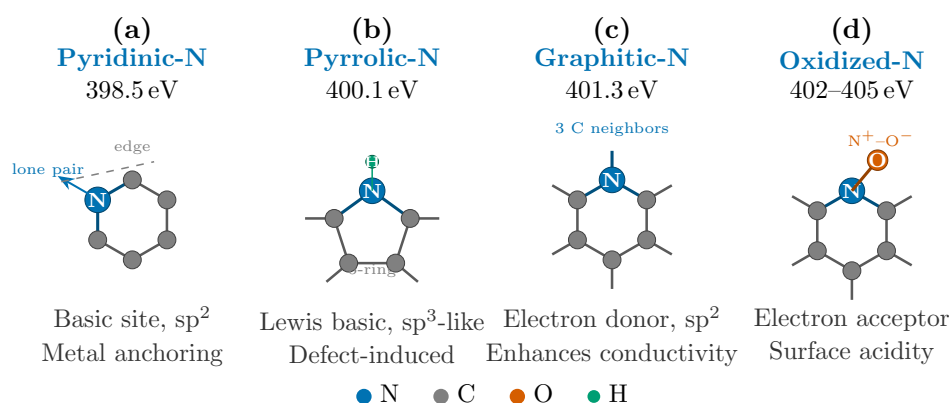


Figure 6.17: Local bonding environments for the four N-doping configurations in CNTs, with characteristic XPS N 1s binding energies for identification.

stacking, surfactants) preserve electronic properties for applications where conductivity is critical.

Remark 6.19 (Concept Check: Functionalization Strategy).

1. A CNT support will be used for a reaction at 350 °C. Which oxygen functional groups from oxidative treatment would survive? *Answer: Carbonyl ($C=O$, stable to $\sim 500^\circ C$). Carboxylic and epoxide groups decompose above 200 °C; hydroxyl above 300 °C.*
2. An XPS spectrum of an N-doped CNT shows a dominant N 1s peak at 401.3 eV. What type of nitrogen is this, and what is its effect on supported metal particles? *Answer: Graphitic-N (substitutional); it acts as an electron donor, modifying the electronic properties of supported metal nanoparticles.*

6.10 CNTs as Catalyst Supports

6.10.1 Comparison with Conventional Supports

Property	CNTs	Activ. C	SiO ₂	Al ₂ O ₃
Surface area (m ² /g)	200–900	500–2000	200–800	100–300
Pore structure	Well-defined	Random	Tunable	Mesoporous
Thermal stab. (°C)	<600 (air)	<400 (air)	>1000	>1000
Elec. conductivity	High	Moderate	Insulator	Insulator
Metal dispersion	Excellent	Good	Good	Good

Remark 6.20 (Surface Area Is Not Everything). Activated carbon has 2–5 times the BET surface area of CNTs, yet CNT-supported catalysts often outperform AC-supported ones. The reasons echo the lesson from Chapter 4: accessible rate depends on transport and kinetics together, not geometric area alone.

- **Metal dispersion:** CNT’s tunable surface chemistry enables smaller, more uniform particles.
- **Transport:** CNT channels provide enhanced diffusion (Section 6.12), whereas AC micropores impose severe Knudsen limitations.
- **Electronic effects:** CNTs modify the metal’s electronic structure, tuning adsorption energies.

6.10.2 Metal Nanoparticle Deposition Methods

Method	Advantages	Limitations
Impregnation	Simple, scalable	Limited size control
Colloidal deposition	Excellent size control	Ligand contamination
In-situ reduction	Good dispersion	Sensitive to conditions
Electrodeposition	Exploits CNT conductivity	Conductive substrates only

6.10.3 Inner vs Outer Surface Loading and Confinement

Metal nanoparticles deposited on the outer surface versus inside nanotube channels show distinct behavior: inner-channel particles resist sintering due to geometric constraint and exhibit modified selectivity through electronic confinement (shifted d-band center, Chapter 2), but diffusion access is limited. Outer-surface loading provides high accessibility and is preferred for fast reactions or large substrates.

Remark 6.21 (Confinement vs Accessibility Trade-off). Confinement enhances stability and selectivity but imposes mass transfer limitations. For small molecules at moderate rates, confinement benefits outweigh diffusion penalties. For fast reactions or large substrates, outer-surface loading is preferred.

Remark 6.22 (Electronic Modification: The d-Band Center Shift). According to the Hammer–Nørskov d-band model (Chapter 2), the position of the metal **d-band center** relative to the Fermi level E_F governs adsorption strength: a higher d-band center (closer to E_F)

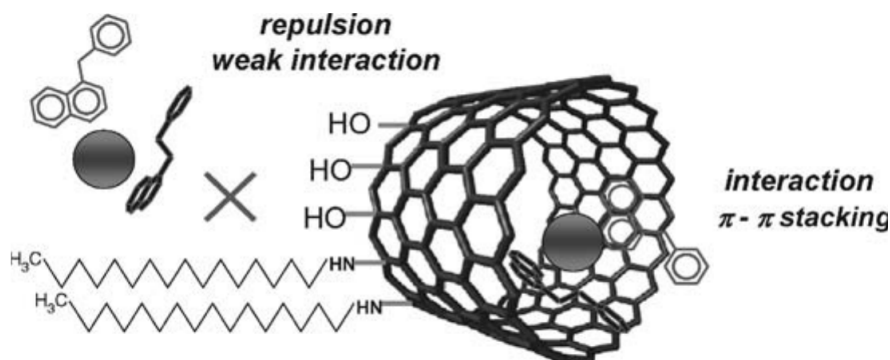


Figure 6.18: Selective confinement of metal nanoparticles inside CNT channels using differential surface chemistry. Adapted from Serp and Figueiredo, Wiley, 2009.

produces stronger adsorbate binding. CNT curvature and π -electron density modify this quantity for supported metal nanoparticles:

- **Inside CNT channels** (concave curvature): increased electron density at the metal–carbon interface shifts the d-band center *down* by 0.1–0.4 eV, weakening CO and H adsorption.
- **Outside CNT surface** (convex curvature): the effect is weaker and in the opposite direction.

The magnitude of the shift scales approximately as $1/d_{\text{CNT}}$: narrower tubes produce larger electronic perturbations. For cinnamaldehyde hydrogenation on Pt/CNT, the downshifted d-band center weakens C=C binding more than C=O binding, redirecting selectivity toward the desired cinnamyl alcohol (80–90% vs. 30–50% on conventional Pt/C).

6.11 Catalytic Applications

CNT-supported catalysts have demonstrated excellent performance across several reaction classes. We highlight two areas where CNT supports offer the most distinct advantages.

6.11.1 Selective Hydrogenation

CNT-supported noble metals (Pt, Pd, Ru) show enhanced selective hydrogenation performance. The benchmark example is cinnamaldehyde hydrogenation, where Pt/CNT achieves 80–90% selectivity to cinnamyl alcohol (C=O reduction preserving C=C), compared to only 30–50% on conventional Pt/C. This selectivity enhancement arises from electronic modifica-

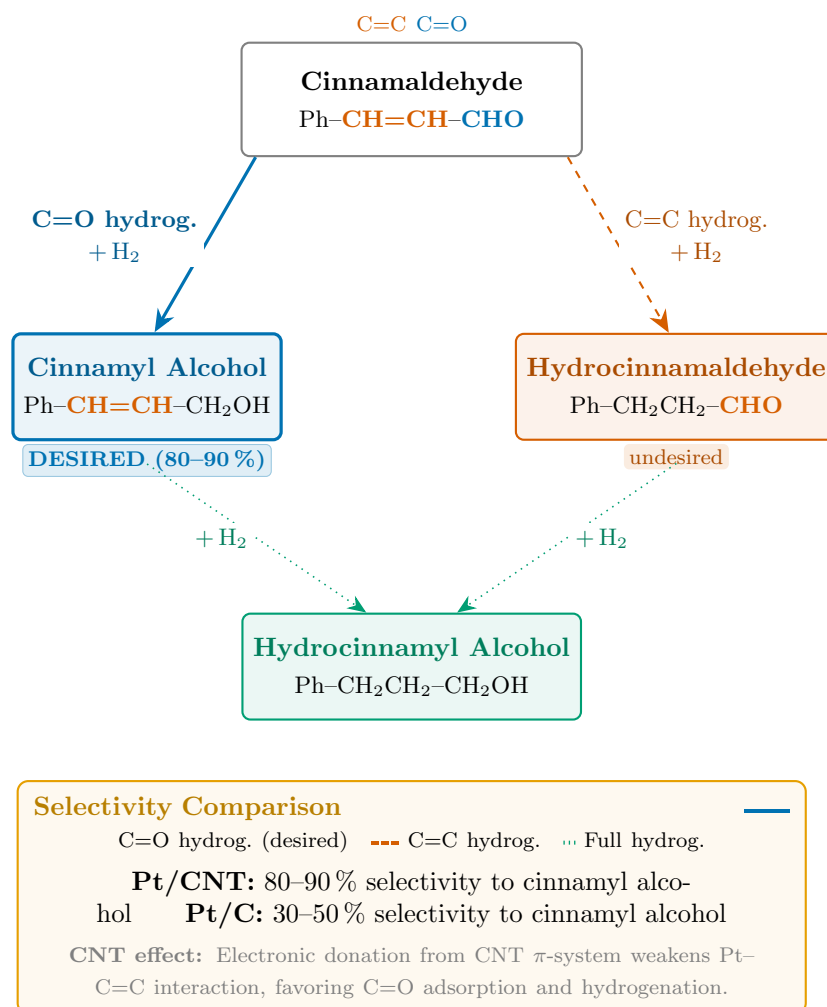


Figure 6.19: Selective hydrogenation of cinnamaldehyde on Pt/CNT. The CNT support enhances selectivity toward cinnamyl alcohol (80–90%) vs. conventional Pt/C (30–50%) by electronically modifying Pt to favor C=O over C=C hydrogenation.

tion of the Pt d-band center by the CNT π -system, controlled metal particle size, and hydrogen spillover. In the language of Chapter 5, the CNT support shifts the ratio $k_{\text{C=O}}/k_{\text{C=C}}$ by modifying the rate constants themselves—an example of catalyst design to control selectivity.

6.11.2 Electrocatalysis

CNTs are particularly attractive for electrocatalysis due to their high electrical conductivity. The TOF concept from Chapter 3 applies: improved electron transport means more catalytic cycles per site per unit time.

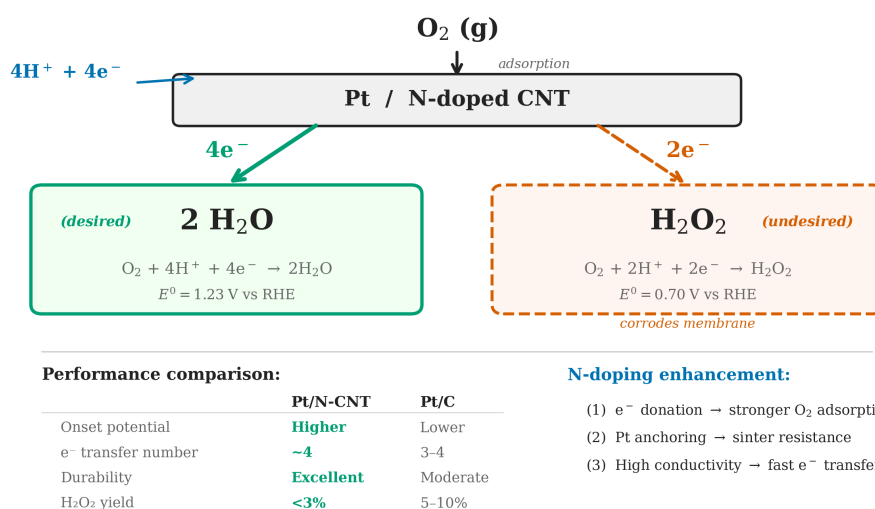


Figure 6.20: Oxygen reduction reaction on Pt/N-CNT. The conductive CNT enables efficient electron transfer, while graphitic-N dopants promote the 4-electron pathway to H₂O.

Key electrocatalytic applications include the oxygen reduction reaction (ORR) for fuel cell cathodes, where Pt/CNT shows improved activity and durability versus Pt/C and N-doped CNTs approach Pt performance as metal-free catalysts; hydrogen evolution (HER) using MoS₂/CNT hybrids; and CO₂ electroreduction on Cu/CNT where confinement modifies product selectivity.

Remark 6.23 (Fischer–Tropsch Synthesis on CNT-Supported Catalysts). Co/CNT and Fe/CNT catalysts achieve higher C₅+ selectivity (long-chain hydrocarbons) than conventional Co/Al₂O₃, making CNTs attractive for Fischer–Tropsch (FT) synthesis. Three effects

contribute:

1. **Confinement geometry**—metal particles inside CNT channels restrict chain-termination pathways, favoring chain growth and raising the Anderson–Schulz–Flory chain-growth probability α .
2. **Electronic modification**—the CNT π -system weakens CO adsorption on the confined metal, increasing the probability of CO insertion (propagation) over hydrogenation (termination).
3. **Enhanced product removal**—the high diffusivity inside CNT channels (Section 6.12.2) evacuates heavy wax products before secondary hydrocracking can occur.

Typical FT conditions on Co/CNT are 200–250°C, 20–30 bar, $\text{H}_2/\text{CO} = 2$. This application unifies transport (Section 6.12.2), confinement (Section 6.10.3), and selectivity engineering (Chapter 5) in a single catalyst system.

Remark 6.24 (When to Choose CNT Supports). CNT supports are most advantageous when selectivity control through electronic effects is desired, high electrical conductivity is needed, confinement can improve performance, or conventional supports suffer from metal sintering. Cost considerations may favor conventional supports for large-scale commodity production.

6.12 Diffusion in Nanoconfined Materials

In Chapter 4, we developed the Thiele modulus and effectiveness factor framework for porous catalysts, focusing on mesoporous and macroporous structures. Both zeolites and carbon nanotubes confine molecules to channels with diameters below ~ 50 nm, but the two materials produce *opposite* deviations from classical Knudsen theory: zeolite micropores *hinder* diffusion far below Knudsen predictions, while smooth CNT channels *enhance* it far above. Understanding both regimes is essential for designing nanoconfined catalysts.

6.12.1 Configurational Diffusion in Zeolites

When pore diameters approach the molecular diameter itself ($d_p < 2$ nm), molecules can no longer move freely. Instead, they must squeeze through constrictions and negotiate a periodic potential energy landscape imposed by the framework walls. This **configurational diffusion** produces diffusivities of 10^{-10} – 10^{-18} m^2/s —up to **13 orders of magnitude** below bulk gas

diffusion. Table 6.2 summarizes the three regimes.

Table 6.2: Comparison of diffusion regimes in porous catalysts

Regime	Pore Size	D_{eff} (m ² /s)	Rate-Limiting Factor
Molecular	>100 nm	10^{-5}	Molecule–molecule collisions
Knudsen	2–100 nm	10^{-6} – 10^{-7}	Molecule–wall collisions
Configurational	<2 nm	10^{-10} – 10^{-18}	Steric constraints

Unlike Knudsen diffusion ($D_K \propto T^{0.5}$), configurational diffusion follows an **Arrhenius-like** temperature dependence:

$$D_{\text{micro}} = D_0 \exp\left(-\frac{E_D}{RT}\right) \quad (6.8)$$

where E_D is the activation energy for diffusion (typically 10–50 kJ/mol, but exceeding 100 kJ/mol when molecular diameter approaches pore diameter). Each molecular “hop” between adsorption sites requires overcoming a steric barrier at a pore window—an activated process governed by $\exp(-E_D/RT)$.

The appropriate **Thiele modulus for zeolite catalysts** uses the crystal radius as the characteristic length:

$$\phi = R_{\text{crystal}} \sqrt{\frac{k_{\text{intrinsic}}}{D_{\text{micro}}}} \quad (6.9)$$

Because D_{micro} is 10^3 – 10^8 times smaller than D_{Knudsen} , even micrometer-sized crystals can give $\phi \gg 1$, leaving interior acid sites underutilized. Fig. 6.21 illustrates how the effectiveness factor drops with increasing crystal radius for three representative diffusivities.

Remark 6.25 (Crystal Size Optimization). Reducing crystal radius by $10\times$ reduces ϕ by $10\times$, potentially shifting from diffusion-limited to kinetic control. Nano-sized zeolites (<100 nm) or hierarchical zeolites with mesopore “highways” can raise η from 0.3 to 0.8–0.9, though mesopores may reduce shape selectivity.

Remark 6.26 (Concept Check: Zeolite Diffusion).

1. A zeolite catalyst gives $\eta = 0.4$ at 400 °C. If the temperature increases to 500 °C, will η increase or decrease? *Answer: It depends on the relative activation energies. If $E_a > E_D$ (common for zeolites), the intrinsic rate increases faster than diffusivity, so ϕ increases and η decreases.*

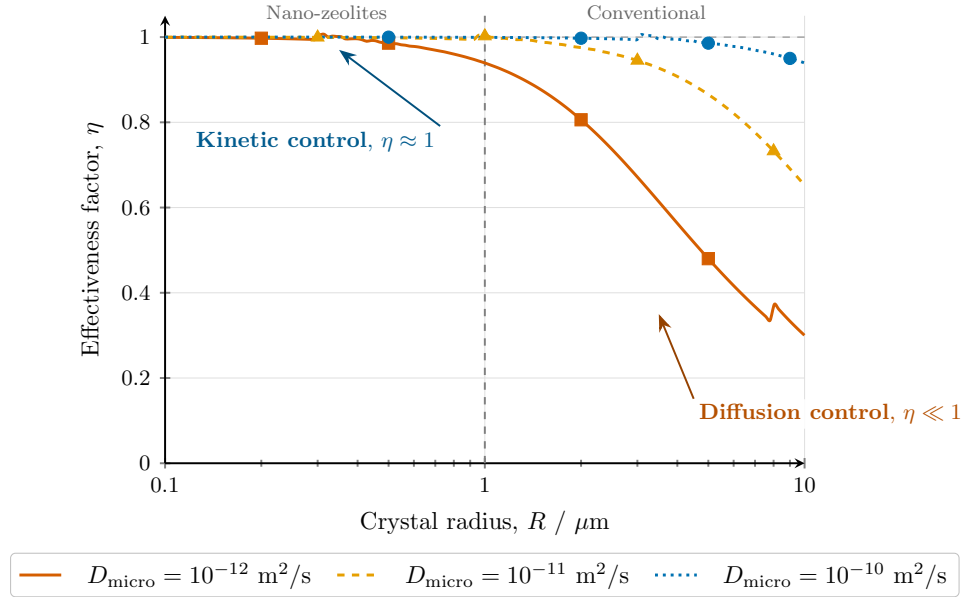


Figure 6.21: Effectiveness factor η vs. crystal radius R for three micropore diffusivities. Slow diffusion ($D_{\text{micro}} = 10^{-12} \text{ m}^2/\text{s}$, solid red): $\eta < 0.5$ above $\sim 0.3 \mu\text{m}$. Intermediate ($10^{-11} \text{ m}^2/\text{s}$, dashed orange): nano-zeolites maintain reasonable η . Fast ($10^{-10} \text{ m}^2/\text{s}$, dotted blue): kinetic control across all practical sizes.

2. Explain in one sentence why the Weisz–Prater criterion uses R_{crystal} rather than the pellet radius for zeolite catalysts. *Answer: The dominant diffusion resistance is inside the microporous crystal, not in the macroporous binder between crystals.*

Remark 6.27 (Applying the Weisz–Prater Criterion to Zeolite Crystals). The Weisz–Prater criterion developed in Eq. (4.26) applies directly to zeolite crystals, but you must substitute the correct length scale and diffusivity:

$$N_{WP} = \frac{r_{\text{obs}} \cdot R_{\text{crystal}}^2}{D_{\text{micro}} \cdot C_s} \quad (6.10)$$

Use the **crystal radius** R_{crystal} (not the pellet radius) and the **micropore diffusivity** D_{micro} (not the effective diffusivity of the binder-bound pellet). The threshold $N_{WP} < 0.3$ for first-order kinetics ensures $\eta > 0.95$. The practical challenge is that D_{micro} spans 10^{-10} to $10^{-18} \text{ m}^2/\text{s}$ and is often the least well-known parameter in zeolite

catalysis. Reliable measurement techniques include the **zero-length column (ZLC)** method, **frequency response**, and **pulsed-field-gradient NMR (PFG-NMR)**. A quick experimental test: crush crystals to different sizes and compare rates—if the rate changes with crystal size, intracrystalline diffusion is limiting.

6.12.2 Enhanced Knudsen Diffusion in Carbon Nanotubes

CNT channels (1–50 nm diameter) operate in the Knudsen regime ($Kn > 10$ at atmospheric conditions), where classical theory predicts:

$$D_K = \frac{d_{\text{pore}}}{3} \sqrt{\frac{8RT}{\pi M}} \quad (6.11)$$

assuming **diffuse reflection**—molecules rebound in random directions after wall collisions. Remarkably, measured diffusivities in CNT channels exceed this prediction by 1–4 orders of magnitude (Fig. 6.22). The physical origin is the atomically smooth sp^2 graphitic wall, which produces **specular (or near-specular) reflection**: molecules preserve their axial momentum upon wall collision rather than randomizing direction. The enhanced diffusivity is:

$$D_{\text{CNT}} = D_K \cdot F \quad (6.12)$$

where the enhancement factor $F = (1 + \sigma)/(1 - \sigma)$ depends on the **specularity parameter** σ ($= 0$ for fully diffuse, $\rightarrow 1$ for fully specular reflection):

$$F = \frac{1 + \sigma}{1 - \sigma} \quad (6.13)$$

Pristine SWCNTs have $\sigma \approx 0.9$ – 0.98 ($F \approx 20$ – 100), while defective or functionalized CNTs drop to $\sigma \approx 0.3$ – 0.5 ($F \approx 2$ – 3). Experimental evidence from membrane permeation, MD simulations, and tracer studies confirms enhancements of 10–1000 $\times$ depending on CNT diameter and molecule.

Remark 6.28 (The Bowling Alley Analogy). Imagine rolling a bowling ball down two different lanes. A **conventional mesopore** behaves like a rough dirt road: each wall collision randomizes the molecule’s direction (**diffuse reflection**), destroying forward momentum. A **CNT channel** behaves like a polished bowling alley lane with smooth bumpers: the ball glances off the surface and keeps rolling forward (**specular reflection**), preserving its axial velocity. The physical basis is that the atomically smooth sp^2 graphitic wall has a surface corrugation height (~ 0.01 nm) far smaller than typical molecular diameters (0.3–0.5 nm), so the gas–wall interaction potential is nearly

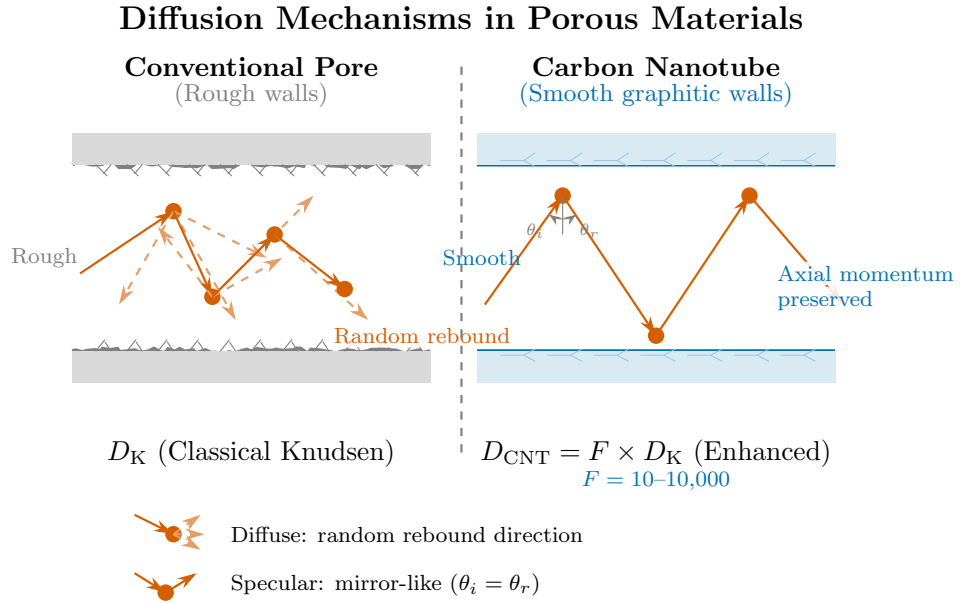


Figure 6.22: Conventional pores (rough walls): diffuse reflection randomizes direction. CNTs (smooth graphitic walls): specular reflection preserves axial momentum, giving $D_{CNT} = F \cdot D_K$ with $F = 10\text{--}10,000$.

flat along the tube axis. Molecular dynamics simulations confirm this picture: the velocity autocorrelation function in CNTs decays $\sim 100\times$ more slowly than in silica or alumina pores of the same diameter, meaning each molecule retains its directed motion over many more collisions before losing memory of its initial trajectory.

For a CNT-supported catalyst, the Thiele modulus becomes:

$$\phi_{CNT} = L \sqrt{\frac{k}{F \cdot D_K}} = \frac{\phi_{\text{classical}}}{\sqrt{F}} \quad (6.14)$$

A 100-fold enhancement reduces ϕ by $10\times$, dramatically expanding the kinetic-control window. Fig. 6.23 shows how η curves shift rightward with increasing F .

Remark 6.29 (Concept Check: CNT Enhanced Diffusion).

1. A pristine SWCNT has $\sigma \approx 0.95$. Calculate the enhancement factor F . *Answer:* $F = (1 + 0.95)/(1 - 0.95) = 1.95/0.05 = 39$.
2. After acid treatment, the specularity drops to $\sigma = 0.5$. What is the new F ? By what factor does ϕ change? *Answer:* $F =$

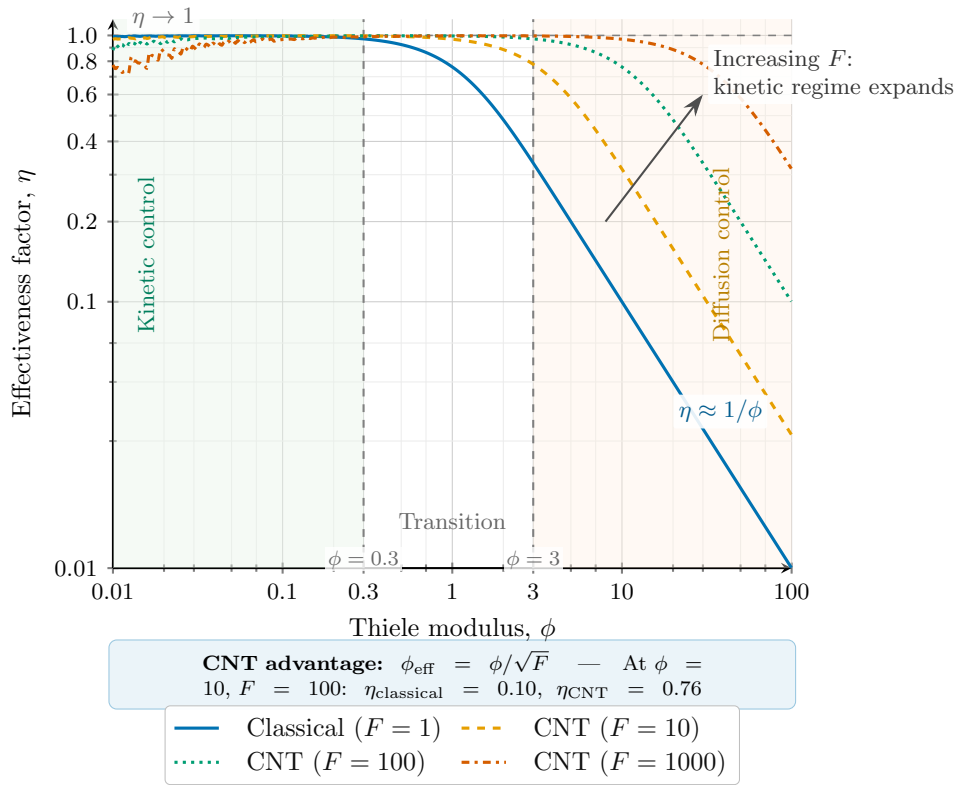


Figure 6.23: η vs. ϕ for classical Knudsen ($F = 1$) and CNT-enhanced transport ($F = 10, 100, 1000$). Enhanced diffusion extends kinetic control to higher intrinsic rate constants.

$1.5/0.5 = 3$. Since $\phi \propto 1/\sqrt{F}$, ϕ increases by $\sqrt{39/3} = 3.6\times$.

- Why would deliberately functionalizing CNT walls worsen their transport advantage? Answer: Functionalization roughens the atomically smooth sp^2 surface, reducing specular reflection and lowering σ and F .

6.12.3 Comparison: Hindered vs. Enhanced Transport

Remark 6.30 (Two Confinement Extremes).

Property	Zeolite Micropores	CNT Channels
Pore diameter	0.3–1.0 nm	1–50 nm
Diffusion regime	Configurational	Enhanced Knudsen
D_{eff} vs. classical	10^{-3} – $10^{-11} \times$ <i>lower</i>	10 – $10^4 \times$ <i>higher</i>
Wall character	Rough (ionic framework)	Smooth (sp^2 graphitic)
T -dependence of D	Arrhenius ($E_D = 10$ – 100 kJ/mol)	$D_K \propto T^{0.5}$
Design lever for η	Reduce crystal size	Increase F (wall quality)

Despite these opposite transport behaviors, both materials benefit from the same Thiele modulus framework (Chapter 4)—only the magnitude and origin of D_{eff} differ.

6.13 Characterization of Zeolite and CNT Catalysts

Building on the general toolkit from Chapter 3, this section focuses on techniques most critical for the two nanoporous materials in this chapter. We emphasize material-specific methods that go beyond the standard approaches covered earlier.

6.13.1 Zeolite-Specific Techniques

XRD for phase identification and crystallinity. Each of the 255+ recognized zeolite frameworks produces a distinct diffraction fingerprint (Section 3.9.1). Beyond phase identification, the unit cell parameter reveals framework composition: for FAU, the Breck–Flanigen relation correlates a_0 with Si/Al, allowing dealumination to be tracked via XRD.

N_2 physisorption and the t-plot method. Standard BET analysis (Section 3.7) gives misleadingly large surface areas (400–800 m²/g) for microporous materials because it interprets pore filling as multilayer adsorption. Instead, the **t-plot method** properly separates micropore volume V_{micro} (from the y -intercept of the high- t linear segment) from external surface area A_{ext} (from its slope). The Type I isotherm—sharp uptake at $P/P_0 < 0.05$ followed by a flat plateau—is the diagnostic hallmark of microporous materials (Fig. 6.24).

NH_3 -TPD for total acidity. Ammonia saturates all acid sites at 100–150°C; temperature-programmed desorption then reveals weak sites (150–300°C peak, Lewis acids and silanols) and strong sites (300–500°C peak, Brønsted acids). Peak areas give total acid site density; peak temperatures provide a coarse strength ranking via the Redhead equation (Eq. (3.38)). Fig. 6.25 compares three zeolites.

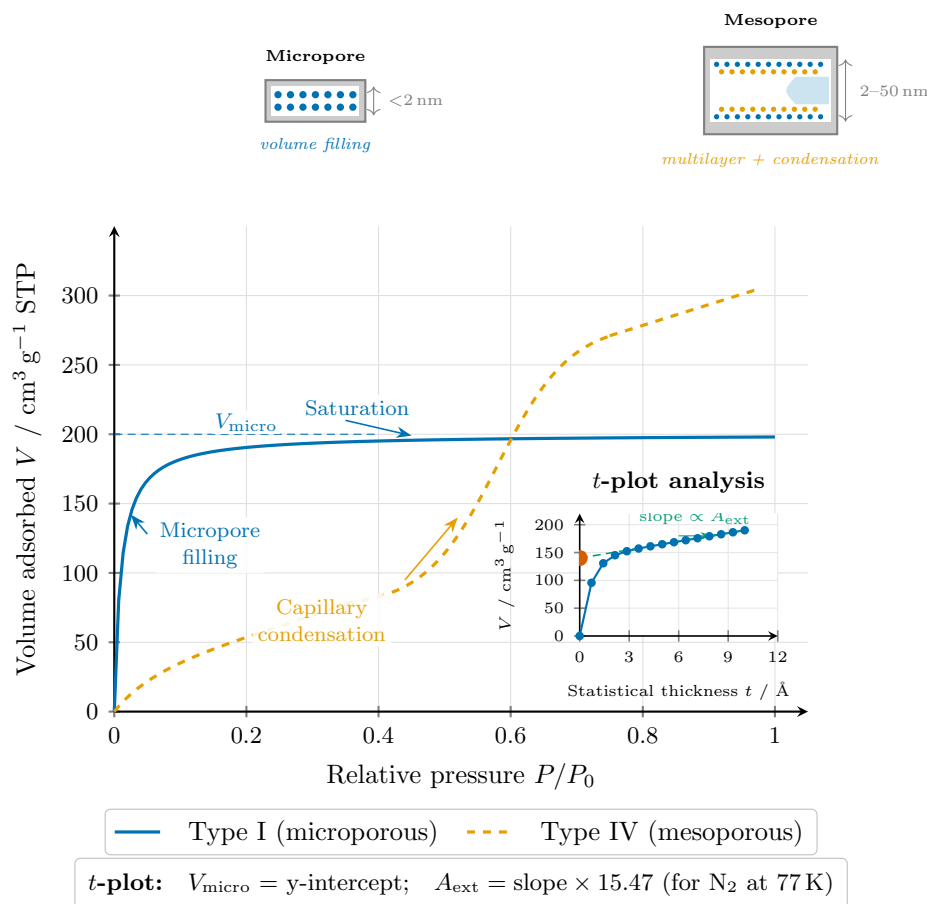


Figure 6.24: Type I isotherm (microporous) vs. Type IV isotherm (mesoporous). Sharp low-pressure uptake indicates micropore filling. Inset: t-plot construction for separating micropore volume from external surface area.

Remark 6.31 (TPD Peak Interpretation). NH_3 -TPD cannot distinguish Brønsted from Lewis acidity. For site-type speciation, use **pyridine-FTIR**: the pyridinium band at $\sim 1545 \text{ cm}^{-1}$ is exclusive to Brønsted sites, while the band at $\sim 1455 \text{ cm}^{-1}$ identifies Lewis sites (Table 6.3).

Pyridine-FTIR for Brønsted/Lewis speciation.

Temperature-resolved evacuation (150, 250, 350, 450°C) after pyridine saturation provides acid strength distributions with site-type discrimination. Note that pyridine ($\sim 5.3 \text{ \AA}$) cannot enter small-pore zeolites (CHA, LTA); CO or acetonitrile must be used instead.

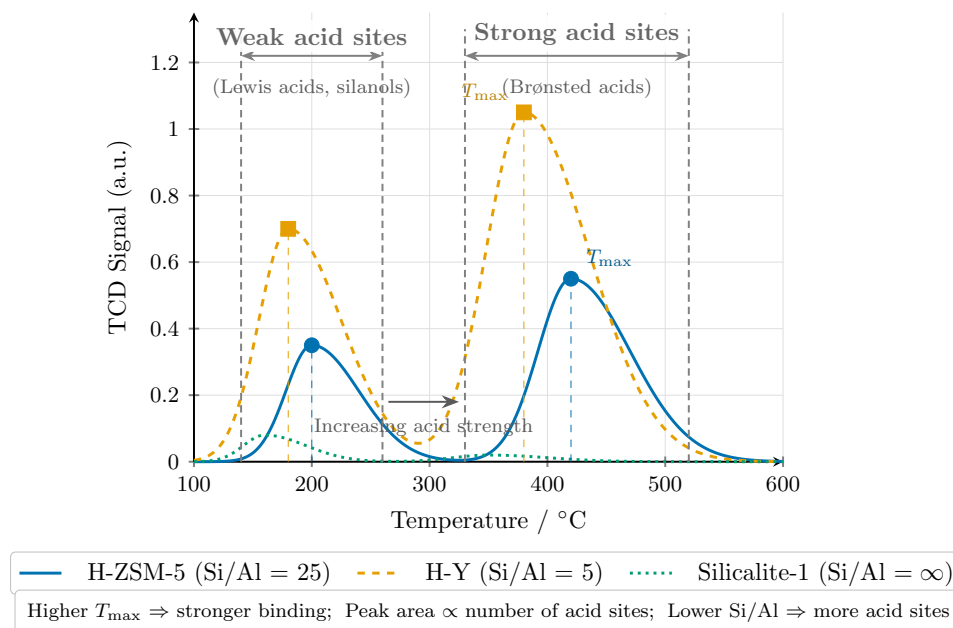


Figure 6.25: NH_3 -TPD profiles comparing zeolite acidity. H-Y (low Si/Al) shows the largest total area. Silicalite-1 (no Al) shows negligible desorption.

Table 6.3: Characteristic pyridine adsorption bands

Band Position	Species	Site Type
$\sim 1545 \text{ cm}^{-1}$	Pyridinium ion (PyH^+)	Brønsted
$\sim 1455 \text{ cm}^{-1}$	Coordinated pyridine (PyL)	Lewis
$\sim 1490 \text{ cm}^{-1}$	Both species	Total

Remark 6.32 (Choosing the Right Probe Molecule). The kinetic diameter of the probe determines which acid sites it can reach. The table below summarizes the four most common FTIR probe molecules:

Probe	Diameter	Accessible pores	Key bands (cm^{-1})	Advantage
Pyridine	5.3 Å	10-MR, 12-MR	1545 (B), 1455 (L)	B/L gold standard
2,6-DTBPpy	7.4 Å	External only	1616	External sites only
CO	3.8 Å	All pore sizes	2170–2230	Strength from $\Delta\nu$
Acetonitrile	4.0 Å	8-MR and larger	2325 (B), 2310 (L)	Small-pore compatible

Practical rule: use at least two probe molecules of different kinetic diameters to separate internal from external acidity. Combining pyridine + 2,6-DTBPpy gives internal Brønsted/Lewis sites by difference.

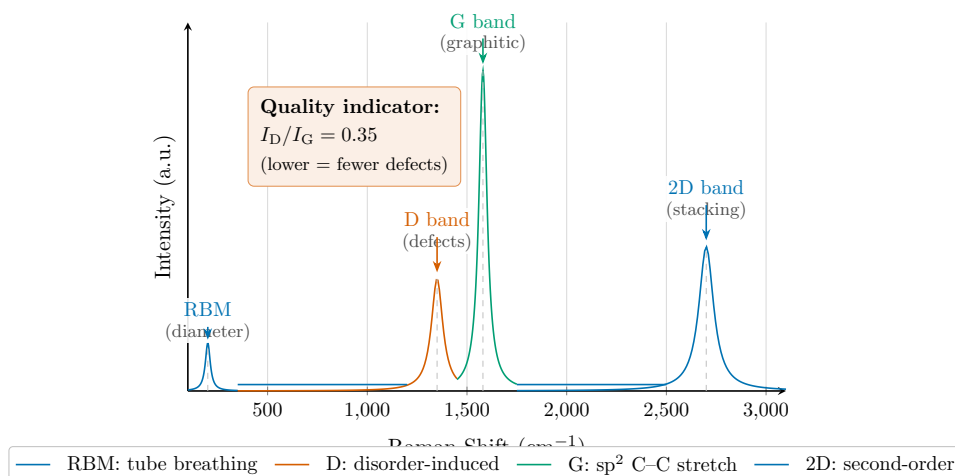


Figure 6.26: Characteristic CNT Raman bands: RBM (diameter-sensitive), D (defects), G (sp^2 carbon), 2D (overtone). The I_D/I_G ratio quantifies structural integrity.

6.13.2 CNT-Specific Techniques

Raman spectroscopy is the primary probe of CNT structural quality (Figs. 6.26 and 6.27). The G band ($\sim 1580\text{ cm}^{-1}$, sp^2 graphitic) and D band ($\sim 1350\text{ cm}^{-1}$, defect-activated) yield the I_D/I_G ratio:

$$\frac{I_D}{I_G} \propto \frac{1}{L_a} \quad (6.15)$$

where L_a is the in-plane crystallite size (Tuinstra–Koenig relation, valid for $L_a > 2\text{ nm}$). High-quality CNTs give $I_D/I_G < 0.1$; heavy functionalization pushes it above 1.0. The radial breathing mode (RBM, $100\text{--}300\text{ cm}^{-1}$) provides SWCNT diameter information.

Remark 6.33 (Tuinstra-Koenig vs Ferrari-Robertson Regimes).

The $I_D/I_G \propto 1/L_a$ relation reverses for highly disordered carbon ($L_a < 2\text{ nm}$), where $I_D/I_G \propto L_a^2$ (Ferrari–Robertson regime). Most CNTs remain in the Tuinstra–Koenig regime.

TGA in air (Fig. 6.29) reveals purity (amorphous carbon oxidizes before CNTs), thermal stability (onset $400\text{--}600^\circ\text{C}$), metal loading (residual mass), and degree of functionalization (weight loss $< 400^\circ\text{C}$).

XPS provides surface-sensitive chemistry (top $5\text{--}10\text{ nm}$): C 1s deconvolution distinguishes sp^2 (284.5 eV), sp^3 (285.1 eV), and oxygen-containing groups (C–O 286.5 , C=O 287.8 , O–C=O 289.0 eV). For N-doped CNTs, the N 1s spectrum resolves pyridinic-N (398.5 eV), pyrrolic-N (400.1 eV), and graphitic-N (401.3 eV).

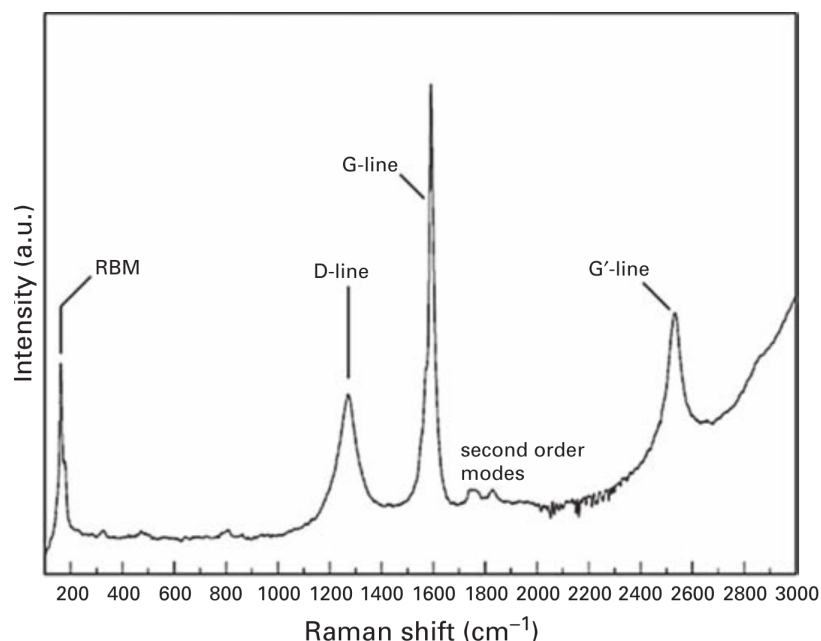


Figure 6.27: Experimental SWCNT Raman spectrum showing RBM, D-line, G-line, and G'-line (2D band). Adapted from Belin and Epron; reproduced from Harris, *Carbon Nanotube Science*, Cambridge University Press, 2009.

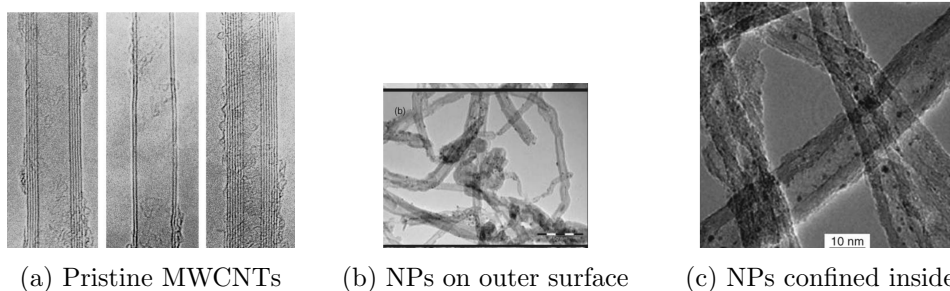


Figure 6.28: TEM images: (a) pristine MWCNTs showing concentric walls; (b) Pt NPs on outer surface; (c) Rh NPs confined inside channels. Adapted from Serp and Machado, RSC, 2015.

TEM (Fig. 6.28) directly visualizes wall number, diameters, metal particle size distributions, and particle location (inner vs. outer surface). Statistical analysis of >200 particles is needed for reliable size distributions.

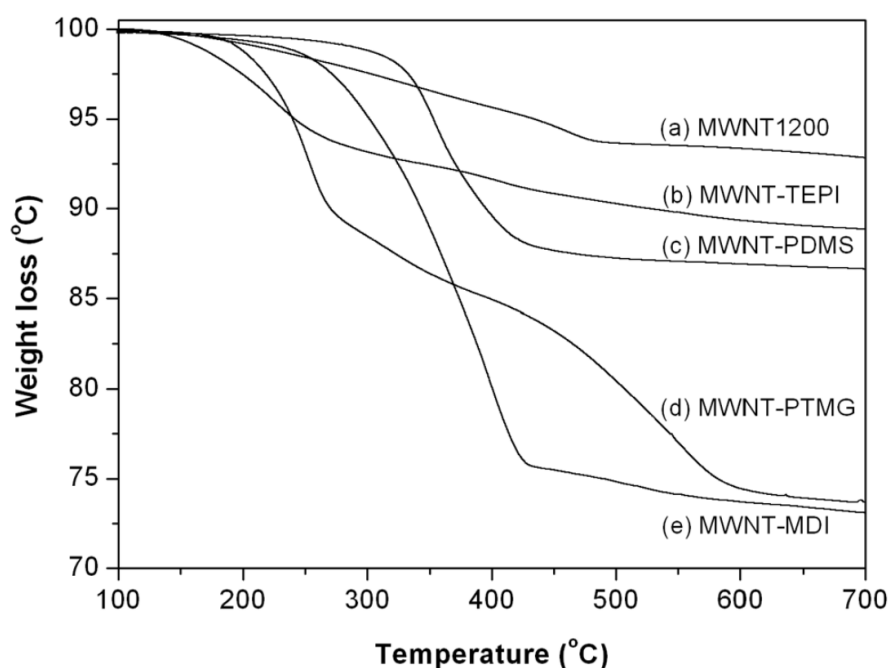


Figure 6.29: TGA traces of MWNTs with different functionalization degrees. Pristine MWNTs are stable until $\sim 500^\circ\text{C}$; heavily functionalized variants show $\sim 25\%$ weight loss below 200°C . Adapted from Lee et al., InTechOpen, 2013.

6.14 Worked Examples

Example 6.1: Acid site density from Si/Al ratio

- Problem:** An H-ZSM-5 zeolite has a Si/Al ratio of 25 (by molar ratio). The MFI framework has a unit cell formula of $[\text{Al}_x\text{Si}_{96-x}\text{O}_{192}]$ with a unit cell mass that depends on x .
- Calculate the number of Al atoms per unit cell.
 - Calculate the theoretical acid site density in mmol/g, assuming each Al generates one Brønsted acid site.
 - If NH_3 -TPD measures only 0.35 mmol/g of acid sites, what fraction of Al atoms are generating accessible Brønsted sites?

Solution (Framework Composition Analysis):

(a) The Si/Al ratio is defined as:

$$\frac{\text{Si}}{\text{Al}} = \frac{96 - x}{x} = 25$$

Solving for x :

$$96 - x = 25x \quad \Rightarrow \quad 96 = 26x \quad \Rightarrow \quad x = 3.69 \approx 3.7 \text{ Al per unit cell}$$

(b) The unit cell mass is:

$$\begin{aligned} M_{\text{UC}} &= 3.7(27) + 92.3(28) + 192(16) \\ &= 99.9 + 2584.4 + 3072 = 5756 \text{ g/mol per unit cell} \end{aligned}$$

The theoretical acid site density is:

$$n_{\text{acid}} = \frac{3.7 \text{ mol Al}}{5756 \text{ g}} = 6.4 \times 10^{-4} \text{ mol/g} = 0.64 \text{ mmol/g}$$

(c) The fraction of Al generating accessible Brønsted sites:

$$f = \frac{0.35}{0.64} = 0.55 = 55\%$$

Answer:

$$x = 3.7 \text{ Al/UC}; n_{\text{acid}} = 0.64 \text{ mmol/g}; 55\% \text{ accessible}$$

Physical interpretation: Only 55% of framework Al generates accessible Brønsted sites. The “missing” acidity may be due to: (1) extra-framework Al (EFAL) that is Lewis acidic rather than Brønsted acidic, (2) inaccessible sites deep within the crystal, or (3) framework defects. This discrepancy is common and highlights why both theoretical calculations and experimental measurements are needed.

Example 6.2: Thiele modulus for zeolite crystal

Problem: Consider a first-order cracking reaction in an H-Y zeolite crystal with the following parameters:

- Crystal radius: $R = 0.5 \mu\text{m} = 5 \times 10^{-7} \text{ m}$
 - Intrinsic rate constant: $k = 10 \text{ s}^{-1}$ at 500°C
 - Micropore diffusivity: $D_{\text{micro}} = 5 \times 10^{-12} \text{ m}^2/\text{s}$
- (a) Calculate the Thiele modulus.
 (b) Calculate the effectiveness factor (spherical geometry).
 (c) What is the maximum crystal radius that still maintains $\eta > 0.9$?

Solution (Thiele Modulus Framework):

(a) The Thiele modulus for a spherical crystal is:

$$\begin{aligned}\phi &= R \sqrt{\frac{k}{D_{\text{micro}}}} = (5 \times 10^{-7}) \sqrt{\frac{10}{5 \times 10^{-12}}} \\ &= (5 \times 10^{-7})(1.41 \times 10^6) = 0.71\end{aligned}$$

(b) For a sphere with first-order kinetics (Eq. (4.25)):

$$\eta = \frac{3}{\phi^2} (\phi \coth \phi - 1)$$

With $\phi = 0.71$: $\coth(0.71) = 1.64$

$$\eta = \frac{3}{(0.71)^2} (0.71 \times 1.64 - 1) = \frac{3}{0.504} (1.164 - 1) = 5.95 \times 0.164 = 0.97$$

(c) For $\eta > 0.9$, we need $\phi < 1.32$ (from numerical solution of the spherical effectiveness factor formula). Solving:

$$R_{\text{max}} = \frac{\phi}{\sqrt{k/D}} = \frac{1.32}{1.41 \times 10^6} = 9.4 \times 10^{-7} \text{ m} = 0.94 \mu\text{m}$$

Answer:

$$\phi = 0.71; \eta = 0.97; \text{ Need } R < 940 \text{ nm for } \eta > 0.9$$

Physical interpretation: With sub-micron crystals ($0.5 \mu\text{m}$), the zeolite operates near kinetic control ($\eta = 0.97$). However, the margin

is narrow: micron-sized crystals would suffer severe diffusion limitations. This sensitivity to crystal size is a direct consequence of the slow micropore diffusivity ($D \sim 10^{-12} \text{ m}^2/\text{s}$), which is $\sim 10^7$ times slower than bulk gas diffusion.

Example 6.3: Shape selectivity: Para-xylene diffusion

Problem: In ZSM-5 (MFI) zeolite at 400°C , the diffusivities of xylene isomers are:

- *para*-xylene: $D_p = 1 \times 10^{-11} \text{ m}^2/\text{s}$
- *ortho*-xylene: $D_o = 5 \times 10^{-14} \text{ m}^2/\text{s}$
- *meta*-xylene: $D_m = 2 \times 10^{-13} \text{ m}^2/\text{s}$

Assuming all three isomers are formed at equal rates inside the zeolite crystal:

- (a) Calculate the ratio of diffusivities D_p/D_o and D_p/D_m .
- (b) If the crystal half-thickness is $L = 0.2 \text{ }\mu\text{m}$, estimate the time for each isomer to diffuse out (using $t \sim L^2/D$).
- (c) Explain qualitatively how this leads to *para*-xylene selectivity.

Solution (Product Shape Selectivity Analysis):

(a) Diffusivity ratios:

$$\frac{D_p}{D_o} = \frac{1 \times 10^{-11}}{5 \times 10^{-14}} = 200$$

$$\frac{D_p}{D_m} = \frac{1 \times 10^{-11}}{2 \times 10^{-13}} = 50$$

(b) Characteristic diffusion time $t \sim L^2/D$ with $L = 2 \times 10^{-7} \text{ m}$:

$$t_p = \frac{(2 \times 10^{-7})^2}{1 \times 10^{-11}} = 4 \times 10^{-3} \text{ s} = 4 \text{ ms}$$

$$t_o = \frac{(2 \times 10^{-7})^2}{5 \times 10^{-14}} = 0.8 \text{ s} = 800 \text{ ms}$$

$$t_m = \frac{(2 \times 10^{-7})^2}{2 \times 10^{-13}} = 0.2 \text{ s} = 200 \text{ ms}$$

(c) The *para*-xylene escapes 50–200× faster than the other isomers. While trapped inside, *ortho*- and *meta*-xylene re-isomerize

on acid sites. Each re-equilibration converts some to *para*, which escapes quickly. This continuous “re-equilibration and selective removal” shifts the product distribution far from equilibrium toward *para*-xylene.

Answer:

$$D_p/D_o = 200, D_p/D_m = 50$$

$$t_p = 4 \text{ ms}, t_o = 800 \text{ ms}, t_m = 200 \text{ ms}$$

Physical interpretation: The 200-fold difference in diffusivity is the molecular basis for shape selectivity. The linear *para*-xylene (kinetic diameter 5.8 Å) fits well in the 10-MR channels, while the bulkier *ortho*- and *meta*-isomers (6.8 Å) experience strong steric hindrance. This explains why ZSM-5 achieves >90% *para*-xylene selectivity despite thermodynamic preference for *meta*-xylene.

Example 6.4: Enhanced Diffusion in CNT Channels

Problem: Hydrogen gas diffuses through a carbon nanotube membrane with inner diameter $d = 2 \text{ nm}$ at $T = 400 \text{ K}$. The measured permeance suggests an effective diffusivity 100 times higher than classical Knudsen theory predicts.

- Calculate the classical Knudsen diffusivity for H_2 ($M = 2 \text{ g/mol}$).
- Estimate the actual diffusivity in the CNT.
- If a first-order reaction with $k = 50 \text{ s}^{-1}$ occurs on confined Pt particles, calculate the Thiele modulus using both classical and enhanced diffusivity (CNT length $L = 5 \text{ }\mu\text{m}$).

Solution (Enhanced Knudsen Analysis):

- (a) Classical Knudsen diffusivity from Eq. (6.11):

$$D_K = \frac{d}{3} \sqrt{\frac{8RT}{\pi M}}$$

With $R = 8.314 \text{ J}/(\text{mol}\cdot\text{K})$, $T = 400 \text{ K}$, $M = 0.002 \text{ kg/mol}$, $d = 2 \times 10^{-9} \text{ m}$:

$$D_K = \frac{2 \times 10^{-9}}{3} \sqrt{\frac{8 \times 8.314 \times 400}{\pi \times 0.002}} \\ = 6.67 \times 10^{-10} \times 2058 = 1.37 \times 10^{-6} \text{ m}^2/\text{s}$$

(b) Enhanced diffusivity using Eq. (6.12) with factor $F = 100$:

$$D_{\text{CNT}} = F \times D_K = 100 \times 1.37 \times 10^{-6} = 1.37 \times 10^{-4} \text{ m}^2/\text{s}$$

(c) Thiele modulus (Eq. (4.11)) with $L = 5 \times 10^{-6} \text{ m}$:
Classical:

$$\phi_{\text{classical}} = L \sqrt{\frac{k}{D_K}} = 5 \times 10^{-6} \sqrt{\frac{50}{1.37 \times 10^{-6}}} = 5 \times 10^{-6} \times 6040 = 0.030$$

Enhanced:

$$\phi_{\text{CNT}} = L \sqrt{\frac{k}{D_{\text{CNT}}}} = 5 \times 10^{-6} \sqrt{\frac{50}{1.37 \times 10^{-4}}} = 5 \times 10^{-6} \times 604 = 0.0030$$

Answer:

$$D_K = 1.37 \times 10^{-6} \text{ m}^2/\text{s}; \quad D_{\text{CNT}} = 1.37 \times 10^{-4} \text{ m}^2/\text{s} \\ \phi = 0.030 \text{ (classical) vs. } 0.003 \text{ (enhanced)}$$

Physical interpretation: Both Thiele moduli are well below unity, indicating kinetic control ($\eta \approx 1$) regardless of diffusion enhancement. However, for faster reactions or longer nanotubes, the 100-fold enhancement would make a significant difference in effectiveness factor.

Remark 6.34 (Units Trap in Knudsen Calculations). The classical Knudsen formula is a frequent source of unit errors:

- **Molar mass:** Use M in kg/mol (not g/mol). $M_{\text{H}_2} = 0.002 \text{ kg/mol}$.
- **Pore diameter:** Use SI units (m), not nm. $d = 2 \text{ nm} = 2 \times 10^{-9} \text{ m}$.

- **Gas constant:** Use $R = 8.314 \text{ J}/(\text{mol}\cdot\text{K})$.
- **Check:** D_K should emerge in m^2/s , typically 10^{-7} – 10^{-5} for gases in nanopores.

Example 6.5: Interpreting Raman and TGA Data

Problem: A batch of CVD-grown MWCNTs is characterized before and after oxidative purification (HNO_3 reflux):

	As-synthesized	After purification
Raman I_D/I_G	0.35	0.85
TGA residue (air, 800°C)	8%	2%
TGA onset temp.	520°C	480°C

Interpret these results in terms of (a) structural changes, (b) purity, and (c) suitability as a catalyst support.

Solution (Data Interpretation):

(a) Structural changes (I_D/I_G ratio):

The I_D/I_G ratio increased from 0.35 to 0.85—defect density increased by $\sim 2.4\times$. HNO_3 treatment attacks CNT sidewalls, creating vacancies and introducing carboxylic, hydroxyl, and carbonyl groups.

(b) Purity (TGA residue):

Residual mass decreased from 8% to 2%, indicating $\sim 75\%$ of metal impurities removed. The original 8% residue (as oxide) corresponds to $\sim 5.6\%$ metallic Fe. Remaining 2% may be encapsulated metal inaccessible to acid.

(c) Suitability assessment:

Oxidation onset decreased by 40°C ($520 \rightarrow 480^\circ\text{C}$) due to introduced defects, but CNTs remain stable for most catalytic applications below 450°C in air. The trade-off is favorable: reduced metal contamination and more defect sites for nanoparticle anchoring outweigh the moderate decrease in thermal stability.

Answer:

75% of metal impurities removed; suitable up to 450°C

Physical interpretation: The moderate I_D/I_G increase (0.85 is acceptable) combined with reduced metal contamination makes these CNTs appropriate catalyst supports for liquid-phase reactions or inert atmosphere conditions. For applications requiring pristine CNTs, milder purification should be considered.

6.15 Computational Exploration: Notebook NB6

Notebook NB6 brings together the transport phenomena analysis from Chapter 4 and applies it to the two confinement regimes studied in this chapter—zeolite micropores that hinder diffusion and CNT channels that enhance it. The notebook contains four computational tasks progressing from individual calculations to a direct comparison of the two platforms.

1. **Zeolite Thiele modulus vs crystal size.** For a first-order reaction in H-ZSM-5 with configurable k and D_{micro} , plot ϕ and η as functions of crystal radius R from 10 nm to 10 μm on a log scale. Identify the critical crystal size at which η drops below 0.9 and annotate it on the plot.
2. **CNT diameter calculator.** Write a function that takes chiral indices (n, m) and returns the nanotube diameter (Eq. (6.7)), chirality type (armchair, zigzag, or chiral), and electronic character (metallic or semiconducting). Generate a scatter plot of diameter vs. chiral angle for all (n, m) pairs with $n + m \leq 30$, coloring points by electronic character.
3. **Enhanced diffusion comparison.** Calculate the classical Knudsen diffusivity and enhanced CNT diffusivity ($D_{\text{CNT}} = F \cdot D_K$) for H_2 , CH_4 , and C_6H_{12} at 400 K in a 3 nm diameter tube. Plot D_K and D_{CNT} versus molar mass for enhancement factors $F = 1, 10, 100$, and 1000. Compare these with the micropore diffusivity range (10^{-10} – 10^{-18} m^2/s) to appreciate the orders-of-magnitude difference.
4. **Effectiveness factor with enhancement.** Plot η vs. ϕ for slab geometry using the standard formula (Eq. (4.21)), then overlay curves for the same physical system (fixed k, L) but with increasing enhancement factors $F = 1, 10, 100, 1000$. Demonstrate how CNT-enhanced diffusion shifts the operating point along the η – ϕ curve toward kinetic control.

Remark 6.35 (Reflection Questions for NB6). After completing the notebook, consider the following questions. Record your answers in a markdown cell at the end of the notebook.

1. At what crystal size does a zeolite-catalyzed reaction with $k = 50 \text{ s}^{-1}$ and $D_{\text{micro}} = 10^{-12} \text{ m}^2/\text{s}$ transition from kinetic control ($\eta > 0.9$) to diffusion limitation ($\eta < 0.5$)? How does this compare to the CNT length at which the same transition occurs for $F = 100$?
2. For a reaction with $k = 100 \text{ s}^{-1}$ in a 5 nm pore, what enhancement factor F is needed to maintain $\eta > 0.95$ over a 50 μm diffusion length?
3. Zeolite micropores *hinder* diffusion while CNT channels *enhance* it relative to Knudsen theory. What structural feature of the CNT wall—absent in zeolite micropores—is responsible for this dramatic difference?

6.16 Summary

Zeolites

- **Zeolite structure:** Crystalline aluminosilicates built from corner-sharing TO_4 tetrahedra. Framework types (FAU, MFI, CHA) define pore geometry: small pore (8-MR, 3–4 Å), medium pore (10-MR, 5–6 Å), large pore (12-MR, 6–8 Å).
- **Brønsted acidity:** Bridging hydroxyl groups (Si–OH–Al) form when H^+ compensates framework negative charge from Al^{3+} substitution. Lower Si/Al gives more acid sites but weaker individual strength and lower hydrothermal stability.
- **Shape selectivity:** Reactant selectivity (large molecules excluded), product selectivity (faster-diffusing isomers escape preferentially), and transition-state selectivity (bulky TS geometries forbidden).
- **Micropore diffusion:** Configurational diffusion ($D \sim 10^{-10}$ to $10^{-18} \text{ m}^2/\text{s}$) follows Arrhenius behavior. Use crystal size as characteristic length in the Thiele modulus: $\phi = R_{\text{crystal}} \sqrt{k/D_{\text{micro}}}$.
- **Post-synthetic methods:** Ion exchange (H-form, Cu-form), dealumination (raises Si/Al, creates EFAL), desilication (adds mesopores).
- **Industrial applications:** FCC (USY), xylene isomerization (ZSM-5), MTO (SAPO-34), SCR (Cu-SSZ-13).

Carbon Nanotubes

- **Structure:** Classified by chiral indices (n, m) ; SWCNTs and MWCNTs. Diameter $d = (a/\pi)\sqrt{n^2 + nm + m^2}$. Metallic if $(n-m) \bmod 3 = 0$, semiconducting otherwise.
- **Surface chemistry:** Pristine CNTs are chemically inert. Oxidative functionalization and heteroatom doping (N, B) create anchoring sites for metal nanoparticles and modify electronic properties.
- **Enhanced diffusion:** Atomically smooth CNT walls enable near-specular molecular reflection, giving diffusion rates 10–10,000 times higher than classical Knudsen predictions. This reduces ϕ and improves η .
- **Confinement effects:** Metal nanoparticles inside CNT channels experience electronic modification (shifted d-band center) and geometric constraint (sintering resistance, shape-selective access).
- **Applications:** CNT supports excel in selective hydrogenation (electronic modification of noble metals), electrocatalysis (high conductivity for ORR, HER), and reactions where conventional supports suffer from metal sintering.

6.17 Notation Introduced in This Chapter

6.18 Problems

Remark 6.36 (Self-Assessment Checklist). Before attempting the problems, verify that you can:

- ☐ Sketch the building hierarchy $\text{TO}_4 \rightarrow \text{SBU} \rightarrow \text{CBU} \rightarrow \text{framework}$ for at least two types
- ☐ Predict how changing Si/Al affects acid site density vs. individual strength
- ☐ Distinguish the three types of shape selectivity from experimental evidence
- ☐ Calculate CNT diameter from chiral indices and predict metallic/semiconducting character
- ☐ Explain why zeolite pores *hinder* diffusion while CNT channels *enhance* it
- ☐ Select appropriate characterization techniques for a given question about a zeolite or CNT catalyst
- ☐ Apply the Thiele modulus framework to both zeolite crystals and CNT-supported catalysts

Table 6.4: Notation introduced in this chapter—Zeolites

Symbol	Description	Typical Units
Si/Al	Silicon-to-aluminum molar ratio	dimensionless
n -MR	n -membered ring (pore aperture)	—
D_{micro}	Micropore (configurational) diffusivity	m^2/s
R_{crystal}	Zeolite crystal radius	m (or nm, μm)
E_D	Activation energy for diffusion	kJ/mol
n_{acid}	Acid site density	mmol/g
θ	Surface coverage	dimensionless
H_0	Hammett acidity function	dimensionless
N_{WP}	Weisz–Prater number (zeolite)	dimensionless
$S_{A/B}$	Shape selectivity index	dimensionless
FD	Framework density	T-atoms/ $1000 \text{ \AA}^3$
EFAL	Extra-framework aluminum	—
PyH ⁺	Pyridinium ion (Brønsted-bound pyridine)	—
PyL	Lewis-coordinated pyridine	—
USY	Ultra-stable Y zeolite (dealuminated FAU)	—

Table 6.5: Notation introduced in this chapter—Carbon Nanotubes

Symbol	Description	Units
(n, m)	Chiral indices	dimensionless
d	Nanotube diameter	nm
a	Graphene lattice constant (0.246 nm)	nm
$\mathbf{C}_h$	Chiral vector	nm
Kn	Knudsen number	dimensionless
λ	Mean free path	nm
D_K	Classical Knudsen diffusivity	m^2/s
D_{CNT}	Enhanced diffusivity in CNT	m^2/s
F	Enhancement factor	dimensionless
σ	Specularity parameter	dimensionless
I_D/I_G	Raman D/G intensity ratio	dimensionless

1. Framework Composition and Acidity

A zeolite Y (FAU framework) sample has a Si/Al ratio of 5.0. The FAU unit cell contains 192 T-atoms.

- Calculate the number of Al atoms per unit cell.
- Calculate the unit cell mass for the anhydrous H-form (atomic masses: Si = 28, Al = 27, O = 16, H = 1 g/mol; the FAU unit cell contains 384 framework oxygens). Then estimate the theoretical Brønsted acid site density in mmol/g.
- NH₃-TPD measurement gives 2.1 mmol/g. What percentage of the theoretical maximum is actually accessible?
- Suggest two reasons why not all framework Al atoms may generate accessible Brønsted acid sites.

Hint: Verify with `acid_site_density()` in NB6 (Part 2), setting `si_al_ratio=5` and `T_atoms_per_UC=192`.

2. Diffusion Limitations in Zeolite Crystals

For a catalytic reaction in H-ZSM-5:

- Intrinsic rate constant: $k = 50 \text{ s}^{-1}$
- Micropore diffusivity of reactant: $D = 2 \times 10^{-12} \text{ m}^2/\text{s}$

- Calculate the Thiele modulus for zeolite crystals of radius 0.1, 0.5, and 2.0 μm .
- Estimate the effectiveness factor for each crystal size (use the spherical pellet formula or charts).
- At what crystal size does the reaction transition from kinetic control ($\eta > 0.9$) to severe diffusion limitation ($\eta < 0.5$)?
- Propose two strategies to improve the effectiveness factor without changing the zeolite framework type.

Hint: Use `effectiveness_sphere()` from NB6 (Part 5) to plot η vs. crystal radius and identify transition sizes.

3. Zeolite Characterization Data Interpretation

A modified zeolite catalyst is characterized by multiple techniques. Interpret the following observations:

- The N₂ adsorption isotherm shows a Type I shape at low P/P_0 but develops a hysteresis loop at $P/P_0 = 0.4\text{--}0.8$. What does this indicate about the pore structure?
- NH₃-TPD shows two peaks: a small peak at 180°C and a large peak at 420°C. What types of acid sites do these represent?

- (c) Pyridine-FTIR shows bands at both 1545 cm^{-1} and 1455 cm^{-1} , with the 1455 cm^{-1} band being stronger. What does this indicate about the Brønsted/Lewis acid site ratio?
- (d) The XRD pattern shows all expected peaks for the framework type, but peak intensities are 70% of the reference material. What could cause this reduced crystallinity?

4. CNT Structure Calculations

For each of the following carbon nanotubes with chiral indices (n, m) : $(6, 6)$, $(10, 0)$, $(8, 4)$, and $(12, 8)$:

- (a) Calculate the diameter using $d = \frac{0.246}{\pi} \sqrt{n^2 + nm + m^2}$ nm.
- (b) Classify each as armchair, zigzag, or chiral.
- (c) Predict whether each is metallic or semiconducting based on $(n - m) \bmod 3$.
- (d) Which nanotube would have the largest accessible internal volume per unit length for hosting metal nanoparticles? *Hint: Use `cnt_diameter(n, m)` from NB6 (Part 8) to compute diameters and check classifications.*

5. Enhanced Knudsen Diffusion

Methane (CH_4 , $M = 16\text{ g/mol}$) diffuses through a MWCNT membrane at 500 K. The nanotubes have inner diameter $d = 5\text{ nm}$.

- (a) Calculate the classical Knudsen diffusivity D_K for CH_4 .
- (b) Experimental measurements indicate $D_{\text{eff}} = 5 \times 10^{-5}\text{ m}^2/\text{s}$. What is the enhancement factor F ?
- (c) The same membrane shows $F = 30$ for N_2 ($M = 28\text{ g/mol}$) at the same temperature. Suggest reasons for the different enhancement factors.
- (d) If the CNT walls are functionalized with oxygen groups (making them rougher), would you expect the enhancement factor to increase, decrease, or remain the same? Explain. *Hint: Use `knudsen_diffusivity()` from NB6 (Part 10) to compute D_K . The enhancement factor is $F = D_{\text{eff}}/D_K$.*

6. Integrated CNT Catalyst Design

You are developing a Ru/CNT catalyst for ammonia synthesis ($\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$) in a fixed-bed reactor operating at 400°C and 100 bar.

- (a) Why might CNT supports be advantageous compared to the traditional iron-based catalyst on alumina support? Consider electronic effects, mass transfer, and thermal stability.

- (b) Would you recommend using pristine or N-doped CNTs? Consider the Ru-N interaction and its effect on N₂ dissociation (the rate-limiting step).
- (c) The synthesis gas contains trace H₂O (100 ppm) as an impurity. Discuss how this might affect the catalyst and propose mitigation strategies.
- (d) Design a characterization protocol (list 4-5 techniques) to evaluate your catalyst before and after 100 hours of operation. What specific changes would indicate catalyst deactivation?
- (e) Estimate the Thiele modulus for your system using reasonable assumptions (state your assumed values for k , D_{eff} , and pellet dimensions). Would you expect internal diffusion limitations? *Hint: Use `effectiveness_cnt()` from NB6 (Parts 11–12) with your assumed parameters to estimate η .*

7. Industrial Application Analysis

For each of the following industrial processes, identify the zeolite or CNT catalyst used, the key structure–property relationship that makes it effective, and the type of selectivity (if applicable):

- (a) Fluid catalytic cracking (FCC) of vacuum gas oil to gasoline-range hydrocarbons.
- (b) Selective production of *para*-xylene from mixed xylene isomers.
- (c) Methanol-to-olefins (MTO) conversion to ethylene and propylene.
- (d) Selective catalytic reduction (SCR) of NO_x in diesel exhaust using NH₃.
- (e) Selective hydrogenation of cinnamaldehyde to cinnamyl alcohol using a CNT-supported Pd catalyst. Explain how the CNT support modifies Pd selectivity compared to Pd on activated carbon, referencing the d-band center concept.

Looking Ahead

In this chapter, we examined two complementary nanoporous platforms for heterogeneous catalysis. Zeolite micropores impose configurational diffusion that enables shape selectivity but limits effectiveness, while carbon nanotube channels provide dramatically enhanced transport that maintains kinetic control even at high reaction rates. These two confinement regimes illustrate a central design principle: nanoscale architecture can be engineered to either restrict or promote molecular transport, depending on the catalytic objective.

The quantitative tools developed throughout this course—Thiele modulus and effectiveness factor (Chapter 4), selectivity analysis (Chapter 5), and the combined adsorption/kinetics/transport framework (Chapters 2 to 3)—apply directly to both platforms. Appendix G brings these tools together in the context of **catalytic organic synthesis**, where rational catalyst and support selection drives performance in industrially important processes such as Pd-catalyzed cross-coupling and asymmetric catalysis.

Appendix A

Notation and Symbol Glossary

This appendix provides a comprehensive reference for all mathematical notation, symbols, and abbreviations used throughout these lecture notes. Consistent notation is essential for clear communication in surface chemistry and catalysis, and this glossary serves as a quick reference when reading equations and working through problems.

A.1 Primary Symbols

The following table lists the primary symbols used in this course, organized alphabetically by the symbol name. Each entry includes the typical units and the chapter where the symbol is first introduced.

Symbol	Meaning	Typical Units	Defined
A	Pre-exponential (frequency) factor	varies with order	Ch. 2
a	Activity of catalyst	dimensionless	Ch. 3
a_s	Specific external surface area	m^2/kg	Ch. 4
C	Concentration (general)	mol/m^3 or mol/L	Ch. 1
C_A	Concentration of species A	mol/m^3	Ch. 1
C_{A0}	Initial/inlet concentration of A	mol/m^3	Ch. 3
C_{As}	Surface concentration of A	mol/m^3	Ch. 4
c	BET constant	dimensionless	Ch. 3
D	Diffusion coefficient (molecular)	m^2/s	Ch. 4
D_{eff}	Effective diffusivity	m^2/s	Ch. 4
D_K	Knudsen diffusivity	m^2/s	Ch. 4
d_p	Particle (pellet) diameter	m or mm	Ch. 4
E_a	Activation energy	kJ/mol	Ch. 2
E_{app}	Apparent activation energy	kJ/mol	Ch. 2
E_{des}	Desorption activation energy	kJ/mol	Ch. 2

Continued on next page

Table A.1 – *Continued from previous page*

Symbol	Meaning	Typical Units	Defined
F	Molar flow rate	mol/s	Ch. 3
F_A	Molar flow rate of species A	mol/s	Ch. 3
h	Planck's constant	J s	Ch. 2
ΔH_{ads}	Enthalpy of adsorption	kJ/mol	Ch. 2
$\Delta H^\ddagger$	Activation enthalpy	kJ/mol	Ch. 2
K	Equilibrium constant (general)	varies	Ch. 2
K_A	Adsorption equilibrium constant for A	Pa^{-1} or bar^{-1}	Ch. 2
k	Rate constant (general)	varies with order	Ch. 2
k_{ads}	Adsorption rate constant	$\text{Pa}^{-1} \text{s}^{-1}$	Ch. 2
k_{des}	Desorption rate constant	s^{-1}	Ch. 2
k_s	Surface reaction rate constant	varies	Ch. 2
k_B	Boltzmann constant	J/K	Ch. 2
k_c	Mass transfer coefficient	m/s	Ch. 4
L	Characteristic length	m	Ch. 4
M	Molecular weight	g/mol	Ch. 1
n	Reaction order	dimensionless	Ch. 2
N_{WP}	Weisz-Prater criterion	dimensionless	Ch. 4
P	Total pressure	Pa or bar	Ch. 1
P_A	Partial pressure of species A	Pa or bar	Ch. 2
Q	Volumetric flow rate	m^3/s or L/min	Ch. 3
R	Universal gas constant	J/(mol K)	Ch. 1
R_p	Particle (pellet) radius	m	Ch. 4
r	Reaction rate (general)	$\text{mol}/(\text{m}^3 \text{s})$	Ch. 2
r_{obs}	Observed reaction rate	$\text{mol}/(\text{m}^3 \text{s})$	Ch. 4
r''	Rate per unit surface area	$\text{mol}/(\text{m}^2 \text{s})$	Ch. 2
S	Selectivity	dimensionless	Ch. 5
S_{BET}	BET surface area	m^2/g	Ch. 3
$\Delta S^\ddagger$	Activation entropy	J/(mol K)	Ch. 2
T	Temperature	K or $^\circ\text{C}$	Ch. 1
T_p	Peak temperature (TPD)	K	Ch. 2
t	Time	s or min	Ch. 1
V	Reactor volume	m^3 or L	Ch. 3
V_m	Monolayer volume (BET)	$\text{cm}^3/\text{g STP}$	Ch. 3
v_0	Volumetric flow rate at inlet	m^3/s	Ch. 3
W	Catalyst mass	kg	Ch. 3
X	Conversion	dimensionless	Ch. 1
X_A	Conversion of species A	dimensionless	Ch. 3
x	Relative pressure (P/P_0)	dimensionless	Ch. 3
Y	Yield	dimensionless	Ch. 5
Y_B	Yield of product B	dimensionless	Ch. 5

Table A.1: Primary symbols used in Surface Phenomena & Catalysis

A.2 Greek Letters

Greek letters carry specific meanings in catalysis and chemical kinetics. The following table summarizes their usage in this course.

Symbol	Name	Meaning	Units
α	alpha	Dosing ratio (NH_3/NO in SCR)	dimensionless
β	beta	Heating rate (TPD)	K/s
γ	gamma	Activity coefficient	dimensionless
δ	delta	Film thickness	m
ε	epsilon	Void fraction (porosity)	dimensionless
ε_p	epsilon-p	Pellet/particle porosity	dimensionless
η	eta	Effectiveness factor	dimensionless
θ	theta	Fractional surface coverage	dimensionless
θ_A	theta-A	Coverage by species A	dimensionless
θ_*	theta-star	Fraction of vacant sites	dimensionless
λ	lambda	Mean free path	m
μ	mu	Dynamic viscosity	Pa s
ν	nu	TPD pre-exponential; stoichiometric coeff.	s^{-1} ; —
ρ	rho	Density	kg/m^3
ρ_b	rho-b	Bulk density of catalyst bed	kg/m^3
ρ_c	rho-c	Catalyst particle density	kg/m^3
σ	sigma	Molecular cross-sectional area	m^2
τ	tau	Space time	s
ϕ	phi	Thiele modulus	dimensionless
τ_p	tau-p	Tortuosity factor	dimensionless
ω	omega	Mass fraction	dimensionless

Table A.2: Greek letters and their meanings in catalysis

Remark A.1 (Sign Convention for Thermodynamic Quantities).

Throughout these notes, we follow the standard thermodynamic sign convention: $\Delta H_{\text{ads}} < 0$ for exothermic adsorption, and activation energies $E_a > 0$ represent energy barriers. The enthalpy of adsorption is typically reported as a negative value (heat released), while the desorption activation energy E_{des} is positive and approximately equal to $|\Delta H_{\text{ads}}|$ for non-activated adsorption.

A.3 Subscripts and Superscripts

Subscripts and superscripts modify base symbols to indicate specific conditions, species, or properties. Understanding these conventions is essential for interpreting equations correctly.

A.3.1 Common Subscripts

Subscript	Meaning
0	Initial condition or inlet (feed) stream
∞	Bulk condition (far from surface) or infinite time
A, B, C, ...	Species identifier
ads	Adsorption process or adsorbed state
app	Apparent (experimentally observed, may include transport effects)
b	Bulk phase or bed (catalyst bed)
c	Catalyst or characteristic
des	Desorption process
e, eq	Equilibrium condition
eff	Effective (accounting for pore structure)
ext	External (outside the catalyst pellet)
f	Forward reaction or final state
g	Gas phase
i, int	Internal (inside catalyst pellet) or intrinsic
K	Knudsen (diffusion regime)
m	Monolayer or mean value
max	Maximum value
obs	Observed (experimentally measured)
opt	Optimum value
p	Particle or pellet; also peak (as in T_p)
r	Reverse reaction or radial direction
rxn	Reaction
s	Surface condition
sat	Saturation condition
t	Total or at time t
*	Active site or vacant site

Table A.3: Standard subscripts used in notation

A.3.2 Common Superscripts

Superscript	Meaning
$\ddagger$	Transition state (activated complex)
$^\circ$	Standard state (usually 1 bar, 298.15 K)
'	Per unit mass (prime)
''	Per unit area (double prime)
'''	Per unit volume (triple prime)
*	Adsorbed species (alternative notation)
n	Reaction order exponent

Table A.4: Standard superscripts used in notation

A.4 Operators and Special Notation

A.4.1 Differential Operators

Mathematical operations on variables follow standard conventions:

Notation	Name	Description
d	Differential	Total differential operator (upright d)
$\frac{dy}{dx}$	Derivative	Total derivative of y w.r.t. x
$\frac{\partial y}{\partial x}$	Partial deriv.	Partial derivative, other variables held constant
∇	Nabla/Del	Gradient operator
∇^2	Laplacian	Second-order differential operator
Δ	Delta	Finite difference or change in quantity
$\sum$	Summation	Sum over specified index
$\prod$	Product	Product over specified index

Table A.5: Differential operators and mathematical notation

A.4.2 Chemical Notation

Species state and location are indicated using standard notation:

Notation	Meaning	Example
$A_{(g)}$	Gas phase species	$CO_{(g)}$, $O_{2(g)}$
$A_{(l)}$	Liquid phase species	$H_2O_{(l)}$
$A_{(s)}$ or (s)	Solid phase	$Fe_{(s)}$
A^* or A^*	Adsorbed species	CO^* , CO^*
$*$ or $*$	Vacant active site	θ_*
$\rightleftharpoons$	Equilibrium (reversible)	$A_{(g)} + * \rightleftharpoons A^*$
$\longrightarrow$	Forward reaction (irreversible)	$A^* + B^* \longrightarrow P_{(g)} + 2*$

Table A.6: Chemical species notation

Remark A.2 (Site Balance Notation). The site balance is fundamental to surface kinetics. For a surface with total site density L_0 (mol sites per unit area), the fractional coverage definition gives $\theta_A + \theta_B + \cdots + \theta_* = 1$, where θ_* represents the fraction of sites that are vacant (unoccupied). This constraint appears in every Langmuir-type derivation.

A.5 Dimensionless Groups

Dimensionless groups are ratios that characterize the relative importance of different physical phenomena. They enable scaling analysis, experimental design, and regime identification in catalytic systems.

Group	Definition	Physical Meaning	Ch.
Internal Transport			
Thiele modulus	$\phi = L\sqrt{\frac{k}{D_{\text{eff}}}}$	$\sqrt{\frac{\text{reaction rate}}{\text{diffusion rate}}}$	4
Weisz-Prater	$N_{\text{WP}} = \frac{r_{\text{obs}}L^2}{C_{As}D_{\text{eff}}}$	$\frac{\text{observed rate}}{\text{max diffusion rate}}$	4
External Transport			
Biot number (mass)	$\text{Bi}_m = \frac{k_c L}{D_{\text{eff}}}$	$\frac{\text{external transfer}}{\text{internal diffusion}}$	4
Damköhler I	$\text{Da}_I = \frac{k\tau}{1}$	$\frac{\text{reaction rate}}{\text{convective transport}}$	3
Damköhler II	$\text{Da}_{II} = \frac{kL^2}{D}$	$\frac{\text{reaction rate}}{\text{diffusive transport}}$	4
Heat Transport			
Prater number	$\beta_P = \frac{(-\Delta H_{\text{rxn}})D_{\text{eff}}C_{As}}{\lambda_{\text{eff}}T_s}$	$\frac{\text{max temp rise}}{\text{surface temperature}}$	4
Arrhenius group	$\gamma = \frac{E_a}{RT}$	$\frac{\text{activation energy}}{\text{thermal energy}}$	2
Fluid Mechanics			
Reynolds number	$\text{Re} = \frac{\rho v d_p}{\mu}$	$\frac{\text{inertial forces}}{\text{viscous forces}}$	4
Schmidt number	$\text{Sc} = \frac{\mu}{\rho D}$	$\frac{\text{momentum diffusivity}}{\text{mass diffusivity}}$	4
Sherwood number	$\text{Sh} = \frac{k_c d_p}{D}$	$\frac{\text{convective mass transfer}}{\text{diffusive transfer}}$	4

Table A.7: Important dimensionless groups in heterogeneous catalysis

A.5.1 Limiting Behavior of Key Dimensionless Groups

Understanding the limiting behavior of dimensionless groups helps identify controlling regimes:

- **Thiele modulus ϕ :**
 - $\phi \ll 1$: Kinetic control, $\eta \approx 1$, entire pellet at bulk concentration

- $\phi \gg 1$: Diffusion control, $\eta \approx 3/\phi$ (sphere), concentration drops to zero at center
- **Weisz–Prater criterion** N_{WP} :
 - $N_{WP} < 0.3$: Internal diffusion limitations negligible
 - $N_{WP} > 3$: Significant internal diffusion limitations
- **Biot number** Bi_m :
 - $Bi_m \gg 1$: External mass transfer fast, internal diffusion controls
 - $Bi_m \ll 1$: External mass transfer controls

A.6 Abbreviations and Acronyms

The following abbreviations and acronyms appear throughout this course. They are organized by category for convenient reference.

A.6.1 Reactor Types and Configurations

Abbreviation	Full Name
BR	Batch Reactor
CSTR	Continuous Stirred-Tank Reactor
PFR	Plug Flow Reactor
PBR	Packed-Bed Reactor
FBR	Fixed-Bed Reactor
FBCR	Fluidized Bed Catalytic Reactor
MR	Membrane Reactor
STR	Stirred Tank Reactor

A.6.2 Reaction Mechanisms

Abbreviation	Full Name and Description
LH	Langmuir–Hinshelwood (both reactants adsorb before reaction)
ER	Eley–Rideal (one adsorbed + one gas-phase reactant)
MVK	Mars–van Krevelen (redox mechanism with lattice oxygen)
LHHW	Langmuir–Hinshelwood–Hougen–Watson (general formalism)
RDS	Rate-Determining Step
QSSA	Quasi-Steady-State Approximation
PSSH	Pseudo-Steady-State Hypothesis
TST	Transition State Theory

A.6.3 Characterization Techniques

Abbreviation	Full Name and Application
BET	Brunauer–Emmett–Teller (surface area measurement)
BJH	Barrett–Joyner–Halenda (pore size distribution)
TPD	Temperature-Programmed Desorption
TPR	Temperature-Programmed Reduction
TPO	Temperature-Programmed Oxidation
XRD	X-ray Diffraction (crystal structure, crystallite size)
XPS	X-ray Photoelectron Spectroscopy (surface composition)
XAS	X-ray Absorption Spectroscopy
XANES	X-ray Absorption Near Edge Structure
EXAFS	Extended X-ray Absorption Fine Structure
TEM	Transmission Electron Microscopy
SEM	Scanning Electron Microscopy
AFM	Atomic Force Microscopy
STM	Scanning Tunneling Microscopy
FTIR	Fourier Transform Infrared Spectroscopy
DRIFTS	Diffuse Reflectance Infrared Fourier Transform Spectroscopy
NMR	Nuclear Magnetic Resonance
TGA	Thermogravimetric Analysis
DSC	Differential Scanning Calorimetry
ICP-OES	Inductively Coupled Plasma Optical Emission Spectroscopy
ICP-MS	Inductively Coupled Plasma Mass Spectrometry
EDX/EDS	Energy-Dispersive X-ray Spectroscopy

A.6.4 Performance Metrics

Abbreviation	Full Name and Definition
TOF	Turnover Frequency (reactions per site per time)
TON	Turnover Number (total reactions per site)
WHSV	Weight Hourly Space Velocity (mass flow / catalyst mass)
GHSV	Gas Hourly Space Velocity (volumetric flow / bed volume)
LHSV	Liquid Hourly Space Velocity
STY	Space-Time Yield (product per reactor volume per time)

A.6.5 Industrial Processes and Applications

Abbreviation	Full Name and Application
SCR	Selective Catalytic Reduction (NO_x removal with NH_3)
FCC	Fluid Catalytic Cracking (petroleum refining)
HDS	Hydrodesulfurization
HDN	Hydrodenitrogenation
WGS	Water–Gas Shift ($\text{CO} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + \text{H}_2$)
FTS	Fischer–Tropsch Synthesis
MTG	Methanol-to-Gasoline

A. Notation and Symbol Glossary

Abbreviation	Full Name and Application
MTO	Methanol-to-Olefins
TWC	Three-Way Catalyst (automotive emission control)
DOC	Diesel Oxidation Catalyst
DPF	Diesel Particulate Filter
ASC	Ammonia Slip Catalyst
VOC	Volatile Organic Compound

A.6.6 Materials and Structures

Abbreviation	Full Name
CNT	Carbon Nanotube
SWCNT	Single-Walled Carbon Nanotube
MWCNT	Multi-Walled Carbon Nanotube
MOF	Metal-Organic Framework
ZIF	Zeolitic Imidazolate Framework
PGM	Platinum Group Metals
TM	Transition Metal
HMS	Hexagonal Mesoporous Silica
SBA-15	Santa Barbara Amorphous type 15 (mesoporous silica)
MCM-41	Mobil Composition of Matter No. 41

A.7 Chapter-by-Chapter Symbol Introduction

This section provides a roadmap showing where each major symbol or concept is first introduced in the lecture notes. Use this table to locate definitions when encountering an unfamiliar symbol.

Chapter	Symbols Introduced	Key Concepts
Ch. 1	X, τ, r, θ, K, k	Baseline notation, units, Python basics
(Bridge)	C, P, T, R, t	
Ch. 2	θ_A, θ_*, K_A	Langmuir/competitive adsorption
(Computational Perspectives)	$k_{\text{ads}}, k_{\text{des}}, \Delta H_{\text{ads}}$ $k_s, r'', n, \text{LH, ER, MVK}$ $E_a, A, \Delta H^\ddagger, \Delta S^\ddagger, E_{\text{app}}$ k_B, h	LH, ER, MVK mechanisms Eyring TST, apparent E_a Arrhenius/Eyring parameters Fundamental constants
Ch. 3	V, F_A, v_0, Q, W	Reactor design variables
(Reactors & Charact.)	$\tau = V/v_0, \text{Da}$ CSTR, PFR, PBR $S_{\text{BET}}, \text{BJH, TOF, TON}$ TPD, TPR, TPO	Space time, Damköhler Reactor types Surface characterization Temperature-programmed methods

Continued on next page

Table A.14 – Continued from previous page

Chapter	Symbols Introduced	Key Concepts
Ch. 4 (Transport)	XRD, XPS, TEM	Structural/surface analysis
	D , D_{eff} , D_K	Diffusion coefficients
	ϕ (Thiele modulus)	Internal diffusion parameter
	η (effectiveness)	Rate penalty factor
	N_{WP} (Weisz–Prater)	Diagnostic criterion
	R_p , L , d_p	Particle geometry
	ε , τ_p (porosity, tortuosity)	Pore structure
	Bi, Sh, Re, Sc, k_c	Transport dimensionless groups
Ch. 5 (Selectivity)	S (selectivity), Y (yield)	Product distribution
	$S_{\text{B/C}}$, Y_{B}	Selectivity definitions
	Parallel, consecutive	Reaction network types
	τ_{opt}	Optimal residence time
Ch. 6 (Zeolites & CNTs)	Shape selectivity factors	Zeolite framework, acidity
	$D_{\text{CNT}} = F \cdot D_K$	Enhanced CNT diffusion
	(n, m) chiral indices	Nanotube classification

Table A.14: Symbol introduction by chapter

Remark A.3 (Building Your Notation Vocabulary). As you progress through the course, you will naturally build familiarity with these symbols. A useful study strategy is to create flashcards for new symbols as they appear, paying attention to both the symbol itself and its physical meaning. Understanding *what* a symbol represents physically—not just its mathematical definition—is the key to intuitive problem-solving.

For quick-reference equation cards covering adsorption isotherms, kinetic rate laws, temperature dependence, and effectiveness factors, see appendix B.

This concludes the Notation and Symbol Glossary. Return to this appendix whenever you encounter an unfamiliar symbol or need to verify standard notation.

Appendix B

Equation Summary

This appendix provides a comprehensive summary of all key equations from the course, organized by chapter for quick reference during problem solving and examinations. The six compiled chapters are: Bridge Module (no equations), Computational Perspectives on Surface Catalysis, Reactor Design and Catalyst Characterization, Transport Limitations, Selectivity Engineering, and Zeolites and Carbon Nanotubes in Catalysis. All equations use the standard notation defined in Chapter 1.

B.1 Computational Perspectives on Surface Catalysis

This section collects the key equations from adsorption isotherms, surface kinetic rate laws, and temperature dependence of rate constants.

B.1.1 Adsorption Isotherms

The following equations describe the equilibrium coverage of species on catalyst surfaces under various conditions.

Langmuir Isotherm (Single Component)

For a single adsorbate A in equilibrium with vacant sites:

$$\theta_A = \frac{K_A P_A}{1 + K_A P_A} \quad (\text{B.1})$$

Limiting cases:

$$\text{Low pressure } (KP \ll 1) : \theta \approx KP \quad (\text{Henry's law}) \quad (\text{B.2})$$

$$\text{High pressure } (KP \gg 1) : \theta \rightarrow 1 \quad (\text{saturation}) \quad (\text{B.3})$$

$$\text{Half-coverage: } \theta = 0.5 \text{ when } P = \frac{1}{K} \quad (\text{B.4})$$

The site balance for single-component adsorption:

$$\theta_A + \theta_* = 1 \quad (\text{B.5})$$

where θ_* represents the fraction of vacant sites.

Competitive Adsorption

For multiple species A, B, C, ... competing for the same sites:

$$\theta_A = \frac{K_A P_A}{1 + \sum_j K_j P_j} \quad (\text{B.6})$$

The general site balance becomes:

$$\sum_j \theta_j + \theta_* = 1 \quad (\text{B.7})$$

Two-component system (A and B):

$$\theta_A = \frac{K_A P_A}{1 + K_A P_A + K_B P_B} \quad (\text{B.8})$$

$$\theta_B = \frac{K_B P_B}{1 + K_A P_A + K_B P_B} \quad (\text{B.9})$$

Poisoning threshold:

The critical inhibitor pressure at which coverage drops significantly:

$$P_I^{\text{crit}} = \frac{1 + K_A P_A}{K_I} \quad (\text{B.10})$$

Dissociative Adsorption

For a diatomic molecule A_2 that dissociates upon adsorption:

$$\theta_A = \frac{\sqrt{K_{A_2} P_{A_2}}}{1 + \sqrt{K_{A_2} P_{A_2}}} \quad (\text{B.11})$$

This arises because dissociative adsorption requires two adjacent vacant sites:



BET Multilayer Adsorption

For physical adsorption allowing multiple layers:

$$\boxed{\frac{P}{V(P_0 - P)} = \frac{1}{V_m c} + \frac{c - 1}{V_m c} \cdot \frac{P}{P_0}} \quad (\text{B.13})$$

where:

- V = volume adsorbed at pressure P
- V_m = monolayer volume
- P_0 = saturation pressure
- c = BET constant related to heat of adsorption

The original BET equation:

$$\frac{V}{V_m} = \frac{c \cdot (P/P_0)}{(1 - P/P_0)[1 + (c - 1)(P/P_0)]} \quad (\text{B.14})$$

Surface area from BET analysis:

$$S_{\text{BET}} = \frac{V_m \cdot N_A \cdot \sigma_m}{V_{\text{mol}}} \quad (\text{B.15})$$

where σ_m is the molecular cross-sectional area (for N_2 : $\sigma_m = 0.162 \text{ nm}^2$).

Temperature Dependence of Adsorption

The van't Hoff relationship for equilibrium constant:

$$\boxed{K(T) = K_0 \exp\left(-\frac{\Delta H_{\text{ads}}}{RT}\right)} \quad (\text{B.16})$$

Linearized form:

$$\ln K = \ln K_0 - \frac{\Delta H_{\text{ads}}}{R} \cdot \frac{1}{T} \quad (\text{B.17})$$

For exothermic adsorption: $\Delta H_{\text{ads}} < 0$, so K decreases with temperature.

Temperature-Programmed Desorption (TPD)

The Redhead equation relating peak temperature to desorption energy:

$$\boxed{E_{\text{des}} \approx RT_p \left[\ln\left(\frac{\nu T_p}{\beta}\right) - 3.64 \right]} \quad (\text{B.18})$$

where:

- T_p = peak desorption temperature
- β = heating rate (K/s)
- ν = pre-exponential factor (typically 10^{13} s^{-1})

First-order desorption rate:

$$r_{\text{des}} = \nu \cdot \theta \cdot \exp\left(-\frac{E_{\text{des}}}{RT}\right) \quad (\text{B.19})$$

B.1.2 Surface Kinetic Rate Laws

Surface reactions follow different rate laws depending on the mechanism. The three main mechanisms are Langmuir–Hinshelwood (LH), Eley–Rideal (ER), and Mars–van Krevelen (MVK).

Langmuir–Hinshelwood Mechanism

Unimolecular Surface Reaction For a single adsorbed species reacting: $A^* \rightarrow \text{products}$

$$r = k\theta_A = \frac{kK_AP_A}{1 + K_AP_A} \quad (\text{B.20})$$

Limiting cases:

$$KP \ll 1 : \quad r \approx kKP \quad (\text{first-order in } P) \quad (\text{B.21})$$

$$KP \gg 1 : \quad r \approx k \quad (\text{zero-order in } P) \quad (\text{B.22})$$

Bimolecular Surface Reaction For two adsorbed species reacting: $A^* + B^* \rightarrow \text{products}$

$$r = k\theta_A\theta_B = \frac{kK_AK_BP_AP_B}{(1 + K_AP_A + K_BP_B)^2} \quad (\text{B.23})$$

This rate expression shows a **rate maximum** as a function of either partial pressure—a characteristic “fingerprint” of the LH mechanism.

LH with Dissociative Adsorption For A_2 dissociatively adsorbed reacting with molecularly adsorbed B:

$$r = \frac{k\sqrt{K_{A_2}P_{A_2}} \cdot K_BP_B}{(1 + \sqrt{K_{A_2}P_{A_2}} + K_BP_B)^2} \quad (\text{B.24})$$

LH with Product Inhibition When product C adsorbs and blocks sites:

$$r = \frac{kK_AK_BP_AP_B}{(1 + K_AP_A + K_BP_B + K_CP_C)^2} \quad (\text{B.25})$$

Eley–Rideal Mechanism

For gas-phase B reacting with adsorbed A: $B_{(g)} + A^* \rightarrow \text{products}$

$$r = k_{\text{ER}}P_B\theta_A = \frac{k_{\text{ER}}P_BK_AP_A}{1 + K_AP_A} \quad (\text{B.26})$$

Key distinction from LH:

- ER rate is first-order in gas-phase reactant B at all pressures
- ER rate saturates but does not decrease at high P_A
- No rate maximum observed (unlike LH bimolecular)

Mars–van Krevelen Mechanism

For oxidation reactions involving lattice oxygen:

$$r = \frac{k_{\text{red}} k_{\text{ox}} P_A P_{O_2}^n}{k_{\text{red}} P_A + k_{\text{ox}} P_{O_2}^n} \quad (\text{B.27})$$

where:

- k_{red} = rate constant for reduction (substrate oxidation)
- k_{ox} = rate constant for reoxidation by O_2
- n = reaction order in oxygen (typically 0.5 or 1)

This behaves like “two resistances in series”:

$$\frac{1}{r} = \frac{1}{k_{\text{red}} P_A} + \frac{1}{k_{\text{ox}} P_{O_2}^n} \quad (\text{B.28})$$

Power-Law Kinetics

Empirical power-law form (useful for fitting but lacks mechanistic insight):

$$r = k P_A^\alpha P_B^\beta \quad (\text{B.29})$$

where α and β are apparent reaction orders that may vary with conditions.

Rate Law Fingerprints Summary

Mechanism	Rate vs P_A	Identifying Feature
LH Bimolecular	Maximum	Inverse order at high P
ER	Saturation	First-order in gas-phase reactant
MVK	Saturation	Two-resistance behavior
Power Law	Monotonic	No saturation

B.1.3 Temperature Dependence

Arrhenius Equation

The temperature dependence of rate constants:

$$k = A \exp\left(-\frac{E_a}{RT}\right) \quad (\text{B.30})$$

Linearized form for data analysis:

$$\ln k = \ln A - \frac{E_a}{R} \cdot \frac{1}{T} \quad (\text{B.31})$$

From linear regression of $\ln k$ vs $1/T$:

$$\text{Slope} = -\frac{E_a}{R} \Rightarrow E_a = -R \times \text{slope} \quad (\text{B.32})$$

$$\text{Intercept} = \ln A \Rightarrow A = \exp(\text{intercept}) \quad (\text{B.33})$$

Eyring Equation (Transition State Theory)

From transition state theory:

$$k = \frac{k_B T}{h} \exp\left(\frac{\Delta S^\ddagger}{R}\right) \exp\left(-\frac{\Delta H^\ddagger}{RT}\right) \quad (\text{B.34})$$

Linearized form:

$$\ln\left(\frac{k}{T}\right) = \ln\left(\frac{k_B}{h}\right) + \frac{\Delta S^\ddagger}{R} - \frac{\Delta H^\ddagger}{R} \cdot \frac{1}{T} \quad (\text{B.35})$$

From linear regression of $\ln(k/T)$ vs $1/T$:

$$\text{Slope} = -\frac{\Delta H^\ddagger}{R} \quad (\text{B.36})$$

$$\text{Intercept} = \ln\left(\frac{k_B}{h}\right) + \frac{\Delta S^\ddagger}{R} \approx 23.76 + \frac{\Delta S^\ddagger}{R} \quad (\text{B.37})$$

Physical interpretation of $\Delta S^\ddagger$:

- $\Delta S^\ddagger < 0$: Tight, ordered transition state (associative)
- $\Delta S^\ddagger > 0$: Loose transition state (dissociative)
- $\Delta S^\ddagger \approx 0$: Little change in molecular freedom

Relationship Between Arrhenius and Eyring Parameters

$$E_a \approx \Delta H^\ddagger + RT \quad (\text{B.38})$$

At typical reaction temperatures ($T \approx 300\text{--}600$ K):

$$RT \approx 2.5\text{--}5 \text{ kJ/mol} \quad (\text{B.39})$$

Free energy of activation:

$$\Delta G^\ddagger = \Delta H^\ddagger - T\Delta S^\ddagger \quad (\text{B.40})$$

Apparent vs Intrinsic Activation Energy

For surface reactions, the observed (apparent) activation energy depends on the coverage regime:

$$E_a^{\text{app}} = -R \cdot \frac{d \ln r}{d(1/T)} \quad (\text{B.41})$$

For LH unimolecular reaction:

$$KP \ll 1 : E_a^{\text{app}} = E_a^{\text{int}} + \Delta H_{\text{ads}} \quad (\text{B.42})$$

$$KP \gg 1 : E_a^{\text{app}} = E_a^{\text{int}} \quad (\text{B.43})$$

Since $\Delta H_{\text{ads}} < 0$ for exothermic adsorption, the apparent activation energy is *lower* in the low-coverage regime.

B.2 Reactor Design and Catalyst Characterization

This section covers ideal reactor design equations and catalyst characterization techniques.

B.2.1 Ideal Reactor Design

Fundamental Definitions

Conversion:

$$X = \frac{C_{A,0} - C_A}{C_{A,0}} = \frac{F_{A,0} - F_A}{F_{A,0}} \quad (\text{B.44})$$

Space time:

$$\tau = \frac{V}{\dot{V}_0} = \frac{\text{Reactor volume}}{\text{Volumetric flow rate}} \quad (\text{B.45})$$

Concentration–conversion relationship (constant density):

$$C_A = C_{A,0}(1 - X) \quad (\text{B.46})$$

Batch Reactor

Design equation:

$$t = C_{A,0} \int_0^X \frac{dX'}{r(X')} \quad (\text{B.47})$$

For first-order kinetics ($r = kC_A$):

$$t = \frac{1}{k} \ln \left(\frac{1}{1 - X} \right) \quad (\text{B.48})$$

Continuous Stirred-Tank Reactor (CSTR)

Design equation (algebraic):

$$\tau = \frac{C_{A,0}X}{r(C_{A,\text{exit}})} \quad (\text{B.49})$$

First-order kinetics:

$$X = \frac{k\tau}{1 + k\tau} \quad (\text{B.50})$$

Rearranged:

$$\tau = \frac{X}{k(1 - X)} \quad (\text{B.51})$$

LH kinetics ($r = kKC_A/(1 + KC_A)$):

Requires solving:

$$\tau = \frac{C_{A,0}X(1 + KC_{A,0}(1 - X))}{kKC_{A,0}(1 - X)} \quad (\text{B.52})$$

Plug Flow Reactor (PFR)

Design equation (differential):

$$\frac{dX}{d\tau} = \frac{r(X)}{C_{A,0}} \quad (\text{B.53})$$

Integral form:

$$\tau = C_{A,0} \int_0^X \frac{dX'}{r(X')} \quad (\text{B.54})$$

First-order kinetics:

$$X = 1 - \exp(-k\tau) \quad (\text{B.55})$$

Second-order kinetics ($r = kC_A^2$):

$$\tau = \frac{1}{kC_{A,0}} \cdot \frac{X}{1 - X} \quad (\text{B.56})$$

CSTR vs PFR Comparison

For positive-order kinetics, PFR requires less volume than CSTR for the same conversion:

$$\frac{\tau_{\text{CSTR}}}{\tau_{\text{PFR}}} > 1 \quad (\text{for positive-order kinetics}) \quad (\text{B.57})$$

The ratio increases with increasing conversion and reaction order.

Turnover Frequency (TOF)

$$\boxed{\text{TOF} = \frac{r}{N_{\text{sites}}} = \frac{\text{molecules converted}}{\text{active site} \cdot \text{time}}} \quad (\text{B.58})$$

Units: s^{-1} or h^{-1}

Turnover number (TON):

$$\text{TON} = \text{TOF} \times t = \frac{\text{total molecules converted}}{\text{active site}} \quad (\text{B.59})$$

B.2.2 Catalyst Characterization

BET Surface Area Analysis

Linear transform (see Eq. (B.13)):

$$\frac{P}{V(P_0 - P)} = \frac{1}{V_m c} + \frac{c - 1}{V_m c} \cdot \frac{P}{P_0} \quad (\text{B.60})$$

From linear regression of $P/[V(P_0 - P)]$ vs P/P_0 :

$$\text{Slope} = s = \frac{c - 1}{V_m c} \quad (\text{B.61})$$

$$\text{Intercept} = i = \frac{1}{V_m c} \quad (\text{B.62})$$

Solving for V_m and c :

$$\boxed{V_m = \frac{1}{s + i}, \quad c = \frac{s}{i} + 1} \quad (\text{B.63})$$

Scherrer Equation (XRD)

Crystallite size from XRD peak broadening:

$$\boxed{D = \frac{K\lambda}{\beta \cos \theta}} \quad (\text{B.64})$$

where:

- D = crystallite size (nm)
- K = shape factor (≈ 0.9 for spherical particles)
- λ = X-ray wavelength (Cu $K\alpha$: 0.154 nm)
- β = full width at half maximum (FWHM) in radians
- θ = Bragg angle

Metal Dispersion

$$D = \frac{N_{\text{surface}}}{N_{\text{total}}} = \frac{\text{surface metal atoms}}{\text{total metal atoms}} \quad (\text{B.65})$$

For spherical particles:

$$D \approx \frac{6V_{\text{atom}}}{d \cdot A_{\text{atom}}} \approx \frac{1}{d/d_0} \quad (\text{B.66})$$

where d is the particle diameter and d_0 is a metal-specific constant.

B.3 Transport Limitations

B.3.1 Thiele Modulus

The Thiele modulus compares intrinsic reaction rate to diffusion rate.

First-order kinetics:

$$\phi = L \sqrt{\frac{k}{D_{\text{eff}}}} \quad (\text{B.67})$$

where L is the characteristic length.

Characteristic lengths for different geometries:

Geometry	Char. Length L	Thiele Modulus
Slab (half-thickness)	L	$\phi = L \sqrt{k/D_{\text{eff}}}$
Cylinder (radius)	$R/2$	$\phi = (R/2) \sqrt{k/D_{\text{eff}}}$
Sphere (radius)	$R/3$	$\phi = (R/3) \sqrt{k/D_{\text{eff}}}$

For spherical pellets with radius R_p :

$$\phi_s = R_p \sqrt{\frac{k}{D_{\text{eff}}}} \quad (\text{B.68})$$

For n th-order kinetics:

$$\phi_n = L \sqrt{\frac{n+1}{2} \frac{kC_{A,s}^{n-1}}{D_{\text{eff}}}} \quad (\text{B.69})$$

B.3.2 Effectiveness Factor

The effectiveness factor relates observed rate to intrinsic rate:

$$\eta = \frac{r_{\text{obs}}}{r_{\text{intrinsic}}} = \frac{\text{actual rate}}{\text{rate at surface conditions}} \quad (\text{B.70})$$

Slab geometry (first-order):

$$\boxed{\eta = \frac{\tanh \phi}{\phi}} \quad (\text{B.71})$$

Spherical geometry (first-order):

$$\boxed{\eta = \frac{3}{\phi^2} (\phi \coth \phi - 1)}$$
(B.72)

Alternative form using the spherical Thiele modulus $\phi_s = R_p \sqrt{k/D_{\text{eff}}}$:

$$\eta = \frac{3}{\phi_s} \left(\frac{1}{\tanh \phi_s} - \frac{1}{\phi_s} \right) \quad (\text{B.73})$$

Limiting cases:

$$\phi \ll 1 : \quad \eta \rightarrow 1 \quad (\text{kinetic control}) \quad (\text{B.74})$$

$$\phi \gg 1 : \quad \eta \rightarrow \frac{1}{\phi} \text{ (slab)}, \frac{3}{\phi} \text{ (sphere)} \quad (\text{diffusion control}) \quad (\text{B.75})$$

B.3.3 Weisz–Prater Criterion

Observable modulus for diagnosing diffusion limitations:

$$\boxed{N_{WP} = \Phi_{\text{obs}} = \frac{r_{\text{obs}} \cdot R_p^2}{C_{A,s} \cdot D_{\text{eff}}} = \eta \cdot \phi^2} \quad (\text{B.76})$$

Criterion:

$$N_{WP} < 0.3 : \quad \text{No significant diffusion limitation} \quad (\text{B.77})$$

$$N_{WP} > 3 : \quad \text{Strong diffusion limitation} \quad (\text{B.78})$$

B.3.4 Effective Diffusivity

$$\boxed{D_{\text{eff}} = \frac{\varepsilon_p}{\tau_p} D} \quad (\text{B.79})$$

where:

- ε_p = pellet porosity (typically 0.3–0.5)
- τ_p = tortuosity factor (typically 2–6)
- D = bulk or Knudsen diffusivity

Combined diffusivity (molecular and Knudsen in series):

$$\frac{1}{D} = \frac{1}{D_{\text{mol}}} + \frac{1}{D_K} \quad (\text{B.80})$$

B.3.5 Knudsen Diffusion

When pore diameter is smaller than mean free path:

$$D_K = \frac{d_p}{3} \sqrt{\frac{8RT}{\pi M}} \quad (\text{B.81})$$

where:

- d_p = pore diameter
- M = molecular weight
- R = gas constant (8.314 J/(mol K))

Simplified form:

$$D_K = 48.5 \cdot d_p \sqrt{\frac{T}{M}} \quad [\text{m}^2/\text{s}, \text{nm}, \text{K}, \text{g/mol}] \quad (\text{B.82})$$

B.3.6 External Mass Transfer

Mass transfer coefficient correlation (Sherwood number):

$$Sh = \frac{k_m d_p}{D} = 2 + 0.6 Re^{1/2} Sc^{1/3} \quad (\text{B.83})$$

External effectiveness factor:

$$\eta_{\text{ext}} = \frac{C_{A,s}}{C_{A,b}} \quad (\text{B.84})$$

Overall effectiveness:

$$\eta_{\text{overall}} = \eta_{\text{ext}} \cdot \eta_{\text{int}} \quad (\text{B.85})$$

B.4 Selectivity Engineering

B.4.1 Definitions

Conversion of A:

$$X_A = \frac{C_{A,0} - C_A}{C_{A,0}} \quad (\text{B.86})$$

Yield of B (from A):

$$Y_B = \frac{C_B}{C_{A,0}} \quad (\text{B.87})$$

Overall selectivity to B:

$$S_{B/A} = \frac{Y_B}{X_A} \quad (\text{B.88})$$

Instantaneous selectivity (rate ratio):

$$s_{B/C} = \frac{r_B}{r_C} \quad (\text{B.89})$$

Relationship:

$$Y_B = S_{B/A} \times X_A \quad (\text{B.90})$$

B.4.2 Parallel Reactions

For the reaction scheme:



First-order parallel reactions:

$$\boxed{S_{B/C} = \frac{k_1}{k_2}} \quad (\text{B.92})$$

Selectivity is independent of conversion and reactor type for first-order parallel reactions.

Different order parallel reactions:

If $r_1 = k_1 C_A^{n_1}$ and $r_2 = k_2 C_A^{n_2}$:

$$S_{B/C} = \frac{k_1}{k_2} C_A^{n_1 - n_2} \quad (\text{B.93})$$

Implications:

- If $n_1 > n_2$: High C_A favors B (use PFR)
- If $n_1 < n_2$: Low C_A favors B (use CSTR)

B.4.3 Consecutive Reactions

For the reaction scheme:



PFR concentration profiles (first-order):

$$C_A = C_{A,0} \exp(-k_1 \tau) \quad (\text{B.95})$$

$$C_B = C_{A,0} \frac{k_1}{k_2 - k_1} [\exp(-k_1 \tau) - \exp(-k_2 \tau)] \quad (\text{B.96})$$

$$C_C = C_{A,0} - C_A - C_B \quad (\text{B.97})$$

Optimal space time for maximum B:

$$\boxed{\tau^* = \frac{\ln(k_2/k_1)}{k_2 - k_1}} \quad (\text{B.98})$$

Maximum yield of B:

$$Y_{B,\max} = \left(\frac{k_1}{k_2}\right)^{k_2/(k_2-k_1)} \quad (\text{B.99})$$

Conversion at maximum yield:

$$X^* = 1 - \left(\frac{k_1}{k_2}\right)^{k_1/(k_2-k_1)} \quad (\text{B.100})$$

B.4.4 Temperature Effects on Selectivity

Using Arrhenius expressions for competing reactions:

$$\frac{k_1}{k_2} = \frac{A_1}{A_2} \exp\left(-\frac{E_a^{(1)} - E_a^{(2)}}{RT}\right) \quad (\text{B.101})$$

Implications:

- If $E_a^{(1)} > E_a^{(2)}$: Higher T favors reaction 1
- If $E_a^{(1)} < E_a^{(2)}$: Lower T favors reaction 1

B.5 Zeolites and Carbon Nanotubes in Catalysis

This section combines the key equations for microporous zeolite catalysts and carbon nanotube catalyst supports.

B.5.1 Zeolites and Microporous Materials

Shape Selectivity

For zeolite-catalyzed reactions, shape selectivity arises from molecular sieving:

- **Reactant selectivity:** Only molecules smaller than pore openings enter
- **Product selectivity:** Only products smaller than pores can exit
- **Transition state selectivity:** Only reactions with sufficiently small TS occur

Configurational Diffusion

In zeolite micropores (pore diameter $\approx$ molecular diameter):

$$D_{\text{config}} = D_0 \exp\left(-\frac{E_D}{RT}\right) \quad (\text{B.102})$$

where E_D is the activation energy for diffusion (typically 10–50 kJ/mol).

The diffusivity is strongly temperature-dependent and much smaller than Knudsen diffusion:

$$D_{\text{config}} \ll D_K \ll D_{\text{mol}} \quad (\text{B.103})$$

Zeolite Effectiveness Factor

For zeolite crystals of characteristic size L :

$$\phi_{\text{zeolite}} = L \sqrt{\frac{k}{D_{\text{config}}}} \quad (\text{B.104})$$

Slow configurational diffusion means zeolites typically operate in the diffusion-limited regime, even for small crystals.

Acidity and Si/Al Ratio

Number of Brønsted acid sites per unit cell:

$$N_{\text{acid}} \propto \frac{1}{\text{Si/Al}} \quad (\text{B.105})$$

Higher Si/Al ratio $\Rightarrow$ fewer but stronger acid sites.

B.5.2 Carbon Nanotubes

Enhanced Knudsen Diffusion

In CNT channels, specular reflection enhances diffusion:

$$\boxed{D_{\text{CNT}} = F \cdot D_K} \quad (\text{B.106})$$

where F is the enhancement factor (can be 10–10,000 times).
Standard Knudsen diffusivity (for comparison):

$$D_K = \frac{d_{\text{CNT}}}{3} \sqrt{\frac{8RT}{\pi M}} \quad (\text{B.107})$$

CNT Effectiveness Factor

For reactions in CNT channels:

$$\phi_{\text{CNT}} = L_{\text{CNT}} \sqrt{\frac{k}{D_{\text{CNT}}}} \quad (\text{B.108})$$

Due to enhanced diffusion ($D_{\text{CNT}} \gg D_K$), CNTs can achieve higher effectiveness factors than conventional mesoporous catalysts.

Effectiveness Factor Comparison

For the same reaction and length scale:

$$\frac{\eta_{\text{CNT}}}{\eta_{\text{conv}}} \approx \sqrt{F} \quad (\text{in diffusion-limited regime}) \quad (\text{B.109})$$

For physical constants, unit conversion tables, and molecular property data, see appendix C. This equation summary is organized to follow the course chapters; for detailed derivations and worked examples, refer to the corresponding sections in the main text.

Appendix C

Physical Constants and Unit Conversions

This appendix provides a reference for physical constants and unit conversions commonly encountered in surface phenomena and catalysis.

C.1 Fundamental Physical Constants

Table C.1 lists the fundamental physical constants most frequently used in heterogeneous catalysis calculations. Pay particular attention to the gas constant R , which appears in multiple unit systems depending on whether you are working with energy, pressure-volume, or thermodynamic calculations.

Remark C.1 (Relationship between R and k_B). The gas constant and Boltzmann constant are related through Avogadro's number:

$$R = N_A \cdot k_B \quad (\text{C.1})$$

This means R gives energy per *mole* of molecules, while k_B gives energy per *single* molecule. When working with activation energies, you must be consistent: use R with kJ mol^{-1} or k_B with eV.

Remark C.2 (Always Use Kelvin in Rate Equations). The Arrhenius equation and all thermodynamic relationships require *absolute* temperature in Kelvin. A common error is using Celsius, which gives nonsensical results. For example, at 25°C :

- Correct: $RT = 8.314 \text{ J mol}^{-1} \text{ K}^{-1} \times 298.15 \text{ K} = 2479 \text{ J mol}^{-1}$
- **Wrong:** $RT = 8.314 \text{ J mol}^{-1} \text{ K}^{-1} \times 25 = 208 \text{ J mol}^{-1}$ ($12\times$ too small!)

Table C.1: Fundamental physical constants for catalysis calculations.

Constant	Symbol	Value	Units
<i>Gas constant in various units</i>			
Gas constant	R	8.314	$\text{J mol}^{-1} \text{K}^{-1}$
		8.314×10^{-2}	$\text{bar L mol}^{-1} \text{K}^{-1}$
		8.206×10^{-2}	$\text{atm L mol}^{-1} \text{K}^{-1}$
		1.987	$\text{cal mol}^{-1} \text{K}^{-1}$
		8.314	$\text{Pa m}^3 \text{mol}^{-1} \text{K}^{-1}$
<i>Other fundamental constants</i>			
Boltzmann constant	k_{B}	1.381×10^{-23}	J K^{-1}
		8.617×10^{-5}	eV K^{-1}
Planck constant	h	6.626×10^{-34}	J s
		4.136×10^{-15}	eV s
Avogadro number	N_{A}	6.022×10^{23}	mol^{-1}
Faraday constant	F	96 485	C mol^{-1}
Speed of light	c	2.998×10^8	m s^{-1}
Standard gravity	g	9.807	m s^{-2}

C.2 Energy Conversions

For molar quantities (which you will use most frequently):

Table C.2: Energy conversions for molar quantities (per mole of substance).

Conversion	Factor	Notes
$\text{kJ mol}^{-1} \rightarrow \text{kcal mol}^{-1}$	$\div 4.184$	Thermochemistry
$\text{kcal mol}^{-1} \rightarrow \text{kJ mol}^{-1}$	$\times 4.184$	Thermochemistry
$\text{kJ mol}^{-1} \rightarrow \text{eV per molecule}$	$\div 96.485$	Surface science
$\text{eV per molecule} \rightarrow \text{kJ mol}^{-1}$	$\times 96.485$	Surface science
$\text{J mol}^{-1} \rightarrow \text{cm}^{-1}$	$\div 11.96$	Spectroscopy

Example C.1: Converting Activation Energy Units

Problem: A research paper reports an activation energy of $E_a = 15.2 \text{ kcal mol}^{-1}$ for a catalytic reaction. Convert this to:
(a) kJ mol^{-1} (SI units)
(b) eV per molecule (for comparison with DFT calculations)

Solution (Direct conversion):

(a) **Conversion to kJ mol^{-1} :**

$$E_a = 15.2 \text{ kcal mol}^{-1} \times \frac{4.184 \text{ kJ}}{1 \text{ kcal}} = 63.6 \text{ kJ mol}^{-1}$$

(b) **Conversion to eV per molecule:**

$$E_a = \frac{63.6 \text{ kJ mol}^{-1}}{96.485 \text{ kJ mol}^{-1} \text{ eV}^{-1}} = 0.659 \text{ eV}$$

Alternatively, in one step from kcal:

$$E_a = 15.2 \text{ kcal mol}^{-1} \times \frac{4.184 \text{ kJ}}{1 \text{ kcal}} \times \frac{1 \text{ eV}}{96.485 \text{ kJ mol}^{-1}} = 0.659 \text{ eV}$$

Answer:

(a) 63.6 kJ mol^{-1} ; (b) 0.659 eV

Physical interpretation: The value of approximately 0.66 eV is typical for moderately activated catalytic processes. DFT calculations often report barriers in eV, while experimental papers may use kJ/mol or kcal/mol —being able to convert quickly helps compare theory and experiment.

C.3 Pressure Unit Conversions

Pressure units vary widely across the catalysis literature. Older literature uses Torr or atm, while modern practice favors Pa or bar. Table C.3 provides conversion factors.

Table C.3: Pressure unit conversion factors.

	Pa	kPa	bar	Torr	psi
1 Pa	1	0.001	10^{-5}	7.501×10^{-3}	1.450×10^{-4}
1 kPa	1000	1	0.01	7.501	0.1450
1 bar	100 000	100	1	750.1	14.50
1 atm	101 325	101.325	1.01325	760	14.70
1 Torr	133.32	0.13332	1.333×10^{-3}	1	1.934×10^{-2}
1 psi	6894.8	6.8948	6.895×10^{-2}	51.72	1

Remark C.3 (Standard Conditions). Be aware of different “standard” conditions in use:

- **STP (old):** $T = 273.15 \text{ K}$ (0°C), $P = 1 \text{ atm} = 101.325 \text{ kPa}$
- **STP (IUPAC, 1982):** $T = 273.15 \text{ K}$, $P = 1 \text{ bar} = 100 \text{ kPa}$
- **Standard state (thermodynamics):** $T = 298.15 \text{ K}$ (25°C), $P = 1 \text{ bar}$
- **NTP (industry):** $T = 293.15 \text{ K}$ (20°C), $P = 1 \text{ atm}$

Always check which standard is being used, especially when reporting volumetric flow rates.

C.4 Flow Rate Notes

Remark C.4 (SCCM and SLPM—“Standard” Flow Rates). SCCM (standard cubic centimeters per minute) and SLPM (standard liters per minute) report volumetric flow rates corrected to standard conditions (typically 0°C , 1 atm). These are useful because they represent *molar* flow rates regardless of actual temperature and pressure:

$$\dot{n} = \frac{P_{\text{std}} \dot{V}_{\text{std}}}{RT_{\text{std}}}$$

Mass flow controllers often display SCCM even when the actual gas flow (at reactor conditions) has a different volume.

C.5 Diffusivity Units

Diffusivities appear in the Thiele modulus and effectiveness factor calculations. The SI unit is $\text{m}^2 \text{s}^{-1}$, but older literature often uses $\text{cm}^2 \text{s}^{-1}$.

$$1 \text{ cm}^2 \text{s}^{-1} = 1 \times 10^{-4} \text{ m}^2 \text{s}^{-1} \quad (\text{C.2})$$

C.5.1 Typical Diffusivity Values

Table C.4: Typical diffusivity values for catalysis calculations.

System	D ($\text{m}^2 \text{s}^{-1}$)	Notes
<i>Bulk diffusion (gases, 1 atm)</i>		
Small molecules in N_2 (25 °C)	1×10^{-5}	$\text{H}_2, \text{CO}, \text{O}_2$
Small molecules in N_2 (400 °C)	5×10^{-5}	$D \propto T^{1.75}$
<i>Knudsen diffusion (porous media)</i>		
10 nm pores, 400 °C	1×10^{-6}	$D_K \propto r_p$
1 nm pores (zeolite), 400 °C	1×10^{-7}	Micropore regime
<i>Effective diffusivity in pellets</i>		
Typical catalyst pellet	1×10^{-6}	$D_{\text{eff}} = D\varepsilon/\tau_p$
Zeolite crystal	10^{-8} – 10^{-10}	Configurational diffusion
<i>Liquid-phase diffusion (25 °C)</i>		
Small molecules in water	1×10^{-9}	1000× slower than gas
Small molecules in organic	1×10^{-9}	Similar to water

Remark C.5 (Knudsen Diffusion Formula). When pore diameters are comparable to the mean free path of gas molecules, Knudsen diffusion dominates:

$$D_K = \frac{d_{\text{pore}}}{3} \sqrt{\frac{8RT}{\pi M}} \quad (\text{C.3})$$

where d_{pore} is the pore diameter and M is the molar mass. This explains why diffusion in microporous catalysts like zeolites can be 10–100 times slower than bulk gas diffusion.

C.6 Surface Area and Pore Dimensions

C.6.1 IUPAC Pore Classification

C.6.2 Surface Area Units

BET surface areas are reported in $\text{m}^2 \text{g}^{-1}$. Typical values:

Table C.5: IUPAC classification of pore sizes.

Classification	Pore Diameter	Examples
Micropores	< 2 nm	Zeolites, activated carbon
Mesopores	2 nm–50 nm	MCM-41, SBA-15
Macropores	> 50 nm	Alumina supports, glass frits

Table C.6: Typical BET surface areas for common catalysts and supports.

Material	S_{BET} ($\text{m}^2 \text{g}^{-1}$)	Notes
γ -Alumina	200	Common support
Silica gel	500	Amorphous
Activated carbon	1000	Highly porous
Zeolite Y	700	FAU framework
ZSM-5	400	MFI framework
MCM-41	1000	Ordered mesoporous
Metal nanoparticles	50	Depends on size

Table C.7: Molecular weights and kinetic diameters of common gases.

Molecule	M (g mol^{-1})	σ (nm)	Notes
H ₂	2.02	0.29	Smallest, fastest diffusion
He	4.00	0.26	Inert tracer
N ₂	28.01	0.36	Air component, inert carrier
O ₂	32.00	0.35	Oxidant
Ar	39.95	0.34	Inert carrier
CO	28.01	0.38	Probe molecule, poison
CO ₂	44.01	0.33	Product, greenhouse gas
CH ₄	16.04	0.38	Smallest hydrocarbon
NH ₃	17.03	0.26	SCR reductant, basic probe
H ₂ O	18.02	0.27	Product, can poison
C ₂ H ₄	28.05	0.39	Ethylene
C ₂ H ₆	30.07	0.44	Ethane
C ₆ H ₆	78.11	0.59	Benzene
NO	30.01	0.32	SCR target pollutant
NO ₂	46.01	0.34	Fast SCR component
SO ₂	64.07	0.36	Catalyst poison

C.7 Commonly Needed Molecular Properties

Remark C.6 (Kinetic Diameter and Shape Selectivity). The kinetic diameter σ determines whether a molecule can enter a micropore. For zeolite ZSM-5 with pore openings of approximately 0.55 nm, linear alkanes ($\sigma \approx 0.43$ nm) can enter, while branched isomers ($\sigma \approx 0.62$ nm) cannot. This is the basis for *shape-selective catalysis*.

C.8 Quick Reference: Conversion Examples

Example C.2: Multi-Step Unit Conversion in Kinetics

Problem: The adsorption enthalpy for CO on Pt(111) is reported as -1.4 eV per molecule. You need to use this value in a van't Hoff equation where $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$. Express ΔH_{ads} in kJ mol^{-1} and calculate K_{350}/K_{300} , the ratio of equilibrium constants at 350 K and 300 K.

Solution:

Step 1: Convert eV to kJ mol^{-1} .

$$\Delta H_{\text{ads}} = -1.4 \text{ eV} \times 96.485 \text{ kJ mol}^{-1} \text{ eV}^{-1} = -135 \text{ kJ mol}^{-1}$$

Step 2: Apply the van't Hoff equation.

$$\ln \left(\frac{K_2}{K_1} \right) = \frac{-\Delta H}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

Substituting values:

$$\begin{aligned} \ln \left(\frac{K_{350}}{K_{300}} \right) &= \frac{-(-135\,000 \text{ J mol}^{-1})}{8.314 \text{ J mol}^{-1} \text{ K}^{-1}} \left(\frac{1}{350 \text{ K}} - \frac{1}{300 \text{ K}} \right) \\ &= 16\,240 \text{ K} \times (-4.76 \times 10^{-4} \text{ K}^{-1}) \\ &= -7.73 \end{aligned}$$

Therefore:

$$\frac{K_{350}}{K_{300}} = e^{-7.73} = 4.4 \times 10^{-4}$$

Answer:

$$\Delta H_{\text{ads}} = -135 \text{ kJ mol}^{-1}; K_{350}/K_{300} = 4.4 \times 10^{-4}$$

Physical interpretation: The equilibrium constant for adsorption decreases by more than three orders of magnitude when temperature increases from 300 K to 350 K. This dramatic decrease reflects the exothermic nature of adsorption: higher temperatures strongly disfavor the adsorbed state, leading to lower coverage at a given pressure.

C.9 Summary of Most Common Conversions

For quick reference, here are the conversions you will use most frequently in this course:

Table C.8: Most frequently used unit conversions in surface phenomena and catalysis.

Conversion	Factor
<i>Energy</i>	
$\text{kJ mol}^{-1} \leftrightarrow \text{kcal mol}^{-1}$	$\div$ or $\times 4.184$
$\text{kJ mol}^{-1} \leftrightarrow \text{eV}$	$\div$ or $\times 96.485$
<i>Pressure</i>	
$\text{bar} \leftrightarrow \text{atm}$	$1 \text{ bar} \approx 0.987 \text{ atm}$
$\text{bar} \leftrightarrow \text{kPa}$	$1 \text{ bar} = 100 \text{ kPa}$
$\text{Torr} \leftrightarrow \text{mbar}$	$1 \text{ Torr} \approx 1.333 \text{ mbar}$
<i>Temperature</i>	
$\text{K} \leftrightarrow ^\circ\text{C}$	$T[\text{K}] = T[^\circ\text{C}] + 273.15$
<i>Diffusivity</i>	
$\text{cm}^2 \text{s}^{-1} \leftrightarrow \text{m}^2 \text{s}^{-1}$	$\times$ or $\div 10^{-4}$
<i>Concentration</i>	
$\text{mol L}^{-1} \leftrightarrow \text{mol m}^{-3}$	$\times$ or $\div 1000$
<i>Length</i>	
$\text{\AA} \leftrightarrow \text{nm}$	$1 \text{ nm} = 10 \text{ \AA}$

Appendix D

Catalyst Characterization Quick Reference

This appendix provides practical quick-reference information for the six characterization techniques covered in Chapter 3. Use this as a laboratory companion when planning experiments or interpreting data.

D.1 BET Surface Area Analysis

D.1.1 Sample Preparation Guidelines

Proper sample preparation is critical for accurate BET measurements. The degassing step removes physisorbed water and other contaminants that would otherwise interfere with nitrogen adsorption.

Table D.1: Recommended degassing conditions for common catalyst types.

Catalyst Type	Temperature (°C)	Time (h)
Alumina, silica	200–300	2–4
Zeolites (hydrophilic)	300–350	4–8
Carbon materials	150–200	2–4
Metal oxides (reducible)	150–200	2–4
Supported noble metals	150–200	2–3
Polymers, MOFs	80–120	4–12

Remark D.1 (Avoid thermal damage). Always verify that your degassing temperature is below the decomposition or phase transition temperature of your catalyst. For temperature-sensitive materials like MOFs or polymers, use extended degassing times at lower tempera-

tures rather than high temperatures.

D.1.2 Data Quality Indicators

The BET C constant (see appendix B) provides insight into measurement quality:

Table D.2: Interpretation of the BET C constant.

C Value	Interpretation
$C < 2$	Invalid—negative intercept in BET plot
$2 < C < 50$	Weak interaction; check for micropores
$50 < C < 200$	Ideal range —reliable measurement
$C > 200$	Strong interaction; possible micropore filling

D.1.3 Common Pitfalls

1. **Micropore correction:** Standard BET analysis assumes multilayer formation on flat surfaces. For microporous materials (zeolites, activated carbons), use the t -plot or α_s -plot method to separate micropore volume from external surface area.
2. **Water adsorption:** Hygroscopic samples can readsorb water during transfer. Use a glove box or quick-transfer techniques for highly hygroscopic materials.
3. **Relative pressure range:** The standard BET range ($P/P_0 = 0.05$ – 0.30) may not apply to all materials. For microporous samples, use the Rouquerol consistency criteria to identify the valid range.
4. **Sample mass:** Ensure sufficient sample mass (typically 50–200 mg) to provide adequate surface area (ideally $> 5 \text{ m}^2$) for precise measurements.

D.1.4 Expected Surface Area Ranges

D.2 Temperature-Programmed Desorption (TPD)

D.2.1 Experimental Setup

A typical TPD experiment involves:

1. **Pretreatment:** Heat sample under inert gas to remove contaminants
2. **Adsorption:** Expose to probe molecule at low temperature until saturation
3. **Purge:** Remove gas-phase and weakly physisorbed species

Table D.3: Typical BET surface areas for common catalyst classes.

Material Class	Surface Area (m ² /g)
Non-porous metal oxides (TiO ₂ , ZnO)	10–50
Porous metal oxides (Al ₂ O ₃ , SiO ₂)	100–300
Zeolites	300–800
Mesoporous silica (MCM-41, SBA-15)	500–1200
Activated carbons	500–2000
Metal-organic frameworks (MOFs)	1000–6000
Carbon nanotubes (as-synthesized)	200–400

4. **Temperature ramp:** Linear heating while monitoring desorbed species
5. **Detection:** Mass spectrometry (MS) or thermal conductivity detector (TCD)

D.2.2 Heating Rate Selection

The choice of heating rate β involves trade-offs:

Table D.4: Effects of heating rate on TPD analysis.

Heating Rate	Resolution	Sensitivity	Mass Transfer
Slow (1–5 K/min)	Excellent	Lower	Minimal
Moderate (5–10 K/min)	Good	Good	Low
Fast (10–20 K/min)	Poor	Higher	Significant

Remark D.2 (Peak shift with heating rate). The peak temperature T_p shifts to higher values with increasing heating rate. This is not an artifact—it is predicted by the Redhead equation. Use consistent heating rates when comparing samples, or explicitly account for the shift in your analysis.

D.2.3 Redhead Analysis Quick Reference

Apply the Redhead equation (Eq. (B.18)) with $\nu = 10^{13} \text{ s}^{-1}$ and your measured T_p and β values.

Remark D.3 (Uncertainty in pre-exponential factor). The assumed value of $\nu = 10^{13} \text{ s}^{-1}$ introduces uncertainty of approximately $\pm 10 \text{ kJ/mol}$ in E_{des} . For more accurate values, perform TPD at mul-

multiple heating rates and use the complete analysis (plotting $\ln(T_p^2/\beta)$ vs. $1/T_p$).

D.2.4 Multiple Peak Interpretation

Multiple peaks in a TPD spectrum indicate:

- **Different binding sites:** Distinct adsorption geometries or surface facets
- **Different coverage regimes:** Lateral interactions causing coverage-dependent E_{des}
- **Decomposition:** Sequential desorption of fragments from a decomposing adsorbate
- **Diffusion effects:** Artifacts from readsorption in porous materials

D.2.5 Typical Desorption Temperatures

Table D.5: Characteristic desorption temperatures for common adsorbate/-surface pairs ($\beta = 10 \text{ K/min}$).

Adsorbate	Surface	T_p (K)	E_{des} (kJ/mol)
CO	Pt(111)	400–480	100–130
CO	Pd(111)	450–520	120–150
H ₂	Pt(111)	300–350	70–85
H ₂	Ni(111)	350–400	85–100
NH ₃	Pt/Al ₂ O ₃	400–550	100–140
H ₂ O	Al ₂ O ₃	350–450	80–110
Pyridine (Brønsted)	H-ZSM-5	570–670	150–180
Pyridine (Lewis)	H-ZSM-5	420–520	100–140

D.3 X-Ray Diffraction (XRD)

D.3.1 Sample Preparation

1. **Particle size:** Grind to $< 50 \mu\text{m}$ for representative sampling
2. **Loading:** Press powder gently into sample holder; avoid preferred orientation
3. **Surface:** Ensure flat, level surface using glass slide
4. **Amount:** Fill holder completely for proper peak intensities

Remark D.4 (Preferred orientation). Plate-like or needle-shaped crystallites may align during sample preparation, causing intensity distortions. For quantitative phase analysis, use side-loading or back-loading techniques to minimize this effect.

D.3.2 Peak Identification Procedure

1. Identify the strongest peaks in your pattern
2. Compare 2θ positions with reference databases (ICDD PDF, COD)
3. Match relative intensities (not just peak positions)
4. Check for systematic presence/absence of peaks (space group)
5. Verify with additional characterization if uncertain

D.3.3 Scherrer Equation for Crystallite Size

Use the Scherrer equation (Eq. (B.64)) with these guidelines:

Remark D.5 (Scherrer equation limitations). The Scherrer equation is valid only for crystallite sizes in the range 2–100 nm. Below 2 nm, peaks become too broad to analyze reliably. Above 100 nm, instrumental broadening dominates. Also note that L represents coherent domain size, which may differ from particle size measured by TEM.

D.3.4 Common Reference Patterns

Table D.6: Principal XRD peaks for common catalyst phases (Cu $K\alpha$ radiation).

Phase	2θ (deg)	(hkl)	d -spacing (Å)
Pt (fcc)	39.8, 46.2, 67.5	(111), (200), (220)	2.26, 1.96, 1.38
Pd (fcc)	40.1, 46.7, 68.1	(111), (200), (220)	2.25, 1.94, 1.37
Au (fcc)	38.2, 44.4, 64.6	(111), (200), (220)	2.35, 2.04, 1.44
Ni (fcc)	44.5, 51.8, 76.4	(111), (200), (220)	2.03, 1.76, 1.25
γ -Al ₂ O ₃	37.6, 45.9, 67.0	(311), (400), (440)	2.39, 1.97, 1.40
TiO ₂ (anatase)	25.3, 37.8, 48.0	(101), (004), (200)	3.52, 2.38, 1.89
TiO ₂ (rutile)	27.4, 36.1, 54.3	(110), (101), (211)	3.25, 2.49, 1.69
ZnO (wurtzite)	31.8, 34.4, 36.3	(100), (002), (101)	2.81, 2.60, 2.47

D.3.5 Amorphous vs. Crystalline Identification

- **Crystalline:** Sharp, well-defined peaks at specific 2θ values
- **Amorphous:** Broad hump (typically $2\theta = 15\text{--}30^\circ$), no sharp peaks
- **Nanocrystalline:** Broadened peaks following Scherrer relationship
- **Mixed:** Broad amorphous background with superimposed crystalline peaks

D.4 X-Ray Photoelectron Spectroscopy (XPS)

D.4.1 Surface Sensitivity

XPS probes only the outermost 5–10 nm of the sample due to the limited escape depth of photoelectrons. Signal intensity decays exponentially with depth, governed by the inelastic mean free path ($\lambda \approx 1\text{--}3\text{ nm}$) and the takeoff angle θ .

Remark D.6 (Angle-resolved XPS). By varying the takeoff angle, you can probe different depths. At grazing angles ($\theta \approx 80^\circ$), sensitivity is enhanced toward the topmost atomic layers. This technique helps distinguish surface segregation from bulk composition.

D.4.2 Common Binding Energy References

D.4.3 Quantification Considerations

Surface atomic concentrations are calculated from peak areas divided by relative sensitivity factors (RSFs). The ratio $n_A/n_B = (I_A/S_A)/(I_B/S_B)$, where I is the peak area and S is the RSF provided by the instrument manufacturer.

Remark D.7 (Quantification accuracy). XPS quantification is typically accurate to $\pm 10\text{--}20\%$ for well-resolved peaks. Accuracy decreases for overlapping peaks, low-intensity signals, or samples with significant charging. Always report atomic ratios rather than absolute concentrations.

D.4.4 Charging Correction Methods

Insulating samples accumulate positive charge during XPS analysis, shifting all peaks to higher binding energies. Common corrections:

1. **C 1s reference:** Shift all peaks so adventitious carbon appears at 284.8 eV
2. **Flood gun:** Use low-energy electrons to neutralize surface charge

Table D.7: Standard XPS binding energies for catalyst analysis.

Element (Line)	Chemical State	BE (eV)
C 1s	Adventitious carbon	284.8
C 1s	C–O	286.3
C 1s	C=O	287.8
C 1s	O–C=O (carbonate)	289.0
O 1s	Lattice oxide	529.5–530.5
O 1s	Surface hydroxyl	531.0–532.0
O 1s	Adsorbed water	532.5–533.5
Pt 4f _{7/2}	Pt ⁰ (metallic)	71.2
Pt 4f _{7/2}	Pt ²⁺ (PtO)	72.4
Pt 4f _{7/2}	Pt ⁴⁺ (PtO ₂)	74.0
Pd 3d _{5/2}	Pd ⁰ (metallic)	335.4
Pd 3d _{5/2}	Pd ²⁺ (PdO)	336.8
Au 4f _{7/2}	Au ⁰ (metallic)	84.0
Ni 2p _{3/2}	Ni ⁰ (metallic)	852.7
Ni 2p _{3/2}	Ni ²⁺ (NiO)	854.0
Ti 2p _{3/2}	Ti ⁴⁺ (TiO ₂)	458.8
Ti 2p _{3/2}	Ti ³⁺ (reduced)	457.2
Al 2p	Al ₂ O ₃	74.5
Si 2p	SiO ₂	103.4

3. **Internal standard:** Reference to a known peak from the support material
4. **Au decoration:** Sputter thin Au layer and reference to Au 4f at 84.0 eV

D.5 Transmission Electron Microscopy (TEM)

D.5.1 Sample Preparation for Supported Catalysts

1. **Suspension:** Disperse catalyst powder in ethanol or water (1 mg/mL)
2. **Sonication:** Ultrasonicate for 5–15 minutes to break up agglomerates
3. **Deposition:** Drop 5–10 μL onto holey carbon TEM grid
4. **Drying:** Allow to air dry or use gentle heating
5. **Storage:** Keep in vacuum desiccator until imaging

Remark D.8 (Grid selection). Use holey or lacey carbon grids to image particles at the edge of holes, where support interference is minimized. For very small particles (< 2 nm), ultrathin carbon or graphene supports provide better contrast.

D.5.2 Image Interpretation Guidelines

Table D.8: Contrast mechanisms in TEM imaging of catalysts.

Contrast Type	Origin	Information
Mass-thickness	Electron scattering	Particle location, size
Diffraction	Bragg scattering	Crystallinity, orientation
Phase (HRTEM)	Interference	Atomic structure, lattice
Z-contrast (STEM)	Z^2 dependence	Composition mapping

D.5.3 Particle Size Distribution Analysis

For statistically meaningful size distributions:

1. **Count at least 200 particles** from multiple images
2. **Sample different areas** to avoid biased sampling
3. **Measure consistently:** Use the same definition (diameter, Feret diameter, equivalent circle)
4. **Report distribution:** Histogram with mean, standard deviation, and distribution type (lognormal, Gaussian)

Report the number-average ($\bar{d}_n = \sum n_i d_i / \sum n_i$), surface-average ($\bar{d}_s = \sum n_i d_i^3 / \sum n_i d_i^2$), and volume-average ($\bar{d}_v = (\sum n_i d_i^3 / \sum n_i)^{1/3}$) diameters from the measured distribution.

Remark D.9 (Which average to use?). For catalysis, the surface-average diameter $\bar{d}_s$ is most relevant because reaction rates scale with surface area. The number-average $\bar{d}_n$ underweights large particles, while the volume-average $\bar{d}_v$ overweights them. Always report the distribution, not just the mean.

D.5.4 Lattice Fringe Interpretation

High-resolution TEM (HRTEM) can resolve atomic lattice planes. Measure the fringe spacing directly from the image and compare with the reference d -spacings in Table D.6 to identify crystal phases and exposed facets.

D.6 FTIR Spectroscopy

D.6.1 CO Probe Molecule

Carbon monoxide is the most widely used probe for metal surface characterization:

Table D.9: CO stretching frequencies for different adsorption geometries.

Adsorption Mode	Wavenumber (cm ⁻¹)	Description
Linear (atop)	2000–2130	One CO per metal site
Bridging (2-fold)	1850–2000	CO shared by 2 metal atoms
3-fold hollow	1700–1850	CO in 3-fold site
Gas-phase CO	2143	Reference

The CO stretching frequency decreases with increasing coordination because:

- More metal atoms donate electron density to CO antibonding orbitals
- Increased backbonding weakens the C–O bond
- Lower frequency indicates stronger metal-CO interaction

Remark D.10 (Coverage effects). At high CO coverage, dipole-dipole coupling between adsorbed CO molecules shifts peaks to higher frequencies (blue shift). Always compare spectra at similar coverages when assessing support effects or particle size effects.

D.6.2 Pyridine Adsorption for Acidity

Pyridine distinguishes between Brønsted and Lewis acid sites:

Table D.10: Pyridine adsorption bands for acid site characterization.

Acid Site Type	Wavenumber (cm^{-1})	Species
Brønsted acid	1540–1550	Pyridinium ion (PyH^+)
Lewis acid	1445–1455	Coordinated pyridine
Both types	1485–1495	Overlapping band
H-bonded	1440–1447	Weakly bound

The ratio of Brønsted to Lewis sites (B/L ratio) is calculated from integrated peak areas using appropriate extinction coefficients.

D.6.3 Common Adsorbate Bands

Table D.11: Characteristic IR bands for species commonly observed on catalyst surfaces.

Species	Assignment	Wavenumber (cm^{-1})
Formate	$\nu_{\text{as}}(\text{OCO})$	1560–1580
Formate	$\nu_{\text{s}}(\text{OCO})$	1360–1380
Acetate	$\nu_{\text{as}}(\text{OCO})$	1540–1560
Acetate	$\nu_{\text{s}}(\text{OCO})$	1420–1450
Carbonate	$\nu(\text{CO}_3)$	1400–1600
Bicarbonate	$\nu(\text{HCO}_3)$	1620–1660
Hydroxyl (isolated)	$\nu(\text{O-H})$	3650–3750
Hydroxyl (H-bonded)	$\nu(\text{O-H})$	3200–3600
Adsorbed water	$\delta(\text{H}_2\text{O})$	1620–1650
NO (linear)	$\nu(\text{N-O})$	1800–1900
NO (bridging)	$\nu(\text{N-O})$	1700–1800
NO ₂ (nitro)	$\nu_{\text{as}}(\text{NO}_2)$	1500–1550
NH ₃ (on Brønsted)	$\delta_{\text{as}}(\text{NH}_4^+)$	1430–1450
NH ₃ (on Lewis)	$\delta_{\text{as}}(\text{NH}_3)$	1600–1630

Table D.12: Comparison of the six catalyst characterization techniques.

Tech.	Information	Det. Limit	Sample	Destr.	Time
BET	Surface area, pore volume, pore size	> 1 m ² /g	50–200 mg powder	No	2–4 h
TPD	Binding en- ergies, site density, acid strength	0.1 mono- layer	10–100 mg	No ^a	2–4 h
XRD	Crystal struc- ture, phase ID, crystallite size	2–5 wt%	50–200 mg	No	0.5–2 h
XPS	Surface com- position, oxi- dation states	0.1–1 at%	1–10 mg	No ^b	1–4 h
TEM	Particle size, morphology, structure	Single particle	< 1 mg	No ^c	1–4 h
FTIR	Adsorbed species, acid sites	0.01 mono- layer	5–50 mg	No	0.5–2 h

^aSample may be modified during heating. ^bSome beam damage possible. ^cElectron beam may cause local heating.

D.7 Method Comparison Summary

Remark D.11 (Choosing the right technique). No single technique provides complete catalyst characterization. Use complementary methods:

- BET + XRD for physical structure
- XPS + TPD for surface chemistry
- TEM for local structure and size distribution
- FTIR for in situ surface species during reaction

Correlate results across techniques to build a complete picture of your catalyst.

Appendix E

Python Cheatsheet for Surface Catalysis

This appendix provides a compact reference for the Python code patterns used in the companion Jupyter notebooks (NB1–NB6, plus NB-S). For Python fundamentals and Jupyter basics, see NB1. All code examples use the colorblind-safe palette defined in the course materials.

E.1 Python Environment Setup

For Jupyter notebook basics and Python fundamentals, see NB1 (Bridge Module notebook). The companion notebooks require the packages listed in Table E.1. Install any missing packages with:

```
1 pip install numpy scipy matplotlib pandas
```

Table E.1: Required Python packages for this course.

Package	Purpose	Typical Import
NumPy	Arrays and operations	<code>import numpy as np</code>
SciPy	ODE solvers, fitting	<code>from scipy import ...</code>
Matplotlib	Plotting	<code>import matplotlib.pyplot as plt</code>
Pandas	CSV and data handling	<code>import pandas as pd</code>

E.2 NumPy Essentials

NumPy is the foundation of scientific computing in Python. It provides fast, vectorized operations on arrays that avoid slow Python loops.

E.2.1 Array Creation

```

1 import numpy as np
2
3 # From a list
4 P = np.array([0.1, 0.5, 1.0, 2.0, 5.0]) # Pressures in bar
5
6 # Evenly spaced values (for plotting)
7 P = np.linspace(0, 10, 100) # 100 points from 0 to 10
8 T = np.linspace(300, 700, 50) # 50 temperatures from 300 to 700 K
9
10 # Logarithmically spaced (for wide ranges)
11 P = np.logspace(-2, 2, 50) # 50 points from 0.01 to 100
12
13 # Arrays of zeros or ones (for initialization)
14 theta = np.zeros(100) # 100 zeros
15 C = np.ones(50) * 0.1 # 50 values of 0.1 mol/L

```

Listing E.1: Creating NumPy arrays.

E.2.2 Element-wise Operations and Broadcasting

NumPy applies operations element-by-element across entire arrays—no loops needed:

```

1 import numpy as np
2
3 P = np.array([0.1, 0.5, 1.0, 2.0])
4 K = 2.0 # bar-1
5
6 # Element-wise Langmuir isotherm (no loop!)
7 theta = K * P / (1 + K * P)
8 # Result: array([0.167, 0.5, 0.667, 0.8])
9
10 # Temperature dependence
11 T = np.linspace(300, 600, 100)
12 Ea = 50000 # J/mol
13 R = 8.314 # J/(mol*K)
14 k = 1e13 * np.exp(-Ea / (R * T)) # Arrhenius over all T

```

Listing E.2: Element-wise operations on arrays.

E.3 SciPy for Solving ODEs

The `scipy.integrate.solve_ivp` function solves initial value problems—essential for PFR design equations and reaction kinetics. The “ivp” stands for Initial Value Problem.

E.3.1 Basic Usage

The general form of an ODE initial value problem is:

$$\frac{dy}{dt} = f(t, y), \quad y(t_0) = y_0 \quad (\text{E.1})$$

```
1 from scipy.integrate import solve_ivp
2
3 def dydt(t, y):
4     """Define the right-hand side of dy/dt = f(t, y)."""
5     return -k * y # Example: first-order decay
6
7 # Solve from t=0 to t=10, starting with y=1.0
8 solution = solve_ivp(
9     dydt,          # The function defining dy/dt
10    [0, 10],        # Time span: [t_start, t_end]
11    [1.0],          # Initial condition(s) as a list
12    t_eval=np.linspace(0, 10, 100) # Times to store solution
13 )
14
15 t = solution.t      # Time points
16 y = solution.y[0]   # Solution values (first variable)
```

Listing E.3: Basic structure of `solve_ivp`.

E.3.2 PFR Design Equation Integration

For a PFR with the design equation:

$$\frac{dX}{d\tau} = \frac{r(C_A)}{C_{A0}}, \quad \text{where } C_A = C_{A0}(1 - X) \quad (\text{E.2})$$

```
1 import numpy as np
2 from scipy.integrate import solve_ivp
3 import matplotlib.pyplot as plt
4
5 # Parameters
6 k = 0.1      # Rate constant (mol/(L*s))
7 K = 2.0      # Adsorption constant (L/mol)
8 C_A0 = 1.0   # Inlet concentration (mol/L)
9
10 def dXdtau(tau, X):
11     """PFR design equation with LH kinetics."""
12     C_A = C_A0 * (1 - X[0]) # Current concentration
13     r = k * K * C_A / (1 + K * C_A) # LH rate (unimolecular)
14     return [r / C_A0]
15
16 # Solve
17 tau_span = [0, 50] # Space time range (s)
18 tau_eval = np.linspace(0, 50, 200)
19
```

```

20 sol = solve_ivp(dXdtau, tau_span, [0.0], t_eval=tau_eval)
21
22 # Plot
23 plt.figure(figsize=(8, 5))
24 plt.plot(sol.t, sol.y[0], color='#0072B2', linewidth=2)
25 plt.xlabel('Space time  $\tau$  (s)')
26 plt.ylabel('Conversion  $X$ ')
27 plt.title('PFR Conversion Profile (LH Kinetics)')
28 plt.grid(True, alpha=0.3)
29 plt.show()

```

Listing E.4: PFR integration for Langmuir-Hinshelwood kinetics.

E.4 SciPy for Optimization and Curve Fitting

E.4.1 Nonlinear Curve Fitting with `curve_fit`

The `scipy.optimize.curve_fit` function fits experimental data to a model function using least-squares regression. This is essential for extracting kinetic parameters from experimental data.

```

1 from scipy.optimize import curve_fit
2 import numpy as np
3
4 def model(x, param1, param2):
5     """Model function: first argument is independent variable,
6     remaining arguments are parameters to fit."""
7     return param1 * x / (1 + param2 * x) # Example
8
9 # Fit the model to data
10 popt, pcov = curve_fit(model, x_data, y_data, p0=[1.0, 1.0])
11
12 # popt: optimal parameter values [param1_fit, param2_fit]
13 # pcov: covariance matrix (for uncertainties)

```

Listing E.5: General structure of `curve_fit`.

Extracting parameter uncertainties:

```

1 # Standard errors from diagonal of covariance matrix
2 perr = np.sqrt(np.diag(pcov))
3 print(f"param1 = {popt[0]:.3f} +/- {perr[0]:.3f}")
4 print(f"param2 = {popt[1]:.3f} +/- {perr[1]:.3f}")

```

E.4.2 Solving Algebraic Equations with `fsolve`

The `scipy.optimize.fsolve` function finds roots of nonlinear equations—necessary for CSTR design equations where you solve for exit conversion algebraically.

```
1 from scipy.optimize import fsolve
2 import numpy as np
3
4 # Parameters
5 k = 0.1      # Rate constant
6 K = 2.0      # Adsorption constant (L/mol)
7 C_A0 = 1.0   # Inlet concentration (mol/L)
8 tau = 20.0   # Space time (s)
9
10 def cstr_equation(X):
11     """CSTR design equation rearranged to  $f(X) = 0$ ."""
12     C_A = C_A0 * (1 - X)
13     r = k * K * C_A / (1 + K * C_A) # LH rate (unimolecular)
14     # Design equation:  $\tau = C_{A0} * X / r$ 
15     # Rearranged:  $C_{A0} * X - \tau * r = 0$ 
16     return C_A0 * X - tau * r
17
18 # Solve (initial guess  $X = 0.5$ )
19 X_solution = fsolve(cstr_equation, 0.5)[0]
20 print(f"CSTR conversion:  $X = \{X\_solution:.3f\}$ ")
```

Listing E.6: CSTR design equation solver.

Remark E.1 (Initial Guesses Matter). Both `curve_fit` and `fsolve` require initial guesses. Poor guesses can lead to convergence to local minima or failure to converge. For kinetic parameters, use physically reasonable estimates. For conversion problems, start with $X = 0.5$ (middle of the valid range).

E.5 Matplotlib for Publication-Quality Plots

Good visualization is essential for understanding kinetic data and communicating results. This section covers the plotting techniques used throughout the notebooks.

E.5.1 Colorblind-Safe Color Palette

Throughout this course, we use the Wong (2011) colorblind-safe palette. Define these colors at the start of your notebooks:

```
1 # Wong (2011) colorblind-safe palette
2 COLORS = {
3     'blue':      '#0072B2', # Primary data
4     'orange':    '#E69F00', # Secondary data
5     'green':     '#009E73', # Tertiary data
6     'vermillion': '#D55E00', # Warnings/errors
7     'skyblue':   '#56B4E9', # Backgrounds
```



```

8     'purple':    '#CC79A7', # Additional series
9     'yellow':    '#F0E442', # Highlights
10    'black':     '#000000'  # Text/axes
11 }

```

Listing E.7: Colorblind-safe color definitions.

E.5.2 Basic Line and Scatter Plots

```

1  import numpy as np
2  import matplotlib.pyplot as plt
3
4  # Generate data
5  P = np.linspace(0, 10, 100)
6  K1, K2 = 0.5, 2.0
7  theta1 = K1 * P / (1 + K1 * P)
8  theta2 = K2 * P / (1 + K2 * P)
9
10 # Create figure
11 fig, ax = plt.subplots(figsize=(8, 5))
12
13 # Plot with colorblind-safe colors and distinct line styles
14 ax.plot(P, theta1, color='#0072B2', linewidth=2,
15         linestyle='--', label=f'$K = {K1}$ bar$^{-1}$')
16 ax.plot(P, theta2, color='#E69F00', linewidth=2,
17         linestyle='--', label=f'$K = {K2}$ bar$^{-1}$')
18
19 # Scatter plot for experimental data
20 P_exp = np.array([1, 2, 4, 6, 8])
21 theta_exp = np.array([0.45, 0.62, 0.78, 0.84, 0.88])
22 ax.scatter(P_exp, theta_exp, color='#D55E00', s=60,
23           marker='o', label='Experimental', zorder=5)
24
25 # Labels and formatting
26 ax.set_xlabel('Pressure $P$ (bar)', fontsize=12)
27 ax.set_ylabel('Coverage $\theta$', fontsize=12)
28 ax.set_title('Langmuir Isotherm Comparison', fontsize=14)
29 ax.legend(loc='lower right', fontsize=11)
30 ax.grid(True, alpha=0.3)
31 ax.set_xlim(0, 10)
32 ax.set_ylim(0, 1)
33
34 plt.tight_layout()
35 plt.show()

```

Listing E.8: Basic plotting with proper styling.

E.6 Common Calculations: Complete Code Examples

This section provides complete, copy-ready code for the most common calculations in this course. Each example is self-contained and can be used as a template for your own analyses.

E.6.1 Langmuir Isotherm Fitting

Fitting experimental coverage data to the Langmuir isotherm to extract the adsorption equilibrium constant K .

```
1 import numpy as np
2 from scipy.optimize import curve_fit
3 import matplotlib.pyplot as plt
4
5 # Experimental data (pressure and coverage)
6 P_exp = np.array([0.1, 0.5, 1.0, 2.0, 5.0, 10.0]) # bar
7 theta_exp = np.array([0.15, 0.48, 0.63, 0.78, 0.90, 0.95])
8
9 def langmuir(P, K):
10     """Langmuir isotherm: theta = KP / (1 + KP)"""
11     return K * P / (1 + K * P)
12
13 # Fit the model
14 popt, pcov = curve_fit(langmuir, P_exp, theta_exp, p0=[1.0])
15 K_fit = popt[0]
16 K_err = np.sqrt(pcov[0, 0])
17
18 print(f"Fitted K = {K_fit:.3f} +/- {K_err:.3f} bar-1")
19
20 # Plot results
21 P_model = np.linspace(0, 12, 100)
22 theta_model = langmuir(P_model, K_fit)
23
24 fig, ax = plt.subplots(figsize=(8, 5))
25 ax.scatter(P_exp, theta_exp, color='#D55E00', s=80,
26           label='Experimental', zorder=5)
27 ax.plot(P_model, theta_model, color='#0072B2', linewidth=2,
28       label=f'Langmuir fit (K = {K_fit:.2f} bar{-1})')
29 ax.set_xlabel('Pressure P$ (bar)', fontsize=12)
30 ax.set_ylabel('Coverage $\theta$', fontsize=12)
31 ax.legend(fontsize=11)
32 ax.grid(True, alpha=0.3)
33 ax.set_xlim(0, 12)
34 ax.set_ylim(0, 1.05)
35 plt.tight_layout()
36 plt.show()
```

Listing E.9: Langmuir isotherm fitting with uncertainties.

E.6.2 Arrhenius Parameter Extraction

Extracting activation energy and pre-exponential factor from temperature-dependent rate constant data. This requires linearizing the Arrhenius equation.

```

1 import numpy as np
2 from scipy.optimize import curve_fit
3 import matplotlib.pyplot as plt
4
5 # Experimental data
6 T = np.array([350, 400, 450, 500, 550]) # K
7 k = np.array([0.012, 0.048, 0.175, 0.52, 1.35]) # s-1
8 R = 8.314 # J/(mol*K)
9
10 # Method 1: Linear regression on ln(k) vs 1/T
11 inv_T = 1 / T
12 ln_k = np.log(k)
13
14 # Linear fit: ln(k) = ln(A) -Ea/(R*T)
15 coeffs = np.polyfit(inv_T, ln_k, 1)
16 slope = coeffs[0] # = -Ea/R
17 intercept = coeffs[1] # = ln(A)
18
19 Ea = -slope * R # J/mol
20 A = np.exp(intercept) # s-1
21
22 print(f"Activation energy: Ea = {Ea/1000:.1f} kJ/mol")
23 print(f"Pre-exponential factor: A = {A:.2e} s-1")
24
25 # Method 2: Nonlinear fit (more rigorous for uncertainty)
26 def arrhenius(T, A, Ea):
27     return A * np.exp(-Ea / (R * T))
28
29 popt, pcov = curve_fit(arrhenius, T, k, p0=[1e10, 50000])
30 A_fit, Ea_fit = popt
31 perr = np.sqrt(np.diag(pcov))
32 print(f"\nNonlinear fit:")
33 print(f"Ea = {Ea_fit/1000:.1f} +/- {perr[1]/1000:.1f} kJ/mol")
34 print(f"A = {A_fit:.2e} +/- {perr[0]:.2e} s-1")
35
36 # Plot: Arrhenius plot
37 fig, ax = plt.subplots(figsize=(8, 5))
38 ax.semilogy(1000/T, k, 'o', color='#0072B2', markersize=10,
39             label='Experimental')
40 T_fit = np.linspace(340, 560, 100)
41 ax.semilogy(1000/T_fit, arrhenius(T_fit, A_fit, Ea_fit),
42             color='#E69F00', linewidth=2,
43             label=f'Fit: $E_a$ = {Ea_fit/1000:.1f} kJ/mol')
44 ax.set_xlabel('$1000/T$ (K$^{-1}$)', fontsize=12)
45 ax.set_ylabel('$k$ (s$^{-1}$)', fontsize=12)
46 ax.set_title('Arrhenius Plot', fontsize=14)
47 ax.legend(fontsize=11)

```

```
48 ax.grid(True, which='both', alpha=0.3)
49 plt.tight_layout()
50 plt.show()
```

Listing E.10: Arrhenius analysis with linear regression.

E.6.3 CSTR Conversion Calculation

Solving the CSTR design equation for different kinetic models.

```
1 import numpy as np
2 from scipy.optimize import fsolve
3 import matplotlib.pyplot as plt
4
5 # Parameters
6 C_A0 = 1.0 # mol/L (inlet concentration)
7 k = 0.1    # Appropriate units for each model
8
9 # First-order kinetics: analytical solution
10 def cstr_first_order(tau, k):
11     """ $X = k\tau / (1 + k\tau)$ """
12     return k * tau / (1 + k * tau)
13
14 # Langmuir-Hinshelwood: numerical solution required
15 def cstr_lh(tau, k, K, C_A0):
16     """Solve CSTR design equation for LH kinetics."""
17     def equation(X):
18         C_A = C_A0 * (1 - X)
19         r = k * K * C_A / (1 + K * C_A) # LH rate (unimolecular)
20         return C_A0 * X - tau * r
21     X_sol = fsolve(equation, 0.5)[0]
22     return np.clip(X_sol, 0, 1) # Ensure physical bounds
23
24 # Calculate conversion vs space time
25 tau_values = np.linspace(0.1, 100, 100)
26 K = 2.0 # L/mol (adsorption constant for LH)
27
28 X_first = cstr_first_order(tau_values, k)
29 X_lh = np.array([cstr_lh(tau, k, K, C_A0) for tau in tau_values])
30
31 # Plot comparison
32 fig, ax = plt.subplots(figsize=(8, 5))
33 ax.plot(tau_values, X_first, color='#0072B2', linewidth=2,
34         linestyle='--', label='First-order')
35 ax.plot(tau_values, X_lh, color='#E69F00', linewidth=2,
36         linestyle='--', label='Langmuir-Hinshelwood')
37 ax.set_xlabel('Space time  $\tau$  (s)', fontsize=12)
38 ax.set_ylabel('Conversion  $X$ ', fontsize=12)
39 ax.set_title('CSTR Conversion Comparison', fontsize=14)
40 ax.legend(fontsize=11)
41 ax.grid(True, alpha=0.3)
42 ax.set_xlim(0, 100)
43 ax.set_ylim(0, 1)
```

```

44 plt.tight_layout()
45 plt.show()

```

Listing E.11: CSTR conversion for various rate laws.

E.6.4 PFR Integration

Integrating the PFR design equation for complex kinetics.

```

1 import numpy as np
2 from scipy.integrate import solve_ivp
3 import matplotlib.pyplot as plt
4
5 # Parameters
6 k = 0.1      # mol/(L*s)
7 K = 2.0      # L/mol
8 C_A0 = 1.0   # mol/L
9
10 def pfr_ode(tau, X, k, K, C_A0):
11     """PFR design equation: dX/dtau = r/C_A0"""
12     C_A = C_A0 * (1 - X[0])
13     # Langmuir-Hinshelwood rate (unimolecular)
14     r = k * K * C_A / (1 + K * C_A)
15     return [r / C_A0]
16
17 # Solve ODE
18 tau_span = [0, 100]
19 tau_eval = np.linspace(0, 100, 500)
20
21 solution = solve_ivp(
22     lambda tau, X: pfr_ode(tau, X, k, K, C_A0),
23     tau_span,
24     [0.0], # X(0) = 0
25     t_eval=tau_eval,
26     method='RK45'
27 )
28
29 tau = solution.t
30 X = solution.y[0]
31
32 # Find space time for 90% conversion
33 X_target = 0.9
34 idx_90 = np.argmin(np.abs(X - X_target))
35 tau_90 = tau[idx_90]
36 print(f"Space time for 90% conversion: tau = {tau_90:.1f} s")
37
38 # Plot
39 fig, ax = plt.subplots(figsize=(8, 5))
40 ax.plot(tau, X, color='#0072B2', linewidth=2)
41 ax.axhline(y=0.9, color='#009E73', linestyle='--', linewidth=1.5,
42           label='90% conversion')
43 ax.axvline(x=tau_90, color='#009E73', linestyle='--', linewidth=1.5)
44 ax.scatter([tau_90], [0.9], color='#D55E00', s=100, zorder=5)

```

```
45 ax.set_xlabel('Space time  $\tau$  (s)', fontsize=12)
46 ax.set_ylabel('Conversion  $X$ ', fontsize=12)
47 ax.set_title('PFR Conversion Profile (LH Kinetics)', fontsize=14)
48 ax.annotate(f' $\tau_{90} = \tau_{90:.1f}$  s',
49            xy=(tau_90, 0.9), xytext=(tau_90+10, 0.75),
50            fontsize=11, arrowprops=dict(arrowstyle='->', color='gray'))
51 ax.legend(fontsize=11)
52 ax.grid(True, alpha=0.3)
53 ax.set_xlim(0, 100)
54 ax.set_ylim(0, 1)
55 plt.tight_layout()
56 plt.show()
```

Listing E.12: PFR integration with conversion profile.

E.6.5 Effectiveness Factor Computation

Calculating effectiveness factors for porous catalysts as a function of Thiele modulus.

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 def eta_slab(phi):
5     """Effectiveness factor for slab geometry (first-order)."""
6     phi = np.asarray(phi, dtype=float)
7     # Handle phi -> 0 limit to avoid division by zero
8     return np.where(phi < 1e-10, 1.0, np.tanh(phi) / phi)
9
10 def eta_sphere(phi):
11     """Effectiveness factor for sphere geometry (first-order)."""
12     phi = np.asarray(phi, dtype=float)
13     # Handle phi -> 0 limit
14     with np.errstate(divide='ignore', invalid='ignore'):
15         coth_phi = 1.0 / np.tanh(phi)
16         result = 3 * (phi * coth_phi - 1) / phi**2
17         result = np.where(phi < 1e-10, 1.0, result)
18     return result
19
20 # Calculate over range of Thiele modulus
21 phi = np.logspace(-1, 2, 200) # 0.1 to 100
22
23 eta_sl = eta_slab(phi)
24 eta_sp = eta_sphere(phi)
25
26 # Asymptotic limits
27 phi_large = phi[phi > 5]
28 asymp_slab = 1 / phi_large
29 asymp_sphere = 3 / phi_large
30
31 # Plot
32 fig, ax = plt.subplots(figsize=(8, 5))
```

```

33 ax.loglog(phi, eta_sl, color='#0072B2', linewidth=2,
34           linestyle='-', label='Slab')
35 ax.loglog(phi, eta_sp, color='#E69F00', linewidth=2,
36           linestyle='--', label='Sphere')
37 ax.loglog(phi_large, asymp_slab, color='#0072B2', linewidth=1,
38           linestyle=':', alpha=0.7)
39 ax.loglog(phi_large, asymp_sphere, color='#E69F00', linewidth=1,
40           linestyle=':', alpha=0.7)
41
42 # Mark regime boundaries
43 ax.axvline(x=1, color='gray', linestyle='--', alpha=0.5)
44 ax.text(0.15, 0.15, 'Kinetic\ncontrol', fontsize=10, ha='center')
45 ax.text(30, 0.15, 'Diffusion\ncontrol', fontsize=10, ha='center')
46
47 ax.set_xlabel('Thiele modulus  $\phi$ ', fontsize=12)
48 ax.set_ylabel('Effectiveness factor  $\eta$ ', fontsize=12)
49 ax.set_title('Effectiveness Factor vs Thiele Modulus', fontsize=14)
50 ax.legend(fontsize=11, loc='lower left')
51 ax.grid(True, which='both', alpha=0.3)
52 ax.set_xlim(0.1, 100)
53 ax.set_ylim(0.01, 1.5)
54 plt.tight_layout()
55 plt.show()
56
57 # Example calculation
58 L = 0.001      # Half-thickness: 1 mm = 0.001 m
59 k_rxn = 10     # Rate constant: 10 s-1
60 D_eff = 1e-9   # Effective diffusivity: 1e-9 m2/s
61
62 phi_calc = L * np.sqrt(k_rxn / D_eff)
63 eta_calc = eta_slab(phi_calc)
64
65 print(f"\nExample calculation:")
66 print(f" Thiele modulus: phi = {phi_calc:.2f}")
67 print(f" Effectiveness factor: eta = {eta_calc:.4f}")
68 print(f" Observed rate is {eta_calc*100:.1f}% of intrinsic rate")

```

Listing E.13: Effectiveness factor vs Thiele modulus.

E.7 Reading Data Files

```

1 import pandas as pd
2 import numpy as np
3
4 # Read CSV file
5 df = pd.read_csv('kinetic_data.csv')
6
7 # View first few rows
8 print(df.head())
9
10 # Access columns as NumPy arrays

```

```

11 T = df['Temperature_K'].values
12 k = df['Rate_constant'].values
13
14 # Or access directly for calculations
15 theta = df['Coverage'].values
16 P = df['Pressure_bar'].values

```

Listing E.14: Loading experimental data from CSV.

E.8 Quick Reference Tables

Table E.2: NumPy array creation functions.

Function	Description
<code>np.array([1, 2, 3])</code>	Create array from list
<code>np.linspace(0, 10, 100)</code>	100 evenly spaced values from 0 to 10
<code>np.logspace(-2, 2, 50)</code>	50 log-spaced values from 10^{-2} to 10^2
<code>np.zeros(100)</code>	Array of 100 zeros
<code>np.ones(50)</code>	Array of 50 ones
<code>np.arange(0, 10, 0.5)</code>	Values from 0 to 10 with step 0.5

Table E.3: Essential SciPy functions for this course.

Function	Module	Purpose
<code>solve_ivp</code>	<code>scipy.integrate</code>	ODE integration (PFR)
<code>curve_fit</code>	<code>scipy.optimize</code>	Nonlinear regression
<code>fsolve</code>	<code>scipy.optimize</code>	Root finding (CSTR)
<code>minimize</code>	<code>scipy.optimize</code>	General optimization

Remark E.2 (Using the Course Utility Module). The companion notebooks include a shared `utils.py` module with pre-built functions for common calculations. Import it at the start of any notebook:

```

from utils import langmuir_coverage, arrhenius, effectiveness_slab
from utils import COLORS, setup_plot_style

```

See the module documentation for the complete list of available functions.

Table E.4: Matplotlib plotting essentials.

Command	Purpose
<code>plt.subplots(figsize=(8,5))</code>	Create figure and axes
<code>ax.plot(x, y, color='...')</code>	Line plot
<code>ax.scatter(x, y, s=50)</code>	Scatter plot with marker size
<code>ax.semilogy(x, y)</code>	Semi-log plot (log y-axis)
<code>ax.loglog(x, y)</code>	Log-log plot
<code>ax.set_xlabel('...')</code>	Set x-axis label
<code>ax.legend(loc='...')</code>	Add legend
<code>ax.grid(True, alpha=0.3)</code>	Add grid
<code>plt.tight_layout()</code>	Adjust spacing
<code>fig.savefig('file.pdf')</code>	Save figure

Table E.5: Colorblind-safe hex codes (Wong 2011 palette).

Color	Hex Code	Recommended Use
Blue	'#0072B2'	Primary data series
Orange	'#E69F00'	Secondary data series
Green	'#009E73'	Tertiary data / fits
Vermillion	'#D55E00'	Experimental points / warnings
Sky blue	'#56B4E9'	Background / fills
Purple	'#CC79A7'	Additional series
Yellow	'#F0E442'	Highlights (use sparingly)

Appendix F

Grading Rubric

This appendix provides detailed grading criteria for all major assignments in the Surface Phenomena & Catalysis course. These rubrics ensure consistent, transparent evaluation and help you understand the expectations for demonstrating mastery of course concepts.

F.1 Capstone Project Rubric

The capstone project requires you to integrate concepts from all preceding chapters into a comprehensive catalytic system design. Your project will be evaluated across three major categories: Technical Content (60%), Computational Work (20%), and Presentation Quality (20%).

F.1.1 Technical Content (60%)

Technical content forms the core of your project evaluation. Each subcategory assesses a distinct aspect of your catalytic chemistry analysis.

F.1.2 Computational Work (20%)

Your Jupyter notebook implementation demonstrates your ability to translate theoretical understanding into working computational tools.

F.1.3 Presentation Quality (20%)

Clear communication of technical results is essential for scientific practice. Your written report should demonstrate professional standards.

F.2 Problem Set Grading Guidelines

Problem sets are graded on a combination of methodology, accuracy, and physical interpretation. The following guidelines establish expectations for

Table F.1: Technical Content Rubric (60 points total)

Criterion (Points)	Excellent	Satisfactory	Needs Work
Problem Formulation & Assumptions (10 pts)	Clear problem statement with all assumptions explicitly stated and justified. Boundary conditions well-defined. (9–10)	Problem stated adequately. Most assumptions identified but some lack justification. (6–8)	Vague problem statement. Key assumptions missing or unjustified. (0–5)
Kinetic Model Selection & Justification (15 pts)	Appropriate mechanism selected (LH, ER, or MVK) with clear rationale based on system characteristics. Rate expression correctly derived with proper coverage dependencies. (13–15)	Reasonable mechanism choice with partial justification. Minor errors in rate expression derivation. (9–12)	Inappropriate mechanism or no justification provided. Significant errors in rate expression. (0–8)
Reactor Design Calculations (15 pts)	Correct application of CSTR or PFR design equations. Space time, conversion, and rate properly related. All calculations dimensionally consistent and numerically accurate. (13–15)	Design equations correctly identified but with minor calculation errors. Dimensional analysis mostly correct. (9–12)	Wrong design equations or major calculation errors. Unit inconsistencies throughout. (0–8)
Transport Limitation Analysis (10 pts)	Thiele modulus and effectiveness factor correctly calculated. Weisz-Prater criterion applied appropriately. Correct identification of rate-limiting regime (kinetic vs. diffusion). (9–10)	Partial transport analysis with minor errors. Regime identification reasonable but incomplete justification. (6–8)	Transport effects ignored or incorrectly analyzed. Wrong regime identification. (0–5)
Selectivity Optimization (10 pts)	Yield and selectivity correctly defined and calculated. Optimal operating conditions identified with clear trade-off analysis. Temperature effects on selectivity properly considered. ³¹¹ (9–10)	Selectivity concepts applied correctly but optimization incomplete. Some trade-offs identified. (6–8)	Selectivity analysis missing or fundamentally flawed. No optimization attempted. (0–5)

Table F.2: Computational Work Rubric (20 points total)

Criterion (Points)	Excellent	Satisfactory	Needs Work
Code Correctness & Functionality (8 pts)	All code executes without errors. Numerical results match hand calculations. ODE solvers and root-finders used appropriately. (7–8)	Code runs with minor warnings. Results reasonable but small numerical discrepancies present. (5–6)	Code fails to execute or produces incorrect results. Major logical errors. (0–4)
Documentation & Comments (4 pts)	Clear markdown cells explain methodology. Code comments describe key operations. Variable names are descriptive. (4)	Adequate documentation but some sections unclear. Comments present but sparse. (2–3)	Minimal or no documentation. Code difficult to follow. (0–1)
Visualization Quality (4 pts)	Professional-quality plots with proper axis labels, units, legends, and titles. Colorblind-safe palette used. Appropriate plot types selected. (4)	Adequate plots with most labels present. Minor formatting issues. (2–3)	Missing labels or illegible plots. Inappropriate visualizations for data type. (0–1)
Reproducibility (4 pts)	Notebook runs top to bottom without manual intervention. Random seeds fixed where applicable. All dependencies stated. (4)	Notebook mostly reproducible with minor issues. Some cells require specific execution order. (2–3)	Notebook cannot be reproduced. Missing data files or undefined variables. (0–1)

Table F.3: Presentation Quality Rubric (20 points total)

Criterion (Points)	Excellent	Satisfactory	Needs Work
Report Structure & Clarity (8 pts)	Logical organization with clear introduction, methodology, results, and conclusions. Professional writing with minimal grammatical errors. Technical terms used correctly. (7–8)	Adequate organization but some sections unclear or poorly transitioned. Minor writing issues. (5–6)	Disorganized structure. Unclear explanations. Significant writing problems. (0–4)
Figure & Table Quality (6 pts)	All figures and tables have descriptive captions, proper numbering, and are referenced in text. Data presented clearly with appropriate precision. (5–6)	Most figures/tables properly formatted. Some missing captions or references. (3–4)	Figures/tables poorly formatted, unlabeled, or not referenced. (0–2)
Conclusions & Recommendations (6 pts)	Clear summary of key findings tied to original objectives. Practical recommendations based on results. Limitations honestly discussed. (5–6)	Adequate conclusions but weak connection to objectives. Recommendations generic or missing. (3–4)	Conclusions absent or unsupported by results. No actionable recommendations. (0–2)

your homework submissions.

F.2.1 Numerical Accuracy Expectations

Table F.4: Numerical Accuracy Standards

Quantity Type	Required Precision	Example
Surface coverage (θ)	3 significant figures	$\theta = 0.847$
Rate constants (k)	2–3 significant figures	$k = 2.5 \times 10^{-3}$
Activation energies (E_a)	3 significant figures	$E_a = 85.2 \text{ kJ/mol}$
Conversion (X)	2–3 significant figures	$X = 0.92$ or 92%
Effectiveness factor (η)	3 significant figures	$\eta = 0.634$
Thiele modulus (ϕ)	2–3 significant figures	$\phi = 3.2$
Space time (τ)	2–3 significant figures	$\tau = 150 \text{ s}$

Remark F.1 (Significant Figures and Error Propagation). When given data has limited precision, your final answer should not claim more significant figures than justified. If input data has 2 significant figures, reporting 5 significant figures in your answer is inappropriate. Similarly, when combining measured values, consider how uncertainties propagate through your calculations.

F.2.2 Unit Consistency Requirements

Dimensional analysis is fundamental to scientific calculations. We expect:

- (i) **Explicit units at every step:** Carry units through all calculations, not just the final answer.
- (ii) **Consistent unit systems:** Convert all quantities to SI units (or stated alternative) before computation.
- (iii) **Dimensional verification:** The dimensions of your final expression must match the expected result.
- (iv) **Conversion factors shown:** When converting units, show the conversion factor explicitly.

Common unit systems in this course:

- Concentrations: mol/L or mol/m³
- Pressures: Pa, bar, or atm (state which)
- Rates: mol/(m³·s) or mol/(kg_{cat}·s)
- Energies: J/mol or kJ/mol
- Temperature: K (always use absolute temperature in rate expressions)

F.2.3 Partial Credit Allocation Principles

Partial credit is awarded based on demonstrated understanding, even when the final answer is incorrect. The general allocation follows this hierarchy:

Table F.5: Partial Credit Allocation Framework

Demonstrated Skill	Credit	Description
Correct approach and setup	40–50%	Problem correctly formulated; appropriate equations identified; variables properly defined
Correct methodology application	30–40%	Equations correctly manipulated; algebra/calculus performed accurately
Correct numerical result	10–20%	Final numerical answer within acceptable tolerance of expected value
Physical interpretation	10–20%	Results discussed in context; reasonableness verified; implications stated

Implications: A student who sets up the problem correctly but makes an arithmetic error will receive substantial partial credit. Conversely, a correct numerical answer obtained through incorrect methodology (e.g., copied from another source without understanding) will receive minimal credit.

F.2.4 Physical Interpretation Requirements

Every quantitative result should be accompanied by physical interpretation. Strong interpretations address:

- (1) **Reasonableness:** Is the magnitude of the result physically sensible? (e.g., coverage $\theta > 1$ is impossible; $E_a < 0$ requires explanation)
- (2) **Limiting behavior:** Does your result approach expected limits? (e.g., $\theta \rightarrow 1$ as $P \rightarrow \infty$ for Langmuir adsorption)
- (3) **Comparison:** How does your result compare to typical values in the literature or to similar systems?
- (4) **Implications:** What does the result tell us about the system? (e.g., “The high Thiele modulus ($\phi = 8.5$) indicates severe diffusion limitation, suggesting smaller pellets would improve performance.”)

F.3 Jupyter Notebook Rubric

The course notebooks (NB1–NB6, plus NB-S) complement lecture material with hands-on computational exercises. This rubric applies to both assigned notebooks and any computational work submitted for the capstone project.

F.3.1 Code Cell Organization

Well-organized notebooks follow a logical flow from setup to analysis to conclusions:

Table F.6: Code Cell Organization Standards

Aspect	Good Practice	Poor Practice
Import statements	Consolidated at notebook top in a single cell	Scattered throughout notebook
Function definitions	Defined before use; reusable functions in <code>utils.py</code>	Functions redefined multiple times; copy-pasted code blocks
Cell length	10–30 lines per cell; one logical operation per cell	Massive cells with hundreds of lines; or single-line cells throughout
Execution order	Cells can run sequentially (1, 2, 3, ...)	Cells require out-of-order execution; hidden state dependencies
Output management	Clear outputs before submission; regenerate by running all	Stale outputs that don't match code

F.3.2 Markdown Explanation Requirements

Markdown cells provide narrative structure and demonstrate understanding:

- Section headers:** Use hierarchical headers (`#`, `##`, `###`) to organize the notebook.
- Theory summaries:** Before computational implementation, briefly state the relevant equations and assumptions.
- Variable definitions:** Define all symbols used in code (e.g., “`KA` represents the adsorption equilibrium constant for species A”).
- Result interpretation:** After generating plots or numerical results, explain what the output means in physical terms.
- LaTeX equations:** Use LaTeX syntax for equations in markdown cells:

$$\theta = \frac{KP}{1 + KP}$$

F.3.3 Plot Labeling Requirements

All plots must include the following elements:

Table F.7: Mandatory Plot Elements

Element	Requirement	Example
x -axis label	Variable name with units	<code>plt.xlabel('Pressure (bar)')</code>
y -axis label	Variable name with units	<code>plt.ylabel(r'\$\theta\$')</code>
Title	Descriptive (optional for publication)	<code>plt.title('Langmuir Isotherm')</code>
Legend	When multiple curves; placed outside data area	<code>plt.legend(loc='best')</code>
Grid lines	Major grid for readability	<code>plt.grid(True, alpha=0.3)</code>
Figure size	Appropriate for content	<code>plt.figure(figsize=(8, 5))</code>
Font sizes	Readable at final display size	<code>fontsize=12</code> in labels

F.3.4 Colorblind-Safe Color Usage

Approximately 8% of men and 0.5% of women have some form of color vision deficiency. Use the Wong (2011) colorblind-safe palette:

Table F.8: Colorblind-Safe Palette (Wong, 2011)

Color Name	RGB Values	Hex Code
Blue	(0, 114, 178)	#0072B2
Orange	(230, 159, 0)	#E69F00
Green	(0, 158, 115)	#009E73
Yellow	(240, 228, 66)	#F0E442
Sky Blue	(86, 180, 233)	#56B4E9
Vermillion	(213, 94, 0)	#D55E00
Purple	(204, 121, 167)	#CC79A7

Best practices:

- Never rely on color alone—use different line styles (solid, dashed, dotted) in addition to colors
 - Use markers (circles, squares, triangles) to distinguish data series
 - Ensure sufficient contrast between adjacent colors
 - Test plots in grayscale to verify distinguishability
- Example Python code for defining this palette:

```

1 # Wong (2011) colorblind-safe palette
2 CB_COLORS = {

```

```
3   'blue': '#0072B2',  
4   'orange': '#E69F00',  
5   'green': '#009E73',  
6   'yellow': '#F0E442',  
7   'skyblue': '#56B4E9',  
8   'vermillion': '#D55E00',  
9   'purple': '#CC79A7'  
10 }
```

Listing F.1: Colorblind-safe palette definition

F.4 Common Deduction Categories

The following table summarizes common errors and their typical point deductions. These serve as guidelines; actual deductions may vary based on problem context and severity.

Remark F.2 (Cascading Errors). When an early error propagates through subsequent calculations, you will not be penalized multiple times for the same mistake. If your methodology is correct but builds on an earlier incorrect value, you will receive credit for the correct approach. This “carry-forward” policy rewards understanding of the method even when numerical accuracy is compromised by an initial error.

F.4.1 Severity Guidelines

Deduction severity depends on several factors:

- (1) **Problem weight:** A 2-point deduction on a 10-point problem is more significant than on a 20-point problem.
- (2) **Error impact:** Does the error fundamentally change the answer, or is it a minor numerical discrepancy?
- (3) **Error visibility:** Should a reasonable student have caught this error through checking (e.g., negative coverage)?
- (4) **Repetition:** Multiple instances of the same error type may incur cumulative deductions, up to a maximum.

Maximum deduction principle: Total deductions for a single error category are typically capped at 50% of the problem value. For example, multiple unit errors on a 10-point problem would not result in more than 5 points deducted for that category alone.

Table F.9: Common Deductions Reference

Error Category	Deduction	Notes
<i>Unit and Dimensional Errors</i>		
Missing units on final answer	−2 to −3 pts	Per instance; more severe for dimensional quantities
Incorrect unit conversion	−2 to −4 pts	Depends on impact on final result
Dimensionally inconsistent equation	−3 to −5 pts	Indicates fundamental misunderstanding
<i>Numerical and Precision Errors</i>		
Incorrect significant figures	−1 pt	Excessive or insufficient precision in final answer
Arithmetic/calculation error	−1 to −3 pts	Minor: −1; propagating through solution: −3
Using wrong constant value	−1 to −2 pts	e.g., wrong R value for units used
<i>Conceptual and Interpretation Errors</i>		
Missing physical interpretation	−2 to −5 pts	Required for most problems; weighted by problem value
Result not checked for reasonableness	−2 pts	e.g., $\theta > 1$, negative rate constant
Wrong mechanism/-model selected	−5 to −10 pts	Depends on whether selection was specified
<i>Presentation and Formatting Errors</i>		
Unreadable/unlabeled plot	−2 pts/figure	Missing axis labels, legend, or title
Missing equation derivation steps	−1 to −3 pts	When derivation explicitly requested
Poor/illegible handwriting	−1 to −2 pts	If answer cannot be read, cannot be graded
<i>Error Propagation</i>		
Not propagating uncertainty	−2 pts	When uncertainty analysis explicitly required
Claiming false precision	−1 pt	Reporting more digits than data supports

Table F.10: Performance Level Descriptions

Grade	Percentage	Performance Description
A	85–100%	Demonstrates mastery of course concepts. Solutions are complete, accurate, and include insightful physical interpretation. Computational work is professional quality.
B+	75–84%	Shows strong understanding with minor gaps. Most solutions correct with adequate interpretation. Minor errors do not reflect fundamental misunderstanding.
B	65–74%	Demonstrates solid foundational knowledge. Solutions show correct approach but may have errors in execution or incomplete interpretation.
C+	55–64%	Basic understanding demonstrated but significant gaps exist. May struggle with multi-step problems or integration of concepts.
C	45–54%	Minimum acceptable performance. Fundamental concepts partially understood but application is inconsistent.
D+	35–44%	Below expectations. Major conceptual gaps evident. Requires substantial improvement to meet course objectives.
F	< 35%	Unsatisfactory. Does not demonstrate understanding of core course material.

F.5 Grade Scale and Performance Levels

F.6 Appeals and Regrade Requests

If you believe an error was made in grading your work, you may submit a regrade request following these guidelines:

- (1) **Timing:** Requests must be submitted within one week of receiving the graded work.
- (2) **Written submission:** Submit a clear, written explanation identifying the specific question(s) and describing why you believe the grading was incorrect.
- (3) **Refer to rubric:** Reference specific rubric criteria when making your case.
- (4) **Complete review:** The entire assignment may be reviewed during the regrade process, which could result in a higher or lower score.
- (5) **Final decision:** The instructor's decision on regrade requests is final.

Appendix G

Catalytic Organic Synthesis: A Brief Overview

This appendix provides a concise overview of catalytic organic synthesis, highlighting connections to the surface chemistry and reactor engineering concepts developed throughout this course. Students interested in deeper treatment should consult the references in appendix [H](#).

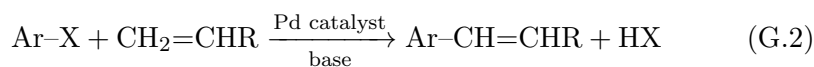
G.1 Pd-Catalyzed Cross-Coupling Reactions

Palladium-catalyzed cross-coupling reactions revolutionized the construction of carbon-carbon bonds in organic synthesis (2010 Nobel Prize to Heck, Negishi, and Suzuki). The three most widely used reactions are:

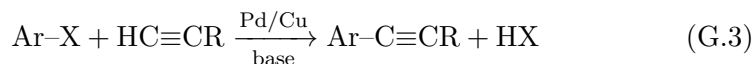
Suzuki-Miyaura coupling — aryl halide + boronic acid:



Heck reaction — aryl halide + alkene:



Sonogashira coupling — aryl halide + terminal alkyne:



All three reactions share a common catalytic cycle consisting of three elementary steps:

1. **Oxidative addition** — Ar-X adds to Pd(0), forming Ar-Pd(II)-X (typically rate-determining)
2. **Transmetalation** — the coupling partner transfers its organic group to Pd(II)

3. **Reductive elimination** — the two organic groups couple, regenerating Pd(0)

Catalyst performance is quantified by two turnover metrics:

$$\text{TON} = \frac{n_{\text{product}}}{n_{\text{catalyst}}} \quad (\text{catalyst durability}) \quad (\text{G.4})$$

$$\text{TOF} = \frac{\text{TON}}{t} \quad (\text{catalyst activity, h}^{-1}) \quad (\text{G.5})$$

Industrial processes typically require $\text{TON} > 10,000$ and $\text{TOF} > 1,000 \text{ h}^{-1}$.

G.2 Asymmetric Catalysis

Many biologically active molecules are chiral—only one enantiomer has the desired activity. Classic examples include thalidomide ((*R*)-sedative vs. (*S*)-teratogenic), naproxen ((*S*)-anti-inflammatory), and carvone ((*R*)-spearmint vs. (*S*)-caraway). Asymmetric catalysis provides access to enantiopure compounds using substoichiometric amounts of a chiral catalyst.

The **enantiomeric excess** (ee) quantifies the degree of enantioselectivity:

$$\text{ee} = \frac{|[R] - [S]|}{[R] + [S]} \times 100\% \quad (\text{G.6})$$

Enantioselectivity arises from the energy difference $\Delta\Delta G^\ddagger$ between diastereomeric transition states. The product ratio follows Boltzmann statistics:

$$\frac{[R]}{[S]} = \exp\left(\frac{\Delta\Delta G^\ddagger}{RT}\right) \quad (\text{G.7})$$

Because the exponential factor $\exp(\Delta\Delta G^\ddagger/RT)$ grows as T decreases, even modest energy differences ($\sim 6 \text{ kJ/mol}$) produce high ee at low temperature. This explains why many asymmetric reactions are conducted at -40 to -78°C .

G.3 Catalyst Selection Guide

Criterion	Homogeneous	Heterogeneous
Active site accessibility	All metal centers	Surface only
Selectivity control	Often superior	Can be challenging
Catalyst separation	Difficult	Easy (filtration)
Recyclability	Often poor	Often good
Metal contamination	Can be problematic	Generally lower
Scalability	Can be challenging	Often easier

Remark G.1 (Heterogeneous Cross-Coupling: From Pd/C to Designed Catalysts).

Pd/C (palladium on activated carbon) is the simplest heterogeneous cross-coupling catalyst and is used industrially for Suzuki and Heck reactions. Its activity depends on Pd dispersion, which can be measured by CO chemisorption, TEM, and XRD (Chapter 3). A key deactivation mechanism is Pd leaching into solution—the “homeopathic” Pd debate centres on whether catalysis actually occurs on the surface or via dissolved Pd species. For large Pd/C particles, the effectiveness factor η (Chapter 4) becomes relevant because the organic substrates must diffuse into the porous carbon support before reaching interior Pd sites.

Green chemistry metrics provide quantitative comparison of process sustainability:

- **Atom economy** = $M_{\text{product}} / \Sigma M_{\text{reactants}} \times 100\%$ — fraction of reactant atoms in the desired product
- **E-factor** = kg waste / kg product — overall waste intensity (pharmaceuticals: 25–100+; bulk chemicals: 1–5)

G.4 Connections to Course Concepts

Remark G.2 (Catalytic Organic Synthesis Draws on Every Chapter).

The topics in this appendix illustrate how fundamental surface chemistry and reactor engineering concepts manifest in real synthetic practice. The table below maps each capstone topic to the chapter where the underlying theory is developed.

Capstone Topic	Chapter	Key Concept
Competitive binding / poisoning	Chapter 2	Competitive Langmuir
Catalytic cycle steps	Chapter 2	LH / ER mechanisms
$\Delta\Delta G^\ddagger$ & selectivity	Chapter 2	Eyring, Boltzmann
TON, TOF, batch analysis	Chapter 3	Reactor design eqns
Catalyst characterization	Chapter 3	BET, XRD, XPS, TEM
Supported catalyst diffusion	Chapter 4	Thiele modulus, η
Parallel / consecutive rxns	Chapter 5	Yield optimization

G.5 Notation Introduced in This Appendix

Table G.1: Notation used in the catalytic organic synthesis appendix.

Symbol	Description	Units
TON	Turnover number	dimensionless
TOF	Turnover frequency	h^{-1} or s^{-1}
ee	Enantiomeric excess	%
er	Enantiomeric ratio	dimensionless
s	Selectivity factor ($k_{\text{fast}}/k_{\text{slow}}$)	dimensionless
$\Delta\Delta G^\ddagger$	Free energy difference between diastereomeric TS	kJ/mol
k_d	Deactivation rate constant	h^{-1} or s^{-1}

Appendix H

Further Reading and Resources

This appendix provides an annotated guide to textbooks, research literature, online databases, and software tools that complement these lecture notes.

H.1 Textbook References

H.1.1 Reactor Engineering Fundamentals

Fogler, H.S. *Elements of Chemical Reaction Engineering*, 6th ed., Pearson, 2020.

The definitive undergraduate text for reaction engineering with extensive worked examples. Now in its sixth edition with online interactive simulations.

Levenspiel, O. *Chemical Reaction Engineering*, 3rd ed., Wiley, 1998.

A classic text renowned for its intuitive graphical approach to reactor design. Levenspiel plots ($1/r$ versus conversion) provide powerful visual insights into reactor selection.

Hill, C.G. and Root, T.W. *Introduction to Chemical Engineering Kinetics and Reactor Design*, 2nd ed., Wiley, 2014.

Bridges chemical kinetics with reactor engineering, providing rigorous kinetic derivations alongside reactor design.

H.1.2 Heterogeneous Catalysis and Surface Science

Chorkendorff, I. and Niemantsverdriet, J.W. *Concepts of Modern Catalysis and Kinetics*, 3rd ed., Wiley-VCH, 2017.

An exceptional bridge between surface science and industrial catalysis.

Covers molecular-level adsorption through microkinetic modeling with extensive experimental illustrations.

Thomas, J.M. and Thomas, W.J. *Principles and Practice of Heterogeneous Catalysis*, 2nd ed., Wiley-VCH, 2015.

Comprehensive coverage from fundamental principles to industrial applications, with excellent discussions of zeolite catalysis and catalyst preparation.

Somorjai, G.A. and Li, Y. *Introduction to Surface Chemistry and Catalysis*, 2nd ed., Wiley, 2010.

Deep insight into molecular-level surface phenomena from a pioneer in surface science. Covers surface structure, bonding, thermodynamics, and kinetics.

Bartholomew, C.H. and Farrauto, R.J. *Fundamentals of Industrial Catalytic Processes*, 2nd ed., Wiley, 2006.

Comprehensive treatment of industrial catalysis spanning petroleum refining, environmental catalysis, and specialty chemicals with detailed case studies.

H.1.3 Transport Phenomena in Catalysis

Froment, G.F., Bischoff, K.B., and De Wilde, J. *Chemical Reactor Analysis and Design*, 3rd ed., Wiley, 2011.

Advanced reactor modeling with rigorous coverage of diffusion-reaction interactions, effectiveness factors, and non-isothermal effects.

Bird, R.B., Stewart, W.E., and Lightfoot, E.N. *Transport Phenomena*, 2nd ed., Wiley, 2002.

The foundational reference for mass, heat, and momentum transfer. Essential background for diffusion limitations in porous media.

H.2 Chapter-Specific References

H.2.1 Chapter 2: Adsorption, Kinetics, and Temperature Dependence

Adsorption – Foundational Papers:

- Langmuir, I. “The adsorption of gases on plane surfaces of glass, mica and platinum,” *J. Am. Chem. Soc.* **40**, 1361–1403 (1918).
- Brunauer, S., Emmett, P.H., and Teller, E. “Adsorption of gases in multimolecular layers,” *J. Am. Chem. Soc.* **60**, 309–319 (1938).

Adsorption – Modern Reviews:

- Masel, R.I. *Principles of Adsorption and Reaction on Solid Surfaces*, Wiley, 1996.

- Campbell, C.T. and Sellers, J.R.V. “Enthalpies and entropies of adsorption on well-defined oxide surfaces,” *Chem. Rev.* **113**, 4106–4135 (2013).

Surface Kinetics – Foundational Papers:

- Hinshelwood, C.N. *The Kinetics of Chemical Change*, Oxford University Press, 1940.
- Mars, P. and van Krevelen, D.W. “Oxidations carried out by means of vanadium oxide catalysts,” *Chem. Eng. Sci.* **3**, 41–59 (1954).
- Hougen, O.A. and Watson, K.M. “Solid catalysts and reaction rates,” *Ind. Eng. Chem.* **35**, 529–541 (1943).

Surface Kinetics – Modern Reviews:

- Dumesic, J.A. et al. *The Microkinetics of Heterogeneous Catalysis*, ACS, 1993.
- Medford, A.J. et al. “From the Sabatier principle to a predictive theory of transition-metal heterogeneous catalysis,” *J. Catal.* **328**, 36–42 (2015).

Temperature Dependence – Foundational Papers:

- Arrhenius, S. “Über die Reaktionsgeschwindigkeit bei der Inversion von Rohrzucker durch Säuren,” *Z. Phys. Chem.* **4**, 226–248 (1889).
- Eyring, H. “The activated complex in chemical reactions,” *J. Chem. Phys.* **3**, 107–115 (1935).

Temperature Dependence – Modern References:

- Laidler, K.J. *Chemical Kinetics*, 3rd ed., Harper & Row, 1987.
- Bligaard, T. et al. “The Brønsted–Evans–Polanyi relation and the volcano curve in heterogeneous catalysis,” *J. Catal.* **224**, 206–217 (2004).

H.2.2 Chapter 4: Transport Limitations

Foundational Papers:

- Thiele, E.W. “Relation between catalytic activity and size of particle,” *Ind. Eng. Chem.* **31**, 916–920 (1939).
- Weisz, P.B. and Prater, C.D. “Interpretation of measurements in experimental catalysis,” *Adv. Catal.* **6**, 143–196 (1954).
- Aris, R. “On shape factors for irregular particles—I,” *Chem. Eng. Sci.* **6**, 262–268 (1957).

Modern Reviews:

- Krishna, R. “Describing the diffusion of guest molecules inside porous structures,” *J. Phys. Chem. C* **113**, 19756–19781 (2009).
- Kärger, J. and Ruthven, D.M. *Diffusion in Nanoporous Materials*, Wiley-VCH, 2012.

H.2.3 Chapter 6: Zeolites and Carbon Nanotubes

Zeolites – Foundational References:

- Breck, D.W. *Zeolite Molecular Sieves*, Wiley, 1974.
- Barrer, R.M. *Zeolites and Clay Minerals as Sorbents and Molecular Sieves*, Academic Press, 1978.

Zeolites – Modern Reviews:

- Corma, A. “Inorganic solid acids and their use in acid-catalyzed hydrocarbon reactions,” *Chem. Rev.* **95**, 559–614 (1995).
- Davis, M.E. “Ordered porous materials for emerging applications,” *Nature* **417**, 813–821 (2002).
- Čejka, J., Corma, A., and Zones, S. (Eds.) *Zeolites and Catalysis*, Wiley-VCH, 2010.

Carbon Nanotubes – Foundational Papers:

- Iijima, S. “Helical microtubules of graphitic carbon,” *Nature* **354**, 56–58 (1991).
- Iijima, S. and Ichihashi, T. “Single-shell carbon nanotubes of 1-nm diameter,” *Nature* **363**, 603–605 (1993).

Carbon Nanotubes – Modern Reviews:

- Serp, P. and Castillejos, E. “Catalysis in carbon nanotubes,” *Chem-CatChem* **2**, 41–47 (2010).
- Su, D.S. et al. “Nanocarbons for the development of advanced catalysts,” *Chem. Rev.* **113**, 5782–5816 (2013).

H.3 Online Resources and Databases

H.3.1 Thermodynamic and Kinetic Data

NIST Chemistry WebBook (<https://webbook.nist.gov/chemistry/>)
Comprehensive database of thermochemical data, IR spectra, and mass spectra for catalyst characterization.

NIST Kinetics Database (<https://kinetics.nist.gov/kinetics/>)
Evaluated rate constant data for gas-phase reactions; useful for validating surface reaction rates.

CRC Handbook of Chemistry and Physics (Online Edition)

Standard reference for physical and chemical property data. University library access typically required.

H.3.2 Catalyst and Materials Databases

Catalysis-Hub (<https://www.catalysis-hub.org/>)
Open database of DFT-calculated reaction energetics, useful for comparing adsorption energies across surfaces.

IZA Structure Database (iza-structure.org/databases/)

Official database of zeolite framework structures with crystallographic data and visualization tools.

Materials Project (<https://materialsproject.org/>)

Open database of computed materials properties including electronic structure and surface energies.

ICSD (Inorganic Crystal Structure Database)

Comprehensive inorganic crystal structure collection for catalyst phase identification by XRD. University subscription required.

H.3.3 Educational Resources

LearnChemE (<https://learncheme.com/>)

Screencasts covering reactor design, kinetics, and transport phenomena with worked examples.

MIT OpenCourseWare – Chemical Engineering

Free lecture notes, problem sets, and exams. Relevant courses: 10.37 (Chemical Kinetics) and 10.65 (Reactor Engineering).

Catalysis by Nanoparticles (Coursera)

Online course covering fundamentals of nanocatalysis. Good supplement to Chapter 6.

H.4 Software and Computational Tools

The companion Jupyter notebooks (NB1–NB6, plus NB-S) use the following Python packages:

NumPy (<https://numpy.org/>)

Fundamental package for numerical computing: array operations, linear algebra, and numerical methods.

SciPy (<https://scipy.org/>)

Scientific computing library for ODE integration, curve fitting, root finding, and statistical analysis.

Matplotlib (<https://matplotlib.org/>)

Comprehensive plotting library. The colorblind-safe palette follows Wong (2011) and is implemented in `utils.py`.

Pandas (<https://pandas.pydata.org/>)

Data analysis library for CSV import/export and experimental data manipulation.

Cantera (<https://cantera.org/>)

Open-source chemical kinetics software for thermodynamic calculations and reaction mechanism simulation.

H.5 Research Journals in Catalysis

H.5.1 Core Catalysis Journals

Journal of Catalysis (Elsevier)

The premier journal for fundamental catalysis research since 1962, covering all aspects of heterogeneous and homogeneous catalysis.

ACS Catalysis (American Chemical Society)

Top-tier venue for catalysis research with strong coverage of computational catalysis and reaction mechanisms.

Nature Catalysis (Springer Nature)

High-impact journal publishing the most significant advances across all catalysis disciplines.

Catalysis Science & Technology (Royal Society of Chemistry)

Good balance of fundamental and applied catalysis research, including characterization methods.

H.5.2 Applied and Environmental Catalysis

Applied Catalysis A: General (Elsevier)

Catalyst preparation, characterization, and evaluation with industrial relevance.

Applied Catalysis B: Environmental (Elsevier)

Environmental applications including emission control, photocatalysis, and water treatment.

Catalysis Today (Elsevier)

Conference proceedings and regular articles; good source for reviews on emerging topics.

Topics in Catalysis (Springer)

Topical collections and review articles on specific catalysis themes.

H.5.3 Reaction Engineering Journals

Chemical Engineering Science (Elsevier)

Leading journal for reaction engineering, reactor modeling, and process intensification.

H. Further Reading and Resources

***AIChE Journal* (Wiley)**

Flagship journal of the American Institute of Chemical Engineers with a strong reaction engineering section.

***Industrial & Engineering Chemistry Research* (ACS)**

Applied research bridging academic findings and industrial practice in reactor design and process development.

Remark H.1 (Staying Current with Literature). Set up Google Scholar alerts or RSS feeds for key authors and topics relevant to your research interests. Most journals offer email alerts for new issues.